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(54) **SYSTEMS AND METHODS FOR RECOMMENDING A FOOD PRODUCT BASED ON CHARACTERISTICS OF AN ANIMAL**

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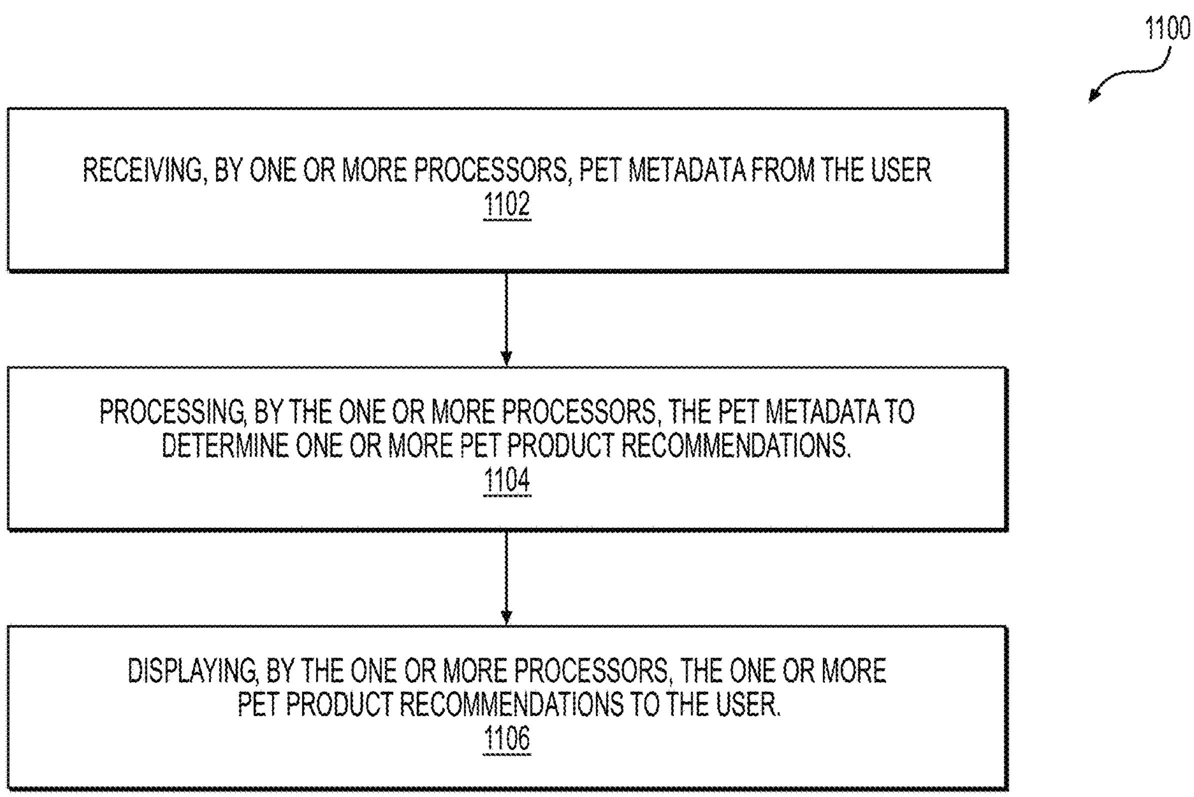
(57) **ABSTRACT**

A computer-implemented method for providing one or more personalized pet product recommendations based on a microbiome result and pet metadata to a user is disclosed. The method includes receiving, by one or more processors, the pet metadata and a pet fecal sample from the user, analyzing, by the one or more processors, the pet fecal sample to determine a microbiome result, processing, by the one or more processors, the pet metadata based on the microbiome result to determine one or more pet product recommendations, and displaying, by the one or more processors, the one or more pet product recommendations to the user.



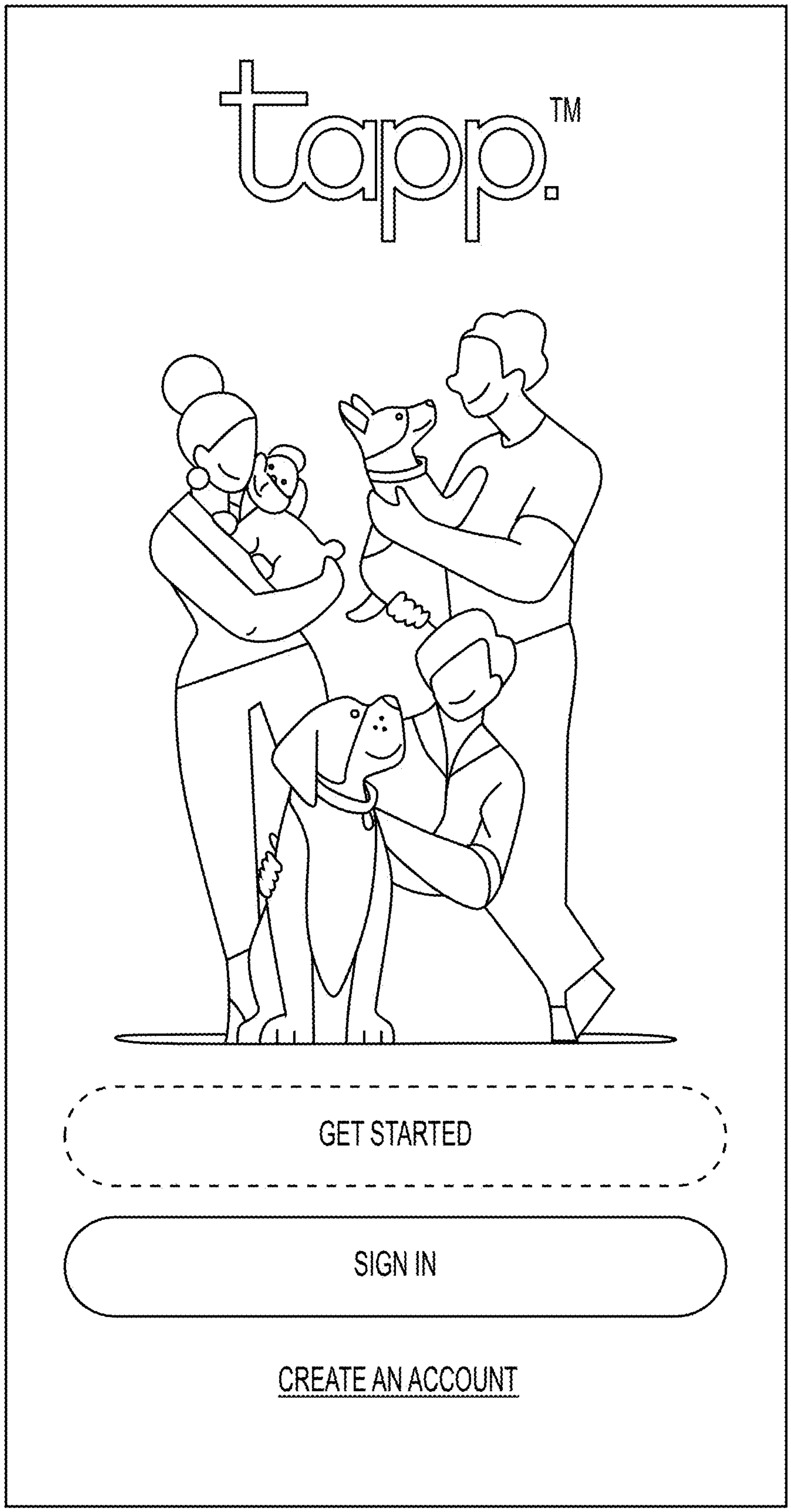
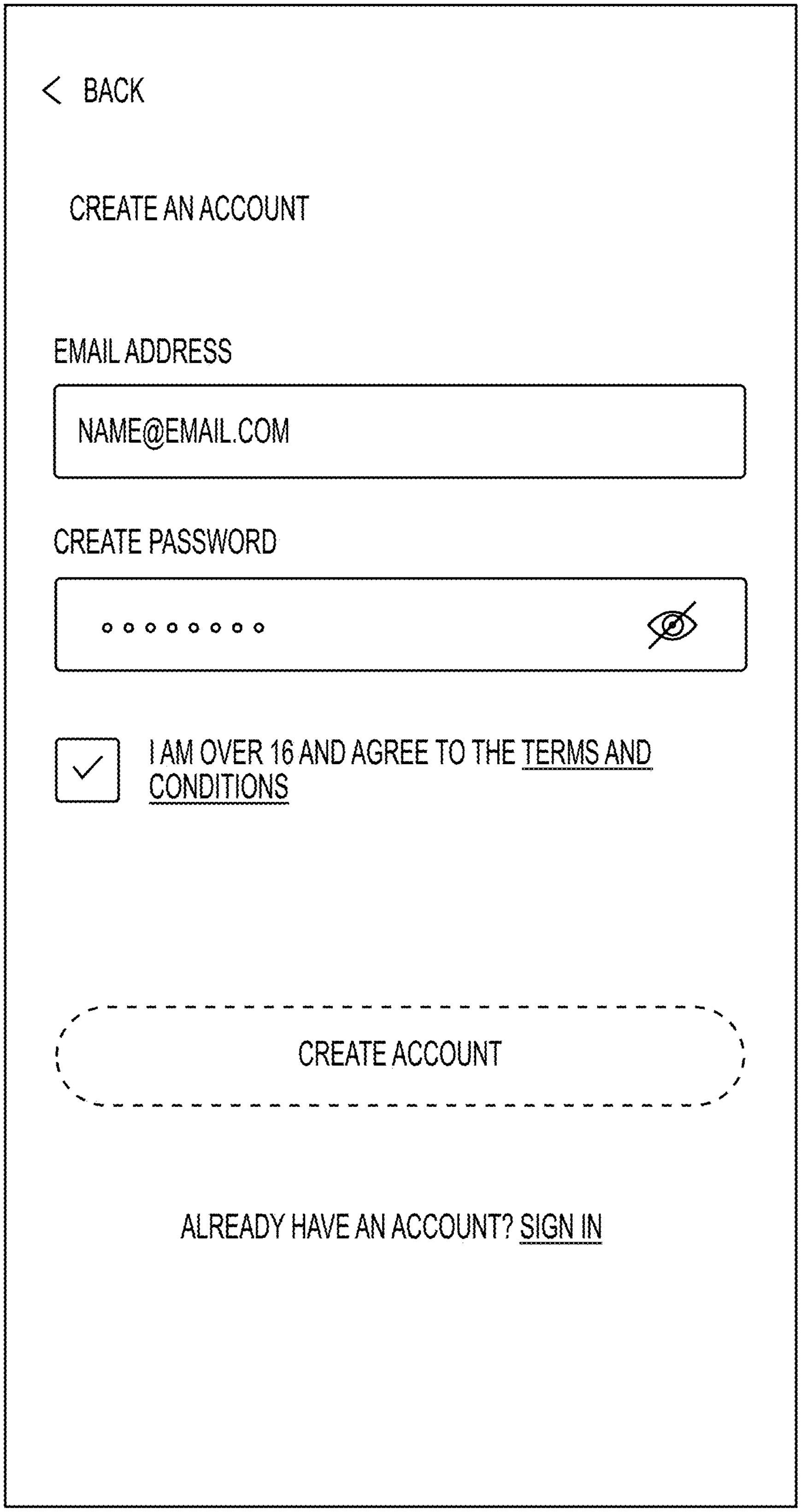


FIG. 1A



< BACK
CREATE AN ACCOUNT
EMAIL ADDRESS
NAME@EMAIL.COM
CREATE PASSWORD
I AM OVER 16 AND AGREE TO THE TERMS AND CONDITIONS
CREATE ACCOUNT
ALREADY HAVE AN ACCOUNT? SIGN IN

FIG. 1B

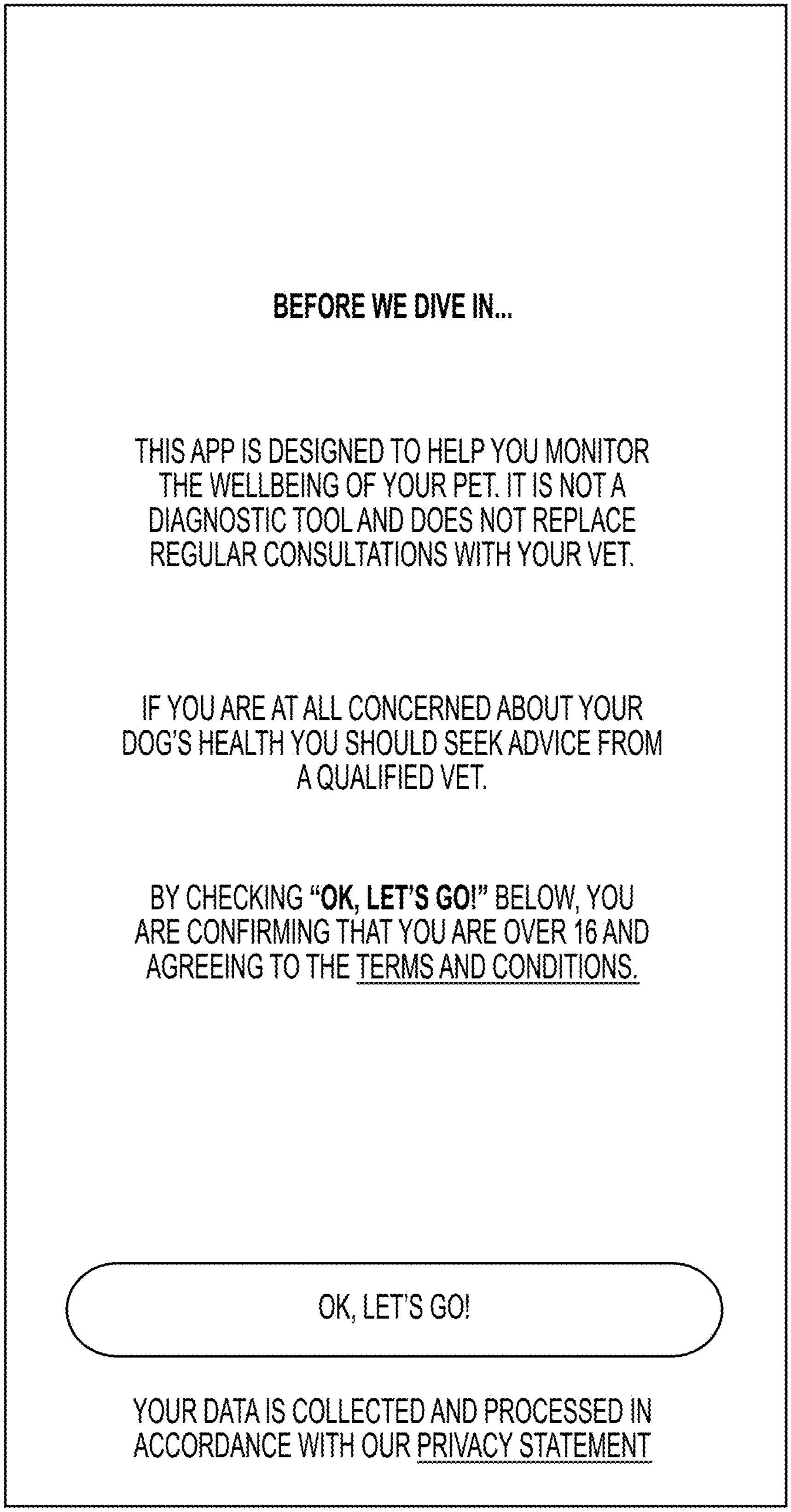

BEFORE WE DIVE IN...
THIS APP IS DESIGNED TO HELP YOU MONITOR THE WELLBEING OF YOUR PET. IT IS NOT A DIAGNOSTIC TOOL AND DOES NOT REPLACE REGULAR CONSULTATIONS WITH YOUR VET.
IF YOU ARE AT ALL CONCERNED ABOUT YOUR DOG'S HEALTH YOU SHOULD SEEK ADVICE FROM A QUALIFIED VET.
BY CHECKING "OK, LET'S GO!" BELOW, YOU ARE CONFIRMING THAT YOU ARE OVER 16 AND AGREEING TO THE TERMS AND CONDITIONS.
OK, LET'S GO!
YOUR DATA IS COLLECTED AND PROCESSED IN ACCORDANCE WITH OUR PRIVACY STATEMENT

FIG. 1C

< BACK

HEALTHY GUT,
HEALTHY BODY

A HEALTHY GUT IS IMPORTANT TO YOUR DOG'S OVERALL HEALTH. BUT THEY CAN'T TELL YOU WHAT THEY NEED. WHAT IS GOING ON IN THERE?

NEXT

FIG. 1D

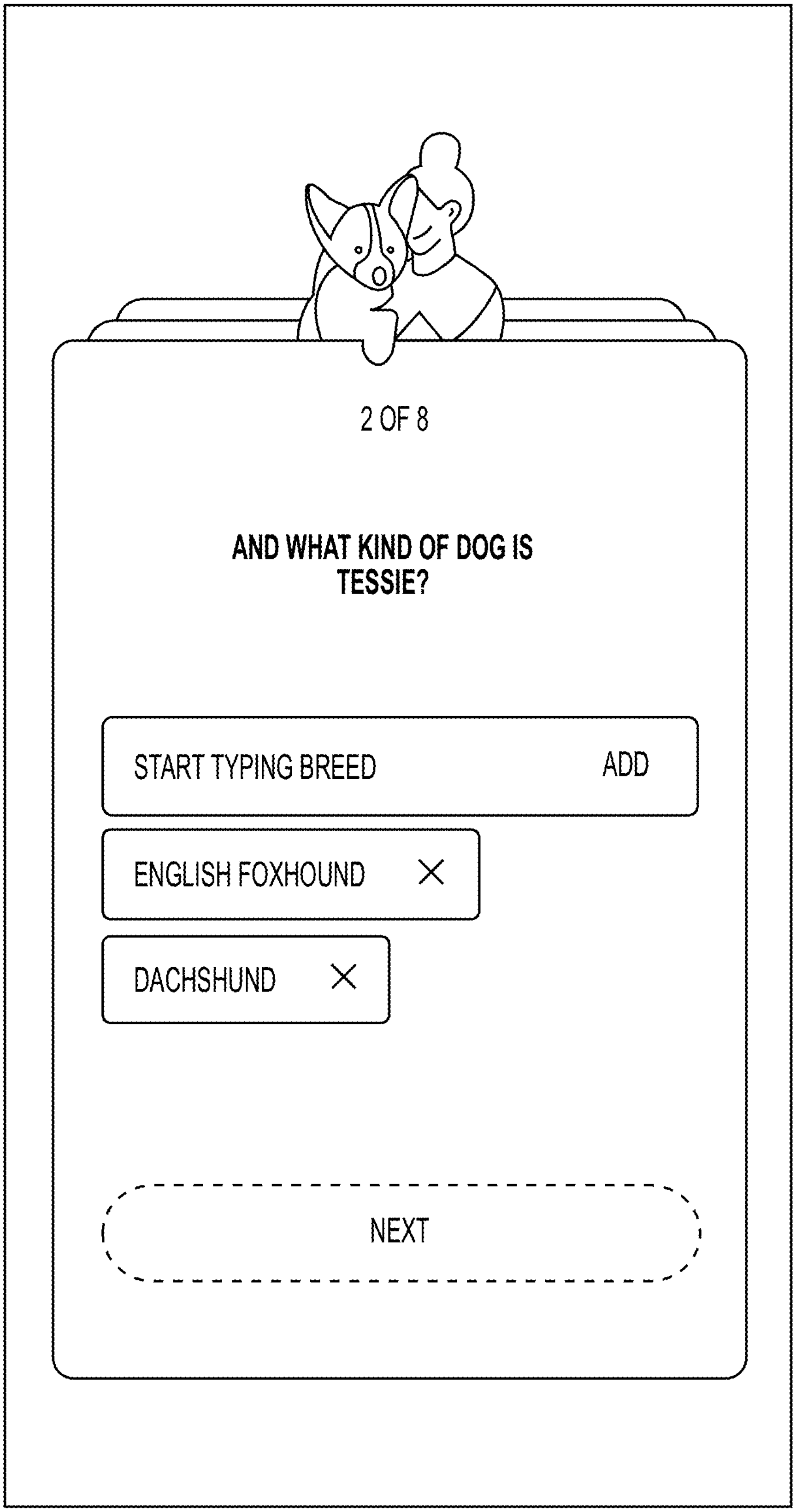

2 OF 8
AND WHAT KIND OF DOG IS TESSIE?
START TYPING BREED
ADD
ENGLISH FOXHOUND
DACHSHUND
NEXT

FIG. 1E

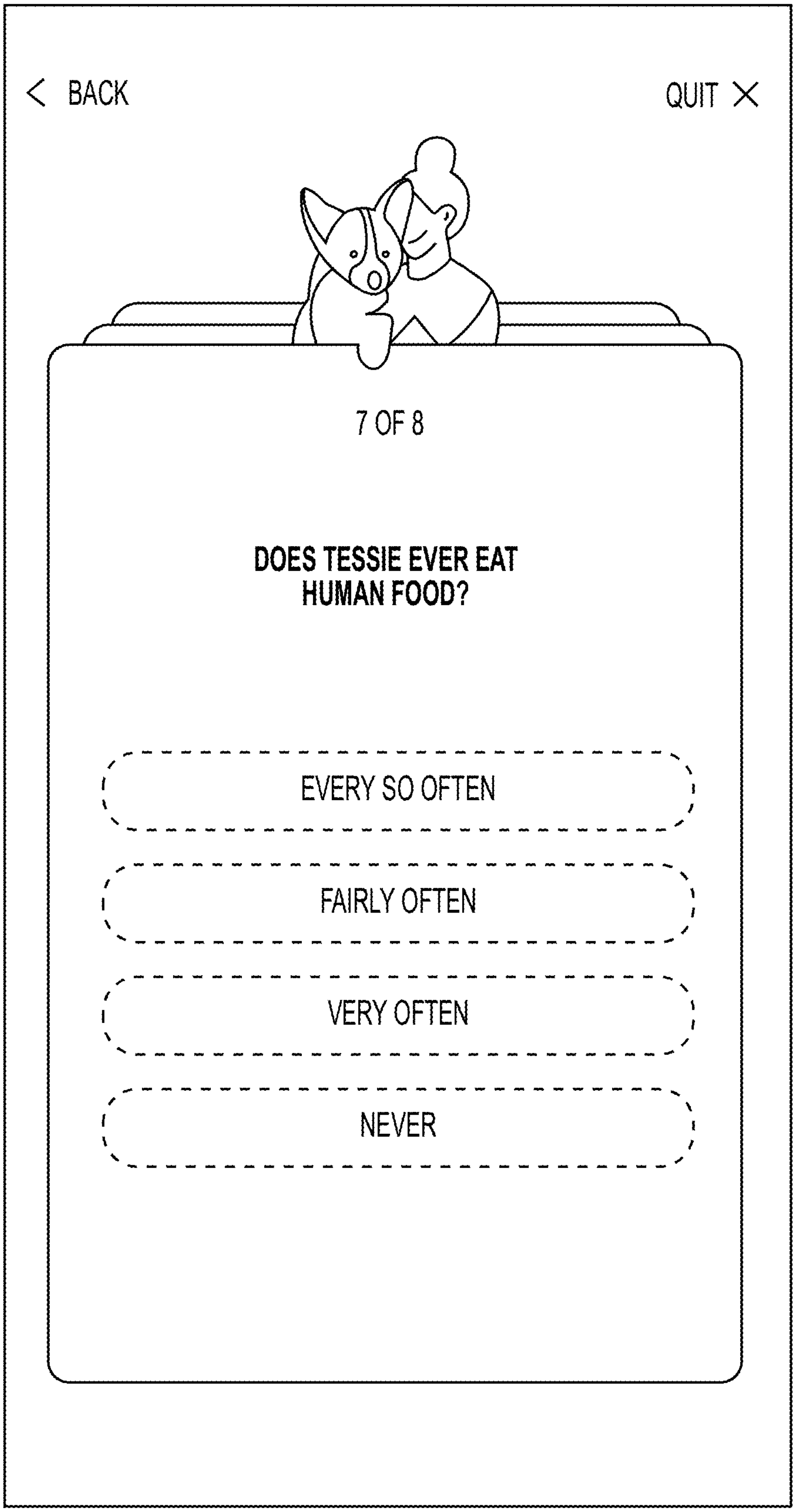

BACK
QUIT
7 OF 8
DOES TESSIE EVER EAT
HUMAN FOOD?
EVERY SO OFTEN
FAIRLY OFTEN
VERY OFTEN
NEVER

FIG. 1F

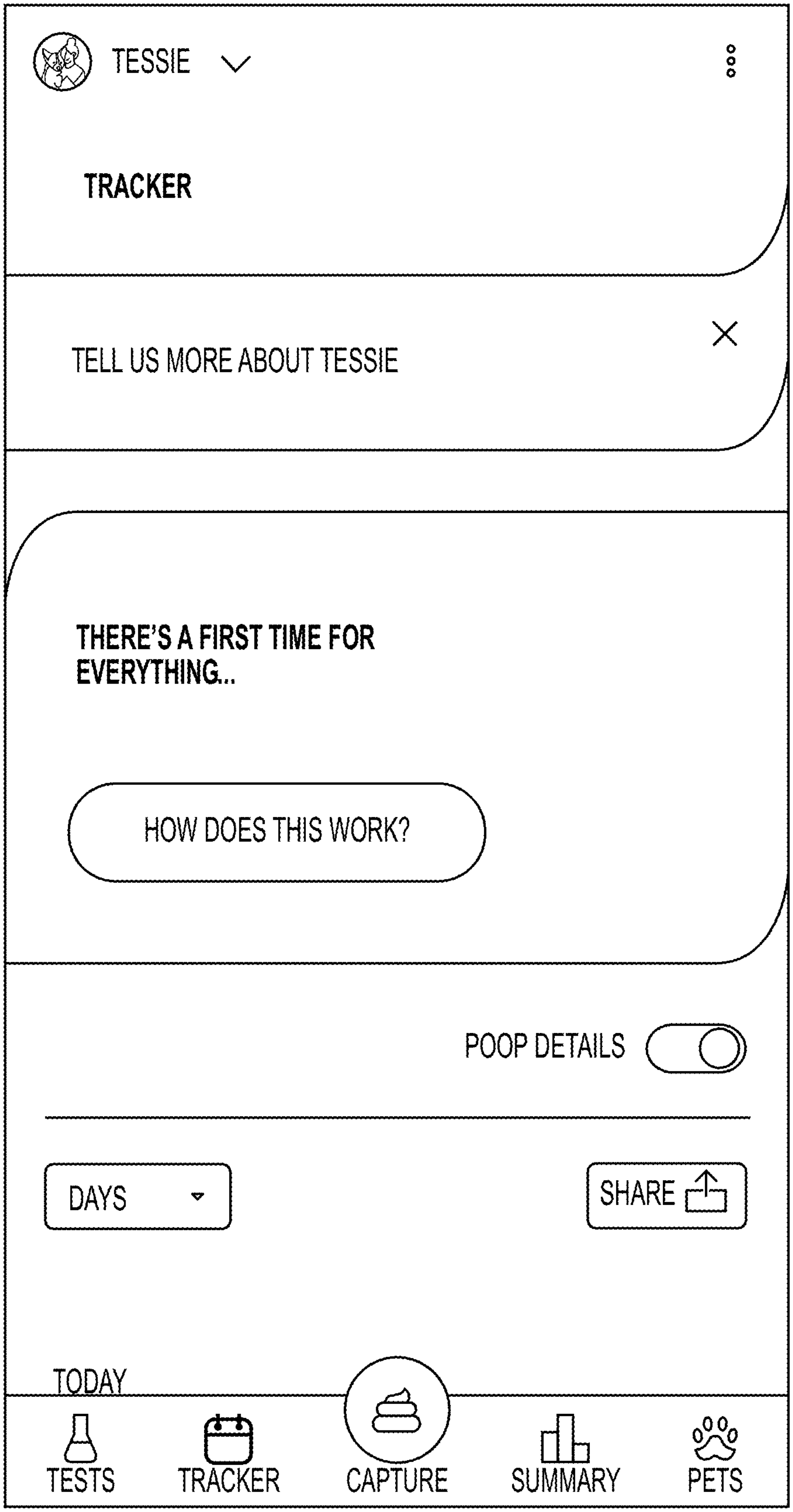

TESSIE
TRACKER
TELL US MORE ABOUT TESSIE
THERE'S A FIRST TIME FOR EVERYTHING...
HOW DOES THIS WORK?
POOP DETAILS
DAYS
SHARE
TODAY
TESTS
TRACKER
CAPTURE
SUMMARY
PETS

FIG. 1G

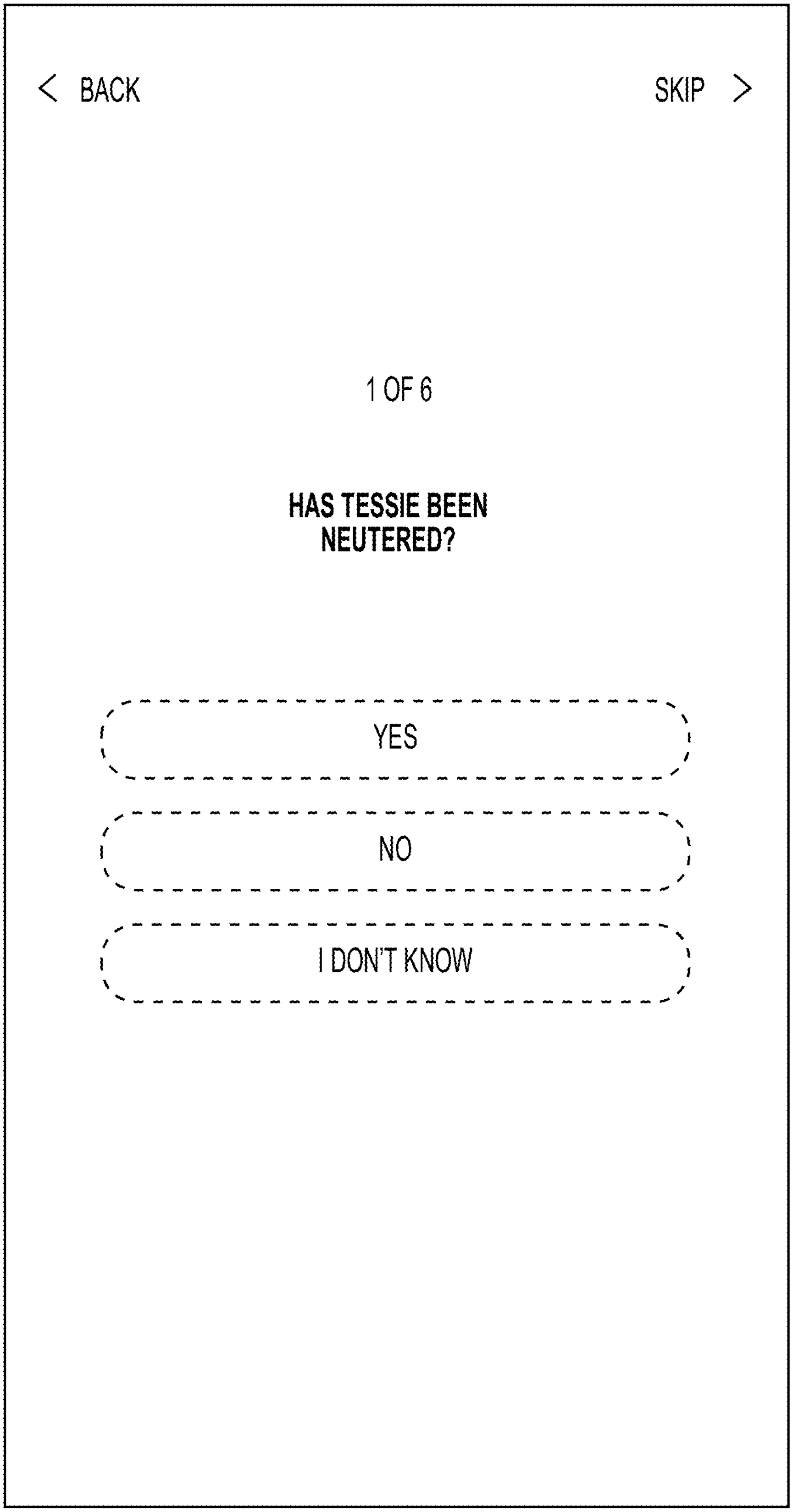


FIG. 1H

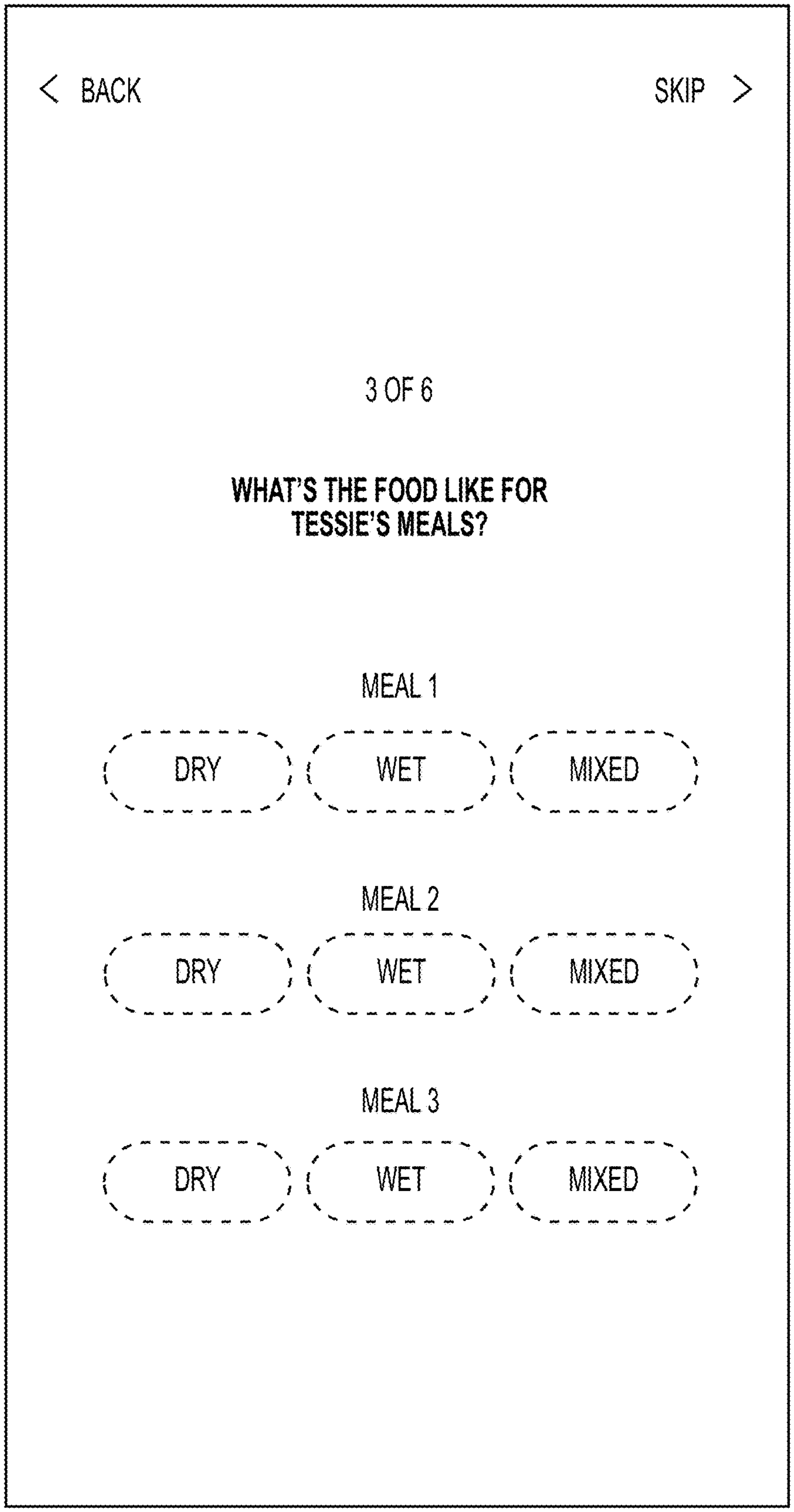
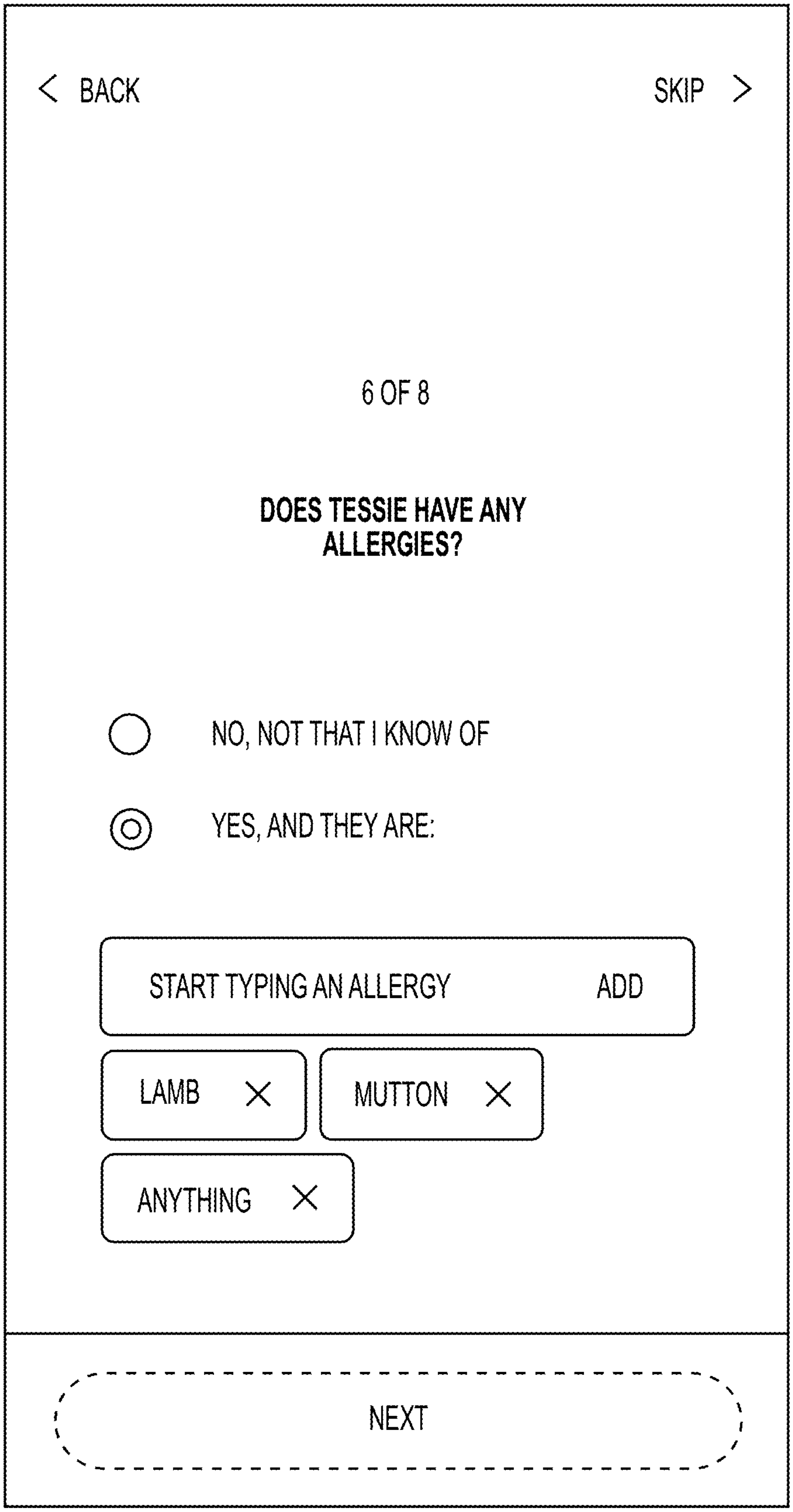

BACK
SKIP
3 OF 6
WHAT'S THE FOOD LIKE FOR TESSIE'S MEALS?
MEAL 1
DRY
WET
MIXED
MEAL 2
DRY
WET
MIXED
MEAL 3
DRY
WET
MIXED

FIG. 1I



< BACK
SKIP >
6 OF 8
DOES TESSIE HAVE ANY ALLERGIES?
NO, NOT THAT I KNOW OF
YES, AND THEY ARE:
START TYPING AN ALLERGY
ADD
LAMB
MUTTON
ANYTHING
NEXT

FIG. 1J

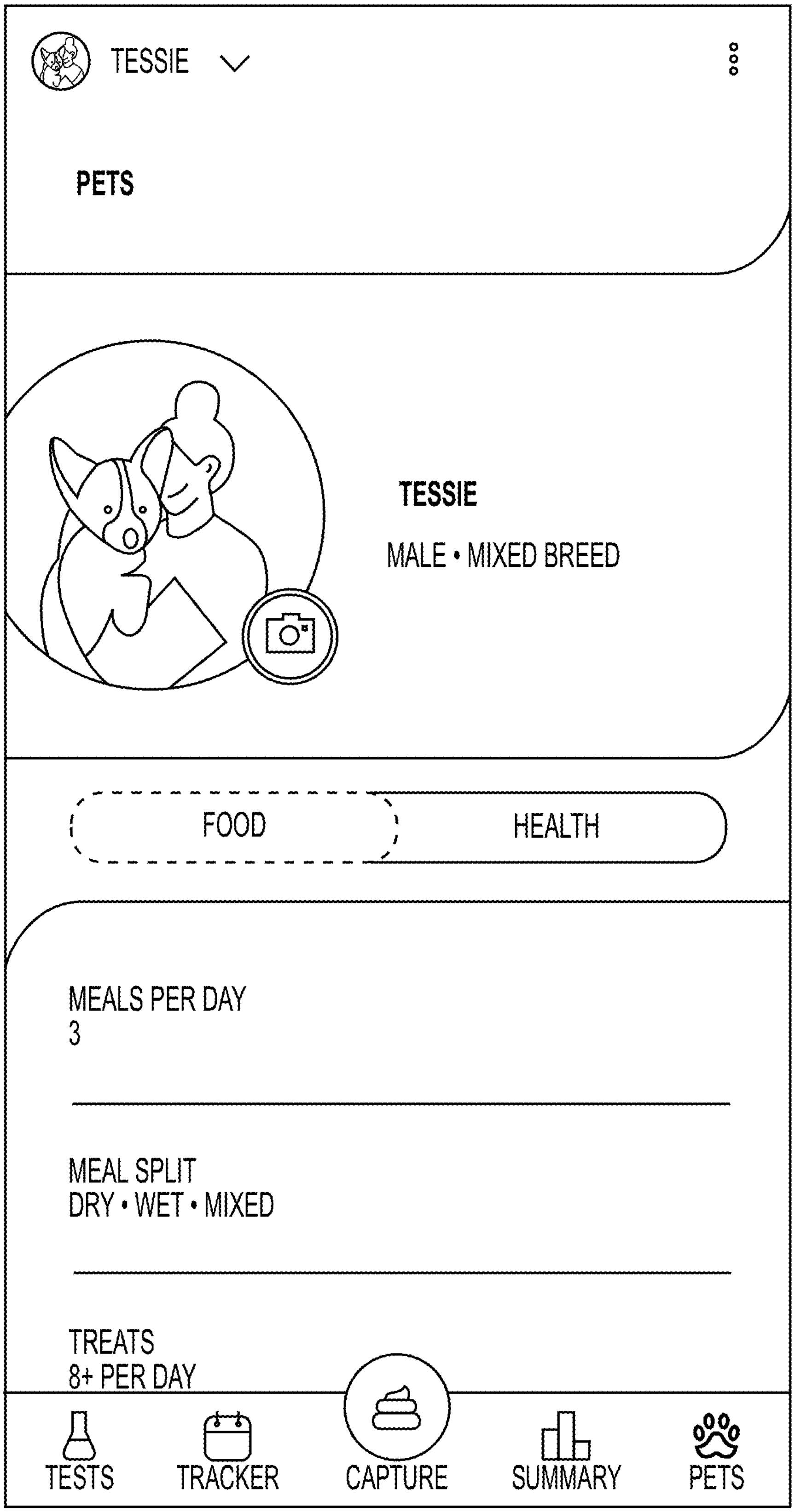

TESSIE
PETS
TESSIE
MALE • MIXED BREED
FOOD
HEALTH
MEALS PER DAY
3
MEAL SPLIT
DRY • WET • MIXED
TREATS
8+ PER DAY
TESTS
TRACKER
CAPTURE
SUMMARY
PETS

FIG. 1K


FOOD
HEALTH
MEALS PER DAY
3
MEAL SPLIT
DRY • WET • MIXED
TREATS
8+ PER DAY
HUMAN TREATS
VERY OFTEN
CURRENT DRY FOOD
BRAND FOR VITALITY - ADULT
CURRENT WET FOOD
BRAND - ADULT
DIET SUPPLEMENT
NONE
TESTS
TRACKER
CAPTURE
SUMMARY
PETS

FIG. 1L

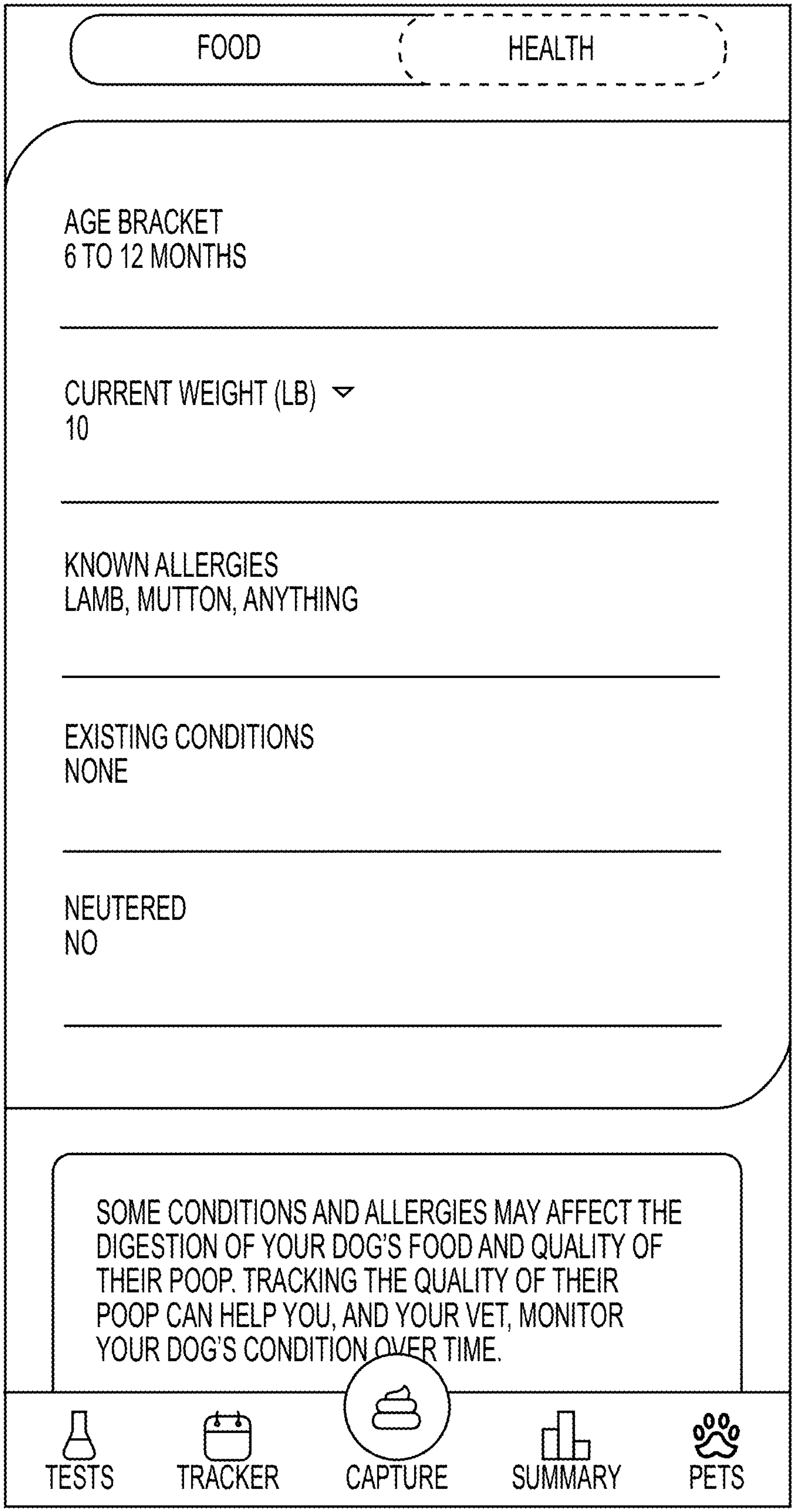

FOOD
HEALTH
AGE BRACKET
6 TO 12 MONTHS
CURRENT WEIGHT (LB)
10
KNOWN ALLERGIES
LAMB, MUTTON, ANYTHING
EXISTING CONDITIONS
NONE
NEUTERED
NO
SOME CONDITIONS AND ALLERGIES MAY AFFECT THE DIGESTION OF YOUR DOG'S FOOD AND QUALITY OF THEIR POOP. TRACKING THE QUALITY OF THEIR POOP CAN HELP YOU, AND YOUR VET, MONITOR YOUR DOG'S CONDITION OVER TIME.
TESTS
TRACKER
CAPTURE
SUMMARY
PETS

FIG. 1M

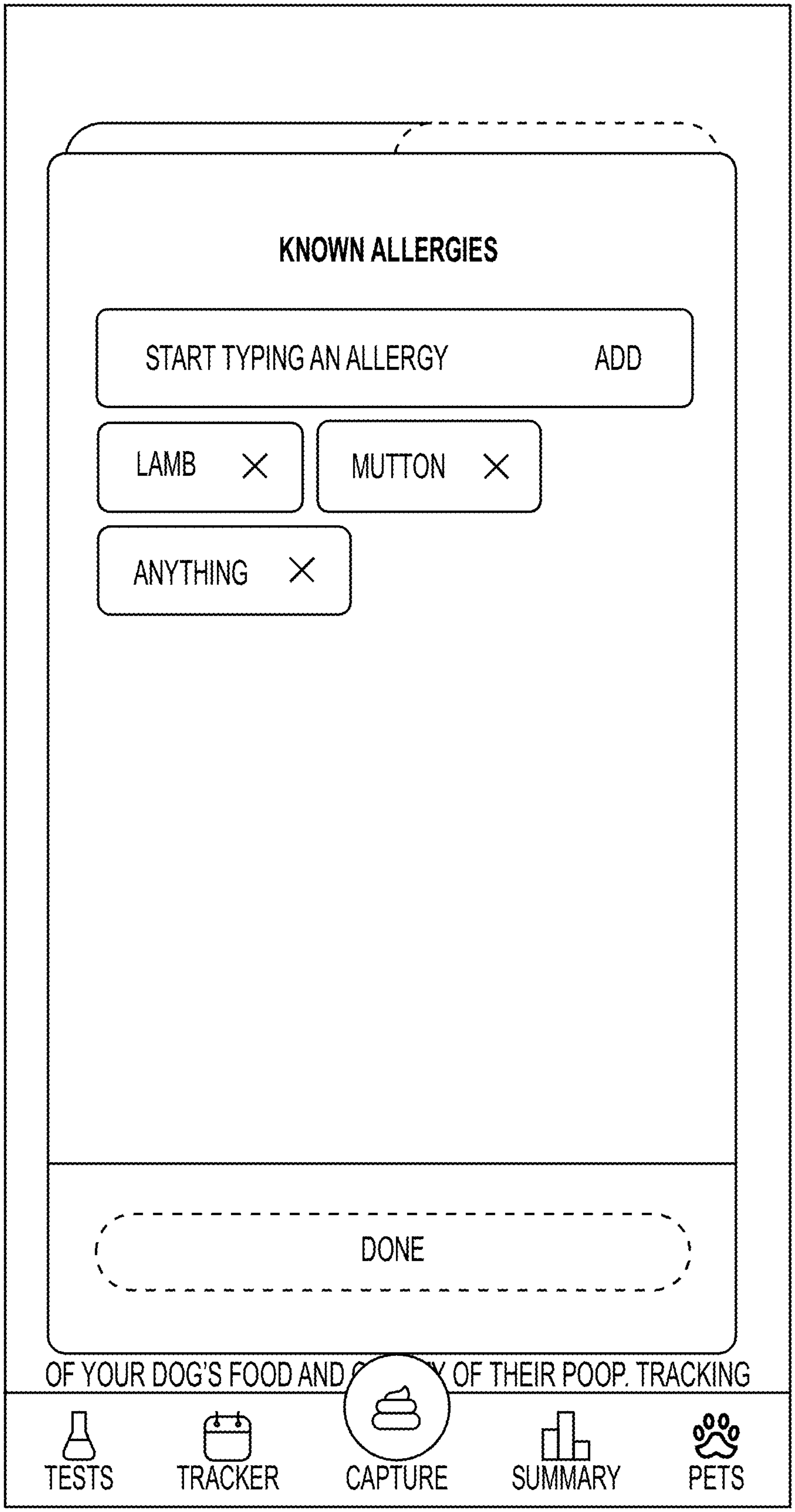

KNOWN ALLERGIES
START TYPING AN ALLERGY
ADD
LAMB
MUTTON
ANYTHING
DONE
OF YOUR DOG'S FOOD AND
OF THEIR POOP. TRACKING
TESTS
TRACKER
CAPTURE
SUMMARY
PETS

FIG. 1N

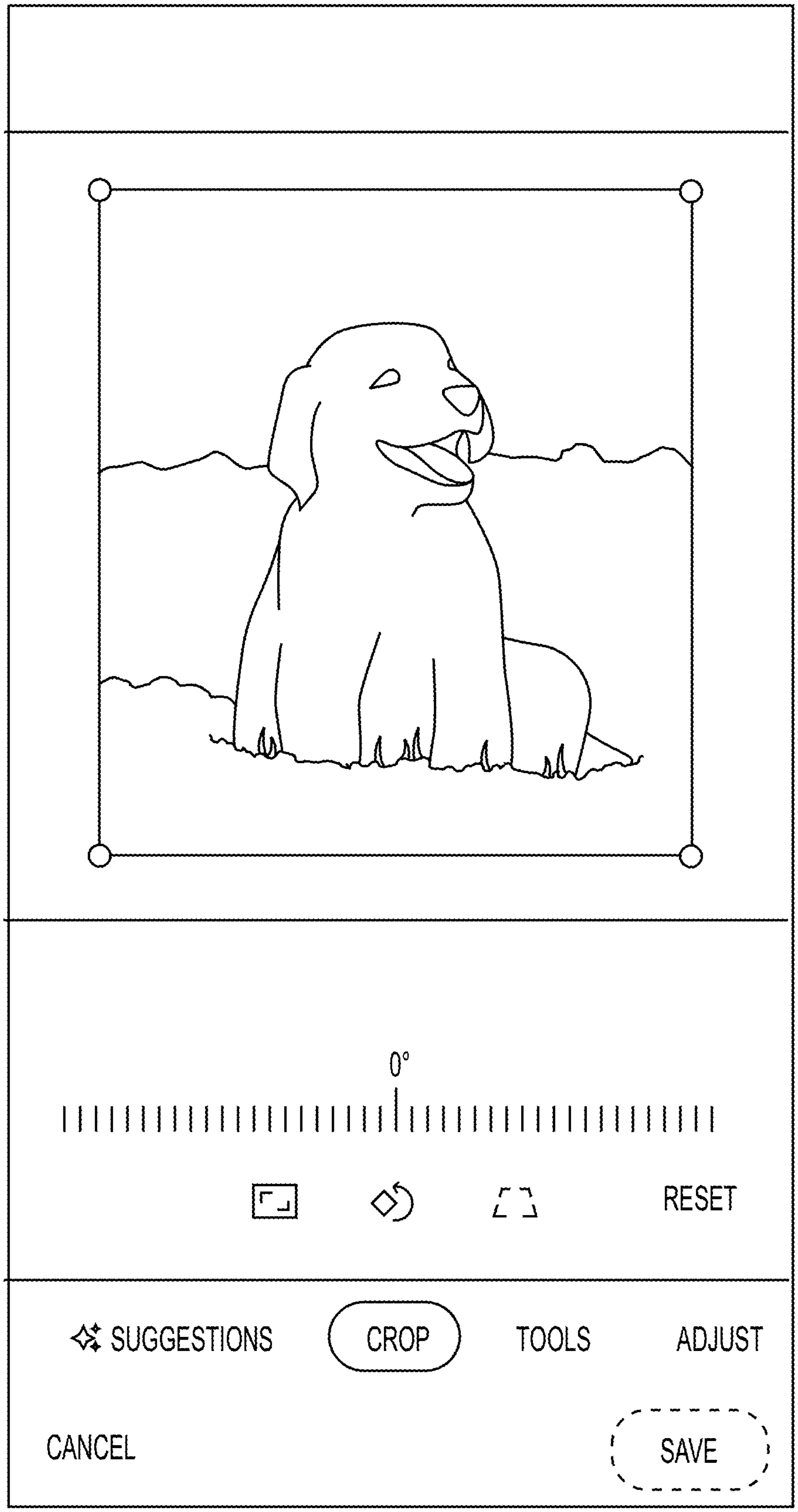


FIG. 1O

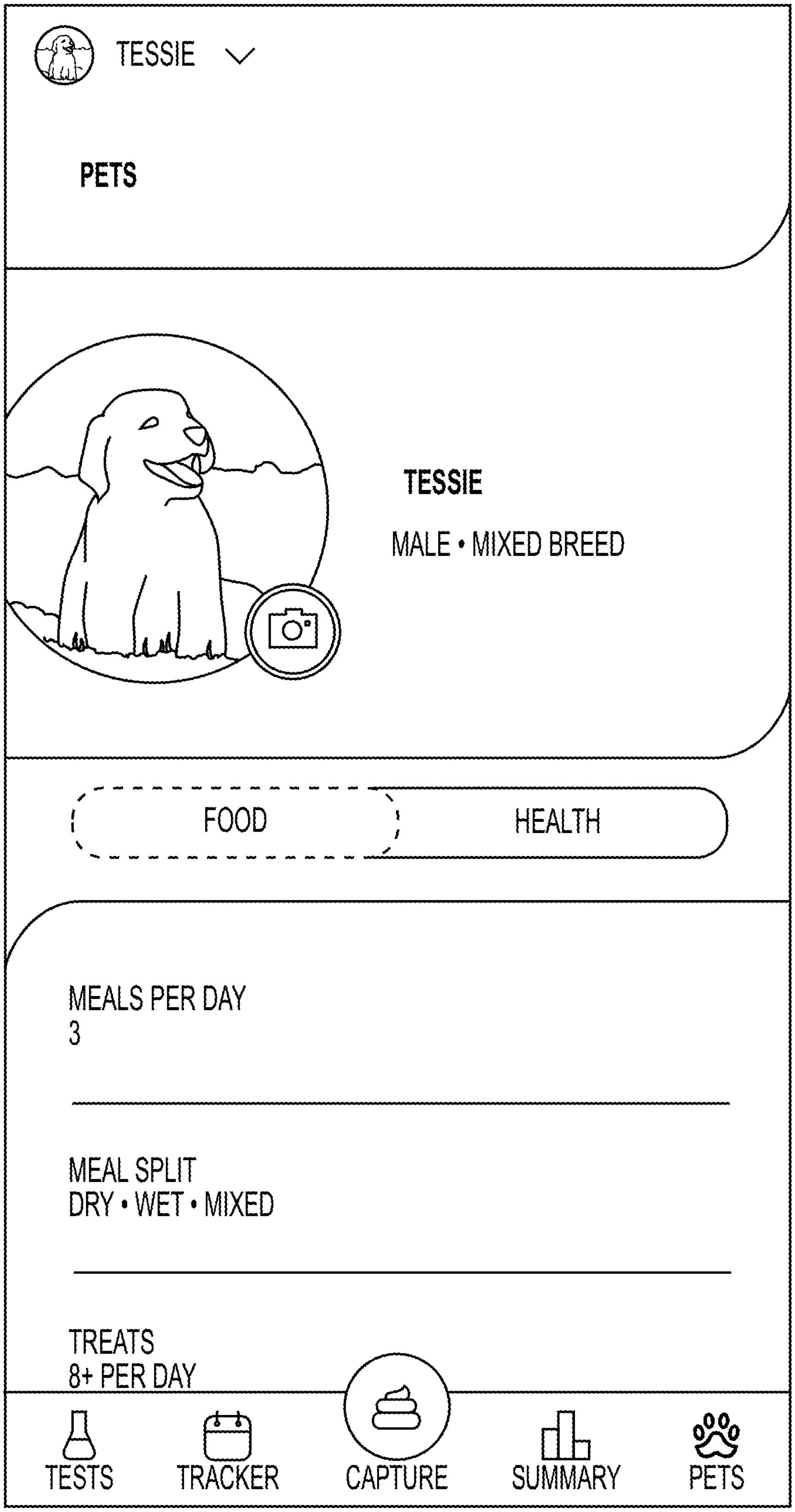

TESSIE
PETS
TESSIE
MALE • MIXED BREED
FOOD
HEALTH
MEALS PER DAY
3
MEAL SPLIT
DRY • WET • MIXED
TREATS
8+ PER DAY
TESTS
TRACKER
CAPTURE
SUMMARY
PETS

FIG. 1P

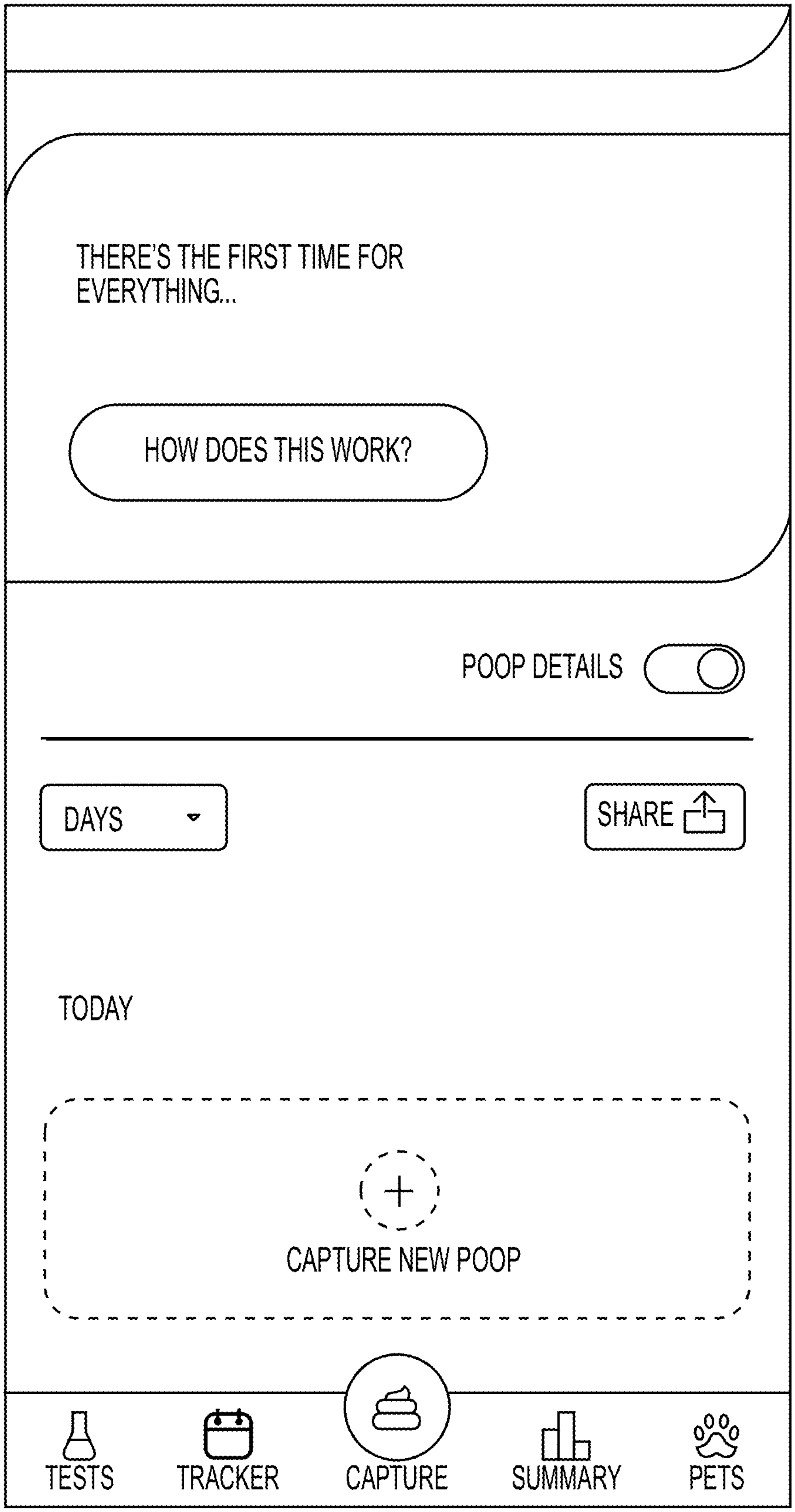

THERE'S THE FIRST TIME FOR EVERYTHING...
HOW DOES THIS WORK?
POOP DETAILS
DAYS
SHARE
TODAY
CAPTURE NEW POOP
TESTS
TRACKER
CAPTURE
SUMMARY
PETS

FIG. 1Q

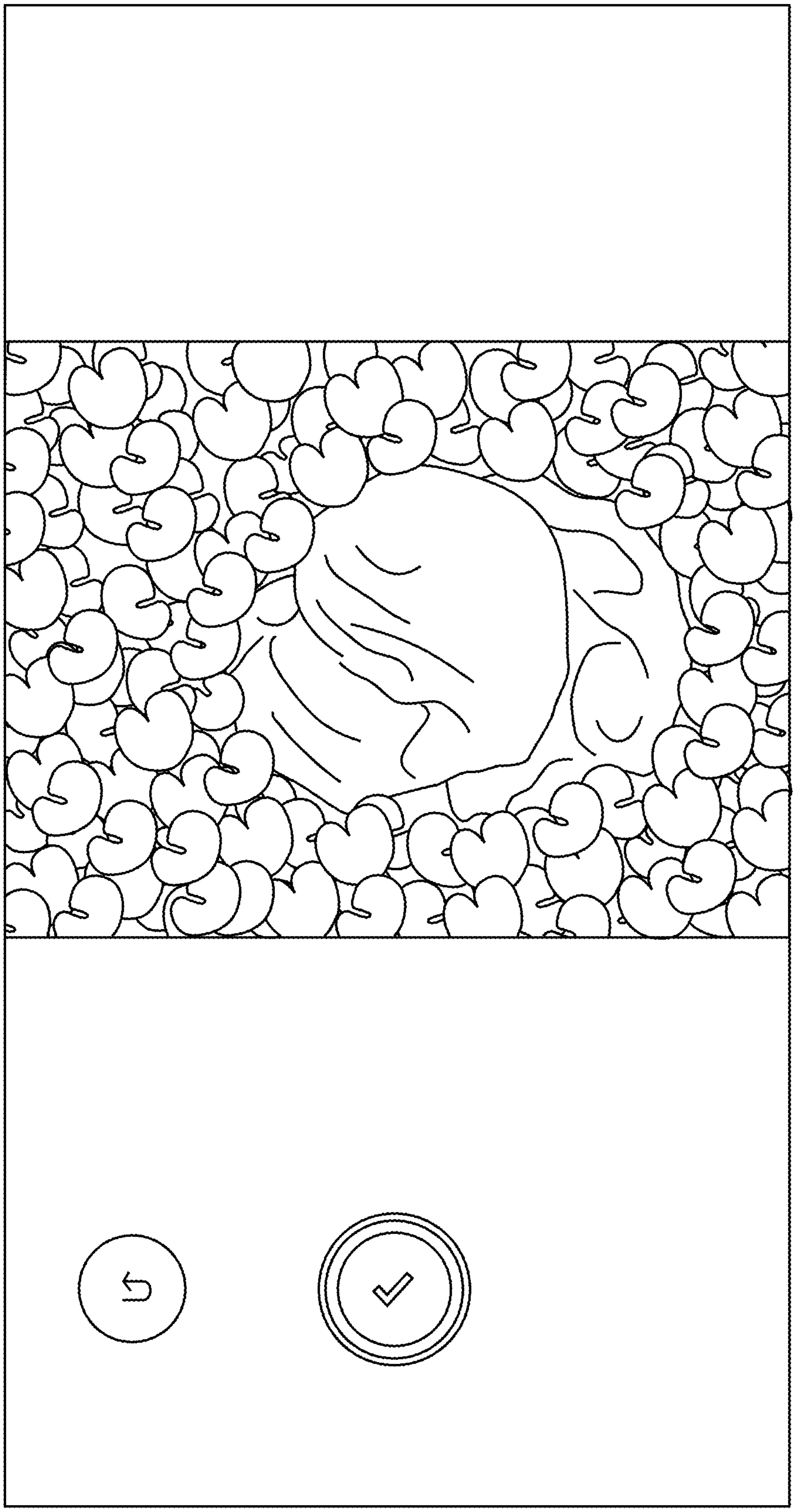


FIG. 1R

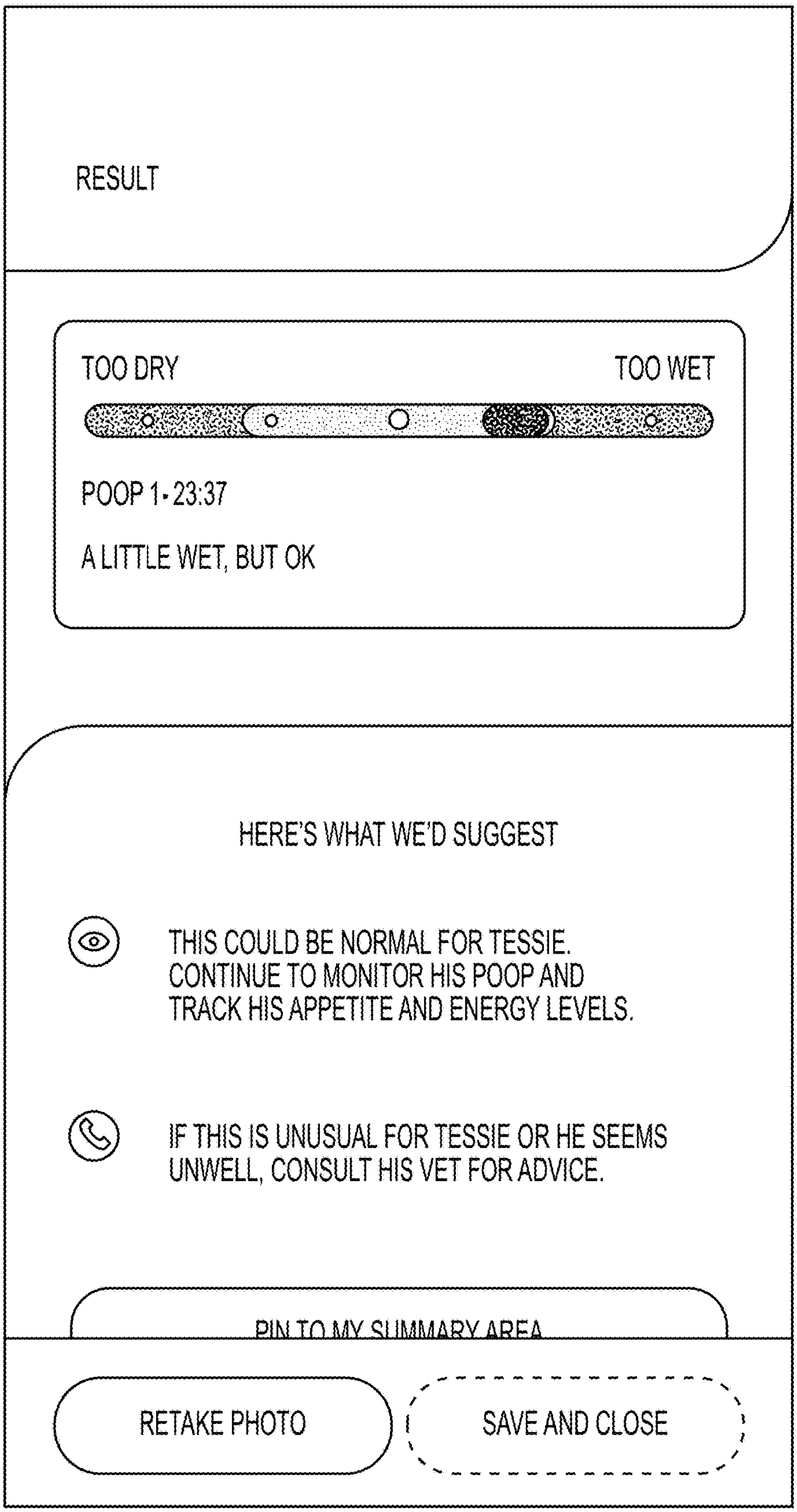

RESULT
TOO DRY
TOO WET
POOP 1 • 23:37
A LITTLE WET, BUT OK
HERE'S WHAT WE'D SUGGEST
THIS COULD BE NORMAL FOR TESSIE.
CONTINUE TO MONITOR HIS POOP AND
TRACK HIS APPETITE AND ENERGY LEVELS.
IF THIS IS UNUSUAL FOR TESSIE OR HE SEEMS
UNWELL, CONSULT HIS VET FOR ADVICE.
RETAKE PHOTO
SAVE AND CLOSE

FIG. 1S

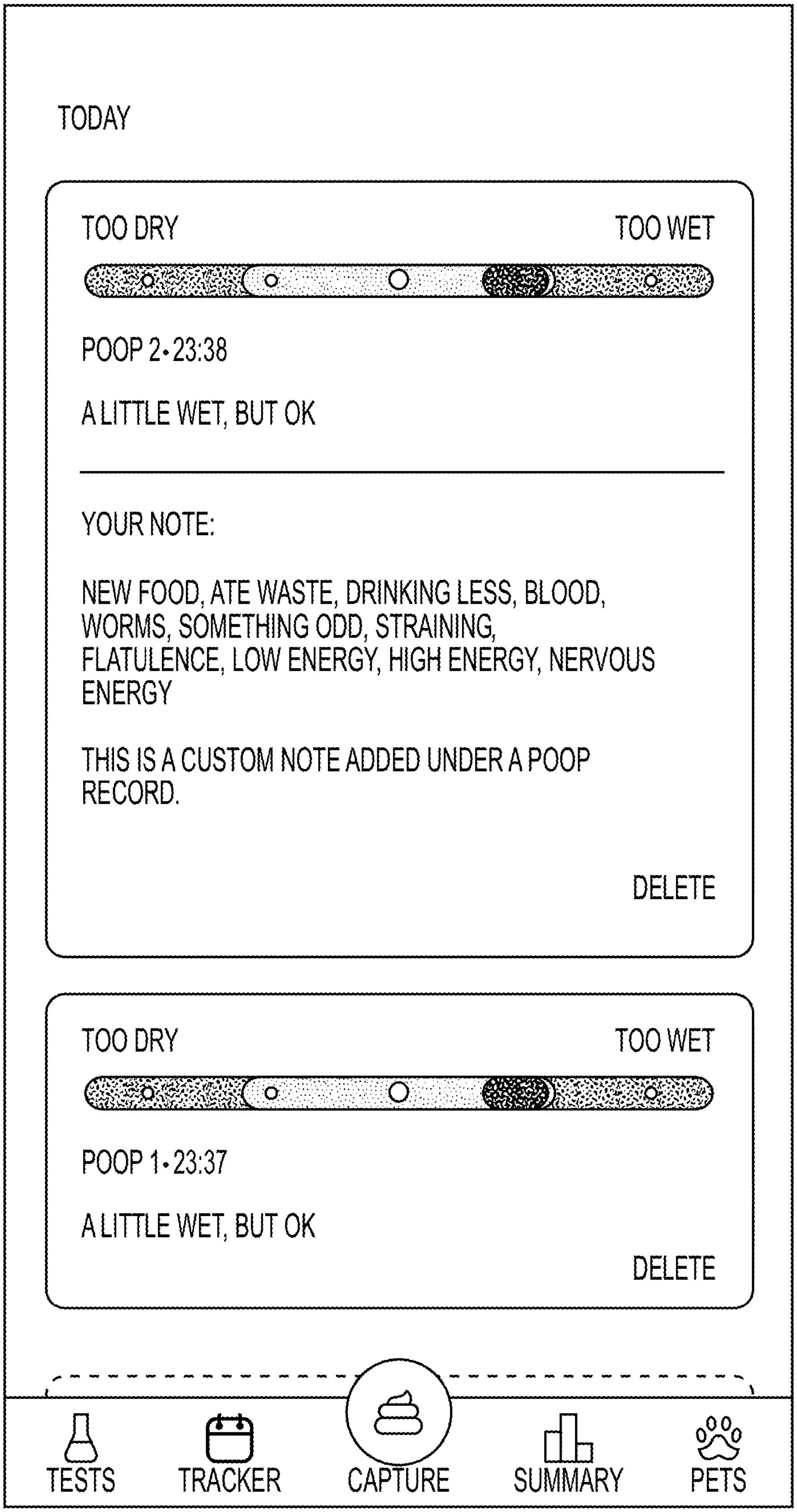

TODAY
TOO DRY
TOO WET
POOP 2•23:38
A LITTLE WET, BUT OK
YOUR NOTE:
NEW FOOD, ATE WASTE, DRINKING LESS, BLOOD, WORMS, SOMETHING ODD, STRAINING, FLATULENCE, LOW ENERGY, HIGH ENERGY, NERVOUS ENERGY
THIS IS A CUSTOM NOTE ADDED UNDER A POOP RECORD.
DELETE
TOO DRY
TOO WET
POOP 1•23:37
A LITTLE WET, BUT OK
DELETE
TESTS
TRACKER
CAPTURE
SUMMARY
PETS

FIG. 1T

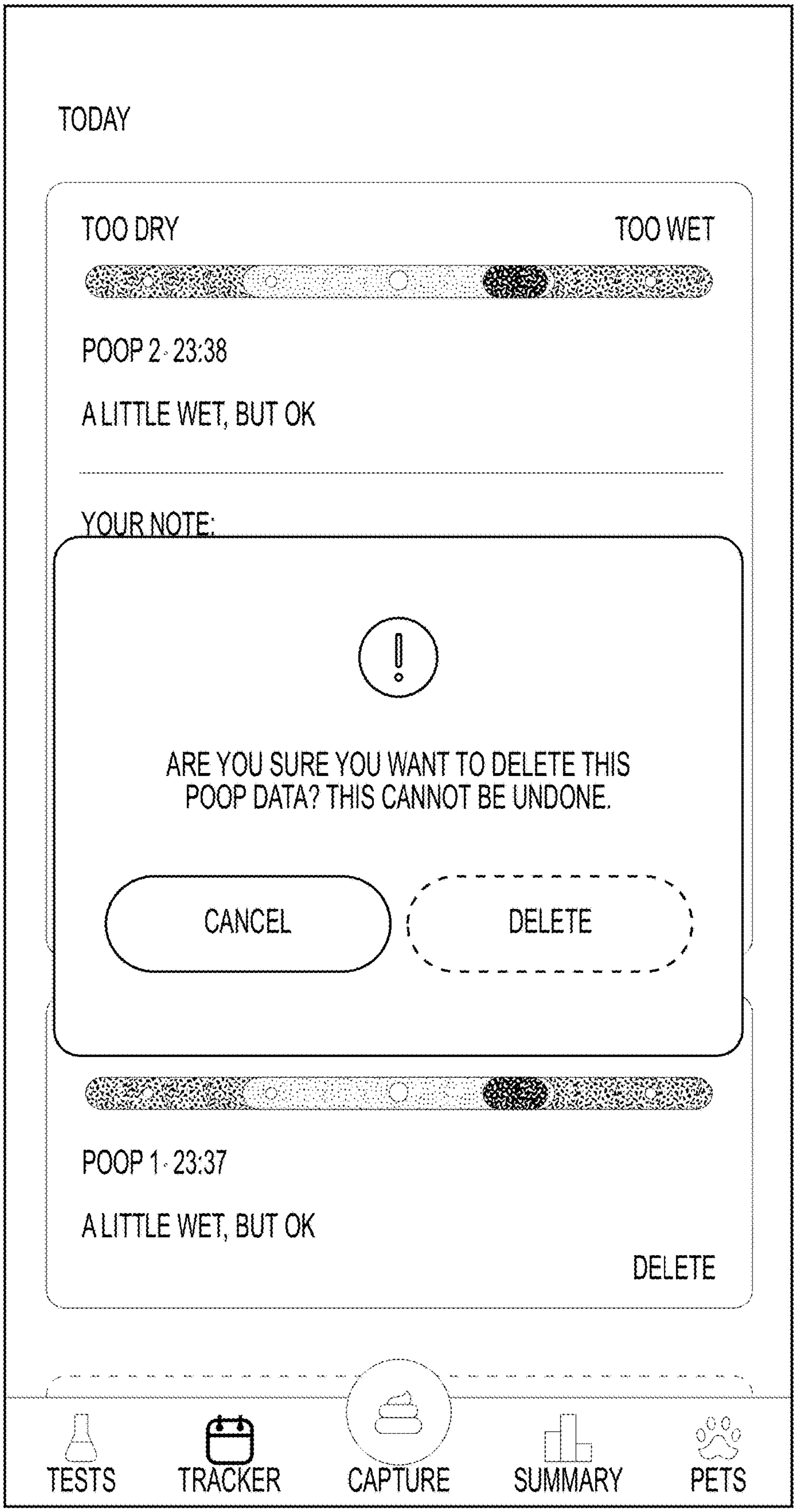

TODAY
TOO DRY
TOO WET
POOP 2 · 23:38
A LITTLE WET, BUT OK
YOUR NOTE:
ARE YOU SURE YOU WANT TO DELETE THIS POOP DATA? THIS CANNOT BE UNDONE.
CANCEL
DELETE
POOP 1 · 23:37
A LITTLE WET, BUT OK
DELETE
TESTS
TRACKER
CAPTURE
SUMMARY
PETS

FIG. 1U

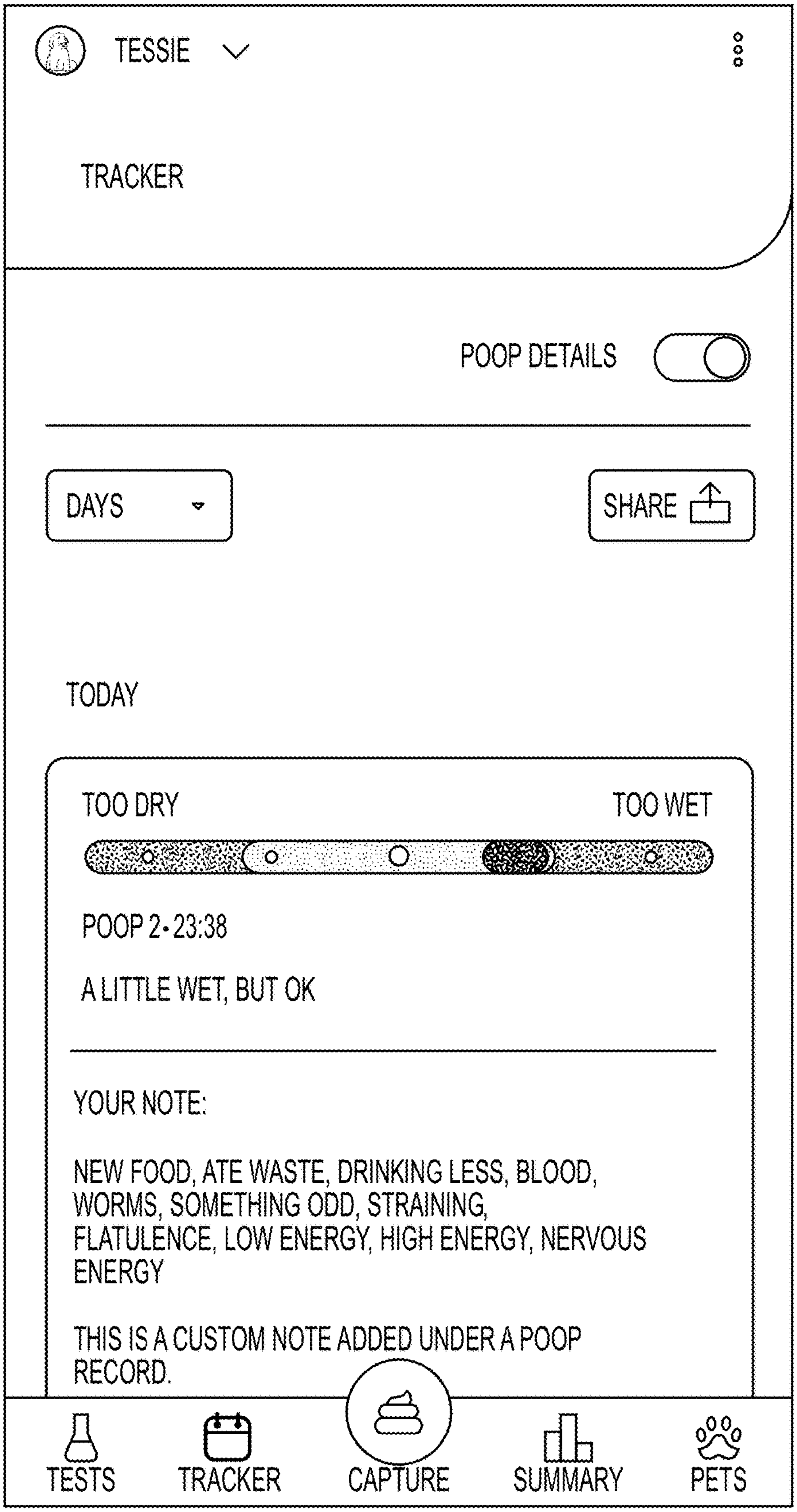

TESSIE
TRACKER
POOP DETAILS
DAYS
SHARE
TODAY
TOO DRY
TOO WET
POOP 2 • 23:38
A LITTLE WET, BUT OK
YOUR NOTE:
NEW FOOD, ATE WASTE, DRINKING LESS, BLOOD, WORMS, SOMETHING ODD, STRAINING, FLATULENCE, LOW ENERGY, HIGH ENERGY, NERVOUS ENERGY
THIS IS A CUSTOM NOTE ADDED UNDER A POOP RECORD.
TESTS
TRACKER
CAPTURE
SUMMARY
PETS

FIG. 1V

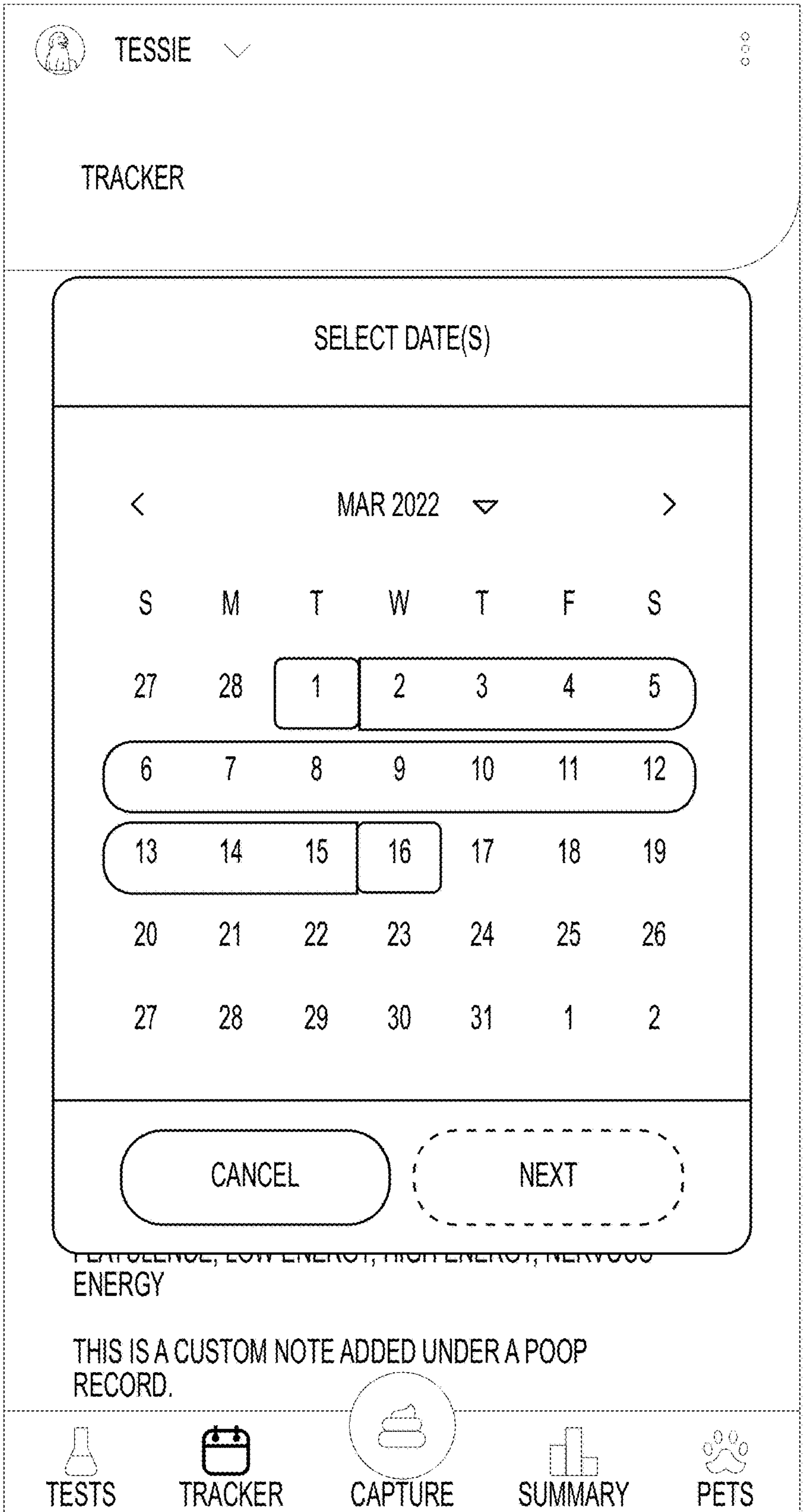

TESSIE
TRACKER
SELECT DATE(S)
MAR 2022
S M T W T F S
27 28 1 2 3 4 5
6 7 8 9 10 11 12
13 14 15 16 17 18 19
20 21 22 23 24 25 26
27 28 29 30 31 1 2
CANCEL
NEXT
ENERGY
THIS IS A CUSTOM NOTE ADDED UNDER A POOP RECORD.
TESTS
TRACKER
CAPTURE
SUMMARY
PETS

FIG. 1W

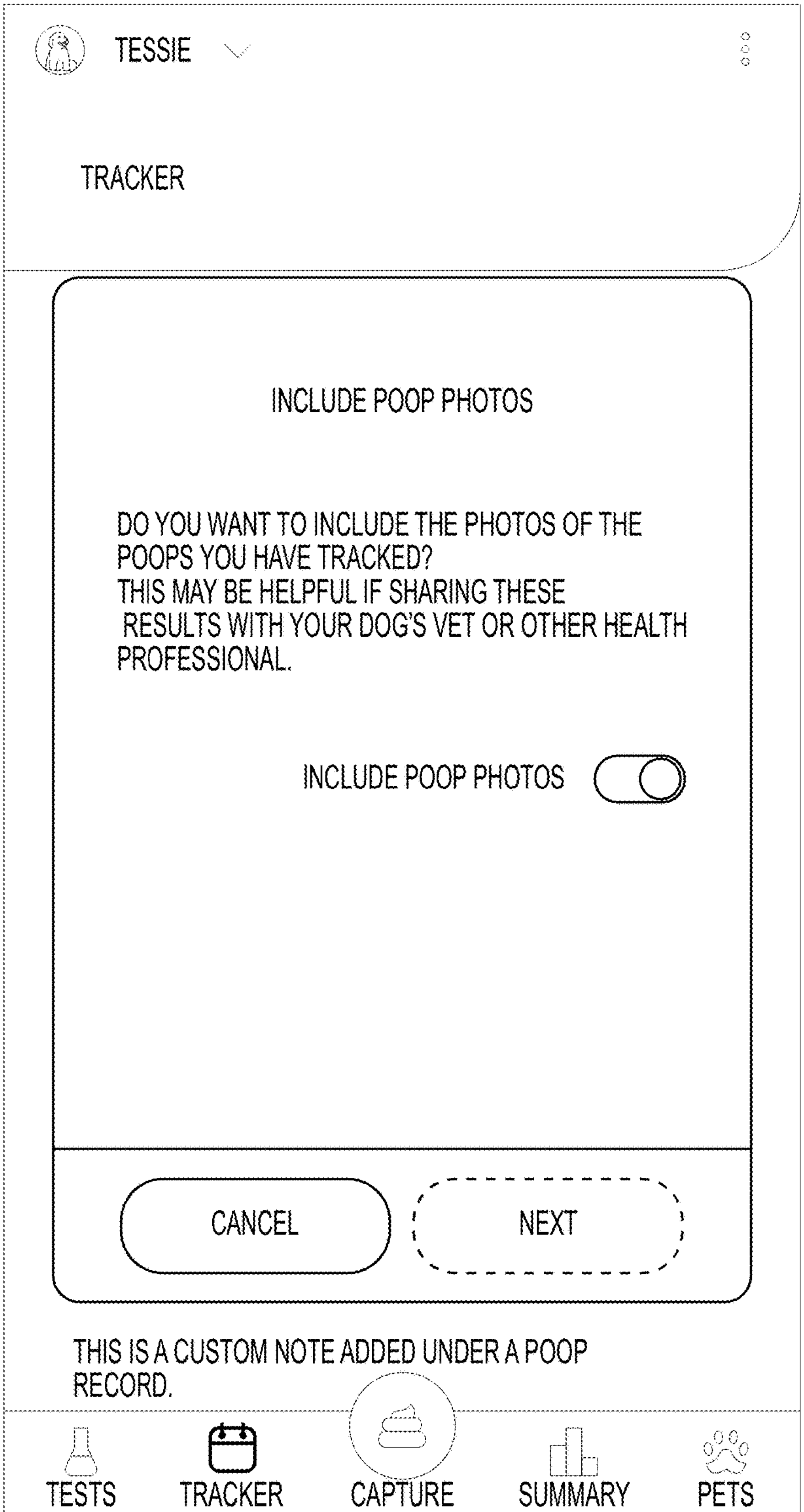

TESSIE
TRACKER
INCLUDE POOP PHOTOS
DO YOU WANT TO INCLUDE THE PHOTOS OF THE POOPS YOU HAVE TRACKED?
THIS MAY BE HELPFUL IF SHARING THESE RESULTS WITH YOUR DOG'S VET OR OTHER HEALTH PROFESSIONAL.
INCLUDE POOP PHOTOS
CANCEL
NEXT
THIS IS A CUSTOM NOTE ADDED UNDER A POOP RECORD.
TESTS
TRACKER
CAPTURE
SUMMARY
PETS

FIG. 1X

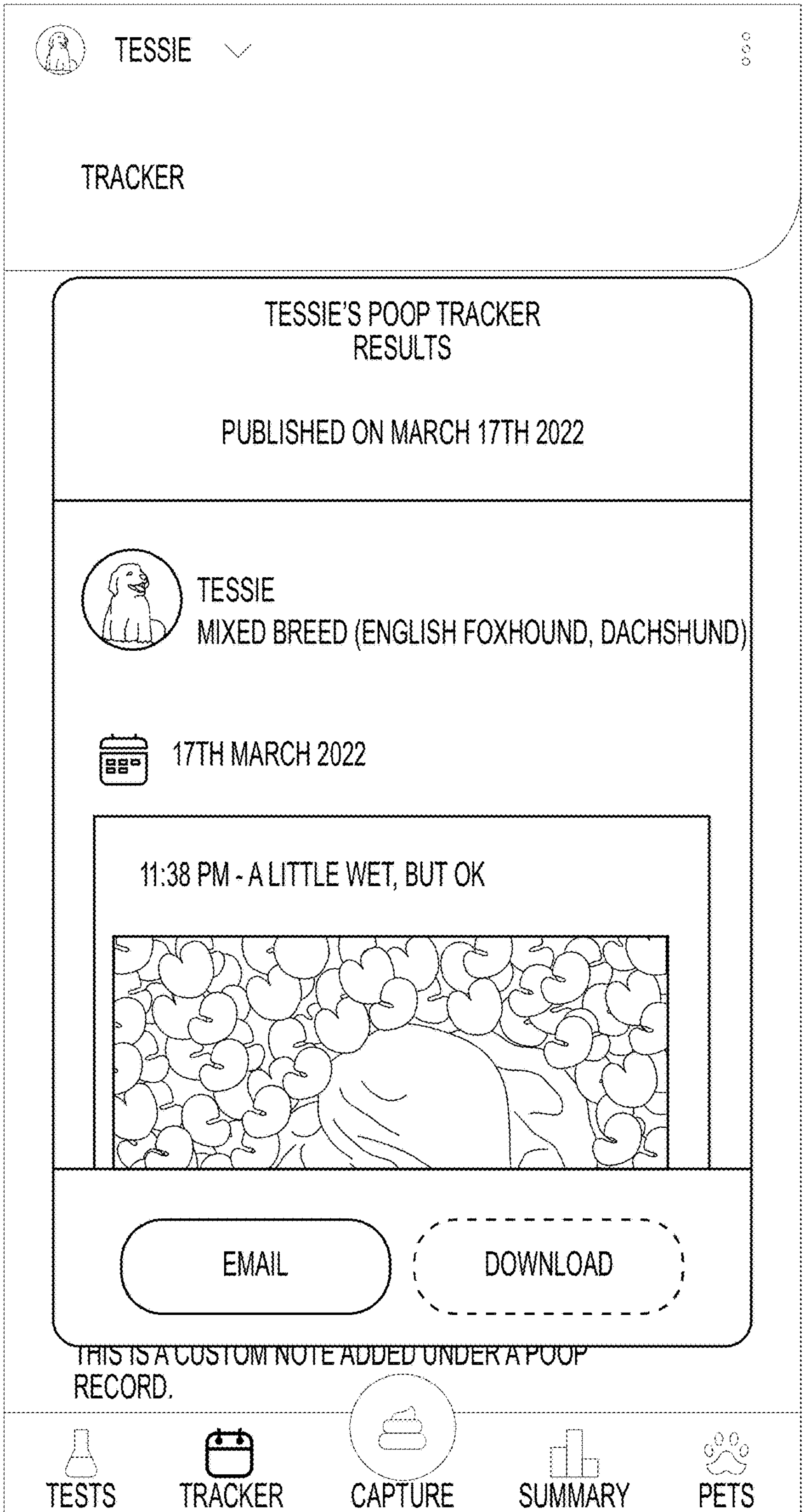

TESSIE
TRACKER
TESSIE'S POOP TRACKER
RESULTS
PUBLISHED ON MARCH 17TH 2022
TESSIE
MIXED BREED (ENGLISH FOXHOUND, DACHSHUND)
17TH MARCH 2022
11:38 PM - A LITTLE WET, BUT OK
EMAIL
DOWNLOAD
THIS IS A CUSTOM NOTE ADDED UNDER A POOP
RECORD.
TESTS
TRACKER
CAPTURE
SUMMARY
PETS

FIG. 1Y

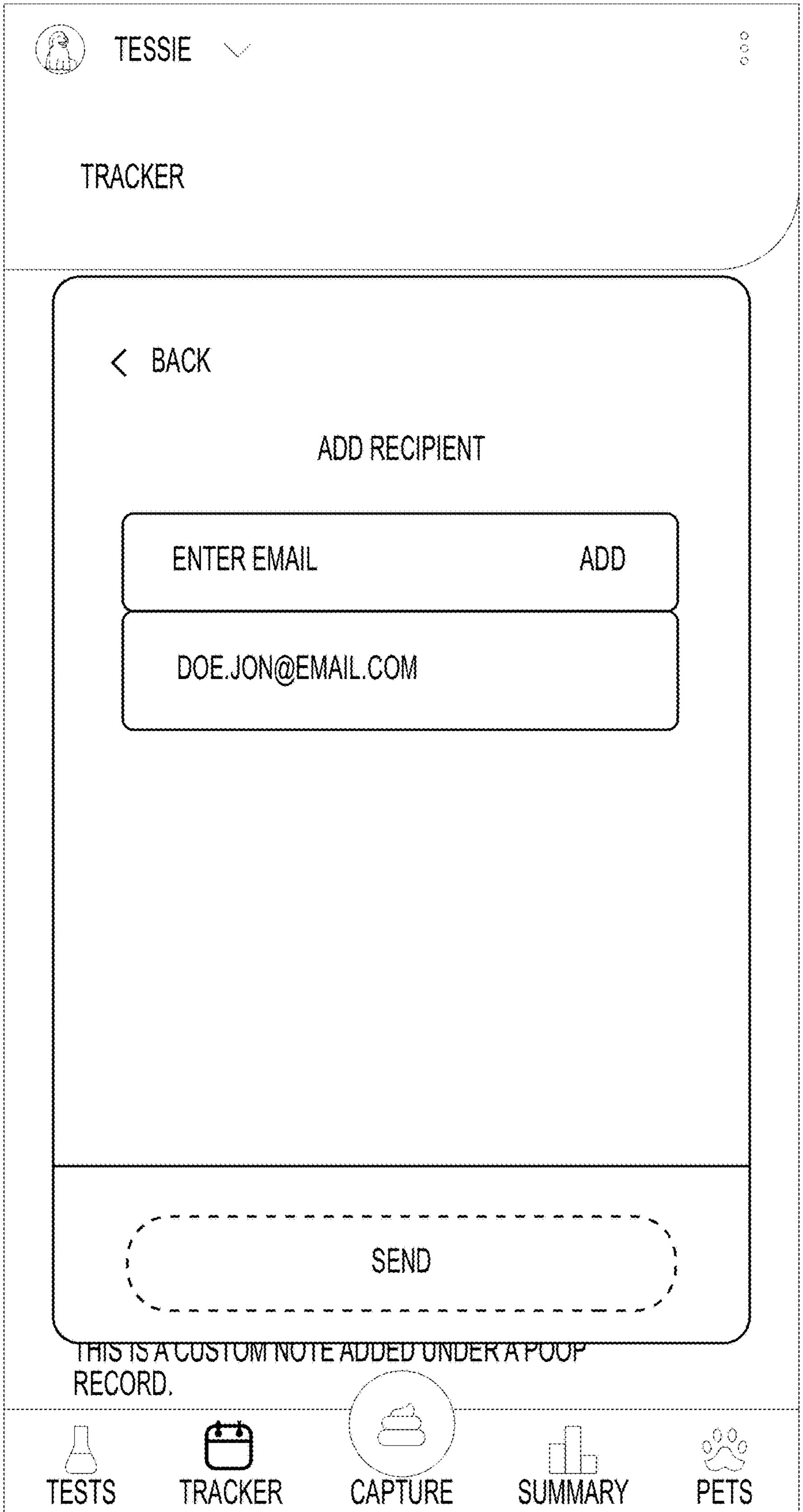
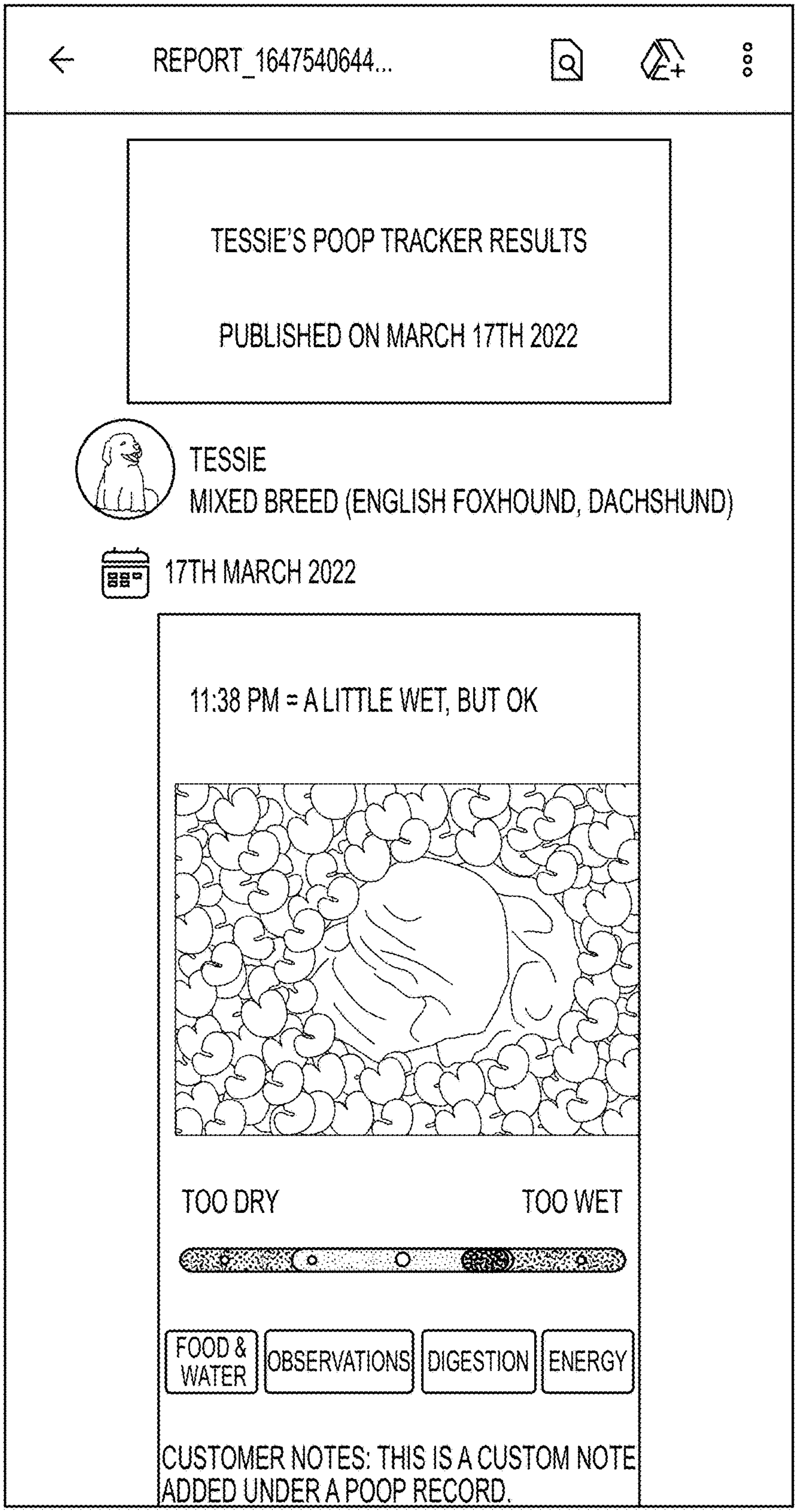

TESSIE
TRACKER
BACK
ADD RECIPIENT
ENTER EMAIL
ADD
DOE.JON@EMAIL.COM
SEND
THIS IS A CUSTOM NOTE ADDED UNDER A POOP
RECORD.
TESTS
TRACKER
CAPTURE
SUMMARY
PETS

FIG. 1Z



REPORT_1647540644...
TESSIE'S POOP TRACKER RESULTS
PUBLISHED ON MARCH 17TH 2022
TESSIE
MIXED BREED (ENGLISH FOXHOUND, DACHSHUND)
17TH MARCH 2022
11:38 PM = A LITTLE WET, BUT OK
TOO DRY
TOO WET
FOOD & WATER
OBSERVATIONS
DIGESTION
ENERGY
CUSTOMER NOTES: THIS IS A CUSTOM NOTE ADDED UNDER A POOP RECORD.

FIG. 1AA

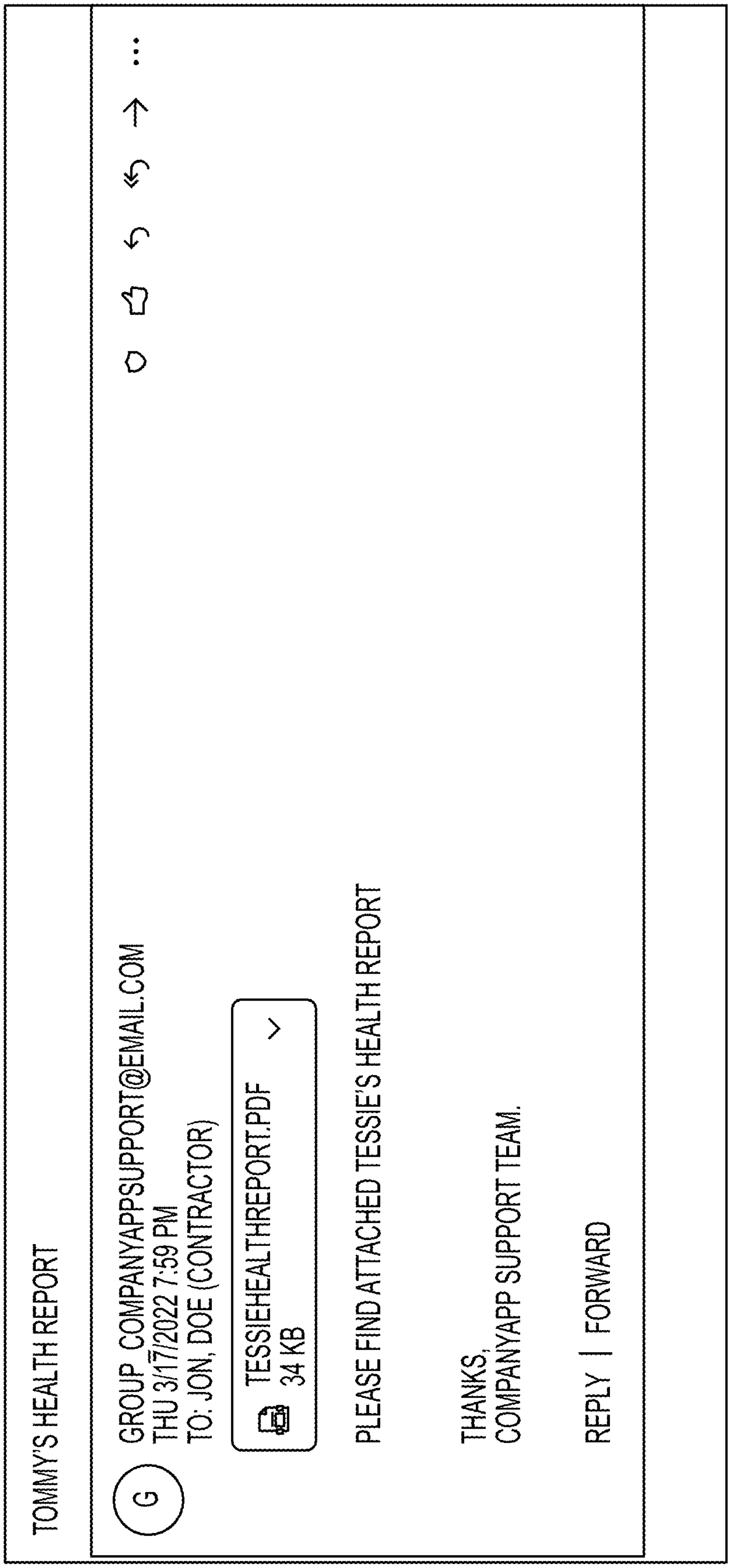

TOMMY'S HEALTH REPORT
G
GROUP_COMPANYAPPSUPPORT@EMAIL.COM
THU 3/17/2022 7:59 PM
TO: JON, DOE (CONTRACTOR)
TESSIEHEALTHREPORT.PDF
34 KB
PLEASE FIND ATTACHED TESSIE'S HEALTH REPORT
THANKS,
COMPANYAPP SUPPORT TEAM.
REPLY | FORWARD

FIG. 1BB

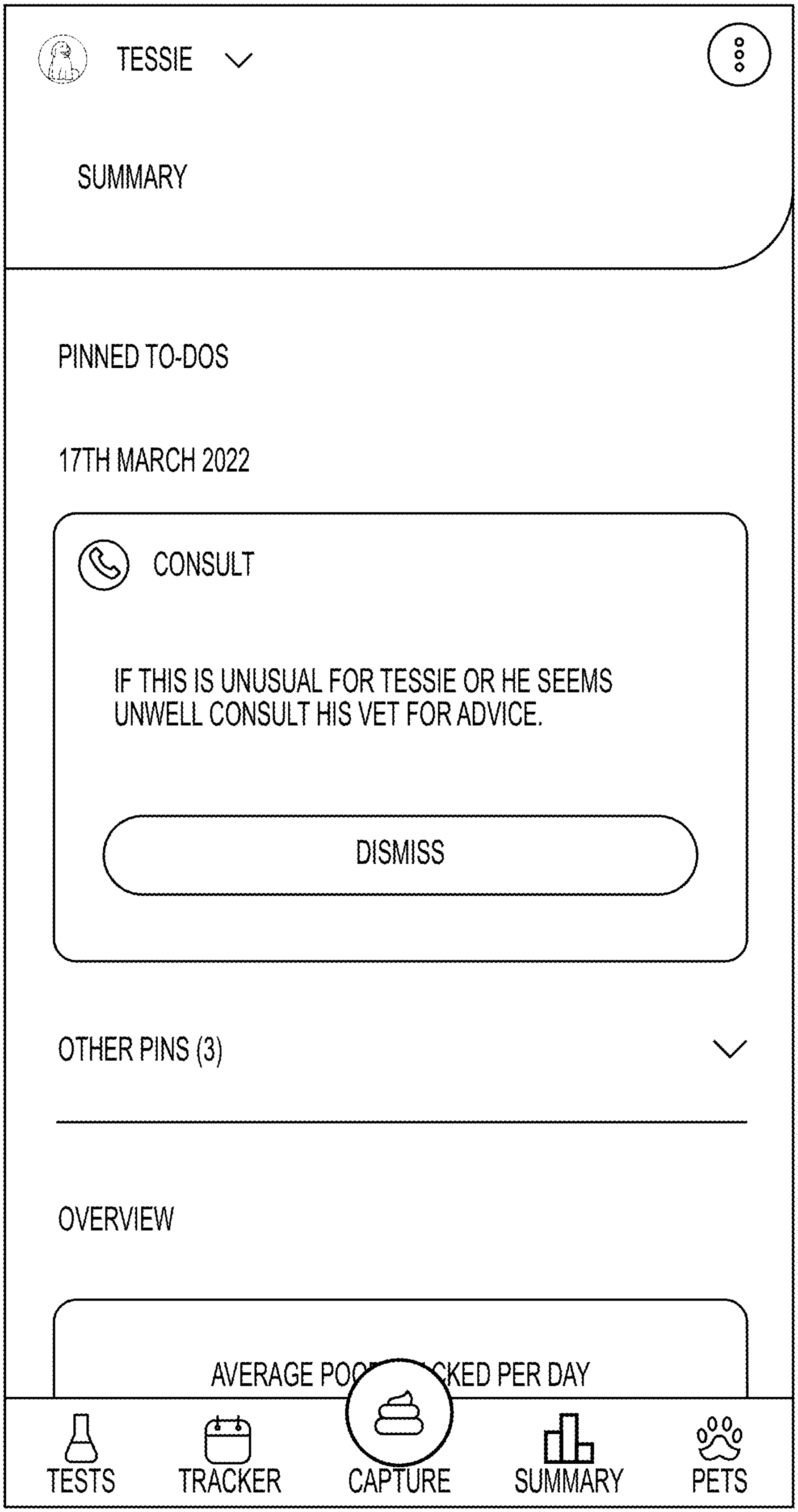

TESSIE
SUMMARY
PINNED TO-DOS
17TH MARCH 2022
CONSULT
IF THIS IS UNUSUAL FOR TESSIE OR HE SEEMS UNWELL CONSULT HIS VET FOR ADVICE.
DISMISS
OTHER PINS (3)
OVERVIEW
TESTS
TRACKER
CAPTURE
SUMMARY
PETS

FIG. 2A

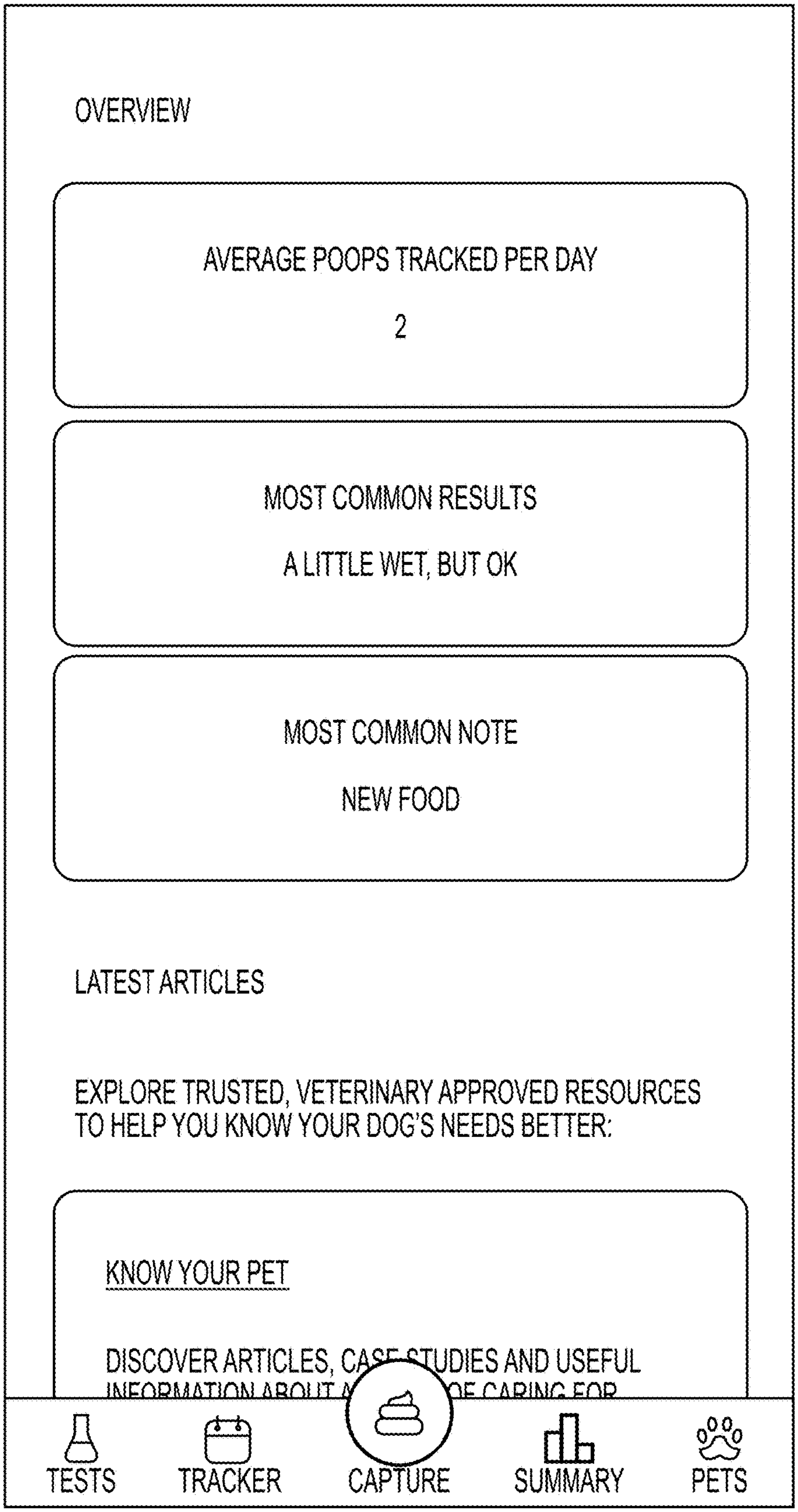

OVERVIEW
AVERAGE POOPS TRACKED PER DAY
2
MOST COMMON RESULTS
A LITTLE WET, BUT OK
MOST COMMON NOTE
NEW FOOD
LATEST ARTICLES
EXPLORE TRUSTED, VETERINARY APPROVED RESOURCES TO HELP YOU KNOW YOUR DOG'S NEEDS BETTER:
KNOW YOUR PET
DISCOVER ARTICLES, CASE STUDIES AND USEFUL
TESTS
TRACKER
CAPTURE
SUMMARY
PETS

FIG. 2B


LATEST ARTICLES
EXPLORE TRUSTED, VETERINARY APPROVED RESOURCES TO HELP YOU KNOW YOUR DOG'S NEEDS BETTER:
KNOW YOUR PET
DISCOVER ARTICLES, CASE STUDIES AND USEFUL INFORMATION ABOUT ALL ASPECTS OF CARING FOR YOUR DOG, FROM THE EXPERTS AT PET AUTHORITY.
DISCOVER NUTRITION FOR PUPPIES AND DOGS
LEARN HOW YOUR DOG'S DIET IMPACTS THEIR OVERALL HEALTH AND WELLBEING FROM LOCAL PET HOSPITAL.
KEEP YOUR DOG AT A HEALTHY WEIGHT
LEARN HOW TO MANAGE YOUR DOG'S FOOD INTAKE TO SUPPORT A HEALTHY WEIGHT FROM PET SCIENCE INSTITUTE.
TESTS
TRACKER
CAPTURE
SUMMARY
PETS

FIG. 2C


HOME / KNOW YOUR PET
KNOW YOUR PET
WE UNDERSTAND YOUR PETS ARE A BIG PART OF YOUR FAMILY. WE ARE HERE FOR YOU EVERY STEP OF THE WAY WHEN IT COMES TO YOUR PET'S HEALTH. OUR VAST LIBRARY IS CONSISTENTLY UPDATED WITH HELPFUL ARTICLES, CASE STUDIES AND INFORMATION ON PET MEDICATION RELEVANT TO YOU AND YOUR PET.
SEARCH
FILTERS
FEATURED ARTICLES

FIG. 2D

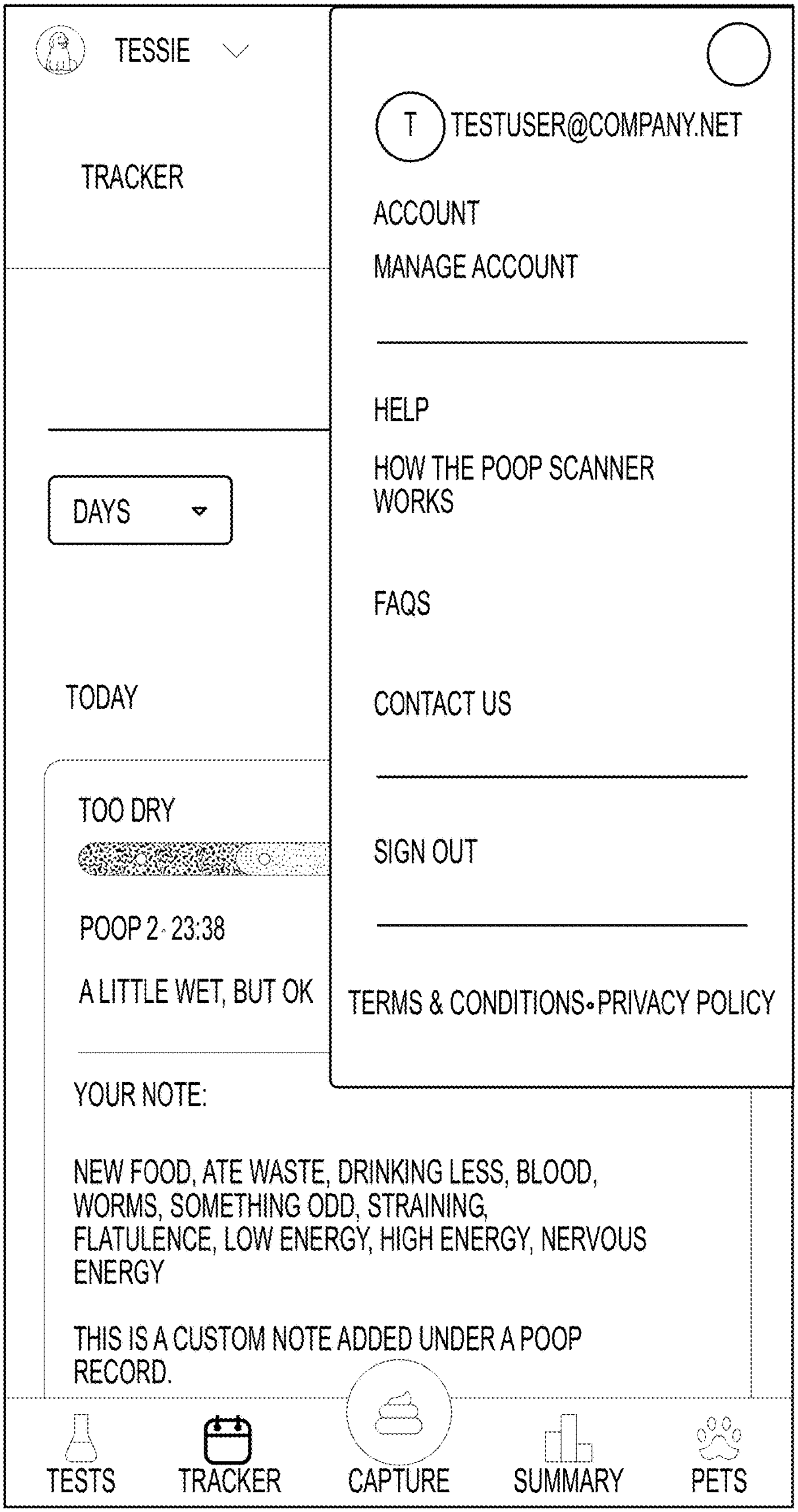

TESSIE
TRACKER
DAYS
TODAY
TOO DRY
POOP 2 · 23:38
A LITTLE WET, BUT OK
YOUR NOTE:
NEW FOOD, ATE WASTE, DRINKING LESS, BLOOD, WORMS, SOMETHING ODD, STRAINING, FLATULENCE, LOW ENERGY, HIGH ENERGY, NERVOUS ENERGY
THIS IS A CUSTOM NOTE ADDED UNDER A POOP RECORD.
T
TESTUSER@COMPANY.NET
ACCOUNT
MANAGE ACCOUNT
HELP
HOW THE POOP SCANNER WORKS
FAQS
CONTACT US
SIGN OUT
TERMS & CONDITIONS • PRIVACY POLICY
TESTS
TRACKER
CAPTURE
SUMMARY
PETS

FIG. 2E

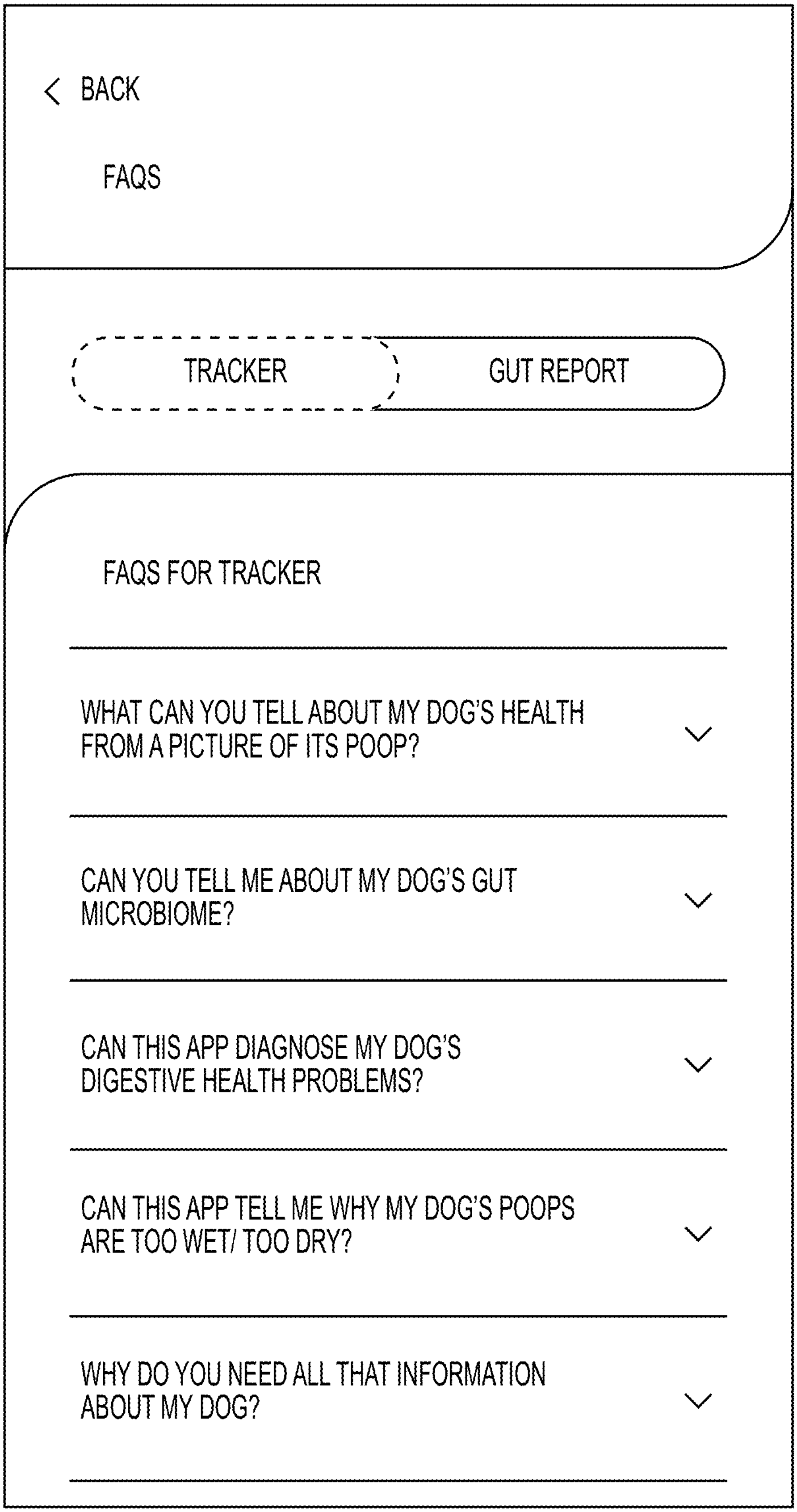

BACK
FAQS
TRACKER
GUT REPORT
FAQS FOR TRACKER
WHAT CAN YOU TELL ABOUT MY DOG'S HEALTH FROM A PICTURE OF ITS POOP?
CAN YOU TELL ME ABOUT MY DOG'S GUT MICROBIOME?
CAN THIS APP DIAGNOSE MY DOG'S DIGESTIVE HEALTH PROBLEMS?
CAN THIS APP TELL ME WHY MY DOG'S POOPS ARE TOO WET/ TOO DRY?
WHY DO YOU NEED ALL THAT INFORMATION ABOUT MY DOG?

FIG. 2F

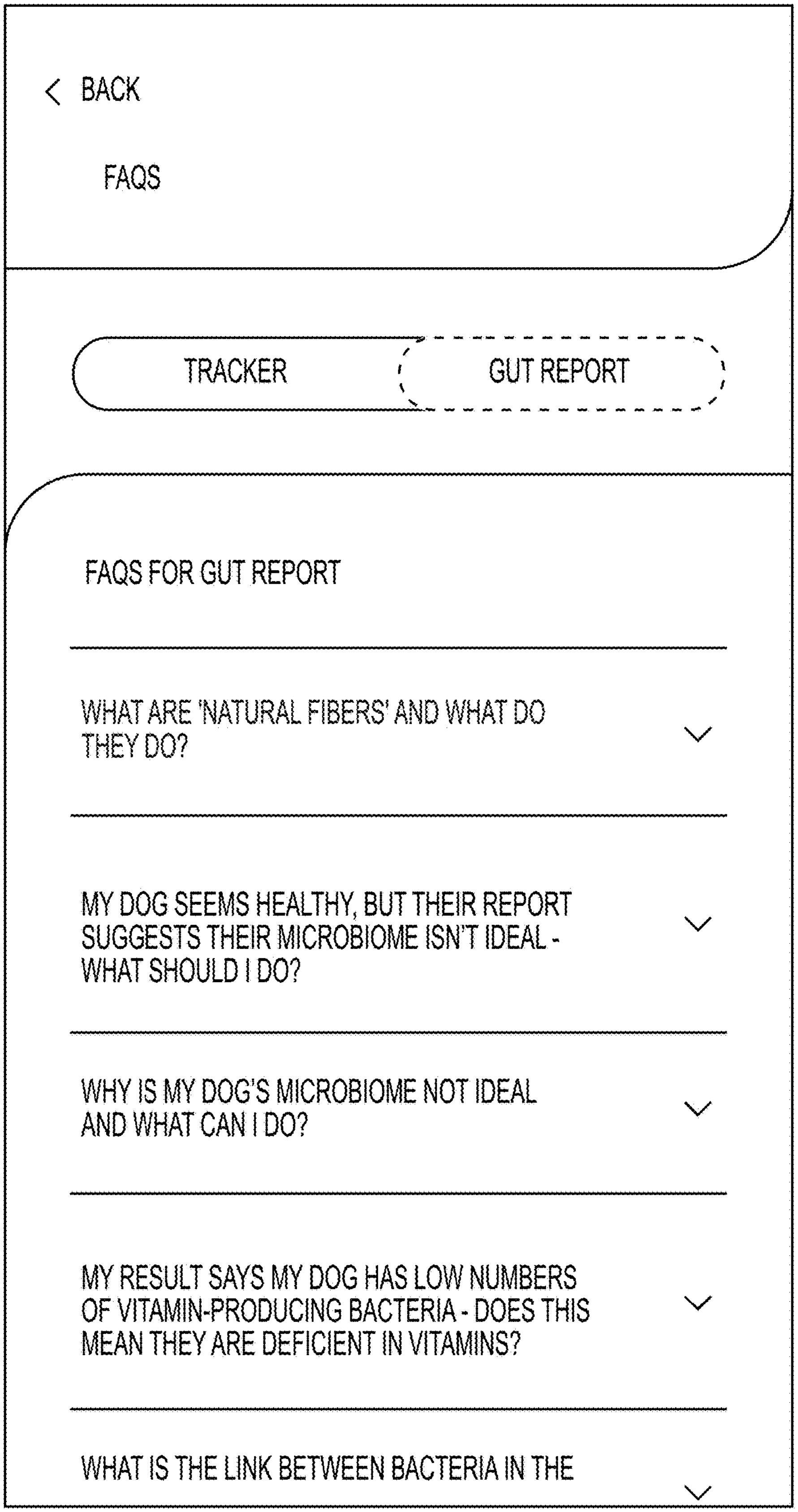

BACK
FAQS
TRACKER
GUT REPORT
FAQS FOR GUT REPORT
WHAT ARE 'NATURAL FIBERS' AND WHAT DO THEY DO?
MY DOG SEEMS HEALTHY, BUT THEIR REPORT SUGGESTS THEIR MICROBIOME ISN'T IDEAL - WHAT SHOULD I DO?
WHY IS MY DOG'S MICROBIOME NOT IDEAL AND WHAT CAN I DO?
MY RESULT SAYS MY DOG HAS LOW NUMBERS OF VITAMIN-PRODUCING BACTERIA - DOES THIS MEAN THEY ARE DEFICIENT IN VITAMINS?
WHAT IS THE LINK BETWEEN BACTERIA IN THE

FIG. 2G

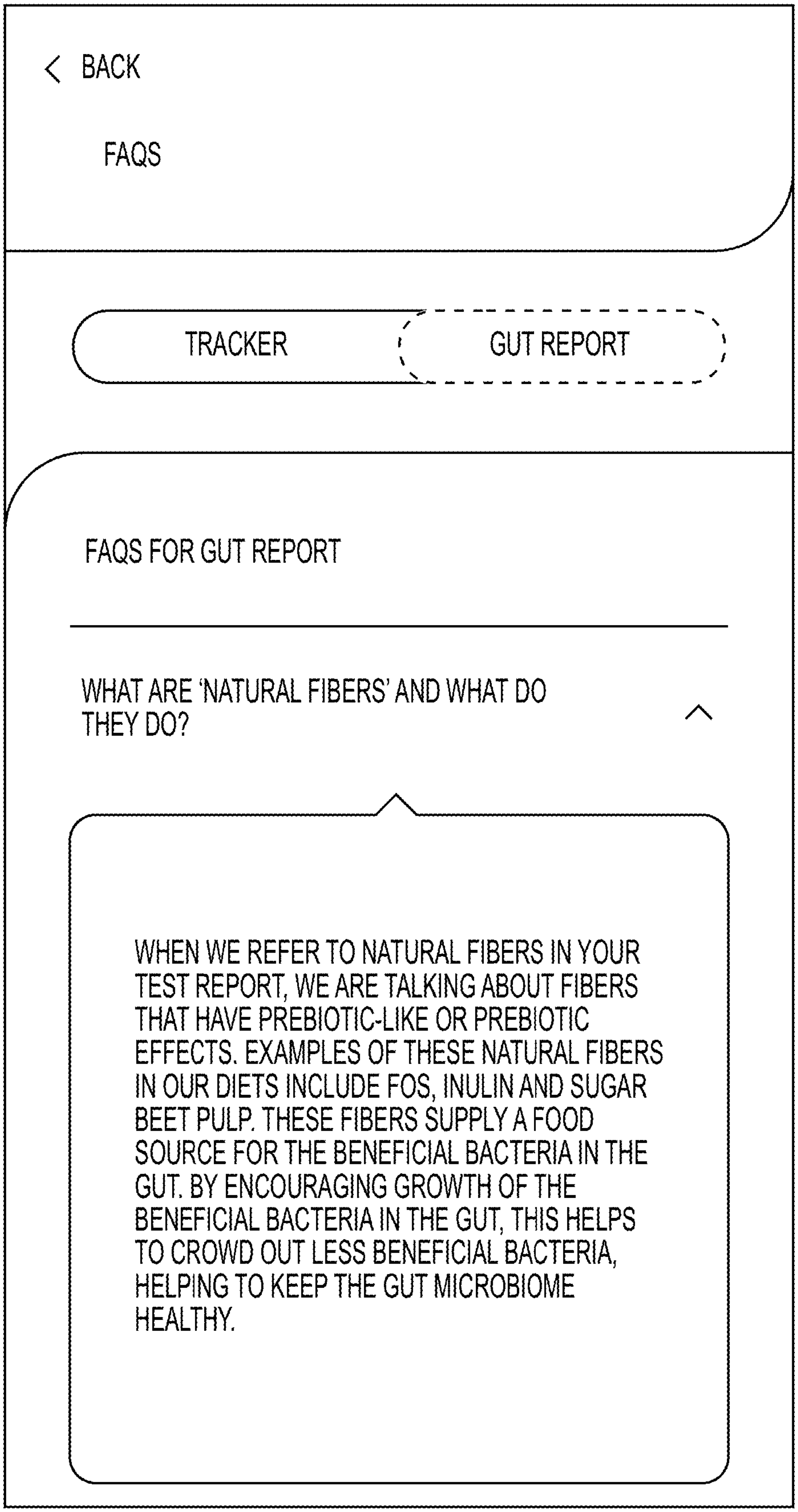

BACK
FAQS
TRACKER
GUT REPORT
FAQS FOR GUT REPORT
WHAT ARE 'NATURAL FIBERS' AND WHAT DO THEY DO?
WHEN WE REFER TO NATURAL FIBERS IN YOUR TEST REPORT, WE ARE TALKING ABOUT FIBERS THAT HAVE PREBIOTIC-LIKE OR PREBIOTIC EFFECTS. EXAMPLES OF THESE NATURAL FIBERS IN OUR DIETS INCLUDE FOS, INULIN AND SUGAR BEET PULP. THESE FIBERS SUPPLY A FOOD SOURCE FOR THE BENEFICIAL BACTERIA IN THE GUT. BY ENCOURAGING GROWTH OF THE BENEFICIAL BACTERIA IN THE GUT, THIS HELPS TO CROWD OUT LESS BENEFICIAL BACTERIA, HELPING TO KEEP THE GUT MICROBIOME HEALTHY.

FIG. 2H

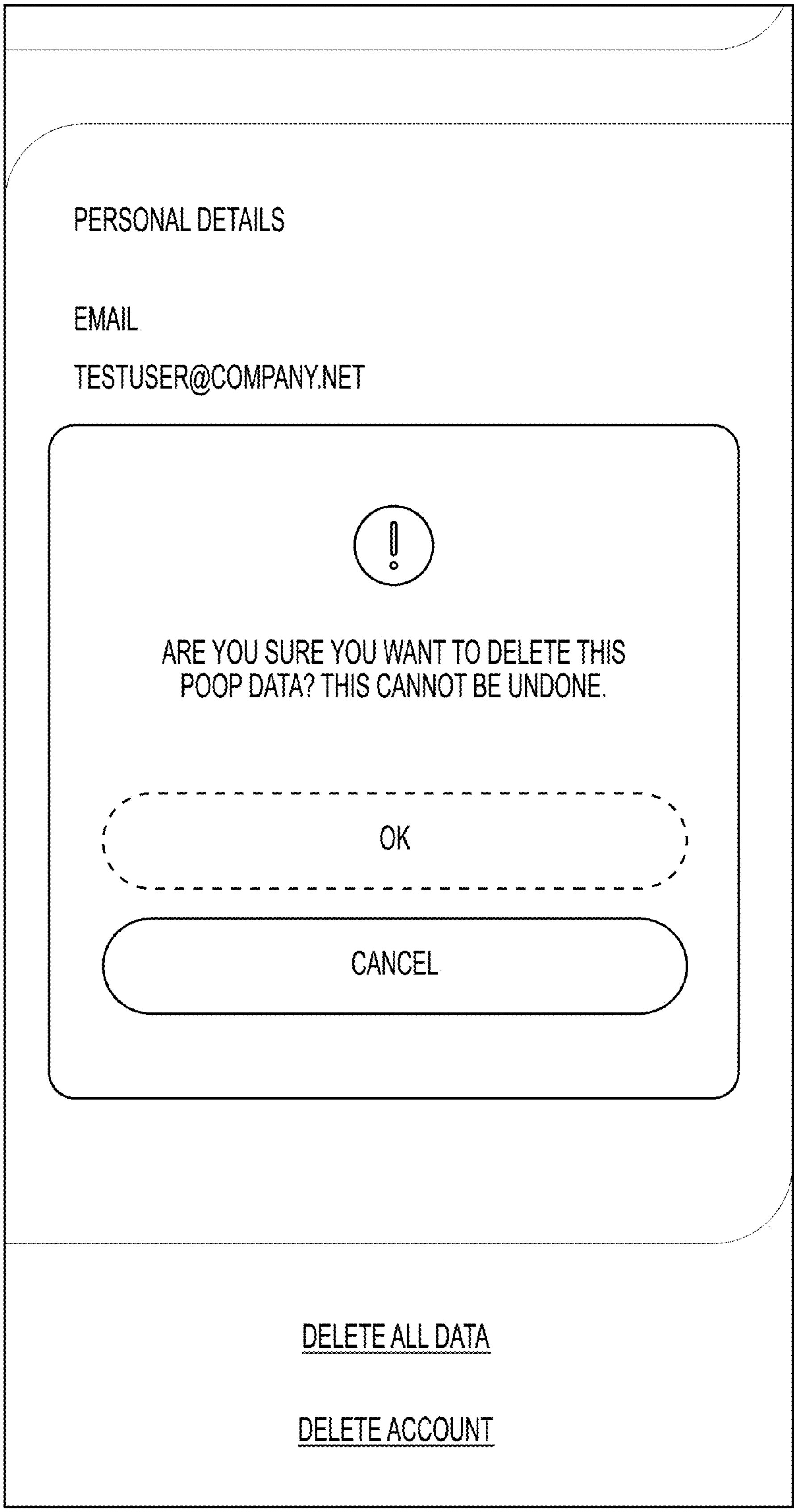

PERSONAL DETAILS
EMAIL
TESTUSER@COMPANY.NET
ARE YOU SURE YOU WANT TO DELETE THIS POOP DATA? THIS CANNOT BE UNDONE.
OK
CANCEL
DELETE ALL DATA
DELETE ACCOUNT

FIG. 2I

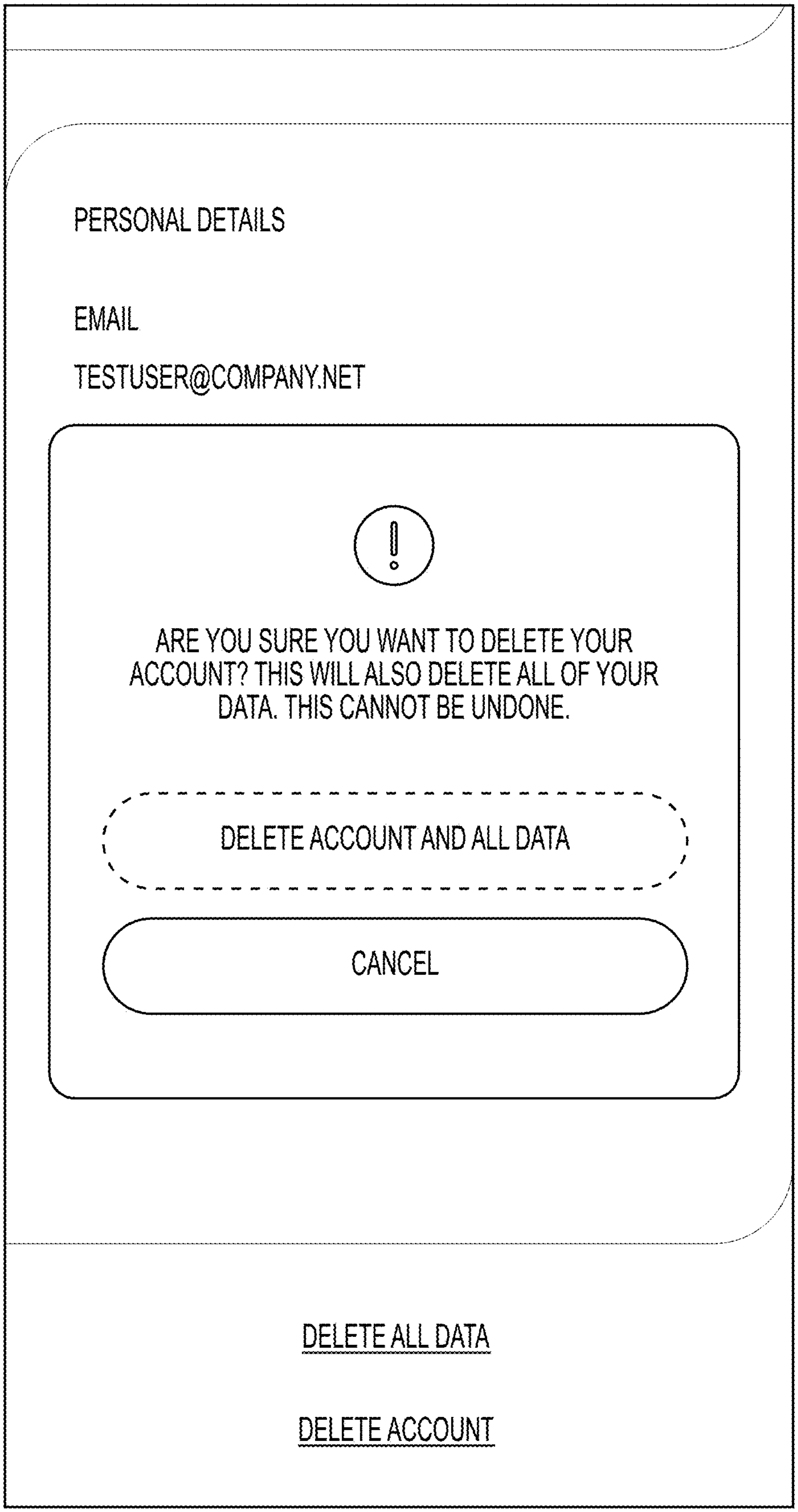

PERSONAL DETAILS
EMAIL
TESTUSER@COMPANY.NET
ARE YOU SURE YOU WANT TO DELETE YOUR ACCOUNT? THIS WILL ALSO DELETE ALL OF YOUR DATA. THIS CANNOT BE UNDONE.
DELETE ACCOUNT AND ALL DATA
CANCEL
DELETE ALL DATA
DELETE ACCOUNT

FIG. 2J

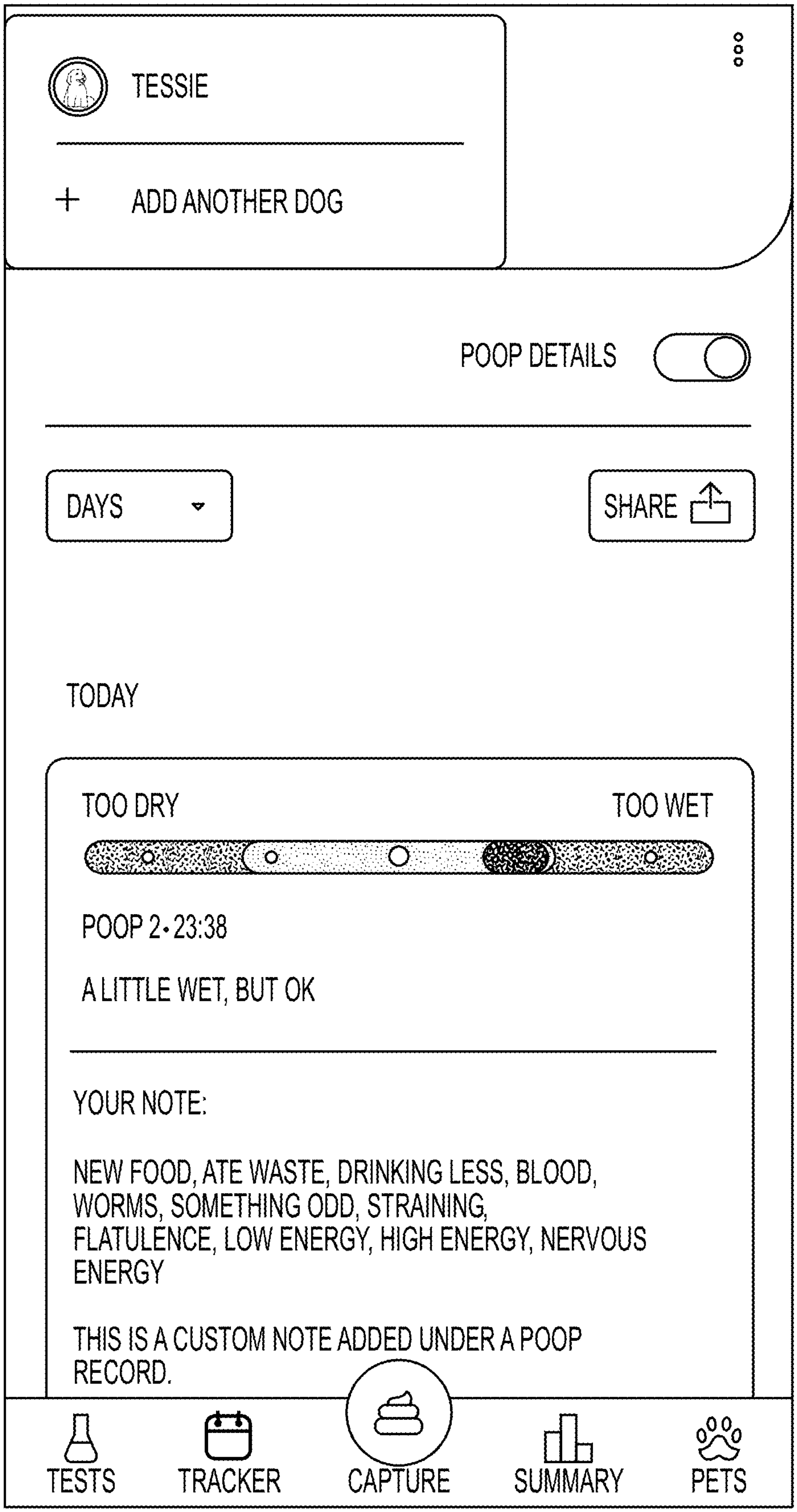

TESSIE
+ ADD ANOTHER DOG
POOP DETAILS
DAYS
SHARE
TODAY
TOO DRY
TOO WET
POOP 2 • 23:38
A LITTLE WET, BUT OK
YOUR NOTE:
NEW FOOD, ATE WASTE, DRINKING LESS, BLOOD, WORMS, SOMETHING ODD, STRAINING, FLATULENCE, LOW ENERGY, HIGH ENERGY, NERVOUS ENERGY
THIS IS A CUSTOM NOTE ADDED UNDER A POOP RECORD.
TESTS
TRACKER
CAPTURE
SUMMARY
PETS

FIG. 3A

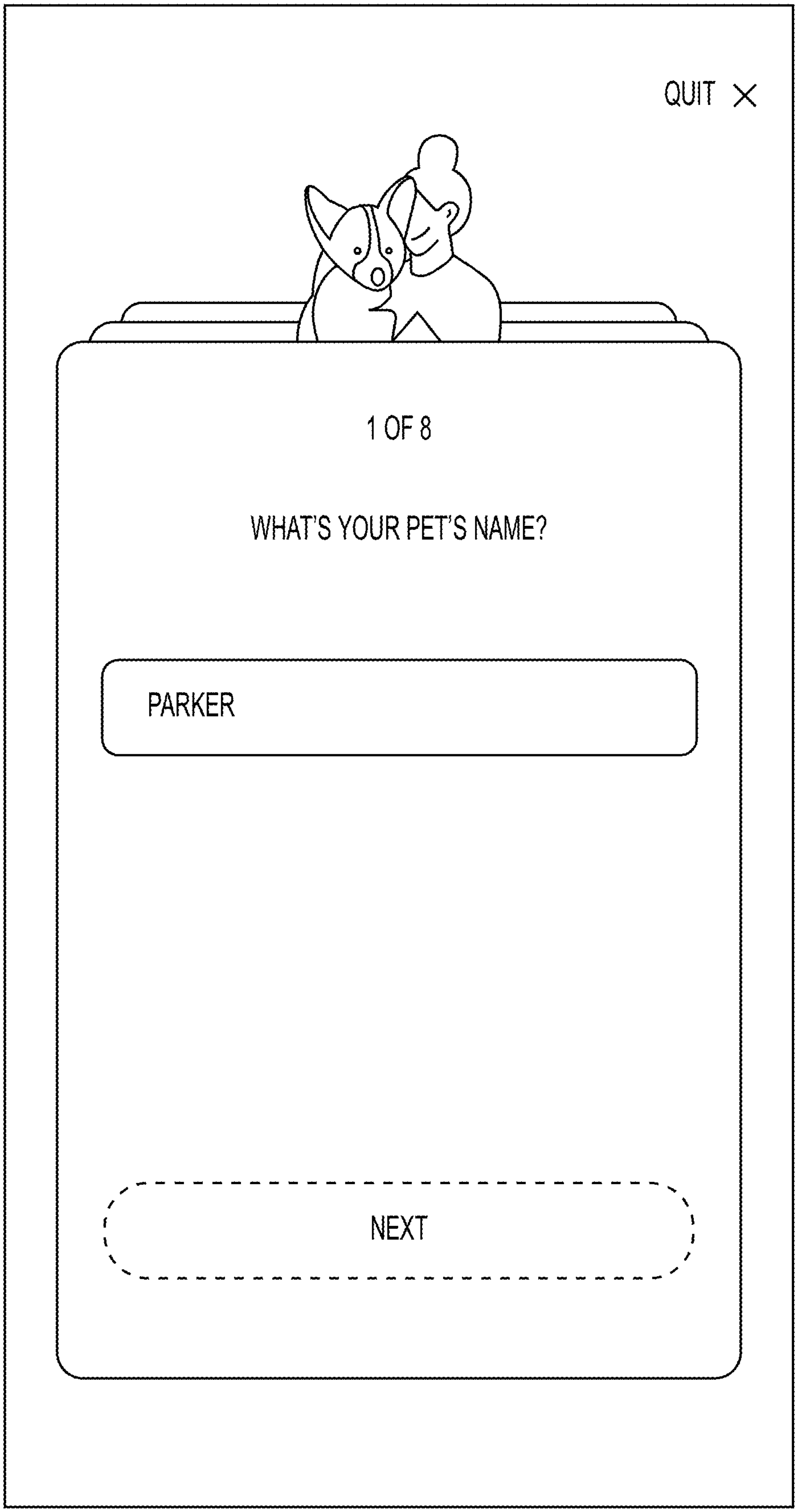


FIG. 3B

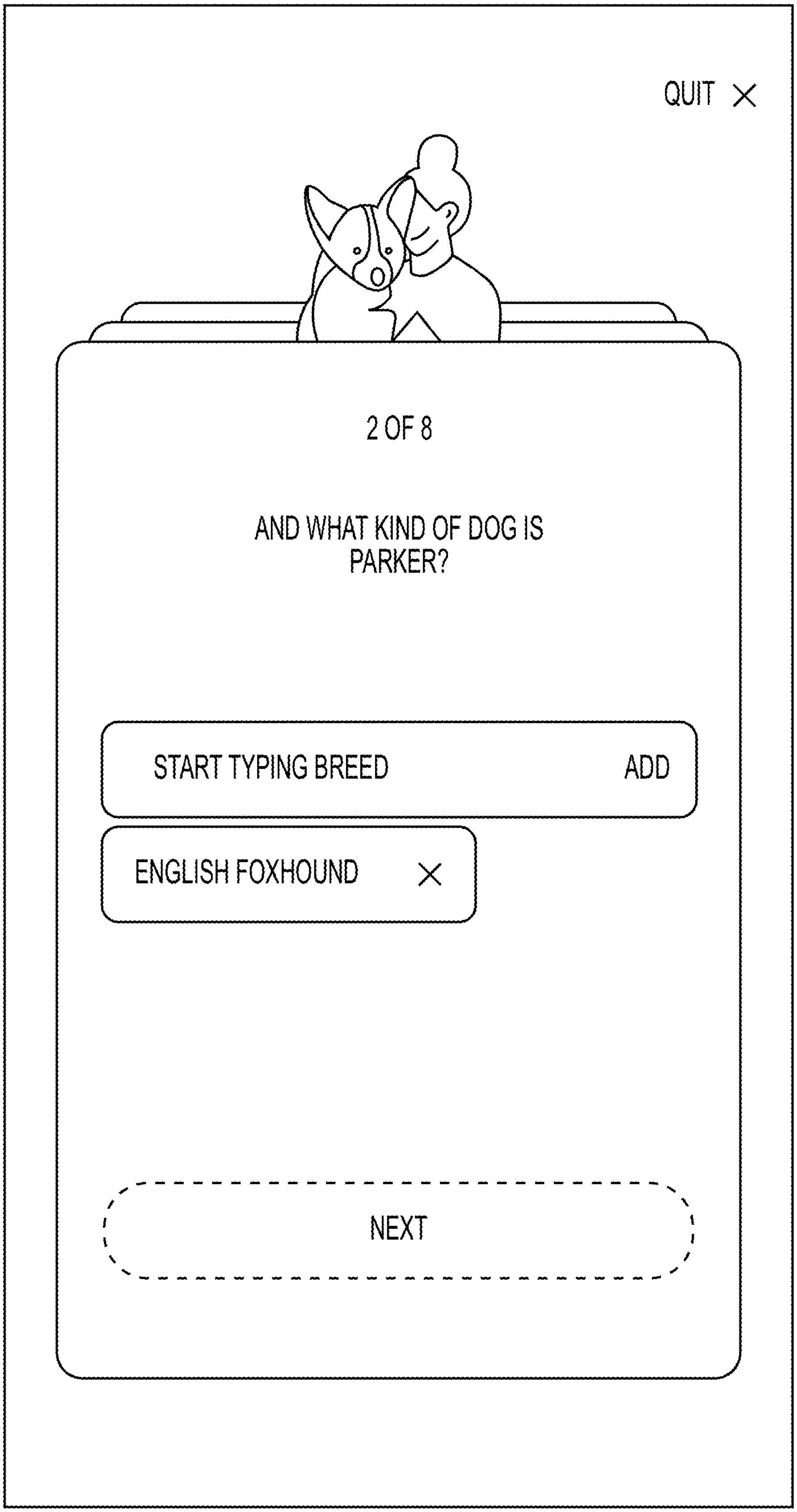

QUIT
2 OF 8
AND WHAT KIND OF DOG IS PARKER?
START TYPING BREED
ADD
ENGLISH FOXHOUND
NEXT

FIG. 3C

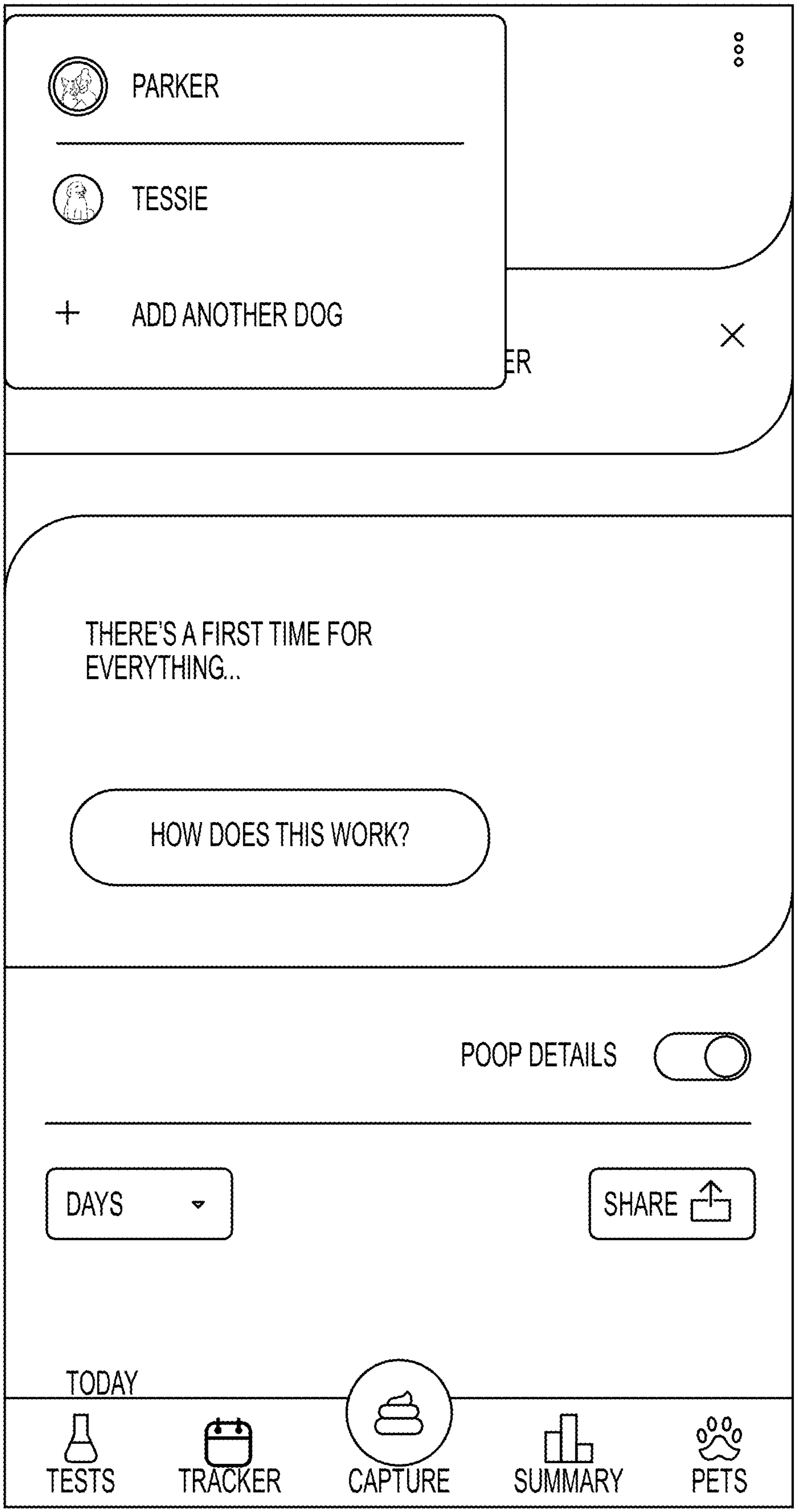

PARKER
TESSIE
+ ADD ANOTHER DOG
ER
THERE'S A FIRST TIME FOR EVERYTHING...
HOW DOES THIS WORK?
POOP DETAILS
DAYS
SHARE
TODAY
TESTS
TRACKER
CAPTURE
SUMMARY
PETS

FIG. 3D

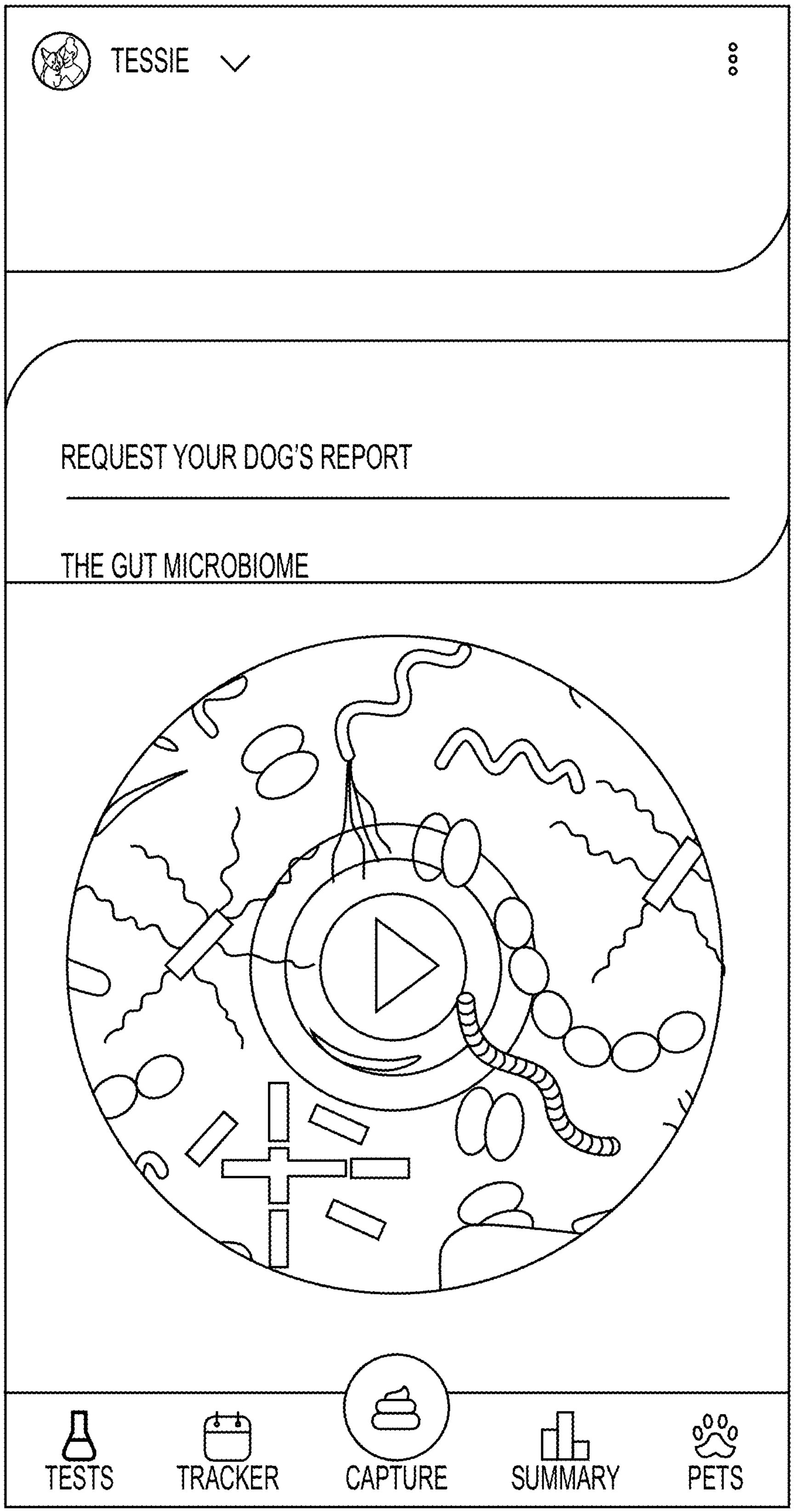

TESSIE
REQUEST YOUR DOG'S REPORT
THE GUT MICROBIOME
TESTS
TRACKER
CAPTURE
SUMMARY
PETS

FIG. 4A

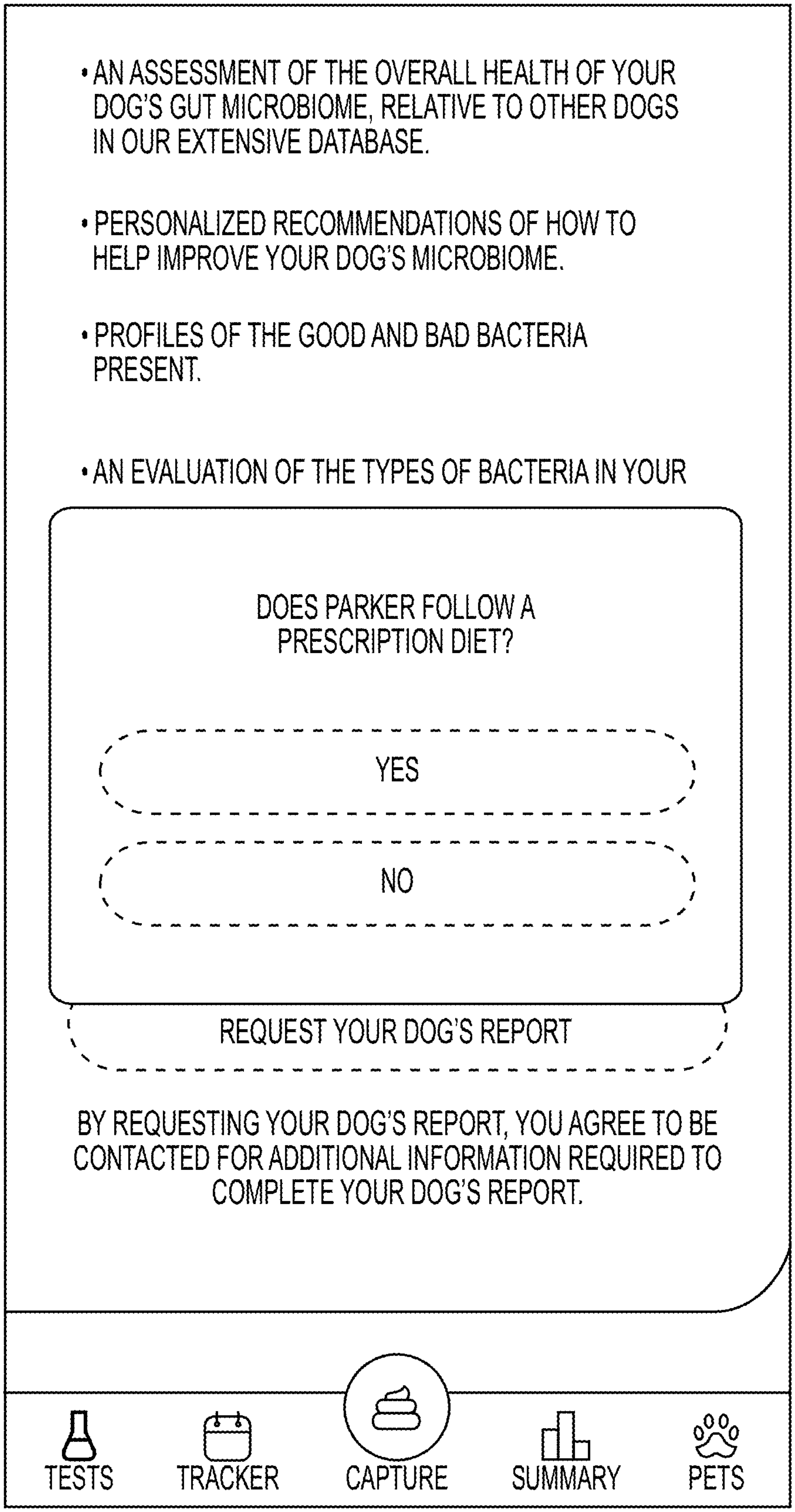

• AN ASSESSMENT OF THE OVERALL HEALTH OF YOUR DOG'S GUT MICROBIOME, RELATIVE TO OTHER DOGS IN OUR EXTENSIVE DATABASE.
• PERSONALIZED RECOMMENDATIONS OF HOW TO HELP IMPROVE YOUR DOG'S MICROBIOME.
• PROFILES OF THE GOOD AND BAD BACTERIA PRESENT.
• AN EVALUATION OF THE TYPES OF BACTERIA IN YOUR
DOES PARKER FOLLOW A PRESCRIPTION DIET?
YES
NO
REQUEST YOUR DOG'S REPORT
BY REQUESTING YOUR DOG'S REPORT, YOU AGREE TO BE CONTACTED FOR ADDITIONAL INFORMATION REQUIRED TO COMPLETE YOUR DOG'S REPORT.
TESTS
TRACKER
CAPTURE
SUMMARY
PETS

FIG. 4B

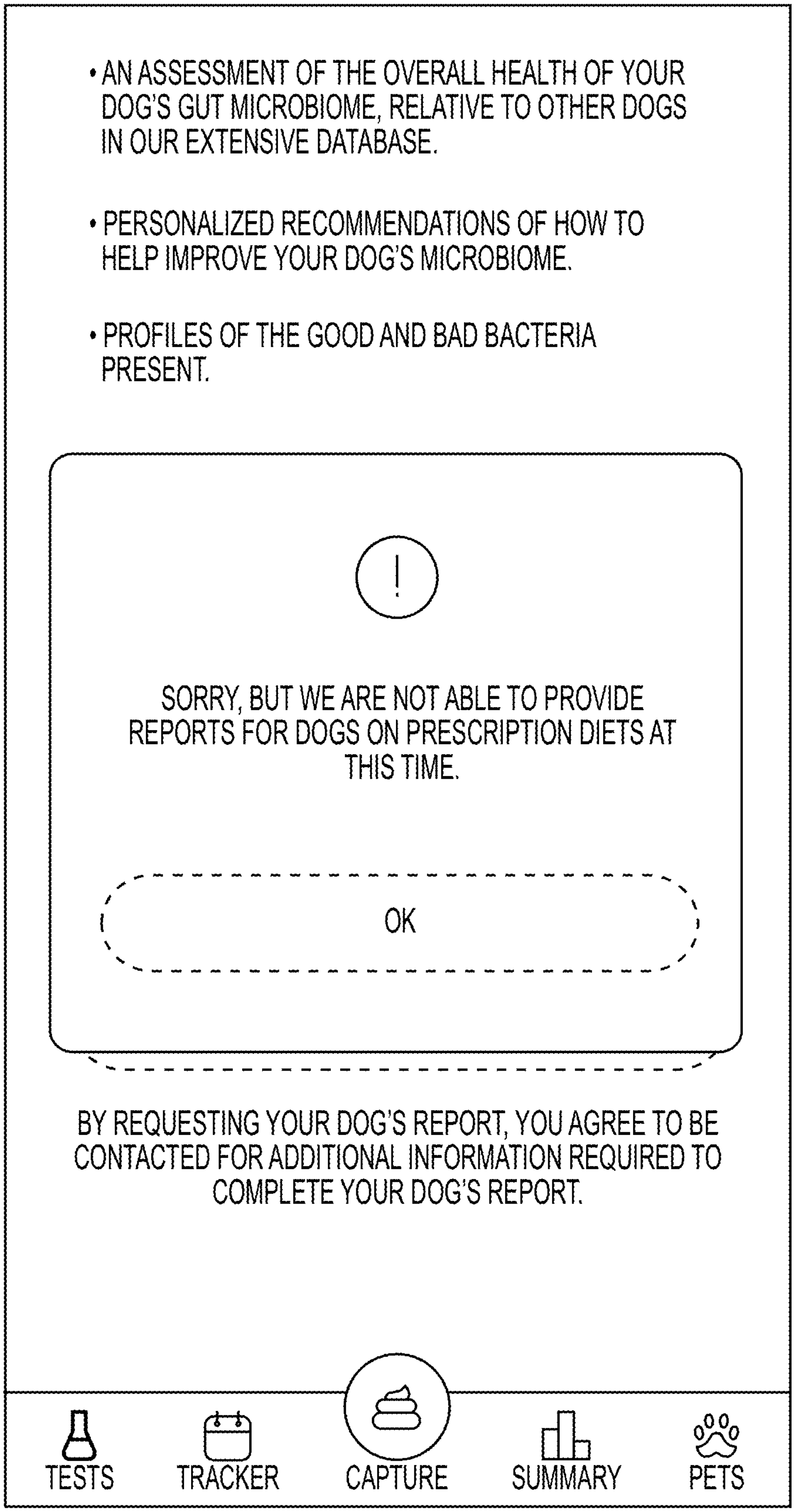

• AN ASSESSMENT OF THE OVERALL HEALTH OF YOUR DOG'S GUT MICROBIOME, RELATIVE TO OTHER DOGS IN OUR EXTENSIVE DATABASE.
• PERSONALIZED RECOMMENDATIONS OF HOW TO HELP IMPROVE YOUR DOG'S MICROBIOME.
• PROFILES OF THE GOOD AND BAD BACTERIA PRESENT.
!
SORRY, BUT WE ARE NOT ABLE TO PROVIDE REPORTS FOR DOGS ON PRESCRIPTION DIETS AT THIS TIME.
OK
BY REQUESTING YOUR DOG'S REPORT, YOU AGREE TO BE CONTACTED FOR ADDITIONAL INFORMATION REQUIRED TO COMPLETE YOUR DOG'S REPORT.
TESTS
TRACKER
CAPTURE
SUMMARY
PETS

FIG. 4C

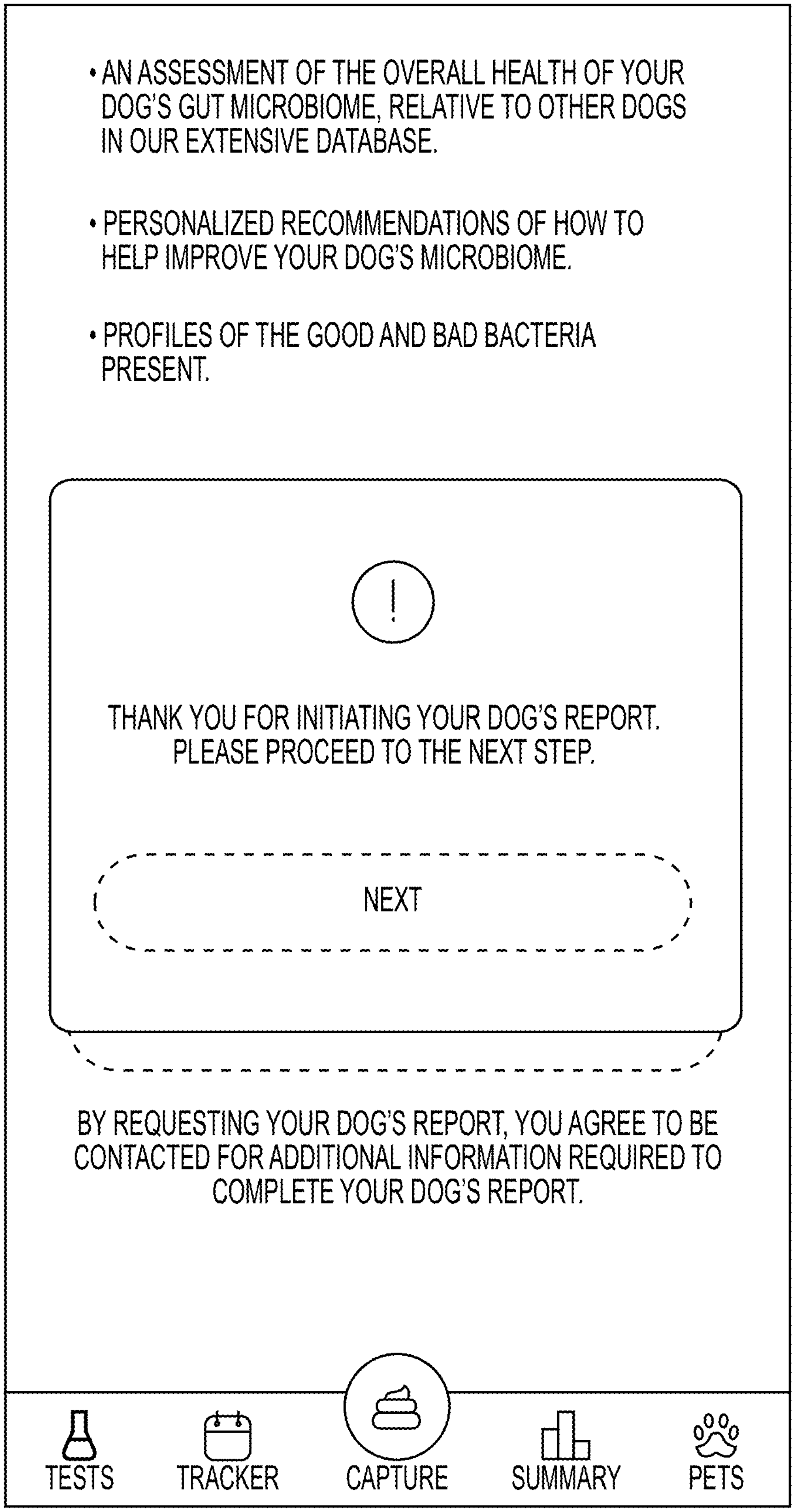

• AN ASSESSMENT OF THE OVERALL HEALTH OF YOUR DOG'S GUT MICROBIOME, RELATIVE TO OTHER DOGS IN OUR EXTENSIVE DATABASE.
• PERSONALIZED RECOMMENDATIONS OF HOW TO HELP IMPROVE YOUR DOG'S MICROBIOME.
• PROFILES OF THE GOOD AND BAD BACTERIA PRESENT.
THANK YOU FOR INITIATING YOUR DOG'S REPORT. PLEASE PROCEED TO THE NEXT STEP.
NEXT
BY REQUESTING YOUR DOG'S REPORT, YOU AGREE TO BE CONTACTED FOR ADDITIONAL INFORMATION REQUIRED TO COMPLETE YOUR DOG'S REPORT.
TESTS
TRACKER
CAPTURE
SUMMARY
PETS

FIG. 4D

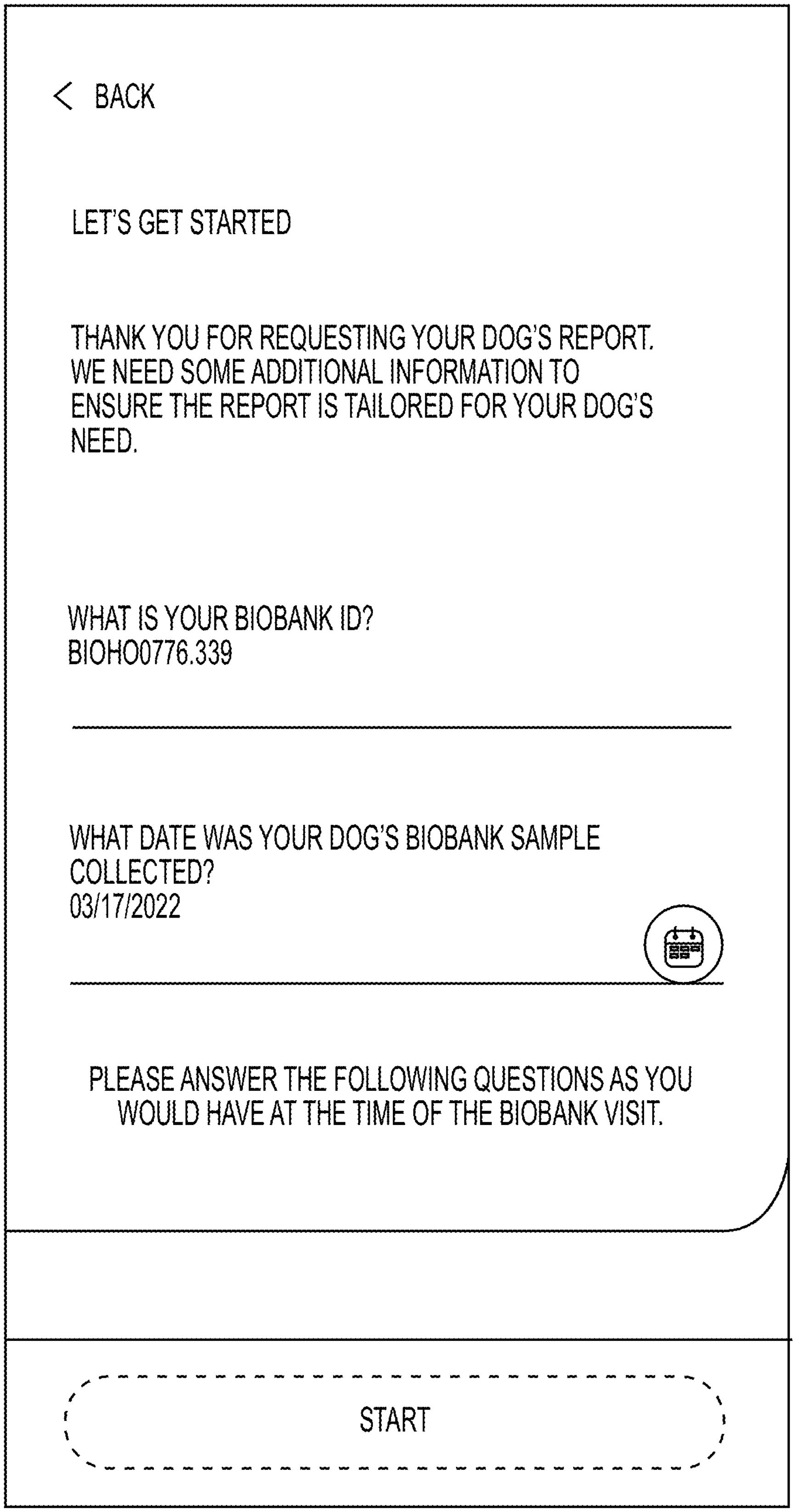

BACK
LET'S GET STARTED
THANK YOU FOR REQUESTING YOUR DOG'S REPORT. WE NEED SOME ADDITIONAL INFORMATION TO ENSURE THE REPORT IS TAILORED FOR YOUR DOG'S NEED.
WHAT IS YOUR BIOBANK ID?
BIOHO0776.339
WHAT DATE WAS YOUR DOG'S BIOBANK SAMPLE COLLECTED?
03/17/2022
PLEASE ANSWER THE FOLLOWING QUESTIONS AS YOU WOULD HAVE AT THE TIME OF THE BIOBANK VISIT.
START

FIG. 4E

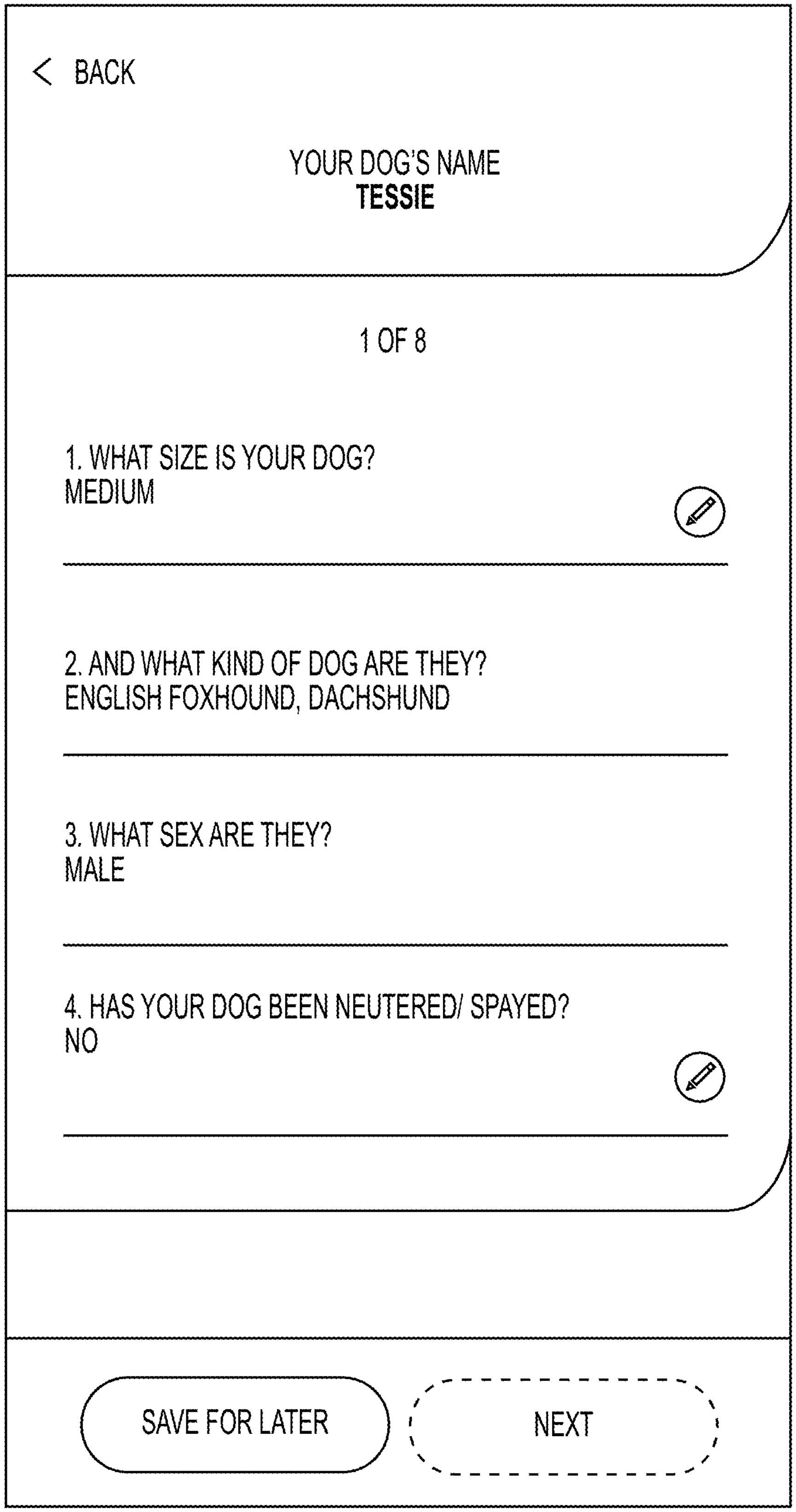

BACK
YOUR DOG'S NAME
TESSIE
1 OF 8
1. WHAT SIZE IS YOUR DOG?
MEDIUM
2. AND WHAT KIND OF DOG ARE THEY?
ENGLISH FOXHOUND, DACHSHUND
3. WHAT SEX ARE THEY?
MALE
4. HAS YOUR DOG BEEN NEUTERED/ SPAYED?
NO
SAVE FOR LATER
NEXT

FIG. 4F

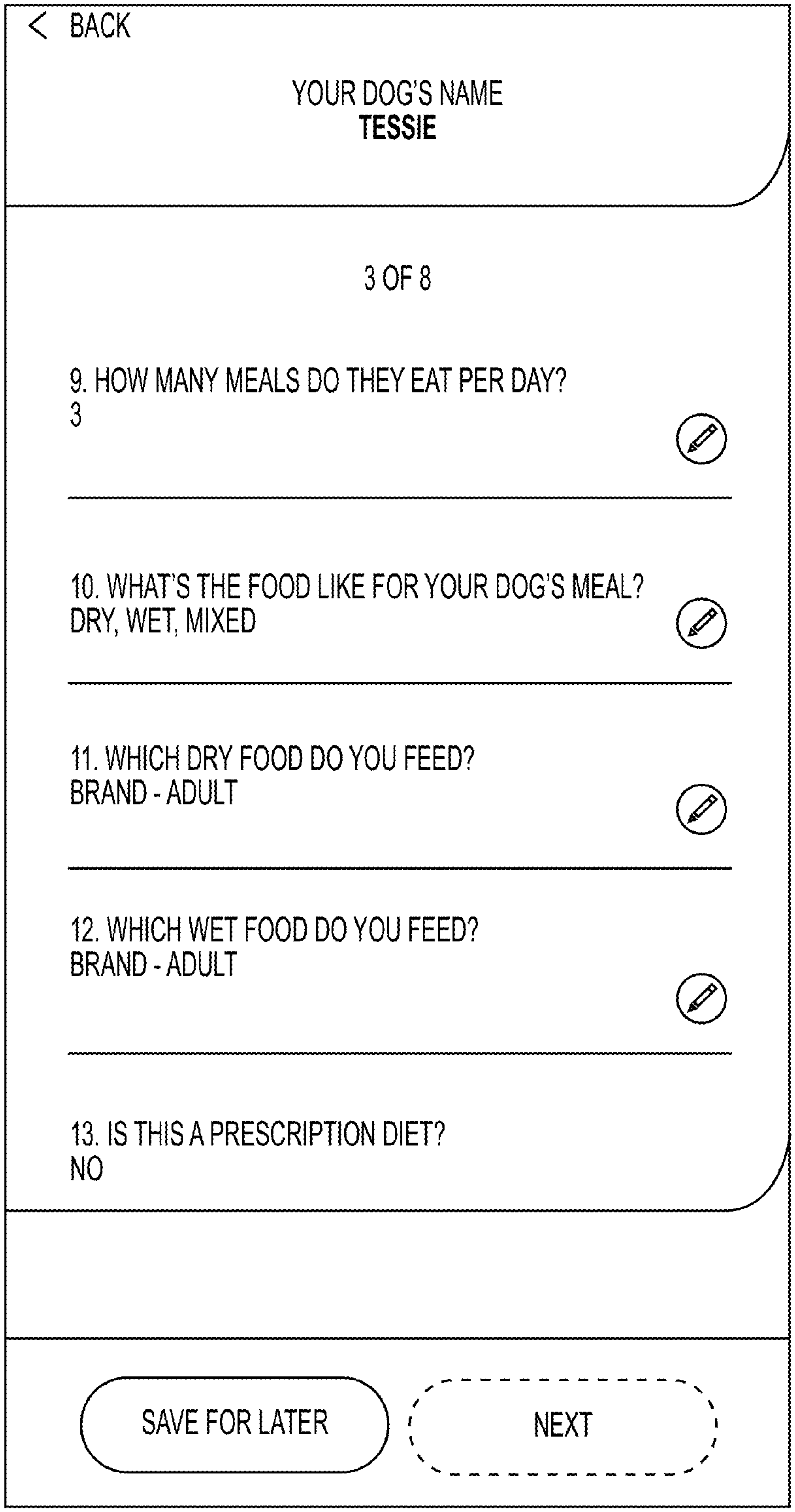

BACK
YOUR DOG'S NAME
TESSIE
3 OF 8
9. HOW MANY MEALS DO THEY EAT PER DAY?
3
10. WHAT'S THE FOOD LIKE FOR YOUR DOG'S MEAL?
DRY, WET, MIXED
11. WHICH DRY FOOD DO YOU FEED?
BRAND - ADULT
12. WHICH WET FOOD DO YOU FEED?
BRAND - ADULT
13. IS THIS A PRESCRIPTION DIET?
NO
SAVE FOR LATER
NEXT

FIG. 4G

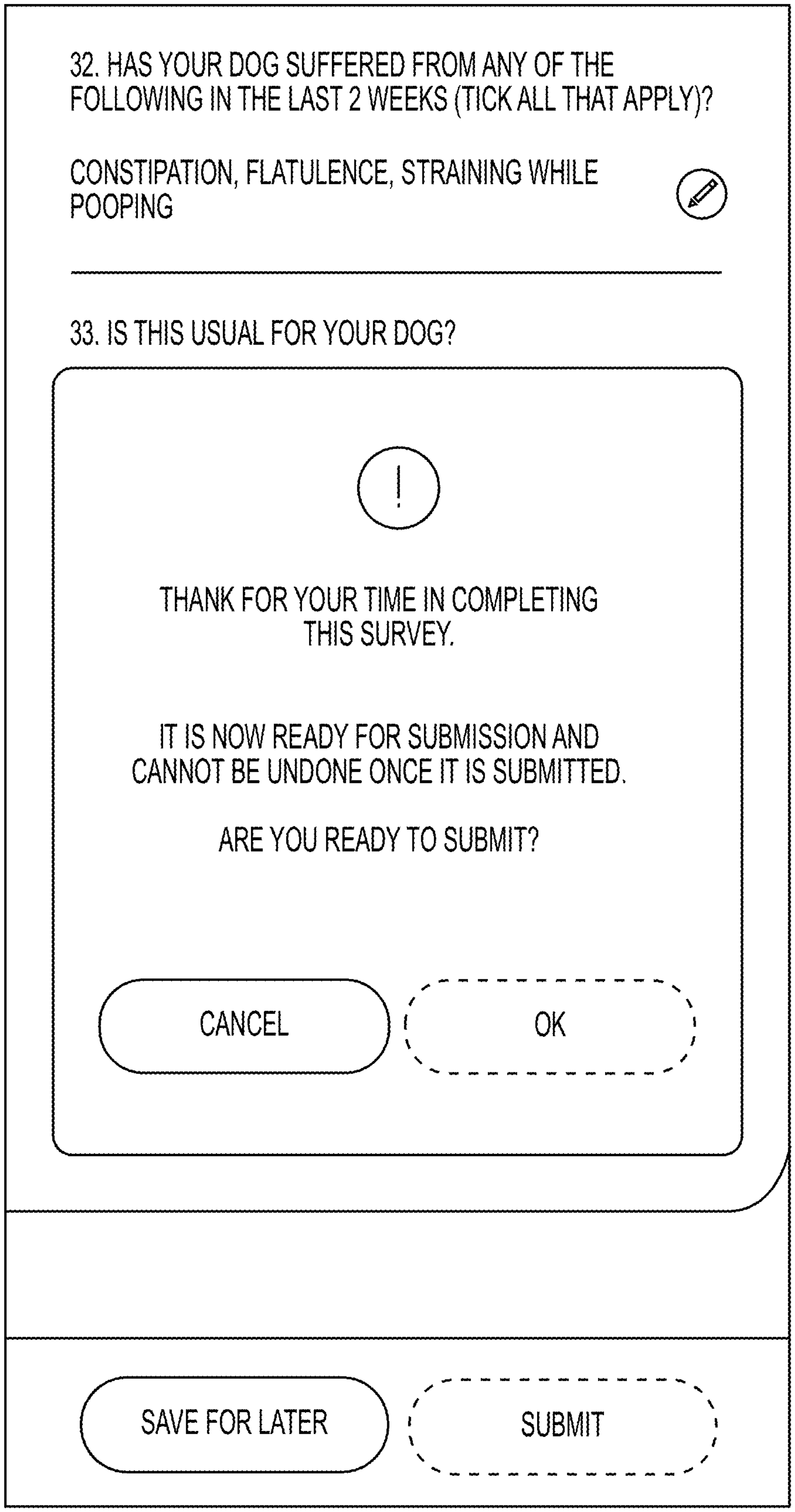

32. HAS YOUR DOG SUFFERED FROM ANY OF THE FOLLOWING IN THE LAST 2 WEEKS (TICK ALL THAT APPLY)?
CONSTIPATION, FLATULENCE, STRAINING WHILE POOPING
33. IS THIS USUAL FOR YOUR DOG?
!
THANK FOR YOUR TIME IN COMPLETING THIS SURVEY.
IT IS NOW READY FOR SUBMISSION AND CANNOT BE UNDONE ONCE IT IS SUBMITTED.
ARE YOU READY TO SUBMIT?
CANCEL
OK
SAVE FOR LATER
SUBMIT

FIG. 4H

• AN ASSESSMENT OF THE OVERALL HEALTH OF YOUR DOG'S GUT MICROBIOME, RELATIVE TO OTHER DOGS IN OUR EXTENSIVE DATABASE.
• PERSONALIZED RECOMMENDATIONS OF HOW TO HELP IMPROVE YOUR DOG'S MICROBIOME.
• PROFILES OF THE GOOD AND BAD BACTERIA PRESENT.
THANK YOU!
WE WILL NOTIFY BY YOUR EMAIL ONCE YOUR DOG'S REPORT IS READY.
OK
BY REQUESTING YOUR DOG'S REPORT, YOU AGREE TO BE CONTACTED FOR ADDITIONAL INFORMATION REQUIRED TO COMPLETE YOUR DOG'S REPORT.
TESTS
TRACKER
CAPTURE
SUMMARY
PETS

FIG. 4I

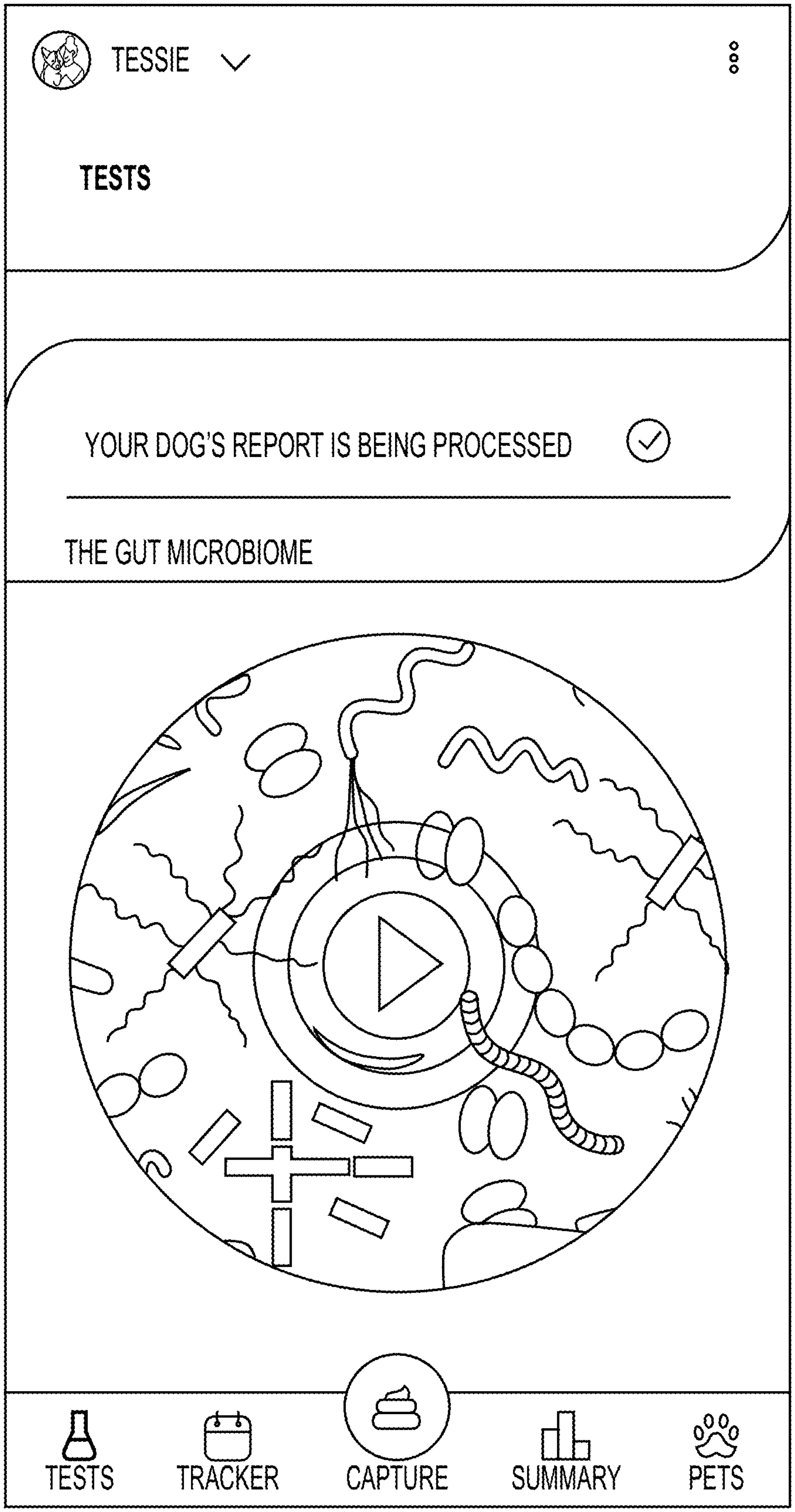

TESSIE
TESTS
YOUR DOG'S REPORT IS BEING PROCESSED
THE GUT MICROBIOME
TESTS
TRACKER
CAPTURE
SUMMARY
PETS

FIG. 4J

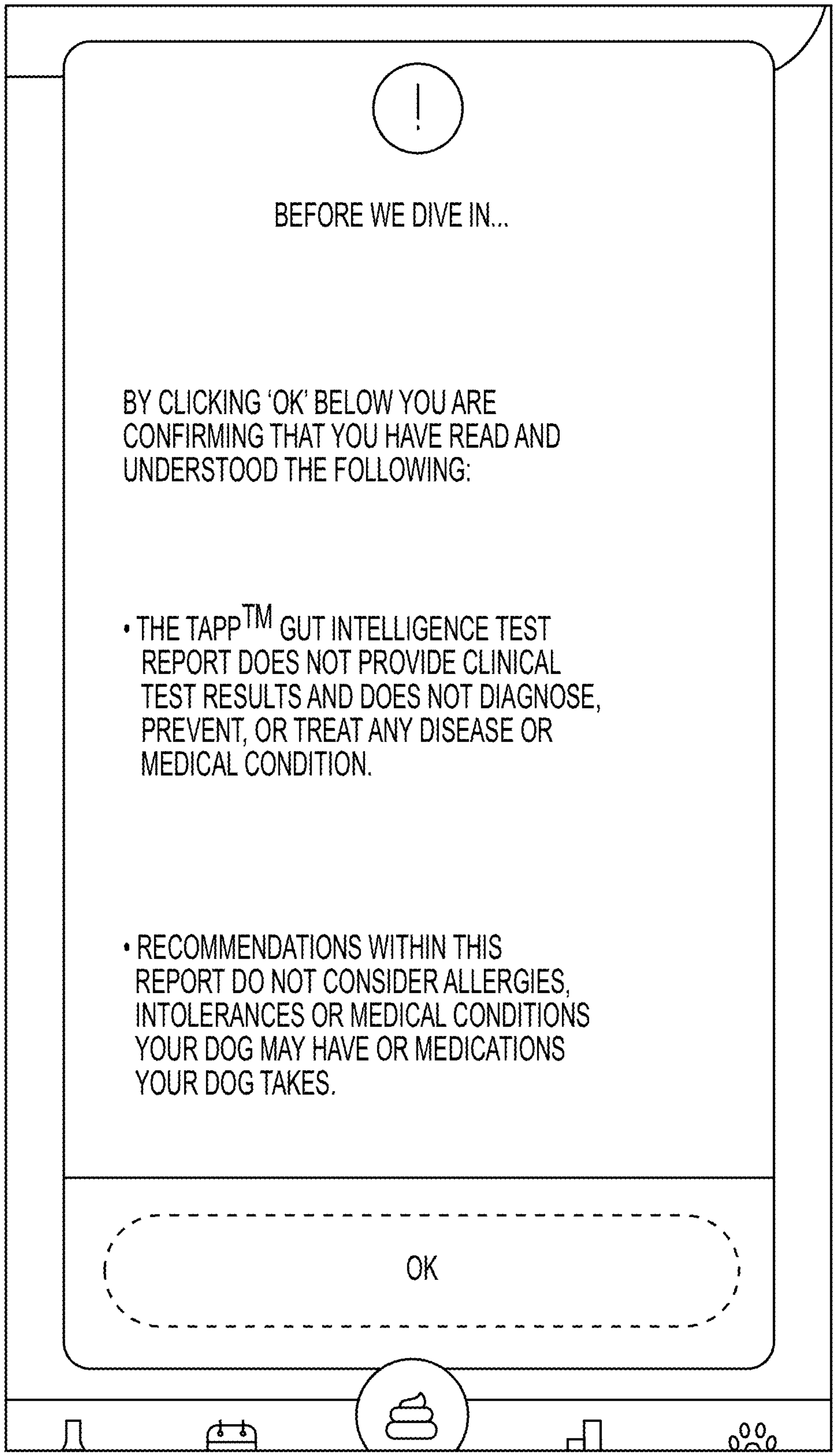

BEFORE WE DIVE IN...
BY CLICKING 'OK' BELOW YOU ARE CONFIRMING THAT YOU HAVE READ AND UNDERSTOOD THE FOLLOWING:
• THE TAPP™ GUT INTELLIGENCE TEST REPORT DOES NOT PROVIDE CLINICAL TEST RESULTS AND DOES NOT DIAGNOSE, PREVENT, OR TREAT ANY DISEASE OR MEDICAL CONDITION.
• RECOMMENDATIONS WITHIN THIS REPORT DO NOT CONSIDER ALLERGIES, INTOLERANCES OR MEDICAL CONDITIONS YOUR DOG MAY HAVE OR MEDICATIONS YOUR DOG TAKES.
OK

FIG. 5A


GUT INTELLIGENCE REPORT
DISCOVER TESSIE'S MICROBIOME
SAMPLE DATE:
SAMPLE ID: HO0776.044
YOUR RESULTS ARE IN!
THE MICROBIOME IS AMAZINGLY COMPLEX, BUT TEST RESULTS DON'T HAVE TO BE.
WE HAVE TAILORED YOUR REPORT TO MAKE IT SIMPLE TO UNDERSTAND, INCLIDING RECOMMENDATIONS OF WAYS TO SUPPORT
CLOSE

FIG. 5B

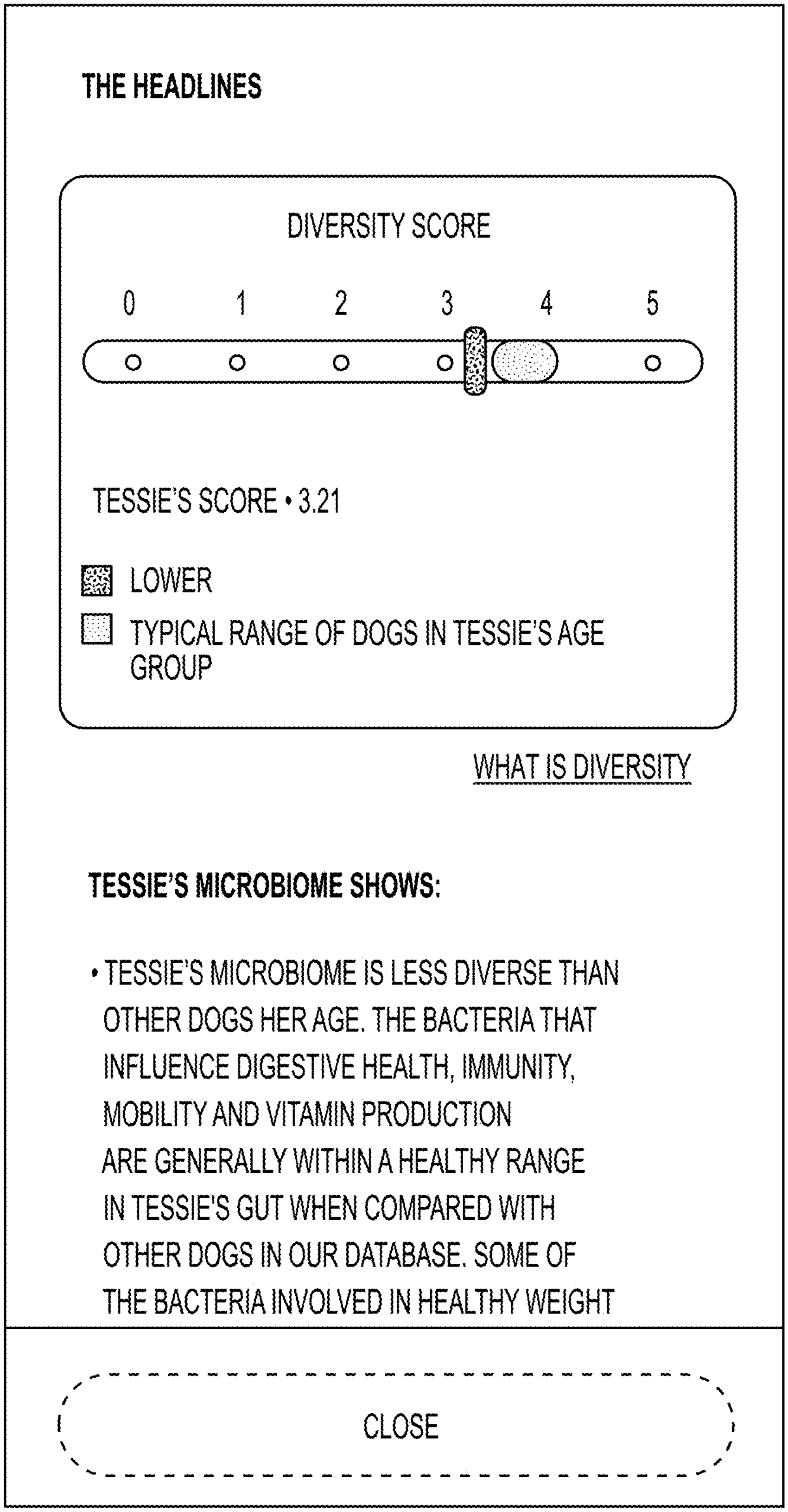

THE HEADLINES
DIVERSITY SCORE
0
1
2
3
4
5
TESSIE'S SCORE • 3.21
LOWER
TYPICAL RANGE OF DOGS IN TESSIE'S AGE GROUP
WHAT IS DIVERSITY
TESSIE'S MICROBIOME SHOWS:
• TESSIE'S MICROBIOME IS LESS DIVERSE THAN OTHER DOGS HER AGE. THE BACTERIA THAT INFLUENCE DIGESTIVE HEALTH, IMMUNITY, MOBILITY AND VITAMIN PRODUCTION ARE GENERALLY WITHIN A HEALTHY RANGE IN TESSIE'S GUT WHEN COMPARED WITH OTHER DOGS IN OUR DATABASE. SOME OF THE BACTERIA INVOLVED IN HEALTHY WEIGHT
CLOSE

FIG. 5C

THE BACTERIA INVOLVED IN HEALTHY WEIGHT AND EXERCISE ARE OUT OF RANGE SO TESSIE MAY NEED SOME SUPPORT IN THESE AREAS.

WHAT THIS MEANS

- WHILST TESSIE IS LIKELY TO BE BENEFITTING FROM THE GOOD THINGS THAT SOME OF THE DIFFERENT BUGS CAN DO FOR HER, SHE MAY NEED SOME HELP TO BALANCE HER MICROBIOME
- AS TESSIE IS A SENIOR AND THE DIVERSITY OF A DOG'S GUT MICROBIOME TENDS TO DECLINE AS A DOG AGES, TESSIE WILL LIKELY NEED SOME HELP TO KEEP IT ON TRACK.

WHY THIS IS IMPORTANT

- THE GUT MICROBIOME IMPACTS MANY AREAS OF HEALTH, INCLUDING DIGESTIVE HEALTH, IMMUNITY, ASPECTS OF HEALTHY LIFESTYLE AND ON-BOARD VITAMIN SUPPLY.
- AS SEVERAL JOBS IN THE GUT ARE SHARED BY DIFFERENT BUGS, IF NUMBERS OF CERTAIN BUGS ARE LOW, OTHER BUGS CAN

CLOSE

FIG. 5D

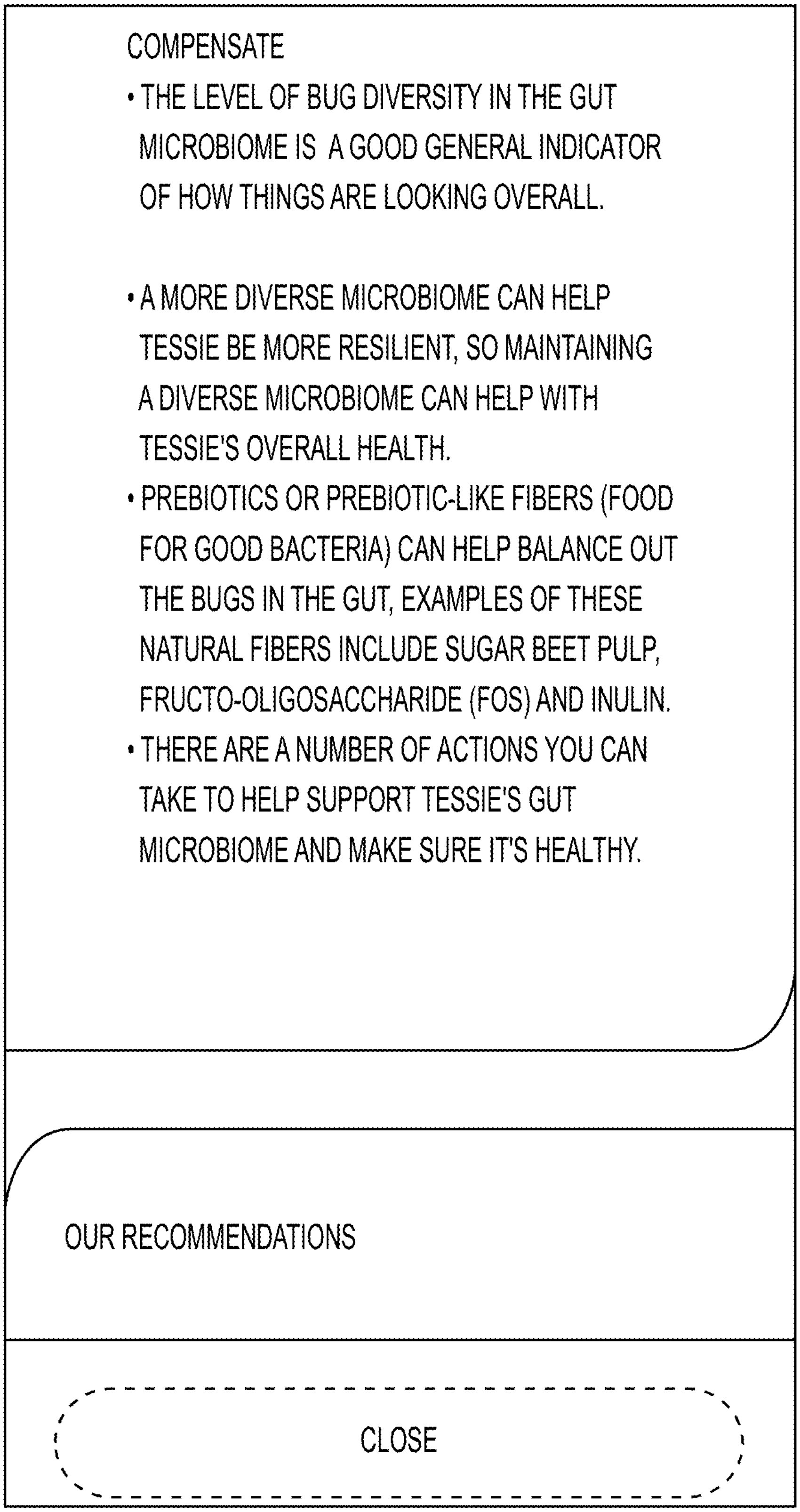

COMPENSATE
• THE LEVEL OF BUG DIVERSITY IN THE GUT MICROBIOME IS A GOOD GENERAL INDICATOR OF HOW THINGS ARE LOOKING OVERALL.
• A MORE DIVERSE MICROBIOME CAN HELP TESSIE BE MORE RESILIENT, SO MAINTAINING A DIVERSE MICROBIOME CAN HELP WITH TESSIE'S OVERALL HEALTH.
• PREBIOTICS OR PREBIOTIC-LIKE FIBERS (FOOD FOR GOOD BACTERIA) CAN HELP BALANCE OUT THE BUGS IN THE GUT, EXAMPLES OF THESE NATURAL FIBERS INCLUDE SUGAR BEET PULP, FRUCTO-OLIGOSACCHARIDE (FOS) AND INULIN.
• THERE ARE A NUMBER OF ACTIONS YOU CAN TAKE TO HELP SUPPORT TESSIE'S GUT MICROBIOME AND MAKE SURE IT'S HEALTHY.
OUR RECOMMENDATIONS
CLOSE

FIG. 5E


OUR RECOMMENDATIONS
CONSIDER THE AMOUNT OF EXERCISE TESSIE GETS. WHY?
ENSURE TESSIE'S DIET CONTAINS NATURAL FIBERS. WHY?
DISCOVER YOUR DOG'S RECOMMENDED DIETS
THE DETAILS
TESSIE'S MICROBIOME CAN GIVE US CLUES ABOUT HER WELLBEING IN 6 IMPORTANT AREAS:
CLOSE

FIG. 5F

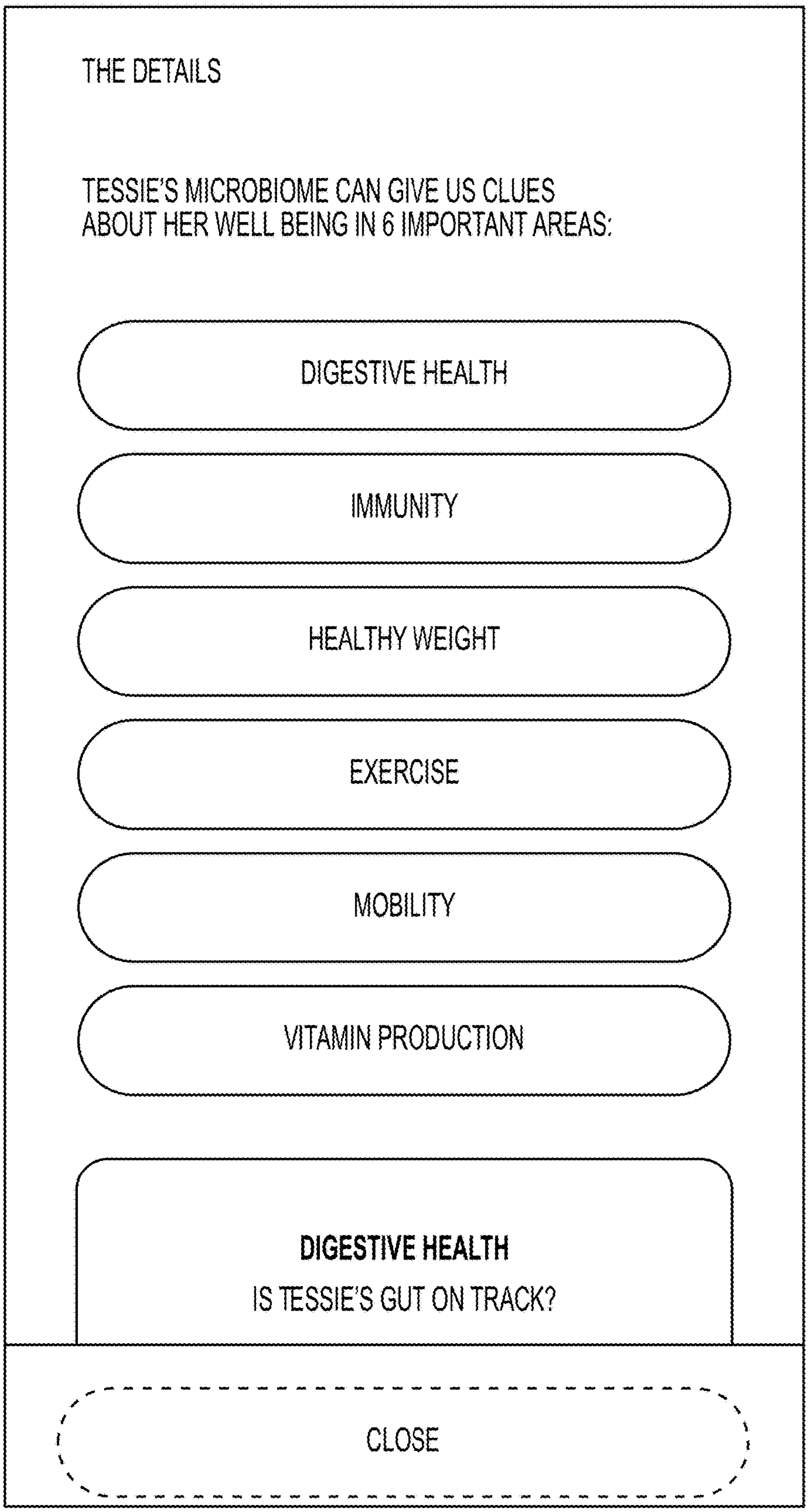

THE DETAILS
TESSIE'S MICROBIOME CAN GIVE US CLUES
ABOUT HER WELL BEING IN 6 IMPORTANT AREAS:
DIGESTIVE HEALTH
IMMUNITY
HEALTHY WEIGHT
EXERCISE
MOBILITY
VITAMIN PRODUCTION
DIGESTIVE HEALTH
IS TESSIE'S GUT ON TRACK?
CLOSE

FIG. 5G

DIGESTIVE HEALTH

IS TESSIE'S GUT ON TRACK?

GOOD

THE GUT MICROBIOME PLAYS AN IMPORTANT PART IN A VARIETY OF DIFFERENT TASKS THAT HELP SUPPORT TESSIE'S OVERALL DIGESTIVE HEALTH.

WHAT YOU CAN DO FOR TESSIE? —

TESSIE'S DIGESTIVE SYSTEM APPEARS TO BE IN PRETTY GOOD SHAPE, BASED ON THE OVERALL MIX OF BACTERIA WE'VE SEEN YOU TOLD US TESSIE HAS HAD SIGNS OF DIGESTIVE UPSET RECENTLY (FLATULENCE) CHECK POOP QUALITY FOR AN OVERALL CURRENT INDICATOR (YOU CAN USE THE IN-APP POOP CHECKER): IF IT'S FREQUENTLY OFF-TRACK. YOU MAY WANT TO TRY FEEDING TESSIE A DIET CONTAINING

CLOSE

FIG. 5H

DEEP DIVE

DIGESTIVE HEALTH

THE GUT MICROBIOME PLAYS AN IMPORTANT PART IN A VARIETY OF DIFFERENT TASKS THAT HELP SUPPORT TESSIE'S OVERALL DIGESTIVE HEALTH.

WE KNOW THAT TRILLIONS OF BACTERIA CONTRIBUTE TO OVERALL DIGESTIVE HEALTH. THAT'S A LOT OF BUGS TO STUDY! OUR RESEARCH SO FAR ALLOWS US TO ZOOM INTO SEVEN BUGS THAT WE KNOW PLAY AN IMPORTANT ROLE IN TESSIE'S DIGESTIVE HEALTH.

WHAT YOU CAN DO FOR TESSIE? —

TESSIE'S DIGESTIVE SYSTEM APPEARS TO BE IN PRETTY GOOD SHAPE, BASED ON THE OVERALL MIX OF BACTERIA WE'VE SEEN YOU TOLD US TESSIE HAS HAD SIGNS OF DIGESTIVE UPSET RECENTLY (FLATULENCE). CHECK POOP QUALITY FOR AN OVERALL CURRENT INDICATOR

BACK TO RESULTS

FIG. 5I

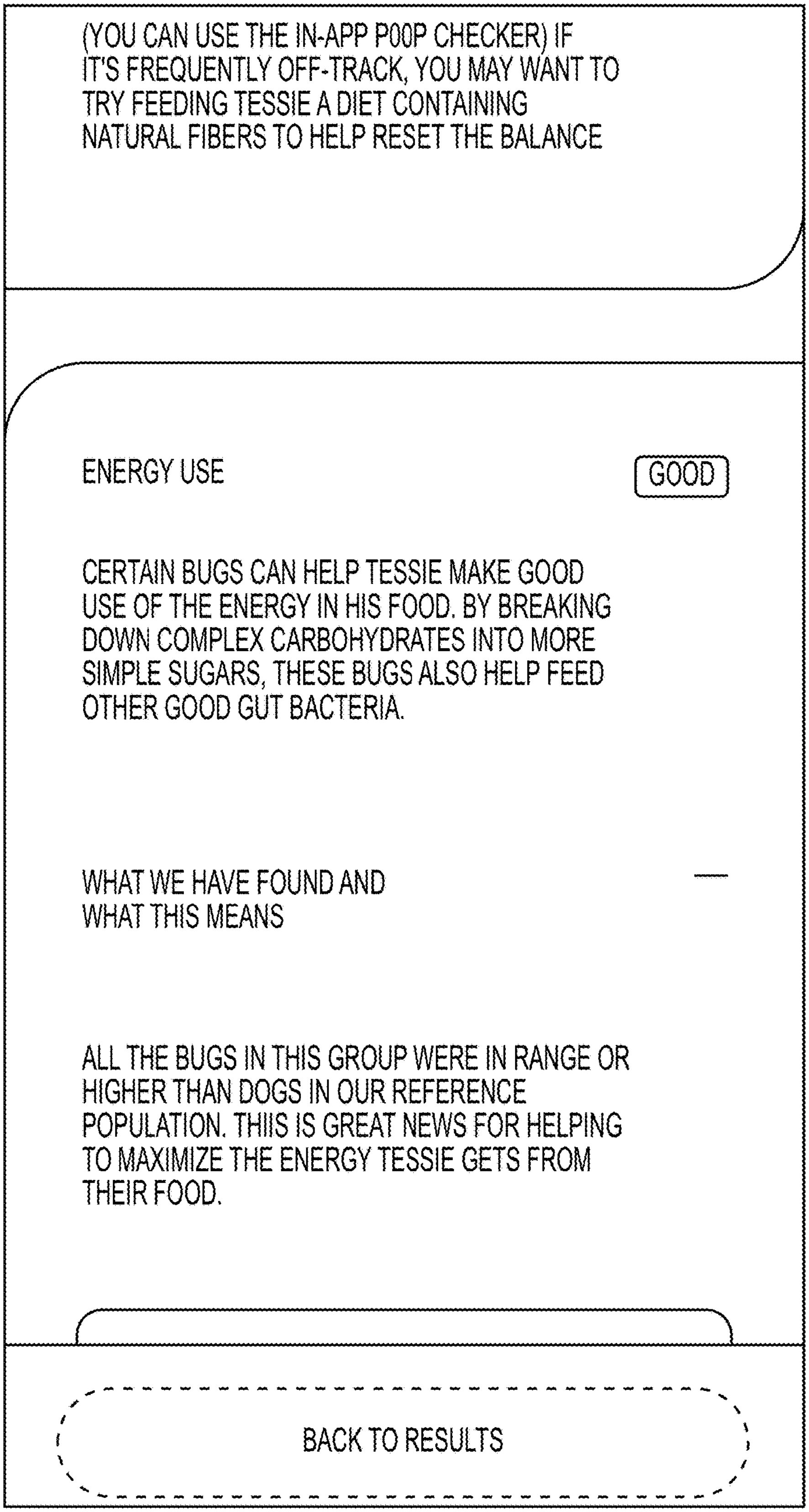

(YOU CAN USE THE IN-APP POOP CHECKER) IF
IT'S FREQUENTLY OFF-TRACK, YOU MAY WANT TO
TRY FEEDING TESSIE A DIET CONTAINING
NATURAL FIBERS TO HELP RESET THE BALANCE
ENERGY USE
GOOD
CERTAIN BUGS CAN HELP TESSIE MAKE GOOD
USE OF THE ENERGY IN HIS FOOD. BY BREAKING
DOWN COMPLEX CARBOHYDRATES INTO MORE
SIMPLE SUGARS, THESE BUGS ALSO HELP FEED
OTHER GOOD GUT BACTERIA.
WHAT WE HAVE FOUND AND
WHAT THIS MEANS
ALL THE BUGS IN THIS GROUP WERE IN RANGE OR
HIGHER THAN DOGS IN OUR REFERENCE
POPULATION. THIIS IS GREAT NEWS FOR HELPING
TO MAXIMIZE THE ENERGY TESSIE GETS FROM
THEIR FOOD.
BACK TO RESULTS

FIG. 5J

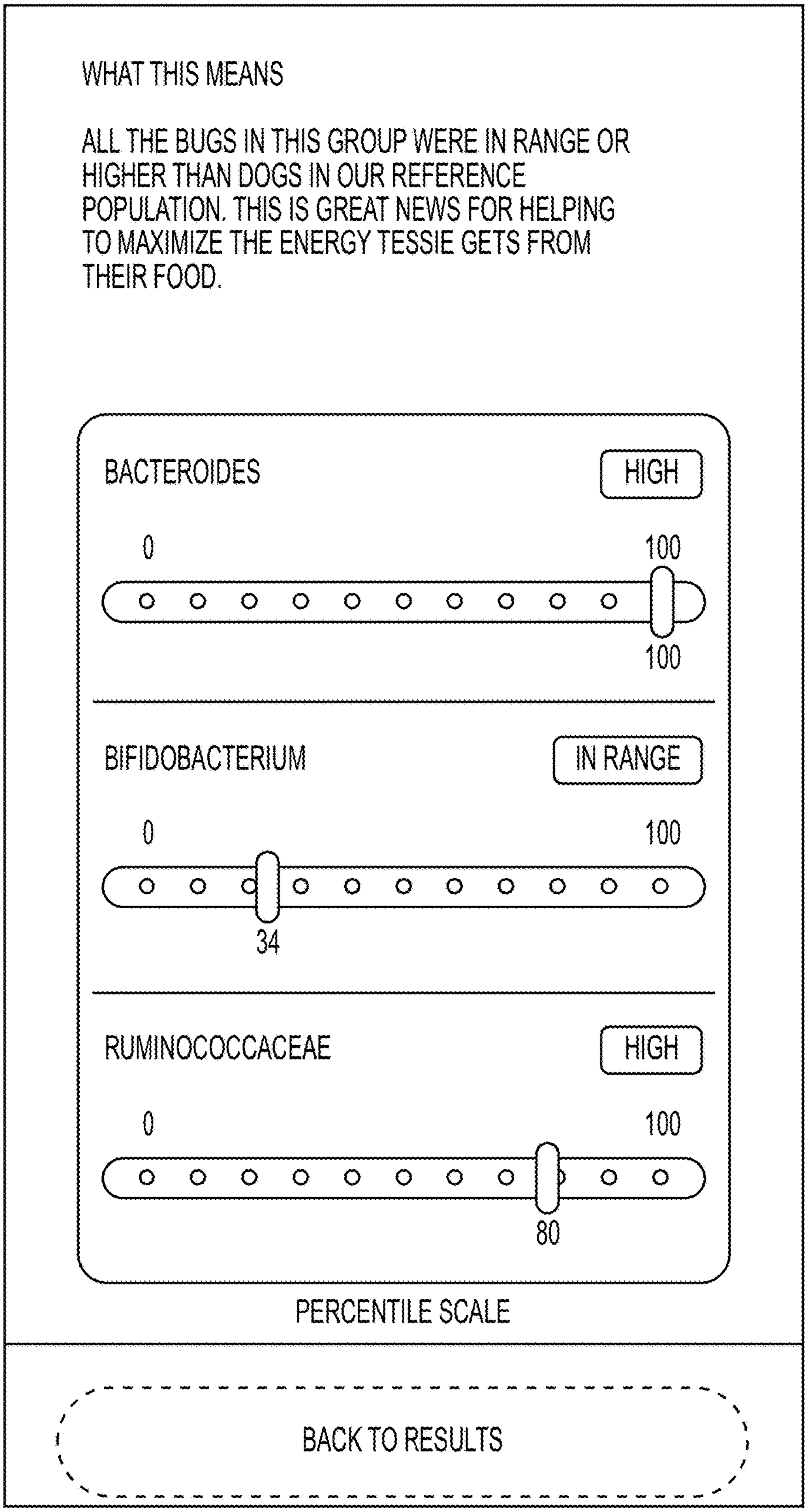

WHAT THIS MEANS
ALL THE BUGS IN THIS GROUP WERE IN RANGE OR HIGHER THAN DOGS IN OUR REFERENCE POPULATION. THIS IS GREAT NEWS FOR HELPING TO MAXIMIZE THE ENERGY TESSIE GETS FROM THEIR FOOD.
BACTEROIDES
HIGH
0
100
100
BIFIDOBACTERIUM
IN RANGE
0
100
34
RUMINOCOCCACEAE
HIGH
0
100
80
PERCENTILE SCALE
BACK TO RESULTS

FIG. 5K

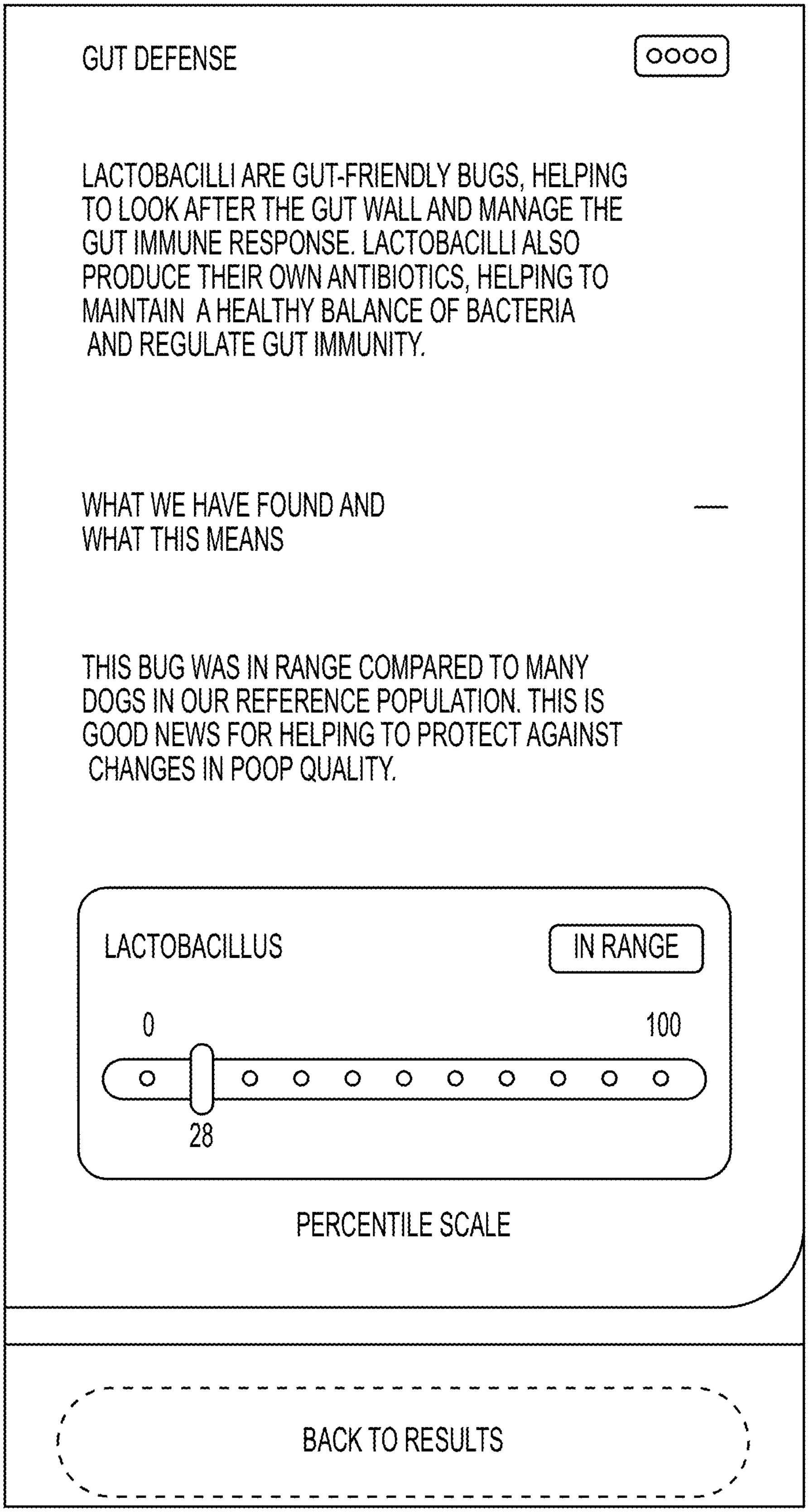

GUT DEFENSE
LACTOBACILLI ARE GUT-FRIENDLY BUGS, HELPING TO LOOK AFTER THE GUT WALL AND MANAGE THE GUT IMMUNE RESPONSE. LACTOBACILLI ALSO PRODUCE THEIR OWN ANTIBIOTICS, HELPING TO MAINTAIN A HEALTHY BALANCE OF BACTERIA AND REGULATE GUT IMMUNITY.
WHAT WE HAVE FOUND AND WHAT THIS MEANS
THIS BUG WAS IN RANGE COMPARED TO MANY DOGS IN OUR REFERENCE POPULATION. THIS IS GOOD NEWS FOR HELPING TO PROTECT AGAINST CHANGES IN POOP QUALITY.
LACTOBACILLUS
IN RANGE
0
100
28
PERCENTILE SCALE
BACK TO RESULTS

FIG. 5L

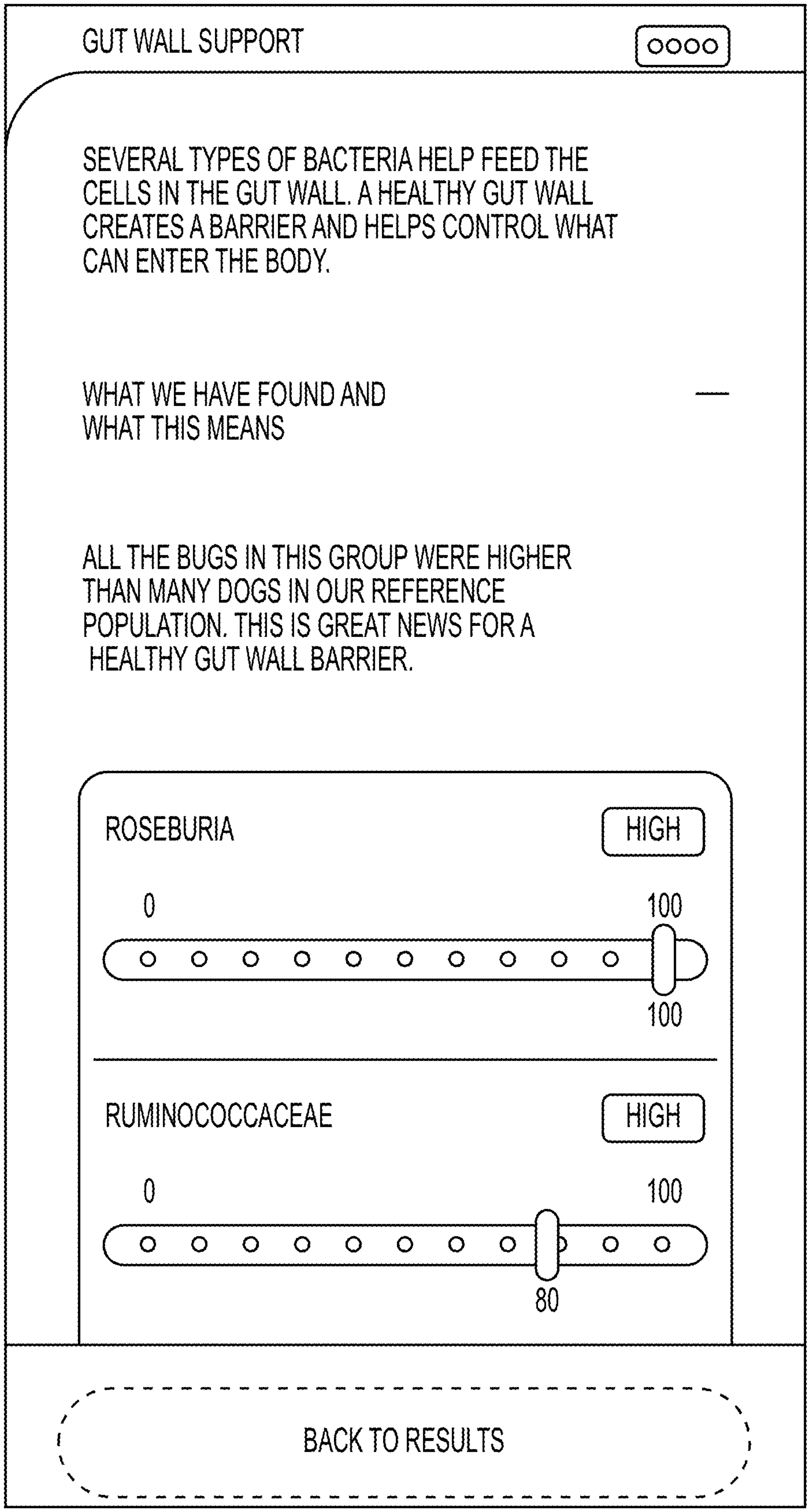

GUT WALL SUPPORT
OOOO
SEVERAL TYPES OF BACTERIA HELP FEED THE CELLS IN THE GUT WALL. A HEALTHY GUT WALL CREATES A BARRIER AND HELPS CONTROL WHAT CAN ENTER THE BODY.
WHAT WE HAVE FOUND AND WHAT THIS MEANS
ALL THE BUGS IN THIS GROUP WERE HIGHER THAN MANY DOGS IN OUR REFERENCE POPULATION. THIS IS GREAT NEWS FOR A HEALTHY GUT WALL BARRIER.
ROSEBURIA
HIGH
0
100
100
RUMINOCOCCACEAE
HIGH
0
100
80
BACK TO RESULTS

FIG. 5M

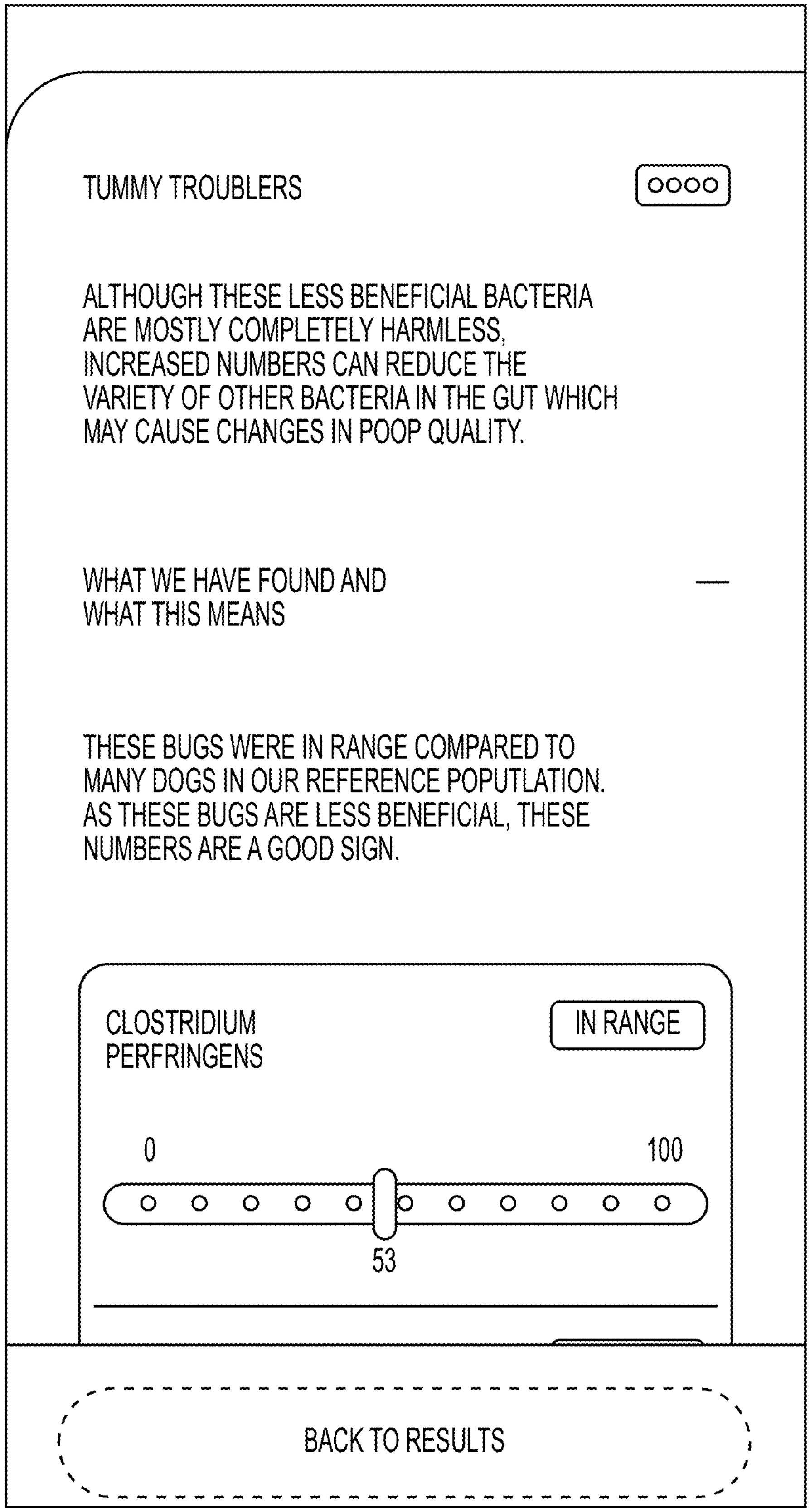

TUMMY TROUBLERS
0000
ALTHOUGH THESE LESS BENEFICIAL BACTERIA ARE MOSTLY COMPLETELY HARMLESS, INCREASED NUMBERS CAN REDUCE THE VARIETY OF OTHER BACTERIA IN THE GUT WHICH MAY CAUSE CHANGES IN POOP QUALITY.
WHAT WE HAVE FOUND AND WHAT THIS MEANS
—
THESE BUGS WERE IN RANGE COMPARED TO MANY DOGS IN OUR REFERENCE POPUTLATION. AS THESE BUGS ARE LESS BENEFICIAL, THESE NUMBERS ARE A GOOD SIGN.
CLOSTRIDIUM PERFRINGENS
IN RANGE
0
100
53
BACK TO RESULTS

FIG. 5N

IMMUNITY

IS TESSIE READY TO FIGHT OFF BAD BUGS?

GOOD

WE ALL KNOW THAT A HEALTHY IMMUNE SYSTEM HELPS FIGHT ILLNESS. DID YOU KNOW THAT SOME GUT BACTERIA HELP KEEP THE IMMUNE SYSTEM IN BALANCE AND HELP DAMPEN DOWN CERTAIN ALLERGIC RESPONSES? LET 'S SEE WHAT TESSIE'S MICROBIOME TELLS US ABOUT HIS IMMUNITY.

WHAT YOU CAN DO FOR TESSIE? —

TESSIE HAS AN EXCELLENT BALANCE OF IMMUNE-SUPPORTING BACTERIA IN HER GUT MICROBIOME. THIS SUGGESTS TESSIE MAY BENEFIT FROM A LOWERED CHANCE OF ALLERGIC RESPONSES AND

CLOSE

FIG. 5O

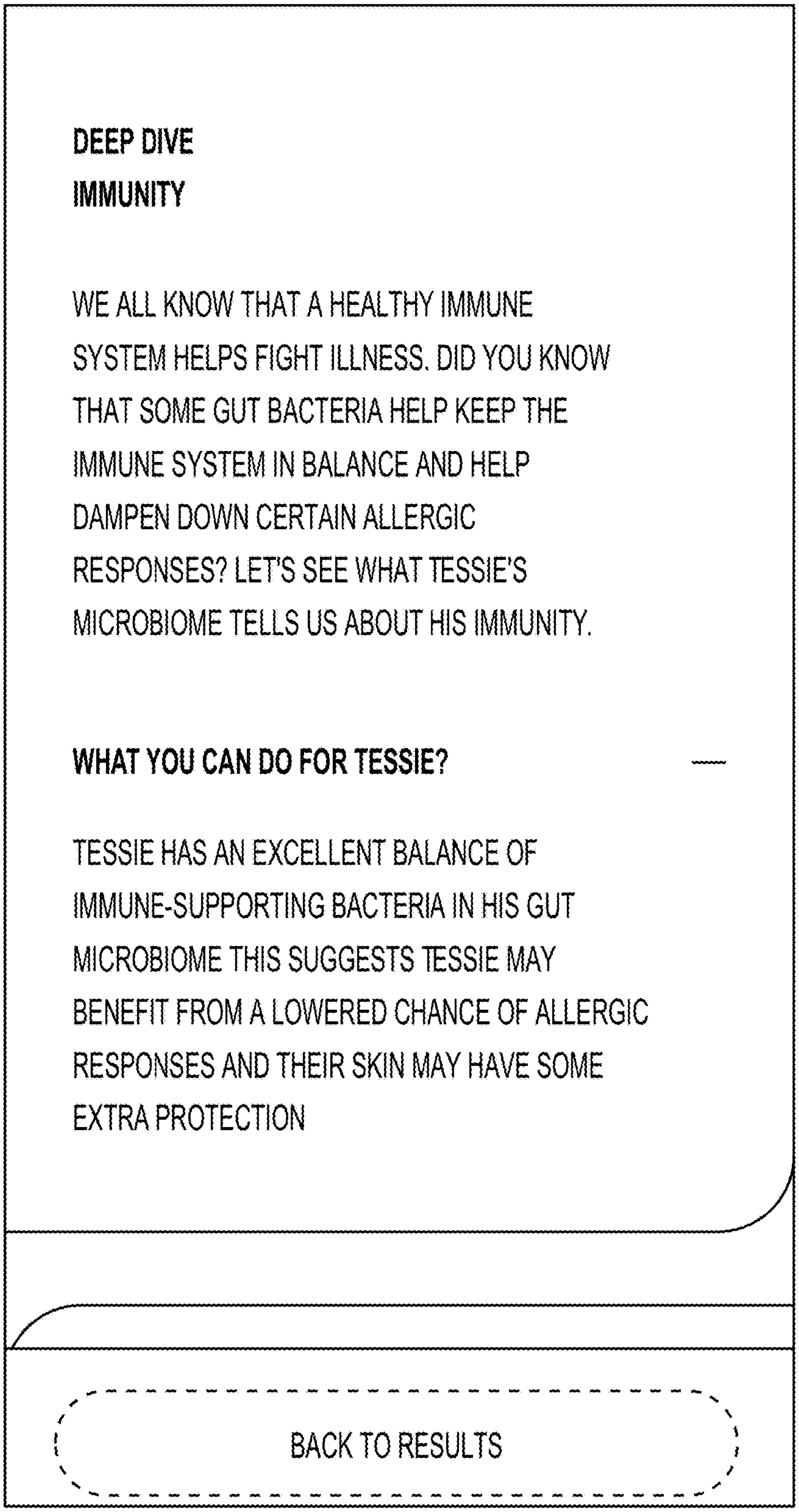

DEEP DIVE
IMMUNITY
WE ALL KNOW THAT A HEALTHY IMMUNE SYSTEM HELPS FIGHT ILLNESS. DID YOU KNOW THAT SOME GUT BACTERIA HELP KEEP THE IMMUNE SYSTEM IN BALANCE AND HELP DAMPEN DOWN CERTAIN ALLERGIC RESPONSES? LET'S SEE WHAT TESSIE'S MICROBIOME TELLS US ABOUT HIS IMMUNITY.
WHAT YOU CAN DO FOR TESSIE?
TESSIE HAS AN EXCELLENT BALANCE OF IMMUNE-SUPPORTING BACTERIA IN HIS GUT MICROBIOME THIS SUGGESTS TESSIE MAY BENEFIT FROM A LOWERED CHANCE OF ALLERGIC RESPONSES AND THEIR SKIN MAY HAVE SOME EXTRA PROTECTION
BACK TO RESULTS

FIG. 5P

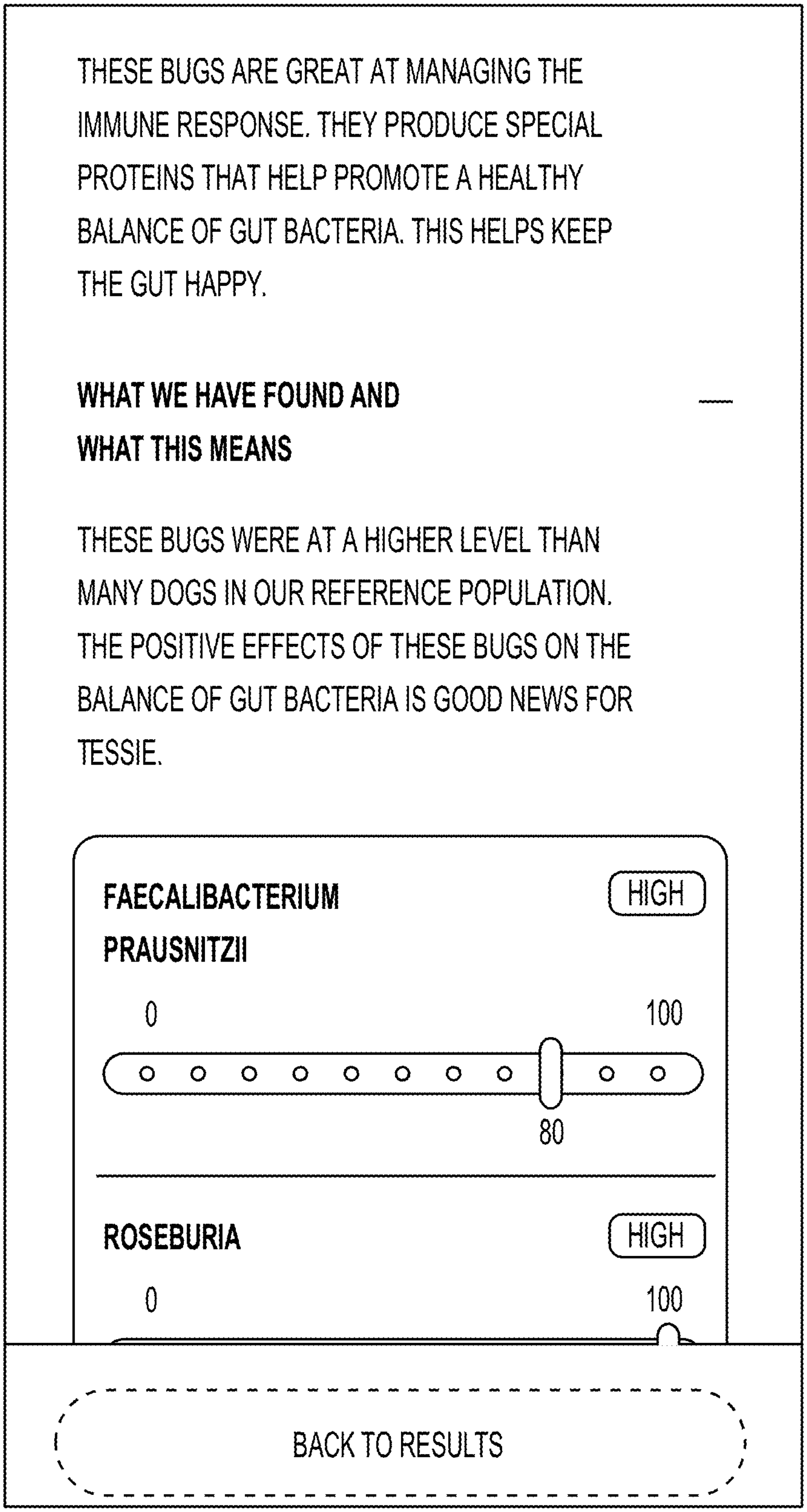

THESE BUGS ARE GREAT AT MANAGING THE IMMUNE RESPONSE. THEY PRODUCE SPECIAL PROTEINS THAT HELP PROMOTE A HEALTHY BALANCE OF GUT BACTERIA. THIS HELPS KEEP THE GUT HAPPY.
WHAT WE HAVE FOUND AND WHAT THIS MEANS
THESE BUGS WERE AT A HIGHER LEVEL THAN MANY DOGS IN OUR REFERENCE POPULATION. THE POSITIVE EFFECTS OF THESE BUGS ON THE BALANCE OF GUT BACTERIA IS GOOD NEWS FOR TESSIE.
FAECALIBACTERIUM PRAUSNITZII
HIGH
0
100
80
ROSEBURIA
HIGH
0
100
BACK TO RESULTS

FIG. 5Q

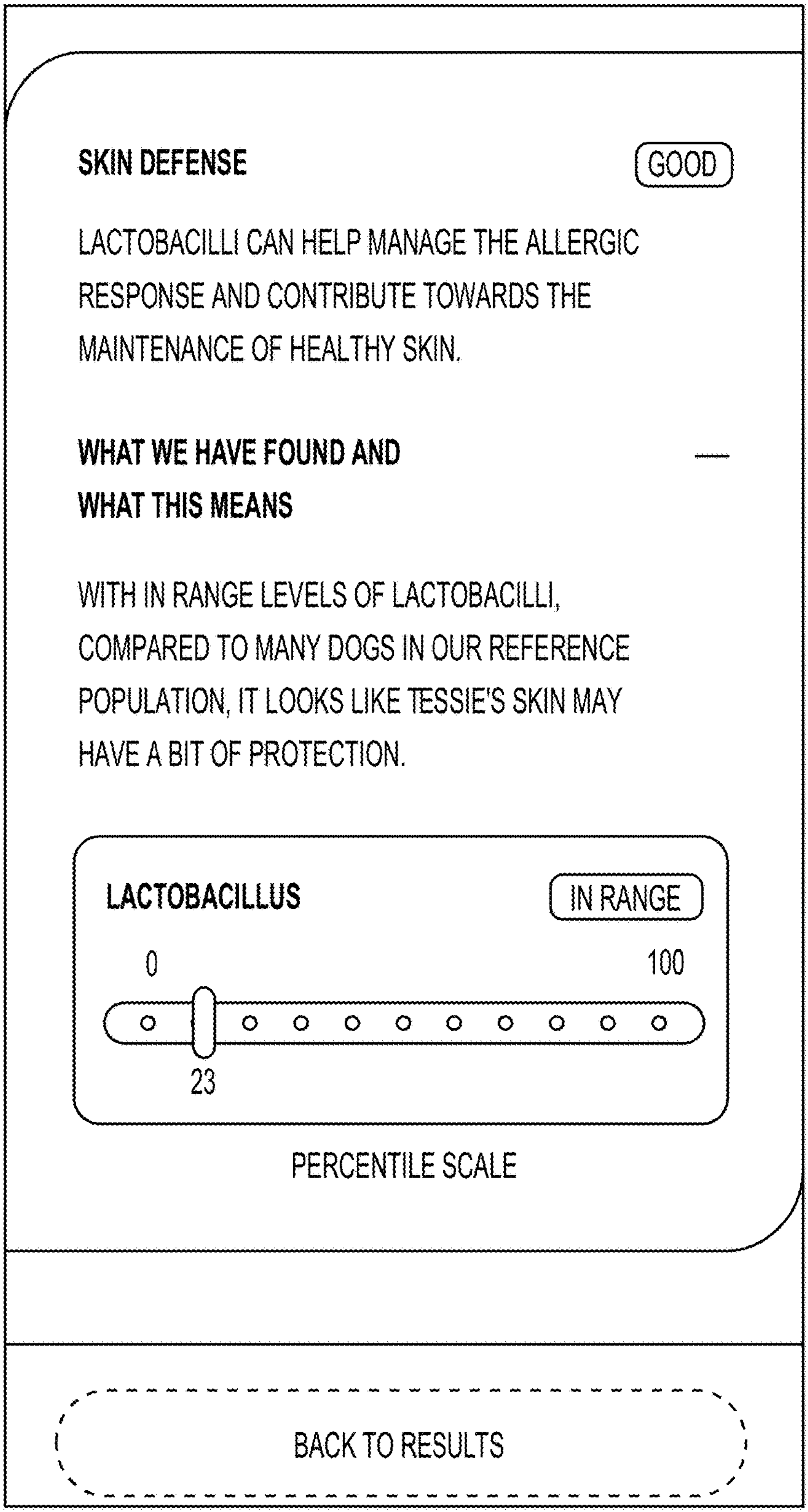

SKIN DEFENSE
GOOD
LACTOBACILLI CAN HELP MANAGE THE ALLERGIC RESPONSE AND CONTRIBUTE TOWARDS THE MAINTENANCE OF HEALTHY SKIN.
WHAT WE HAVE FOUND AND WHAT THIS MEANS
—
WITH IN RANGE LEVELS OF LACTOBACILLI, COMPARED TO MANY DOGS IN OUR REFERENCE POPULATION, IT LOOKS LIKE TESSIE'S SKIN MAY HAVE A BIT OF PROTECTION.
LACTOBACILLUS
IN RANGE
0
100
23
PERCENTILE SCALE
BACK TO RESULTS

FIG. 5R

HEALTHY WEIGHT

IS TESSIE HAVING HEALTHY WEIGHT?

OKAY

READING THE GUT MICROBIOME PROFILE CAN TELL US A LITTLE ABOUT TESSIE'S INBUILT ABILITY TO MAINTAIN A HEALTHY WEIGHT.

WHAT YOU CAN DO FOR TESSIE? —

1. CLUES IN THE MICROBIOME INDICATE TESSIE COULD FIND IT A LITTLE MORE DIFFICULT TO MAINTAIN A HEALTHY WEIGHT
2. FIGURING OUT HOW MUCH TO FEED YOUR DOG CAN BE TRICKY AND DEPENDS ON VARIABLES INCLUDING ACTIVITY LEVEL AND AGE USE THE FEEDING GUIDE ON OUR PRODUCTS TO HELP TESSIE MAINTAIN A HEALTHY

CLOSE

FIG. 5S

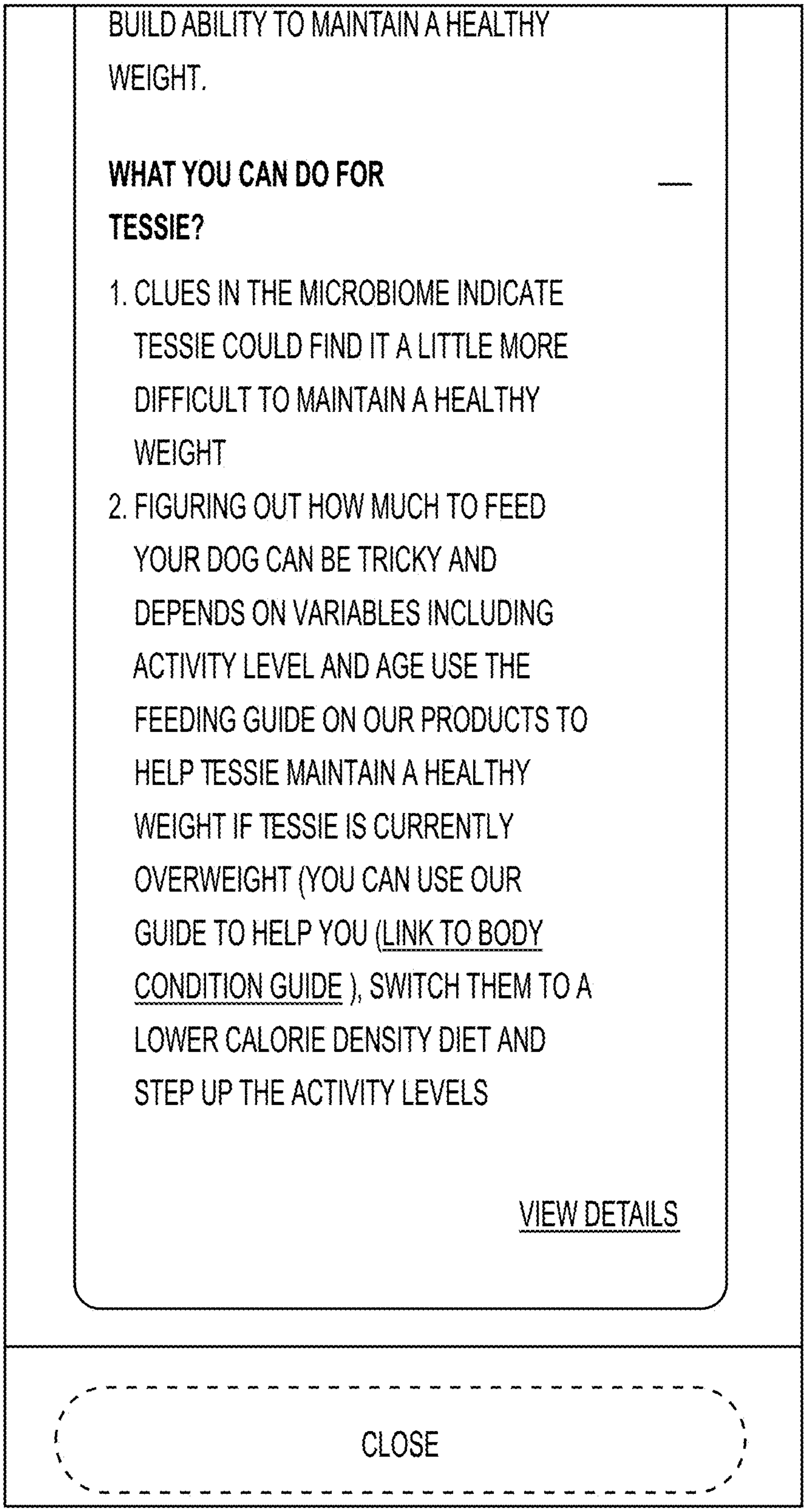

BUILD ABILITY TO MAINTAIN A HEALTHY WEIGHT.
WHAT YOU CAN DO FOR TESSIE?
—
1. CLUES IN THE MICROBIOME INDICATE TESSIE COULD FIND IT A LITTLE MORE DIFFICULT TO MAINTAIN A HEALTHY WEIGHT
2. FIGURING OUT HOW MUCH TO FEED YOUR DOG CAN BE TRICKY AND DEPENDS ON VARIABLES INCLUDING ACTIVITY LEVEL AND AGE USE THE FEEDING GUIDE ON OUR PRODUCTS TO HELP TESSIE MAINTAIN A HEALTHY WEIGHT IF TESSIE IS CURRENTLY OVERWEIGHT (YOU CAN USE OUR GUIDE TO HELP YOU (LINK TO BODY CONDITION GUIDE), SWITCH THEM TO A LOWER CALORIE DENSITY DIET AND STEP UP THE ACTIVITY LEVELS
VIEW DETAILS
CLOSE

FIG. 5T

DEEP DIVE

HEALTHY WEIGHT

READING THE GUT MICROBIOME PROFILE CAN TELL US A LITTLE ABOUT TESSIE'S IN-BUILT ABILITY TO MAINTAIN A HEALTHY WEIGHT.

WHAT YOU CAN DO FOR TESSIE? —

1. CLUES IN THE MICROBIOME INDICATE TESSIE COULD FIND IT A LITTLE MORE DIFFICULT TO MAINTAIN A HEALTHY WEIGHT
2. FIGURING OUT HOW MUCH TO FEED YOUR DOG CAN BE TRICKY AND DEPENDS ON VARIABLES INCLUDING ACTIVITY LEVEL AND AGE USE THE FEEDING GUIDE ON OUR PRODUCTS TO HELP TESSIE MAINTAIN A HEALTHY WEIGHT IF TESSIE IS CURRENTLY OVERWEIGHT (YOU CAN USE OUR GUIDE TO HELP YOU (LINK TO BODY CONDITION GUIDE), SWITCH THEM TO A LOWER CALORIE DENSITY DIET AND STEP UP THE ACTIVITY LEVELS

BACK TO RESULTS

FIG. 5U

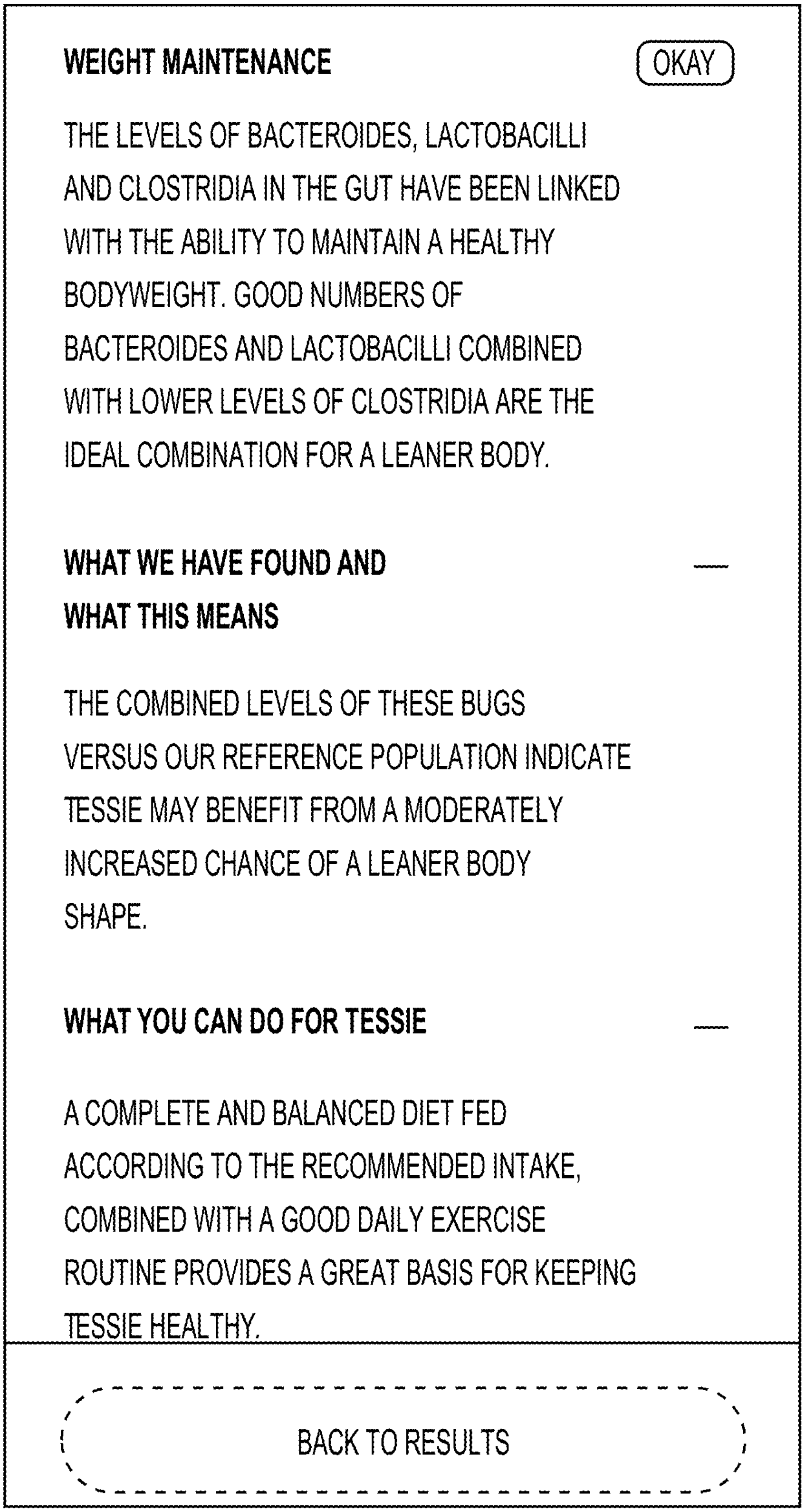

WEIGHT MAINTENANCE
OKAY
THE LEVELS OF BACTEROIDES, LACTOBACILLI AND CLOSTRIDIA IN THE GUT HAVE BEEN LINKED WITH THE ABILITY TO MAINTAIN A HEALTHY BODYWEIGHT. GOOD NUMBERS OF BACTEROIDES AND LACTOBACILLI COMBINED WITH LOWER LEVELS OF CLOSTRIDIA ARE THE IDEAL COMBINATION FOR A LEANER BODY.
WHAT WE HAVE FOUND AND WHAT THIS MEANS
THE COMBINED LEVELS OF THESE BUGS VERSUS OUR REFERENCE POPULATION INDICATE TESSIE MAY BENEFIT FROM A MODERATELY INCREASED CHANCE OF A LEANER BODY SHAPE.
WHAT YOU CAN DO FOR TESSIE
A COMPLETE AND BALANCED DIET FED ACCORDING TO THE RECOMMENDED INTAKE, COMBINED WITH A GOOD DAILY EXERCISE ROUTINE PROVIDES A GREAT BASIS FOR KEEPING TESSIE HEALTHY.
BACK TO RESULTS

FIG. 5V

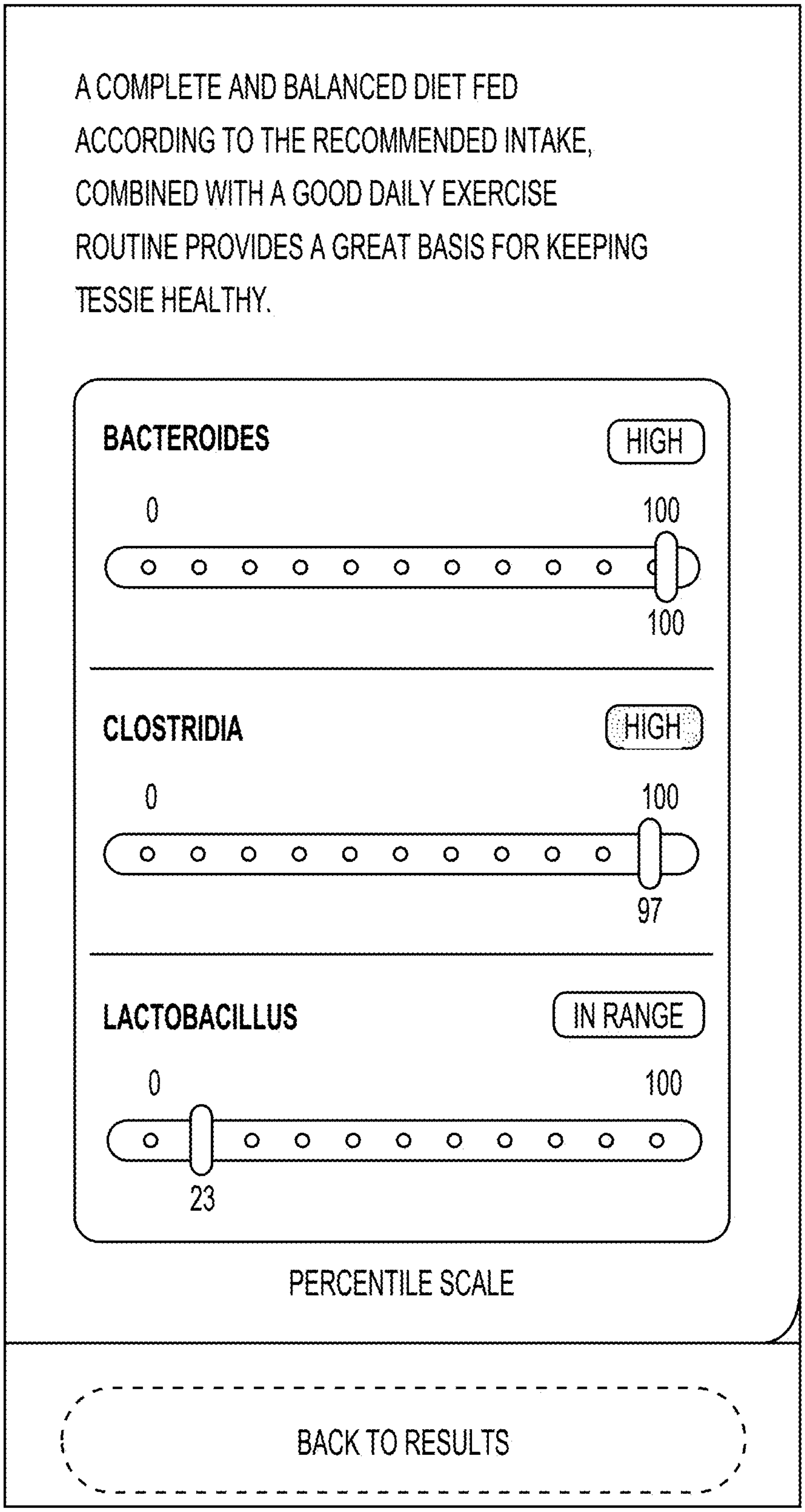

A COMPLETE AND BALANCED DIET FED ACCORDING TO THE RECOMMENDED INTAKE, COMBINED WITH A GOOD DAILY EXERCISE ROUTINE PROVIDES A GREAT BASIS FOR KEEPING TESSIE HEALTHY.
BACTEROIDES
HIGH
0
100
100
CLOSTRIDIA
HIGH
0
100
97
LACTOBACILLUS
IN RANGE
0
100
23
PERCENTILE SCALE
BACK TO RESULTS

FIG. 5W

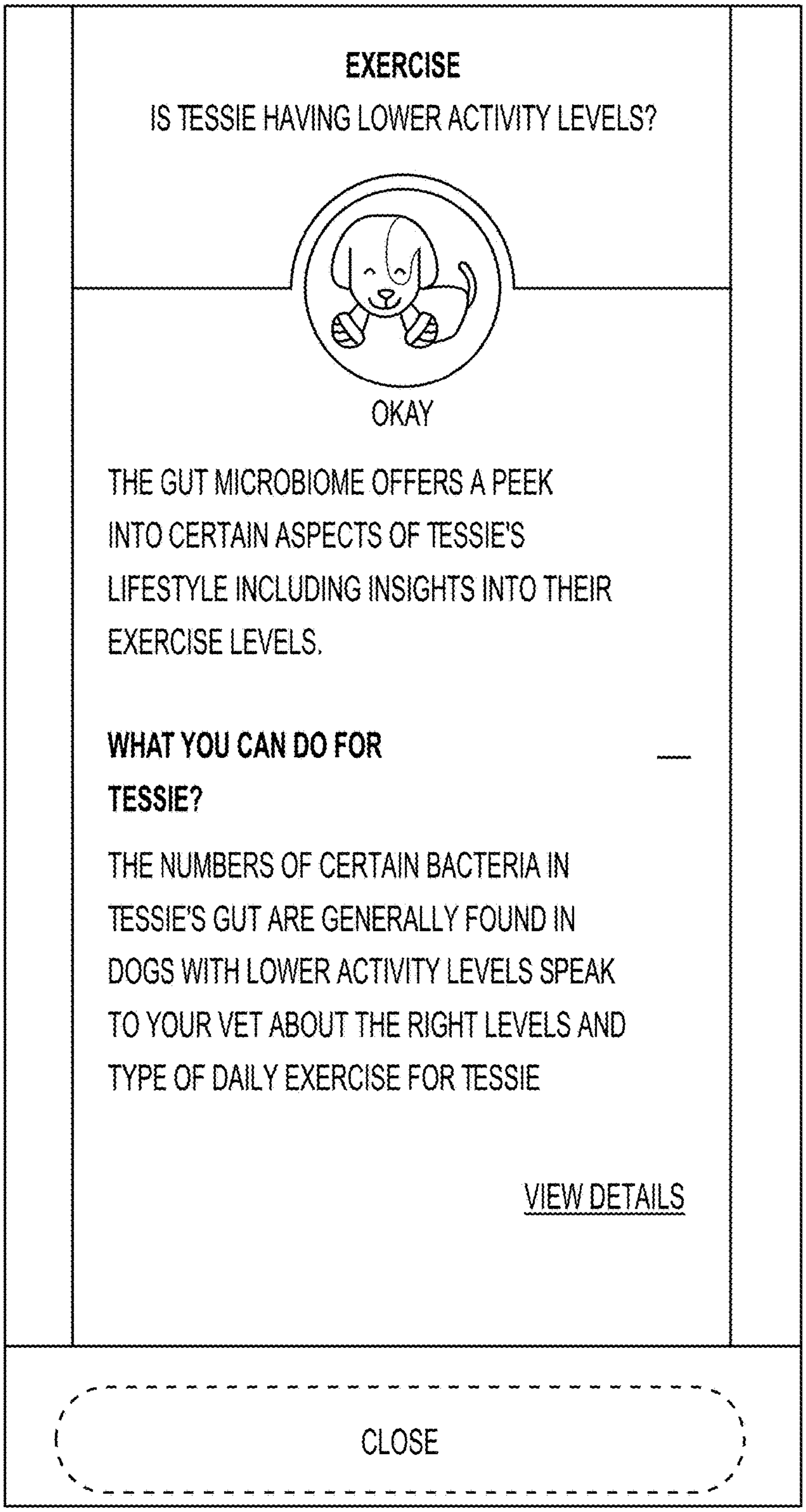

EXERCISE
IS TESSIE HAVING LOWER ACTIVITY LEVELS?
OKAY
THE GUT MICROBIOME OFFERS A PEEK INTO CERTAIN ASPECTS OF TESSIE'S LIFESTYLE INCLUDING INSIGHTS INTO THEIR EXERCISE LEVELS.
WHAT YOU CAN DO FOR TESSIE?
THE NUMBERS OF CERTAIN BACTERIA IN TESSIE'S GUT ARE GENERALLY FOUND IN DOGS WITH LOWER ACTIVITY LEVELS SPEAK TO YOUR VET ABOUT THE RIGHT LEVELS AND TYPE OF DAILY EXERCISE FOR TESSIE
VIEW DETAILS
CLOSE

FIG. 5X

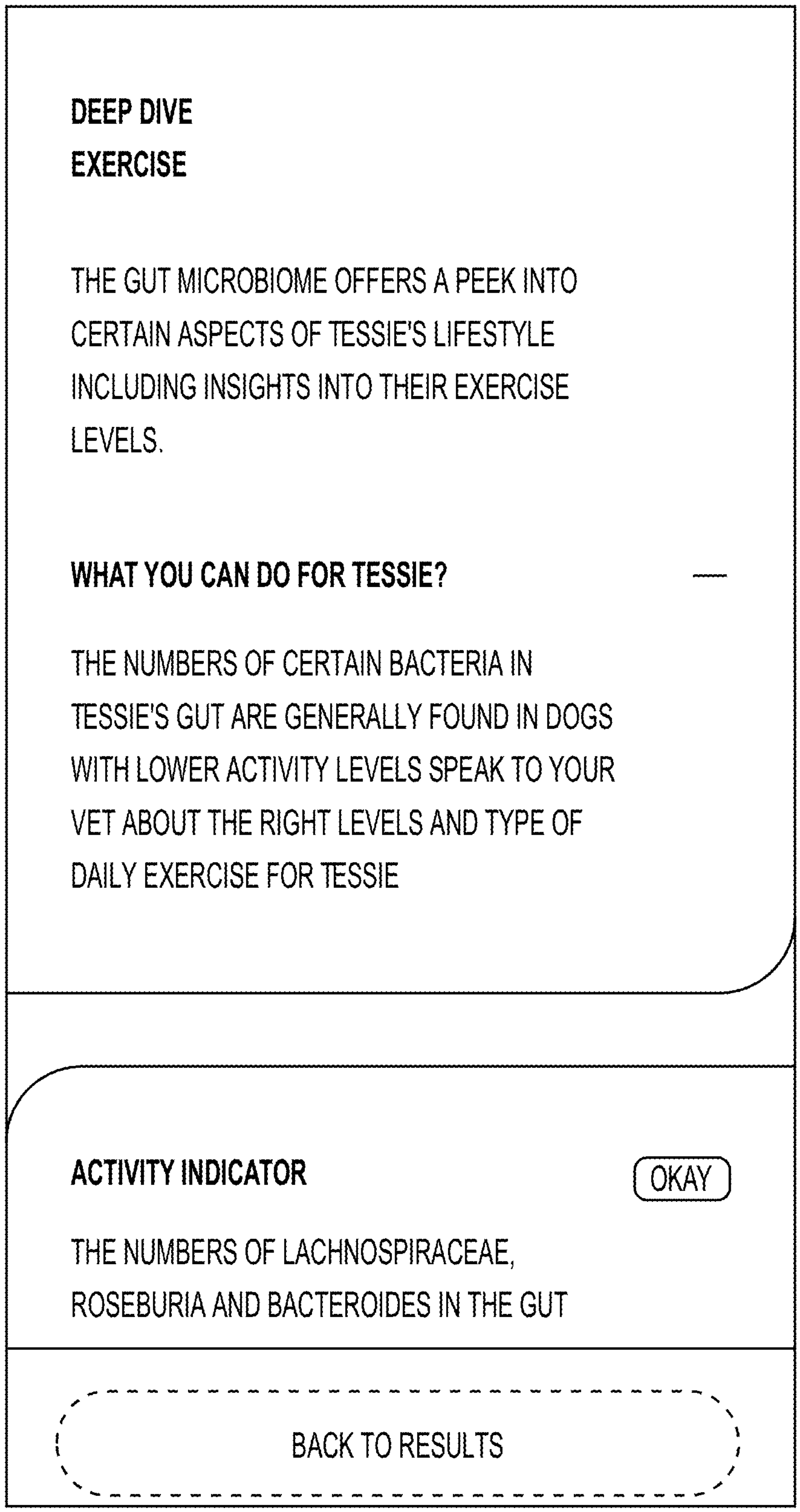

DEEP DIVE
EXERCISE
THE GUT MICROBIOME OFFERS A PEEK INTO CERTAIN ASPECTS OF TESSIE'S LIFESTYLE INCLUDING INSIGHTS INTO THEIR EXERCISE LEVELS.
WHAT YOU CAN DO FOR TESSIE?
THE NUMBERS OF CERTAIN BACTERIA IN TESSIE'S GUT ARE GENERALLY FOUND IN DOGS WITH LOWER ACTIVITY LEVELS SPEAK TO YOUR VET ABOUT THE RIGHT LEVELS AND TYPE OF DAILY EXERCISE FOR TESSIE
ACTIVITY INDICATOR
OKAY
THE NUMBERS OF LACHNOSPIRACEAE, ROSEBURIA AND BACTEROIDES IN THE GUT
BACK TO RESULTS

FIG. 5Y

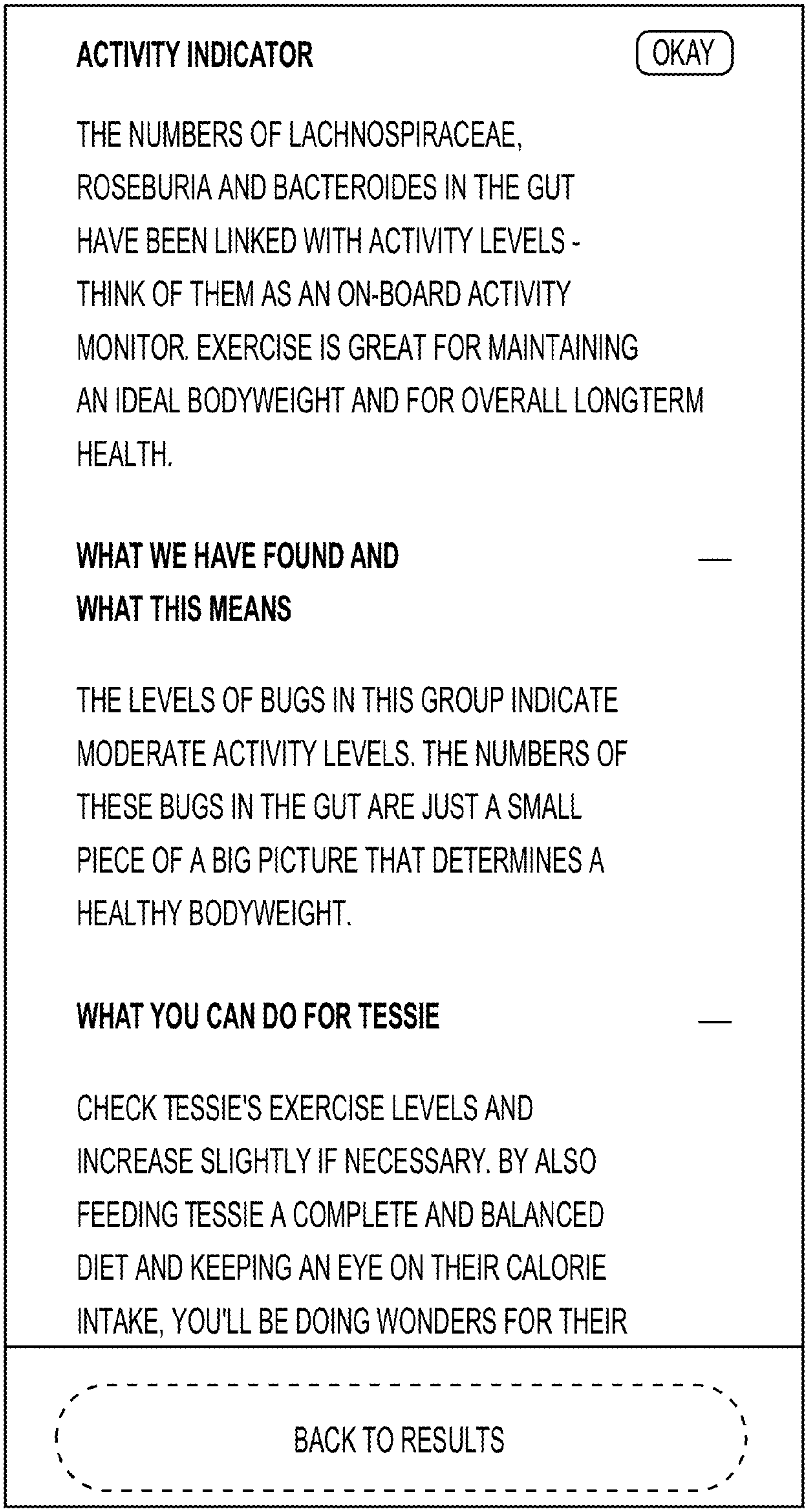

ACTIVITY INDICATOR
OKAY
THE NUMBERS OF LACHNOSPIRACEAE, ROSEBURIA AND BACTEROIDES IN THE GUT HAVE BEEN LINKED WITH ACTIVITY LEVELS - THINK OF THEM AS AN ON-BOARD ACTIVITY MONITOR. EXERCISE IS GREAT FOR MAINTAINING AN IDEAL BODYWEIGHT AND FOR OVERALL LONGTERM HEALTH.
WHAT WE HAVE FOUND AND WHAT THIS MEANS
THE LEVELS OF BUGS IN THIS GROUP INDICATE MODERATE ACTIVITY LEVELS. THE NUMBERS OF THESE BUGS IN THE GUT ARE JUST A SMALL PIECE OF A BIG PICTURE THAT DETERMINES A HEALTHY BODYWEIGHT.
WHAT YOU CAN DO FOR TESSIE
CHECK TESSIE'S EXERCISE LEVELS AND INCREASE SLIGHTLY IF NECESSARY. BY ALSO FEEDING TESSIE A COMPLETE AND BALANCED DIET AND KEEPING AN EYE ON THEIR CALORIE INTAKE, YOU'LL BE DOING WONDERS FOR THEIR
BACK TO RESULTS

FIG. 5Z

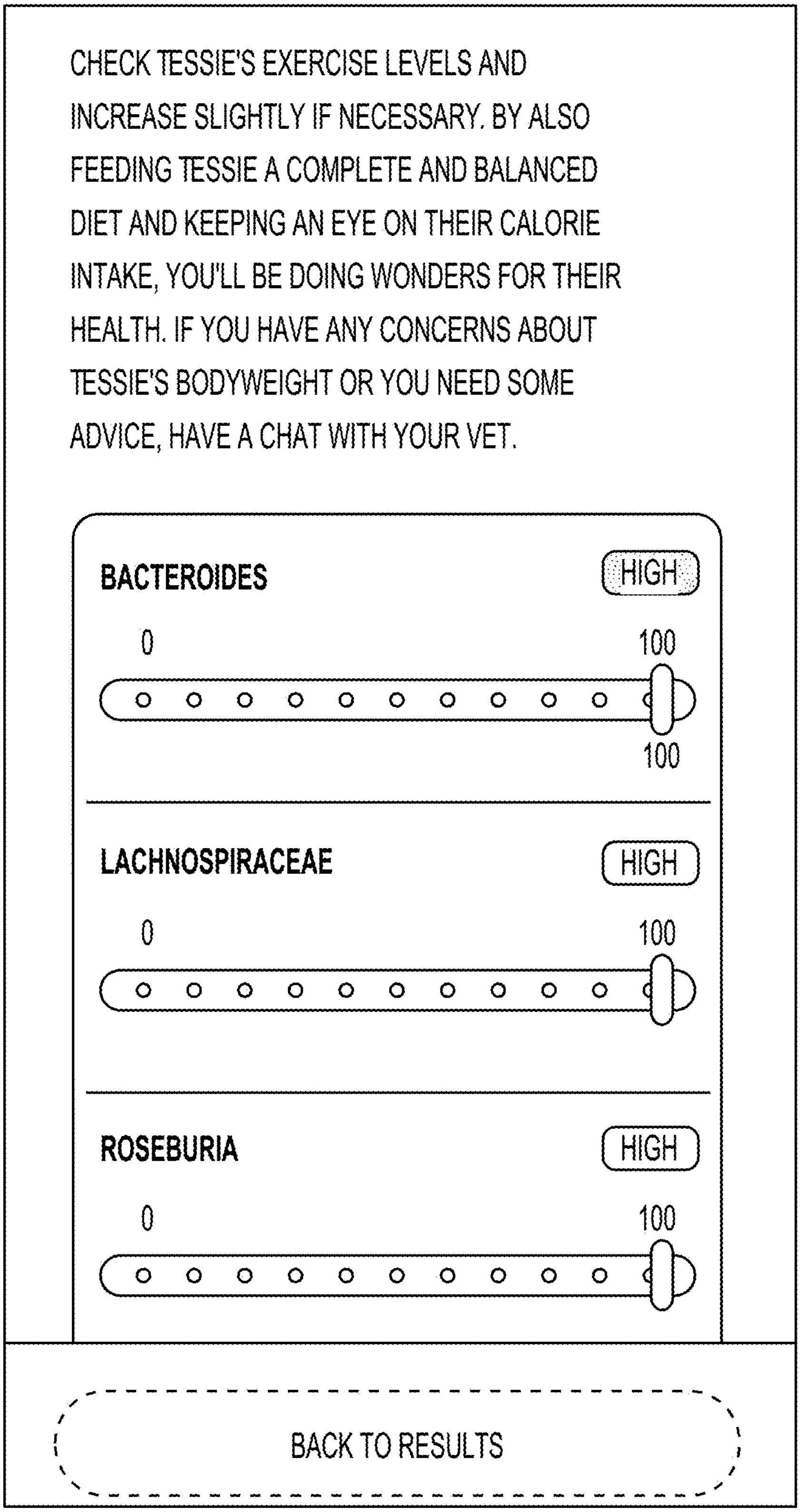

CHECK TESSIE'S EXERCISE LEVELS AND INCREASE SLIGHTLY IF NECESSARY. BY ALSO FEEDING TESSIE A COMPLETE AND BALANCED DIET AND KEEPING AN EYE ON THEIR CALORIE INTAKE, YOU'LL BE DOING WONDERS FOR THEIR HEALTH. IF YOU HAVE ANY CONCERNS ABOUT TESSIE'S BODYWEIGHT OR YOU NEED SOME ADVICE, HAVE A CHAT WITH YOUR VET.
BACTEROIDES
HIGH
0
100
100
LACHNOSPIRACEAE
HIGH
0
100
ROSEBURIA
HIGH
0
100
BACK TO RESULTS

FIG. 5AA

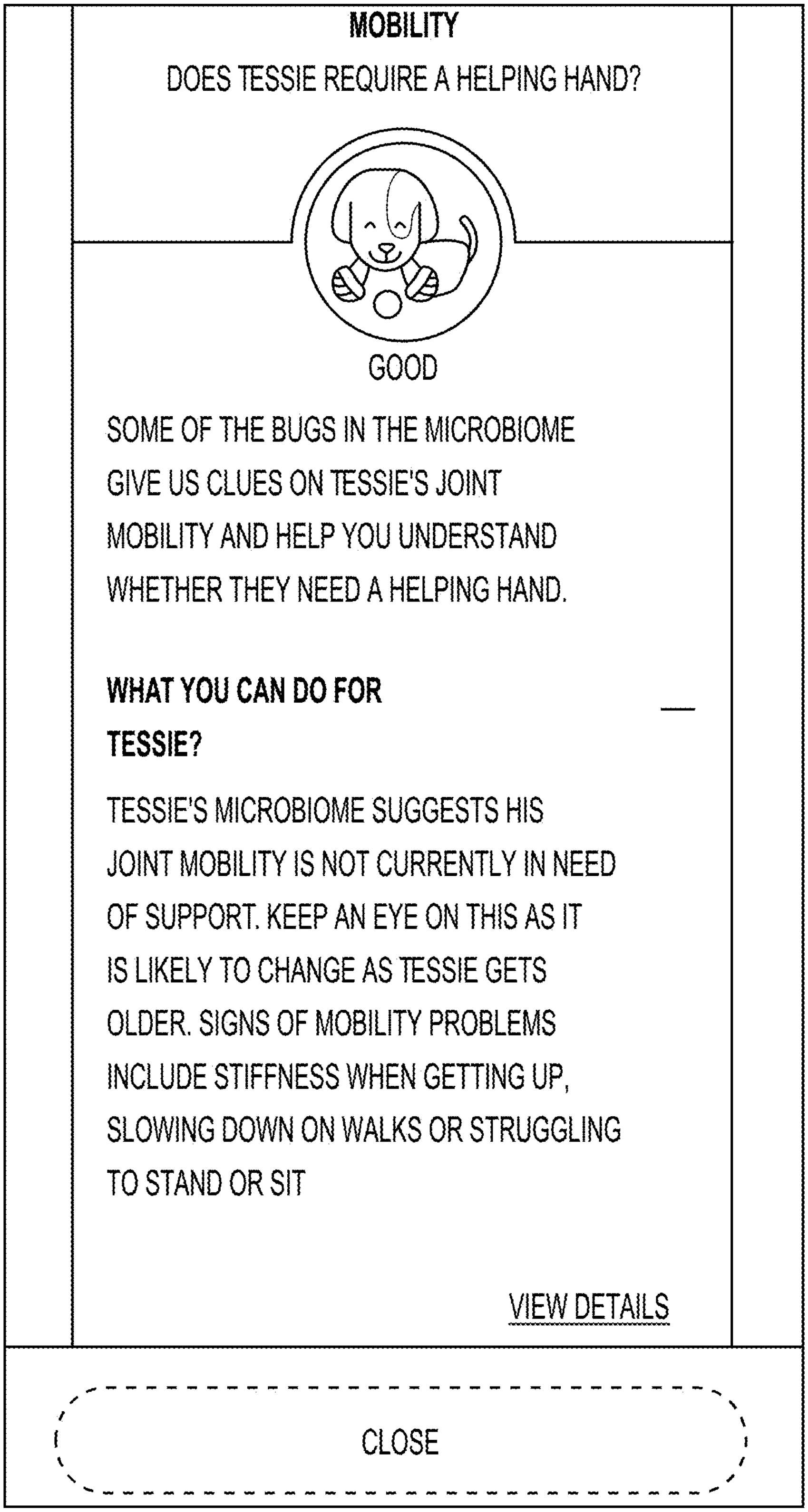

MOBILITY
DOES TESSIE REQUIRE A HELPING HAND?
GOOD
SOME OF THE BUGS IN THE MICROBIOME GIVE US CLUES ON TESSIE'S JOINT MOBILITY AND HELP YOU UNDERSTAND WHETHER THEY NEED A HELPING HAND.
WHAT YOU CAN DO FOR TESSIE?
—
TESSIE'S MICROBIOME SUGGESTS HIS JOINT MOBILITY IS NOT CURRENTLY IN NEED OF SUPPORT. KEEP AN EYE ON THIS AS IT IS LIKELY TO CHANGE AS TESSIE GETS OLDER. SIGNS OF MOBILITY PROBLEMS INCLUDE STIFFNESS WHEN GETTING UP, SLOWING DOWN ON WALKS OR STRUGGLING TO STAND OR SIT
VIEW DETAILS
CLOSE

FIG. 5BB

DEEP DIVE

MOBILITY

SOME OF THE BUGS IN THE MICROBIOME GIVE US CLUES ON TESSIE'S JOINT MOBILITY AND HELP YOU UNDERSTAND WHETHER THEY NEED A HELPING HAND.

WHAT YOU CAN DO FOR TESSIE? —

TESSIE'S MICROBIOME SUGGESTS HIS JOINT MOBILITY IS NOT CURRENTLY IN NEED OF SUPPORT. KEEP AN EYE ON THIS AS IT IS LIKELY TO CHANGE AS TESSIE GETS OLDER. SIGNS OF MOBILITY PROBLEMS INCLUDE STIFFNESS WHEN GETTING UP, SLOWING DOWN ON WALKS OR STRUGGLING TO STAND OR SIT

JOINT SUPPORT OKAY

BACK TO RESULTS

FIG. 5CC

JOINT SUPPORT OKAY

OLDER DOGS MAY EXPERIENCE A DECLINE IN THEIR JOINT MOBILITY. ACCORDING TO DATA FROM HUMANS, GUT LEVELS OF FAECALIBACTERIUM AND PREVOTELLA HELP GIVE US CLUES ABOUT TESSIE'S JOINT MOBILITY.

WHAT WE HAVE FOUND AND WHAT THIS MEANS —

THERE ARE INDICATIONS FROM THE LEVELS OF THESE BUGS THAT TESSIE'S JOINTS MAY NEED A LITTLE SUPPORT. IF TESSIE IS SHOWING RELUCTANCE TO EXERCISE, THIS MAY ALSO BE A SIGN OF DECLINING JOINT MOBILITY. THERE ARE MANY FACTORS THAT CONTRIBUTE TOWARDS TESSIE STAYING ACTIVE AND HEALTHY SO CHECK OUT WHAT YOU CAN DO.

WHAT YOU CAN DO FOR TESSIE —

MAINTAINING A GOOD DAILY EXERCISE ROUTINE IS AN EXCELLENT WAY FOR HELPING TESSIE TO STAY ACTIVE AND HEALTHY. BY ALSO FEEDING TESSIE A COMPLETE AND BALANCED DIET AND

BACK TO RESULTS

FIG. 5DD

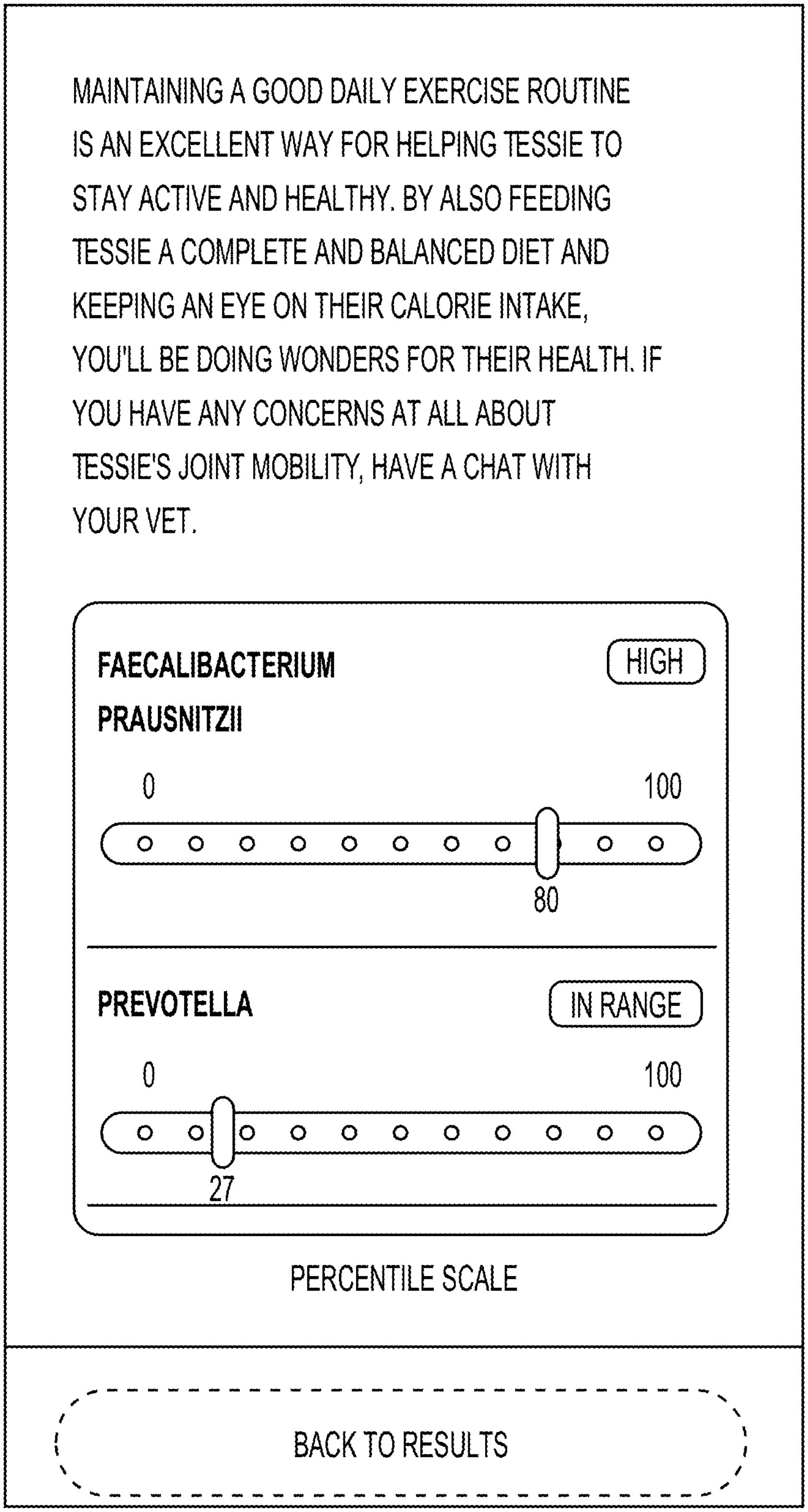

MAINTAINING A GOOD DAILY EXERCISE ROUTINE IS AN EXCELLENT WAY FOR HELPING TESSIE TO STAY ACTIVE AND HEALTHY. BY ALSO FEEDING TESSIE A COMPLETE AND BALANCED DIET AND KEEPING AN EYE ON THEIR CALORIE INTAKE, YOU'LL BE DOING WONDERS FOR THEIR HEALTH. IF YOU HAVE ANY CONCERNS AT ALL ABOUT TESSIE'S JOINT MOBILITY, HAVE A CHAT WITH YOUR VET.
FAECALIBACTERIUM PRAUSNITZII
HIGH
0
100
80
PREVOTELLA
IN RANGE
0
100
27
PERCENTILE SCALE
BACK TO RESULTS

FIG. 5EE

VITAMIN PRODUCTION

IS TESSIE GETTING THE RIGHT VITAMIN?

GOOD

FEEDING A COMPLETE AND BALANCED DIET IS NO DOUBT THE BEST WAY TO ENSURE TESSIE GETS ALL THE VITAMINS THEY NEED IN THE RIGHT AMOUNTS. GUT BACTERIA HAVE A PART TO PLAY IN TESSIE'S VITAMIN SUPPLY TOO, HELPING THEM STAY HEALTHY AND ACTIVE. THIS SECTION GIVES INSIGHTS INTO TESSIE'S ON-BOARD BACTERIAL PRODUCTION OF B VITAMINS AND VITAMIN K AND OFFERS TIPS ON HOW TO BOOST THEIR SUPPLIES IF NEEDED.

WHAT YOU CAN DO FOR TESSIE? —

TESSIE HAS GREAT LEVELS OF VITAMIN-PRODUCING BACTERIA VITAMINS ARE

CLOSE

FIG. 5FF

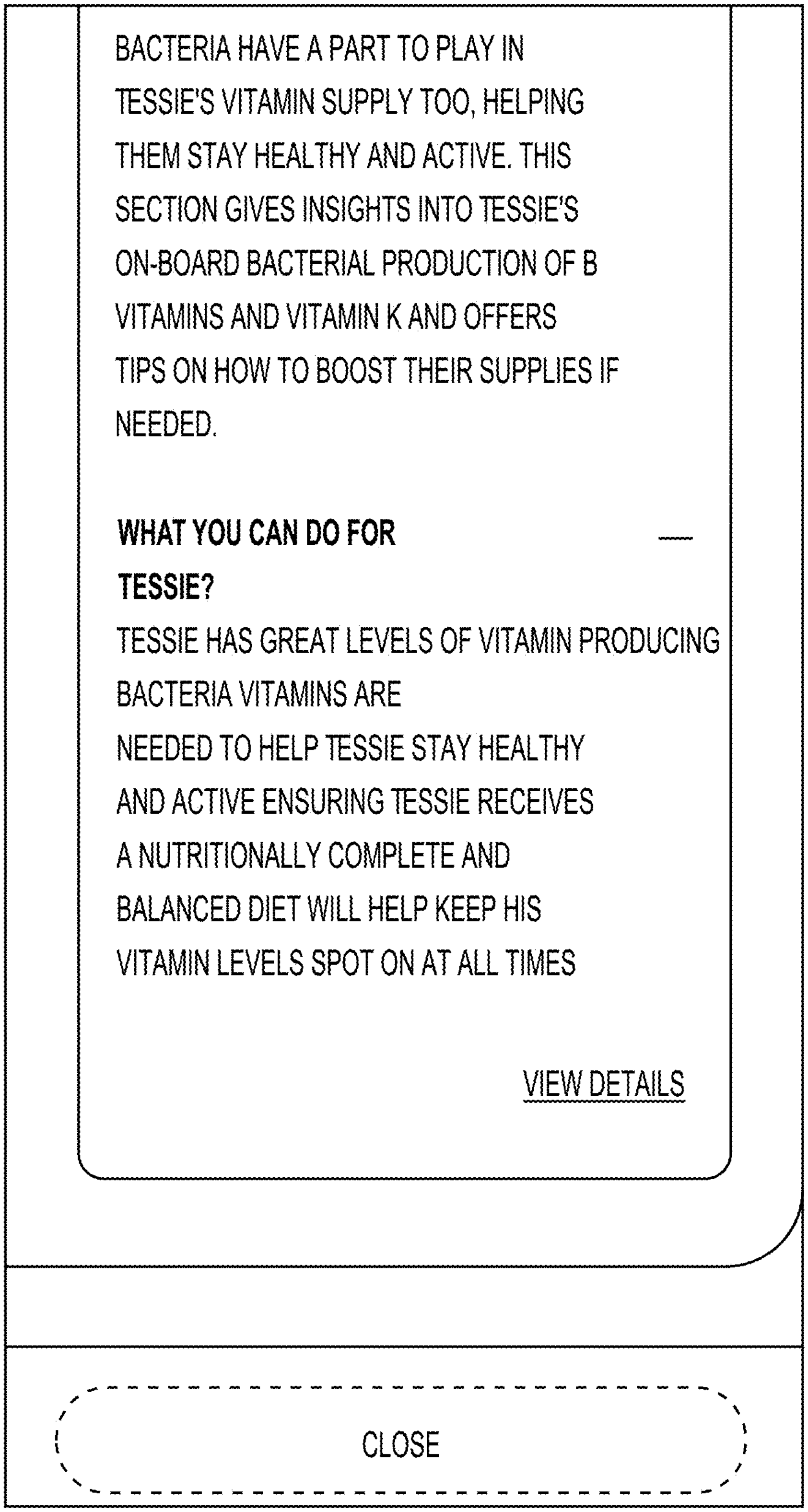

BACTERIA HAVE A PART TO PLAY IN TESSIE'S VITAMIN SUPPLY TOO, HELPING THEM STAY HEALTHY AND ACTIVE. THIS SECTION GIVES INSIGHTS INTO TESSIE'S ON-BOARD BACTERIAL PRODUCTION OF B VITAMINS AND VITAMIN K AND OFFERS TIPS ON HOW TO BOOST THEIR SUPPLIES IF NEEDED.
WHAT YOU CAN DO FOR TESSIE?
—
TESSIE HAS GREAT LEVELS OF VITAMIN PRODUCING BACTERIA VITAMINS ARE NEEDED TO HELP TESSIE STAY HEALTHY AND ACTIVE ENSURING TESSIE RECEIVES A NUTRITIONALLY COMPLETE AND BALANCED DIET WILL HELP KEEP HIS VITAMIN LEVELS SPOT ON AT ALL TIMES
VIEW DETAILS
CLOSE

FIG. 5GG

DEEP DIVE

VITAMIN PRODUCTION

FEEDING A COMPLETE AND BALANCED DIET IS NO DOUBT THE BEST WAY TO ENSURE TESSIE GETS ALL THE VITAMINS THEY NEED IN THE RIGHT AMOUNTS. GUT BACTERIA HAVE A PART TO PLAY IN TESSIE'S VITAMIN SUPPLY TOO, HELPING THEM STAY HEALTHY AND ACTIVE. THIS SECTION GIVES INSIGHTS INTO TESSIE'S ONBOARD BACTERIAL PRODUCTION OF B VITAMINS AND VITAMIN K AND OFFERS TIPS ON HOW TO BOOST THEIR SUPPLIES IF NEEDED.

WHAT YOU CAN DO FOR TESSIE? —

TESSIE HAS GREAT LEVELS OF VITAMIN PRODUCING BACTERIA VITAMINS ARE NEEDED TO HELP TESSIE STAY HEALTHY AND ACTIVE ENSURING TESSIE RECEIVES A NUTRITIONALLY COMPLETE AND BALANCED DIET WILL HELP KEEP HIS VITAMIN LEVELS SPOT ON AT ALL TIMES

BACK TO RESULTS

FIG. 5HH

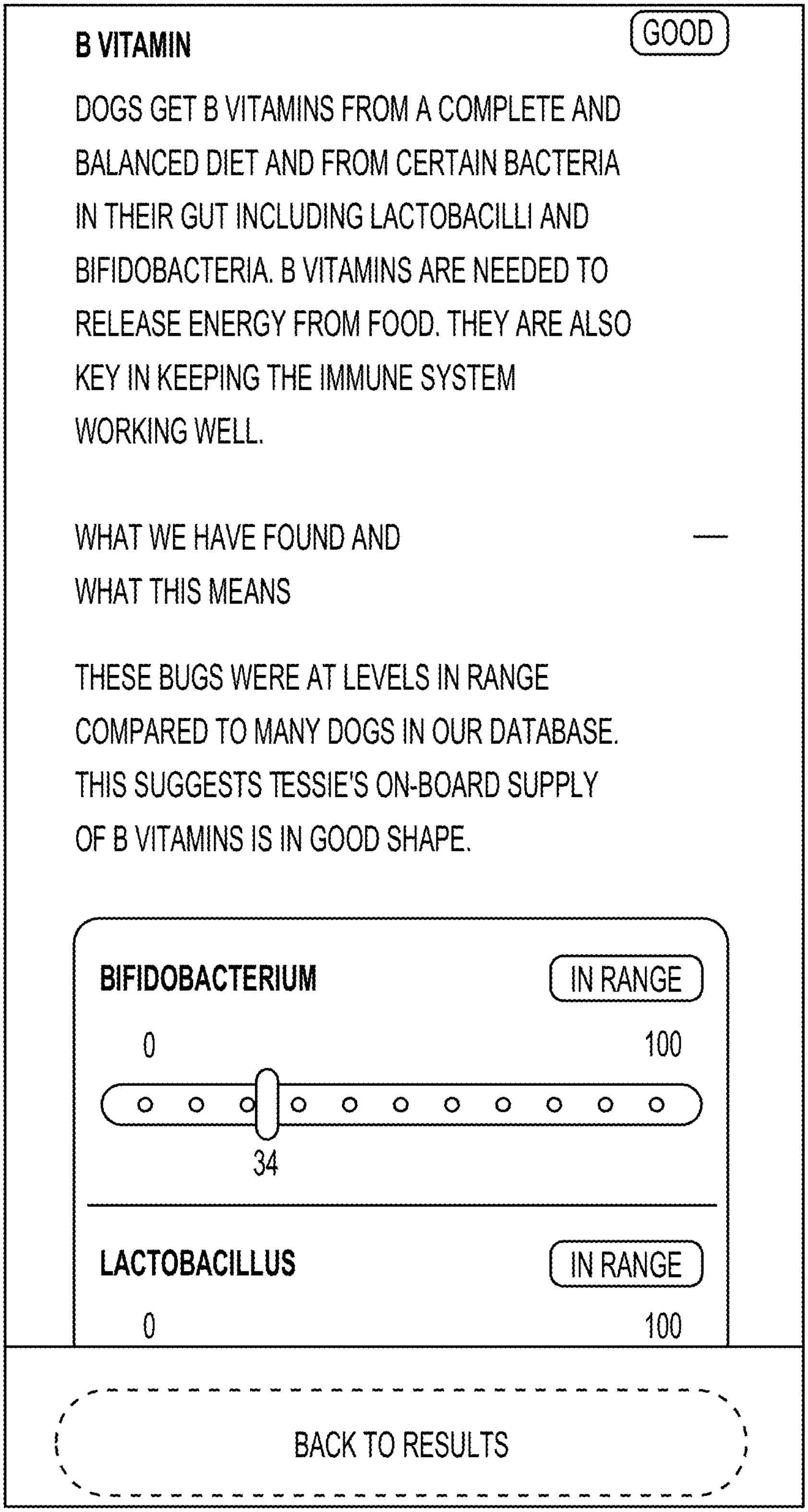

B VITAMIN
GOOD
DOGS GET B VITAMINS FROM A COMPLETE AND BALANCED DIET AND FROM CERTAIN BACTERIA IN THEIR GUT INCLUDING LACTOBACILLI AND BIFIDOBACTERIA. B VITAMINS ARE NEEDED TO RELEASE ENERGY FROM FOOD. THEY ARE ALSO KEY IN KEEPING THE IMMUNE SYSTEM WORKING WELL.
WHAT WE HAVE FOUND AND WHAT THIS MEANS
THESE BUGS WERE AT LEVELS IN RANGE COMPARED TO MANY DOGS IN OUR DATABASE. THIS SUGGESTS TESSIE'S ON-BOARD SUPPLY OF B VITAMINS IS IN GOOD SHAPE.
BIFIDOBACTERIUM
IN RANGE
0
100
34
LACTOBACILLUS
IN RANGE
0
100
BACK TO RESULTS

FIG. 5II

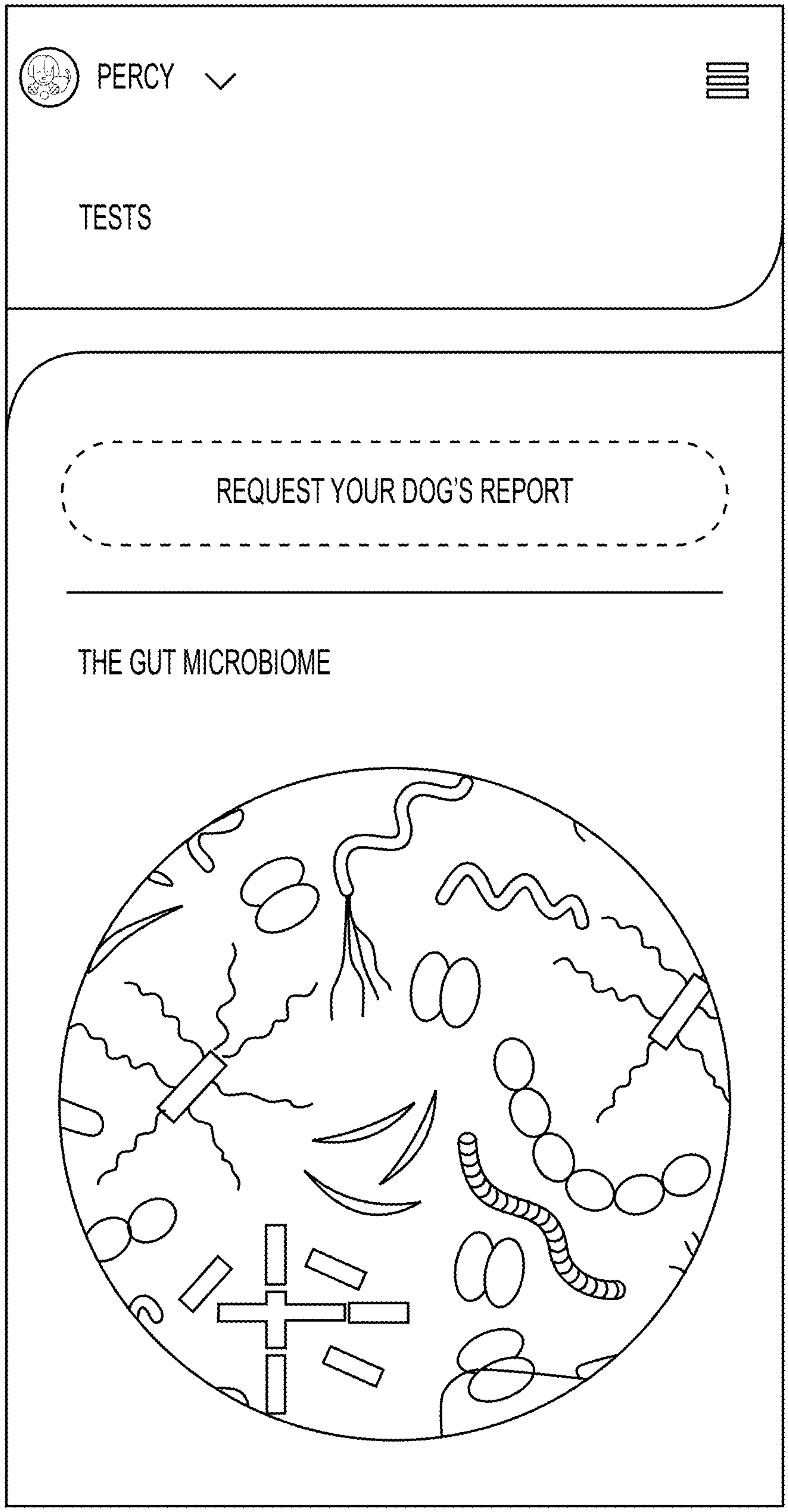


FIG. 5JJ

TAPP INTO WHAT YOUR DOG
CAN'T TELL YOU.

WE LOVE OUR DOGS SO MUCH THAT WE KNOW AT THESE LITTLE QUIRKS, BUT WHILE WE KNOW THEM ON THE OUTSIDE, IT'S NOT ALWAYS EASY TO KNOW WHAT'S GOING ON INSIDE.

WE CAN'T SEE IT, BUT THERE'S A TINY UNIVERSE OF TRILLIONS OF BACTERIA AND OTHER MICROBES WHICH TOGETHER WITH THEIR GENES ARE CALLED THE MICROBIOME. IT IS AN INCREDIBLE LIVING ECOSYSTEM MADE UP FROM BUGS, OR MICROBES IN THE GUT WHICH HELP OUR DOGS STAY HEALTHY.

THANKS TO SCIENTIFIC RESEARCH LIKE COMPANY PETCARE BIOBANK, SCIENTISTS ARE CONTINUOUSLY DISCOVERING WAYS THE MICROBIOME IMPACTS DOGS' HEALTH. BY PARTICIPATING IN RESEARCH LIKE THIS, YOU NOT ONLY HELP US DELVE DEEPER THAN EVER BEFORE - YOU GET TO DISCOVER MORE ABOUT YOUR DOG'S GUT MICROBIOME AND IT'S POTENTIAL IMPACT ON THEIR HEALTH.

✓ COMPLEMENTARY TO COMPANY PETCARE BIOBARK PARTICIPANTS

FIG. 5JJ
(CONT.)

✓ CONVENIENT - USING THE POOP SAMPLE TAKEN AT YOUR BIOBARK APPOINTMENT

✓ DETAILED, PERSONALIZED REPORT

✓ RECEIVED IN-APP - ALWAYS KNOW WHERE TO FIND IT

KNOW YOUR PET'S MICROBIOME AS WELL AS YOU KNOW THEM

TESTING THE GUT MICROBIOME HELPS YOU UNDERSTAND WHAT'S GOING ON INSIDE YOUR DOG AND TO TAKE POSITIVE STEPS TOWARD IMPROVING YOUR PET'S WELLBEING.

WHAT WILL THE TEST COVER?

- AN ASSESSMENT OF THE OVERALL HEALTH OF YOUR DOG'S GUT MICROBIOME, RELATIVE TO OTHER DOGS IN OUR EXTENSIVE DATABASE.
- PERSONALIZED RECOMMENDATIONS OF HOW TO HELP IMPROVE YOUR DOG'S MICROBIOME

FIG. 5JJ (CONT.1)

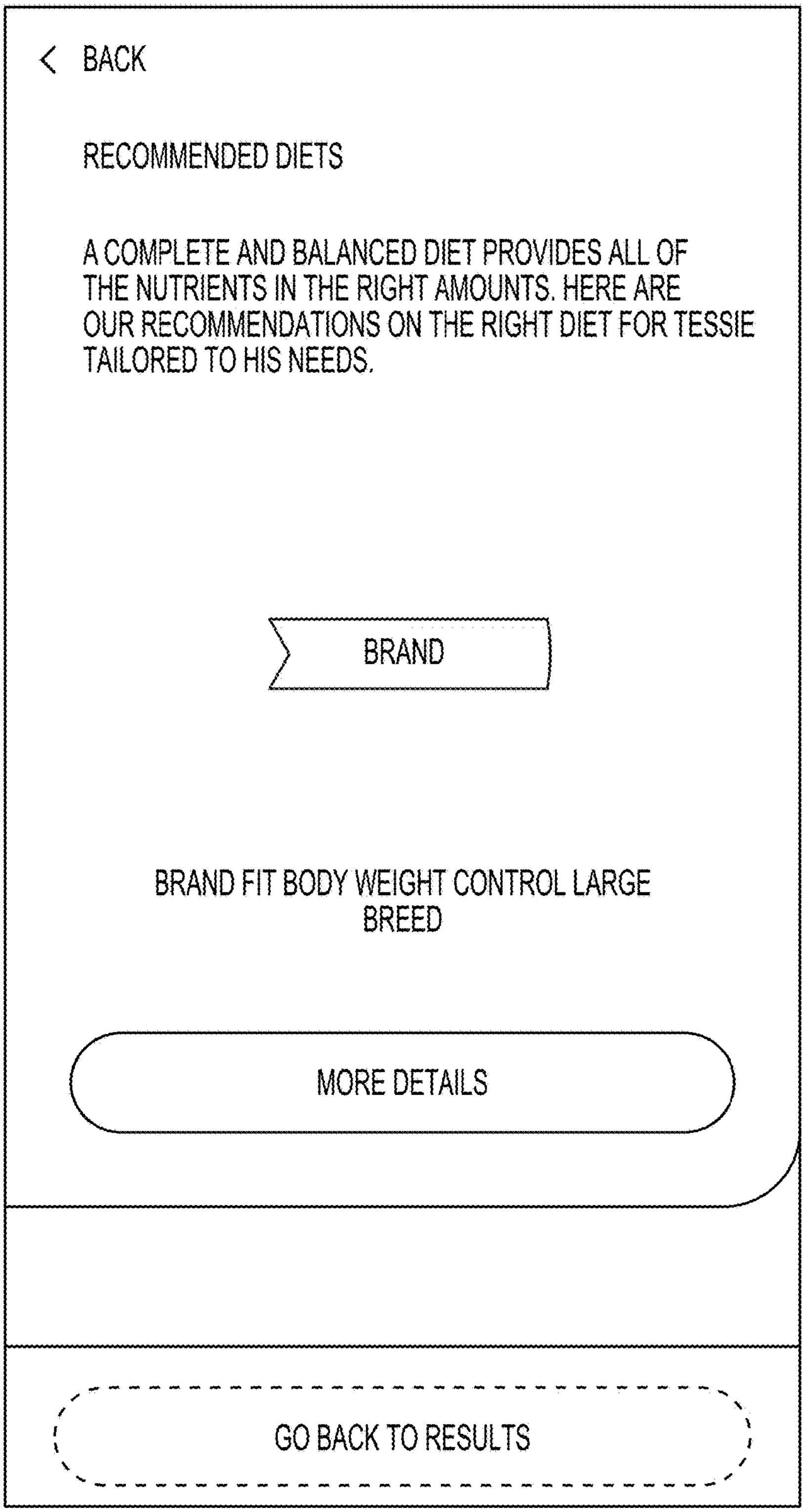

BACK
RECOMMENDED DIETS
A COMPLETE AND BALANCED DIET PROVIDES ALL OF THE NUTRIENTS IN THE RIGHT AMOUNTS. HERE ARE OUR RECOMMENDATIONS ON THE RIGHT DIET FOR TESSIE TAILORED TO HIS NEEDS.
BRAND
BRAND FIT BODY WEIGHT CONTROL LARGE BREED
MORE DETAILS
GO BACK TO RESULTS

FIG. 6A

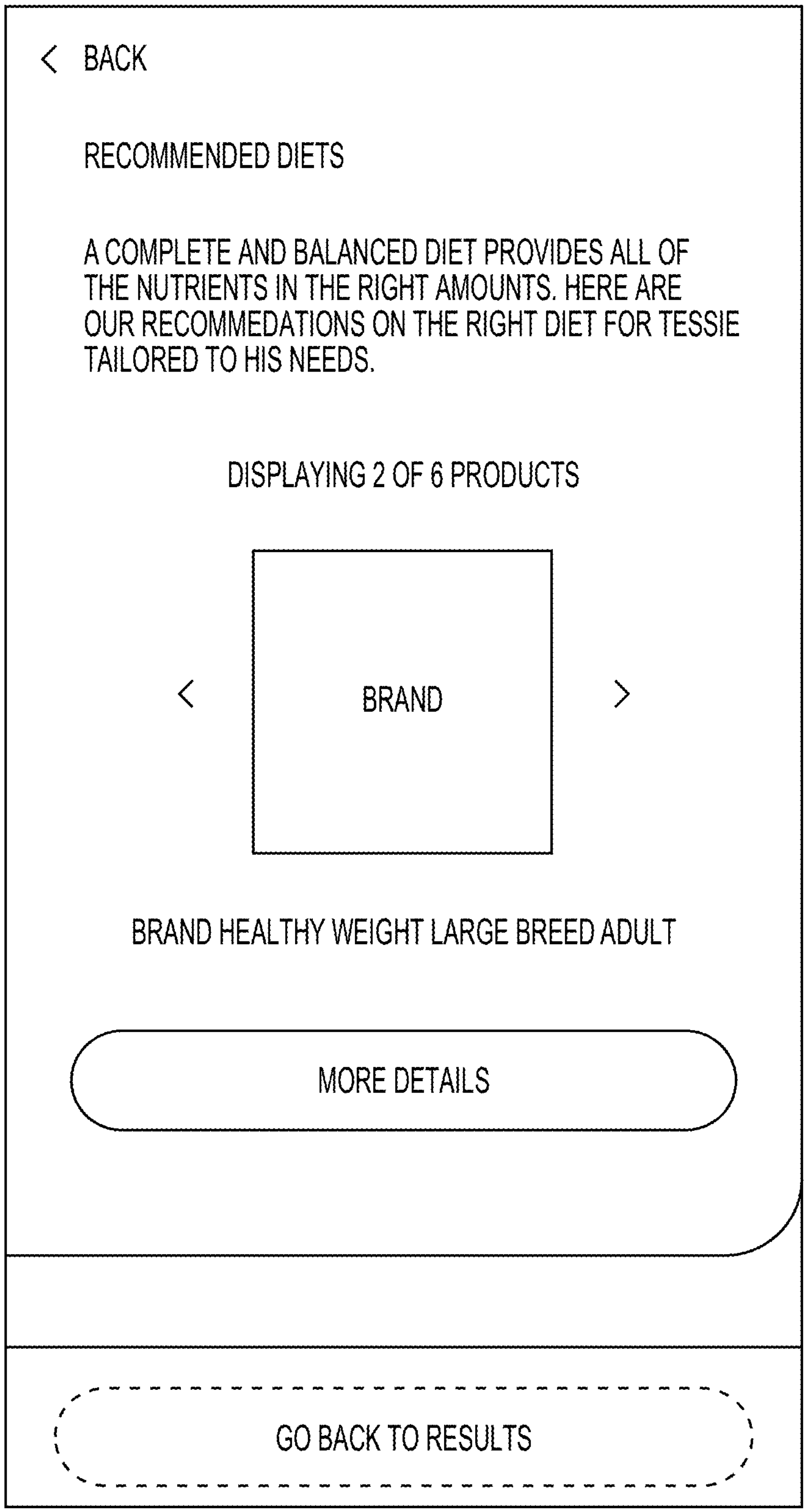

BACK
RECOMMENDED DIETS
A COMPLETE AND BALANCED DIET PROVIDES ALL OF THE NUTRIENTS IN THE RIGHT AMOUNTS. HERE ARE OUR RECOMMEDATIONS ON THE RIGHT DIET FOR TESSIE TAILORED TO HIS NEEDS.
DISPLAYING 2 OF 6 PRODUCTS
BRAND
BRAND HEALTHY WEIGHT LARGE BREED ADULT
MORE DETAILS
GO BACK TO RESULTS

FIG. 6B

< BACK
RECOMMENDED DIETS
A COMPLETE AND BALANCED DIET PROVIDES ALL OF THE NUTRIENTS IN THE RIGHT AMOUNTS. HERE ARE OUR RECOMMEDATIONS ON THE RIGHT DIET FOR TESSIE TAILORED TO HIS NEEDS.
DISPLAYING 3 OF 6 PRODUCTS
< BRAND >
BRAND NATURAL CHOICE™ HEALTHY WEIGHT ADULT CHICKEN & BROWN RICE RECIPE
MORE DETAILS
GO BACK TO RESULTS

FIG. 6C

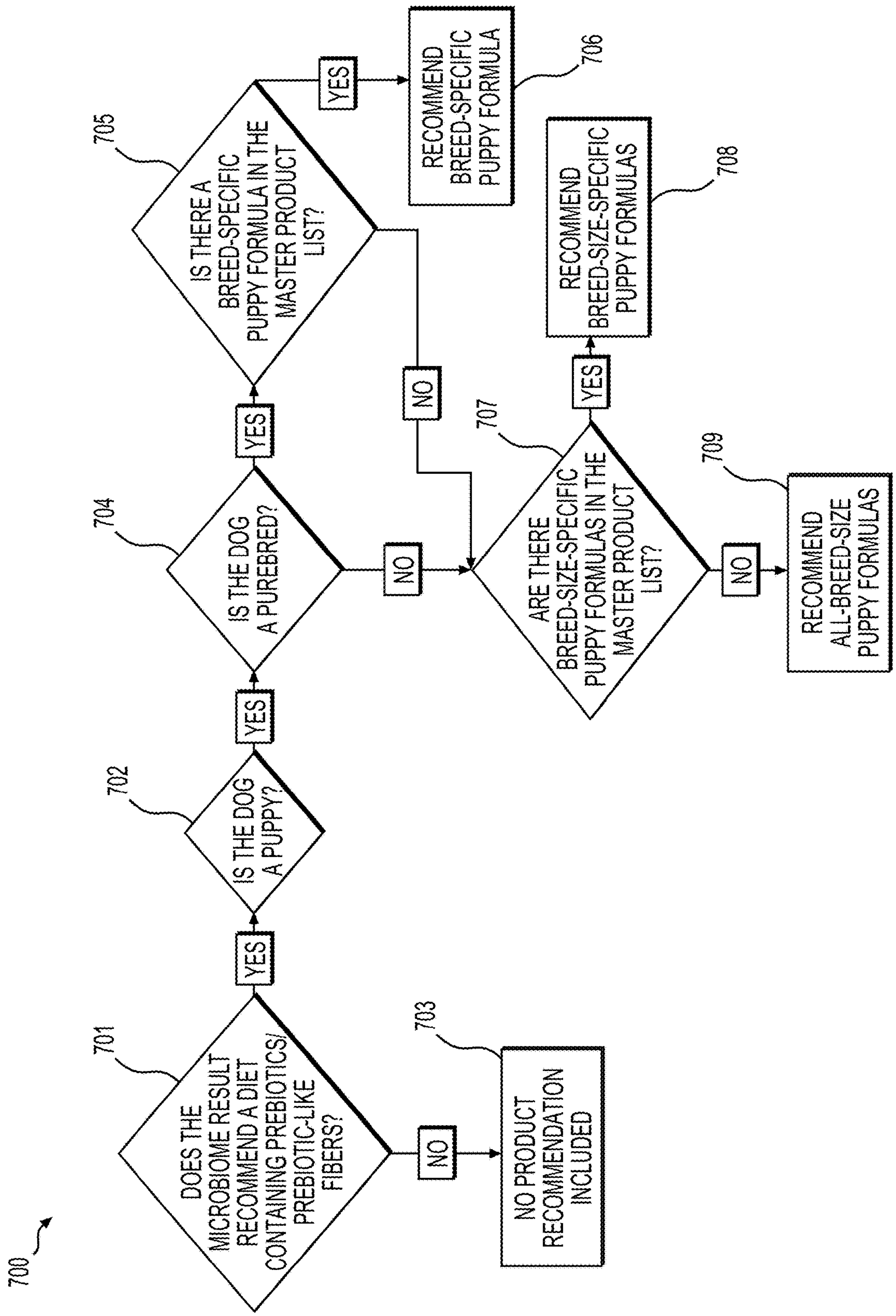

700
701
DOES THE MICROBIOME RESULT RECOMMEND A DIET CONTAINING PREBIOTICS/ PREBIOTIC-LIKE FIBERS?
NO
703
NO PRODUCT RECOMMENDATION INCLUDED
YES
702
IS THE DOG A PUPPY?
YES
704
IS THE DOG A PUREBRED?
YES
705
IS THERE A BREED-SPECIFIC PUPPY FORMULA IN THE MASTER PRODUCT LIST?
YES
RECOMMEND BREED-SPECIFIC PUPPY FORMULA
706
NO
NO
707
ARE THERE BREED-SIZE-SPECIFIC PUPPY FORMULAS IN THE MASTER PRODUCT LIST?
YES
RECOMMEND BREED-SIZE-SPECIFIC PUPPY FORMULAS
708
NO
709
RECOMMEND ALL-BREED-SIZE PUPPY FORMULAS

FIG. 7

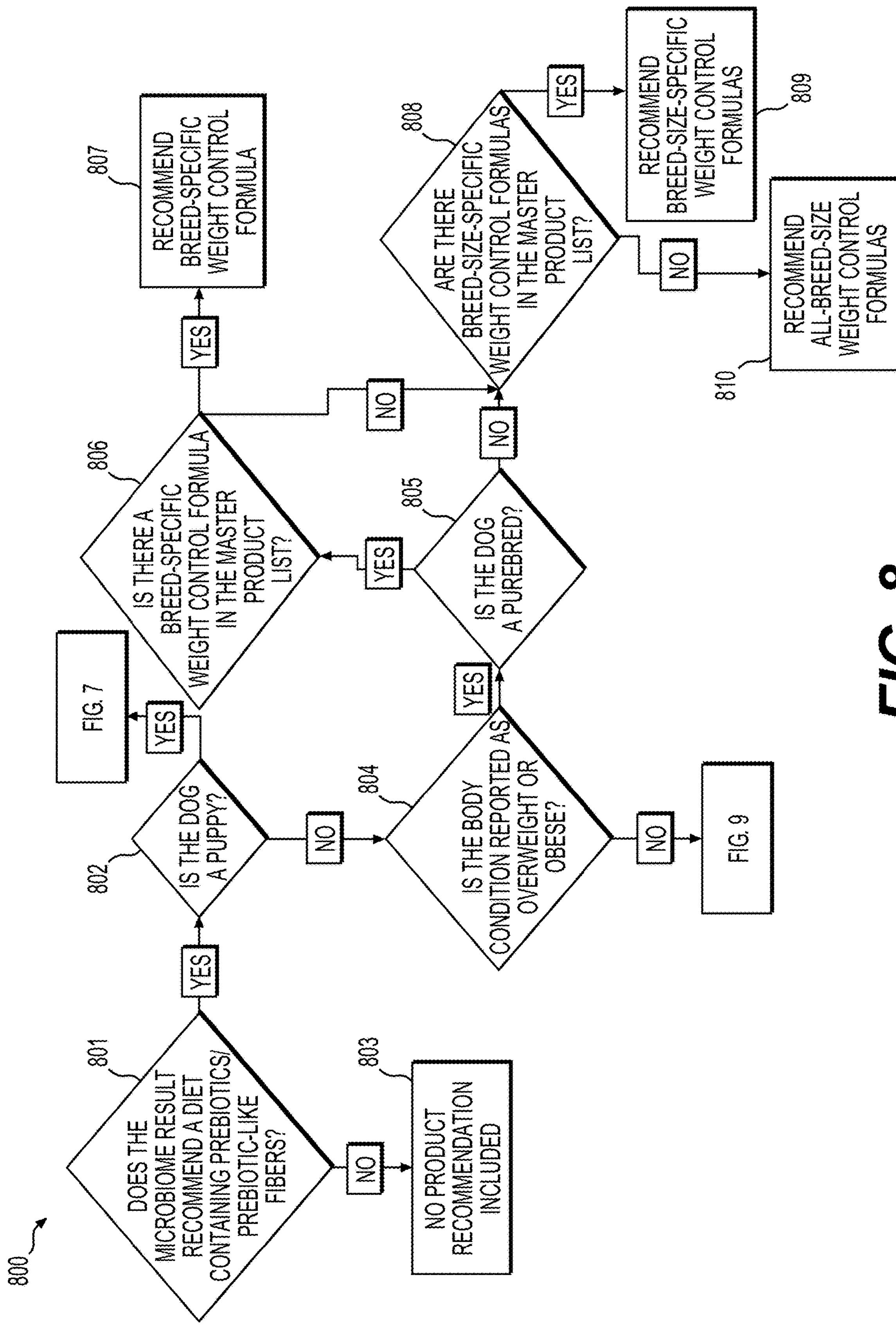

800
801
DOES THE MICROBIOME RESULT RECOMMEND A DIET CONTAINING PREBIOTICS/ PREBIOTIC-LIKE FIBERS?
NO
803
NO PRODUCT RECOMMENDATION INCLUDED
YES
802
IS THE DOG A PUPPY?
YES
FIG. 7
NO
804
IS THE BODY CONDITION REPORTED AS OVERWEIGHT OR OBESE?
NO
FIG. 9
YES
805
IS THE DOG A PUREBRED?
YES
806
IS THERE A BREED-SPECIFIC WEIGHT CONTROL FORMULA IN THE MASTER PRODUCT LIST?
YES
807
RECOMMEND BREED-SPECIFIC WEIGHT CONTROL FORMULA
NO
NO
808
ARE THERE BREED-SIZE-SPECIFIC WEIGHT CONTROL FORMULAS IN THE MASTER PRODUCT LIST?
YES
809
RECOMMEND BREED-SIZE-SPECIFIC WEIGHT CONTROL FORMULAS
NO
810
RECOMMEND ALL-BREED-SIZE WEIGHT CONTROL FORMULAS

FIG. 8

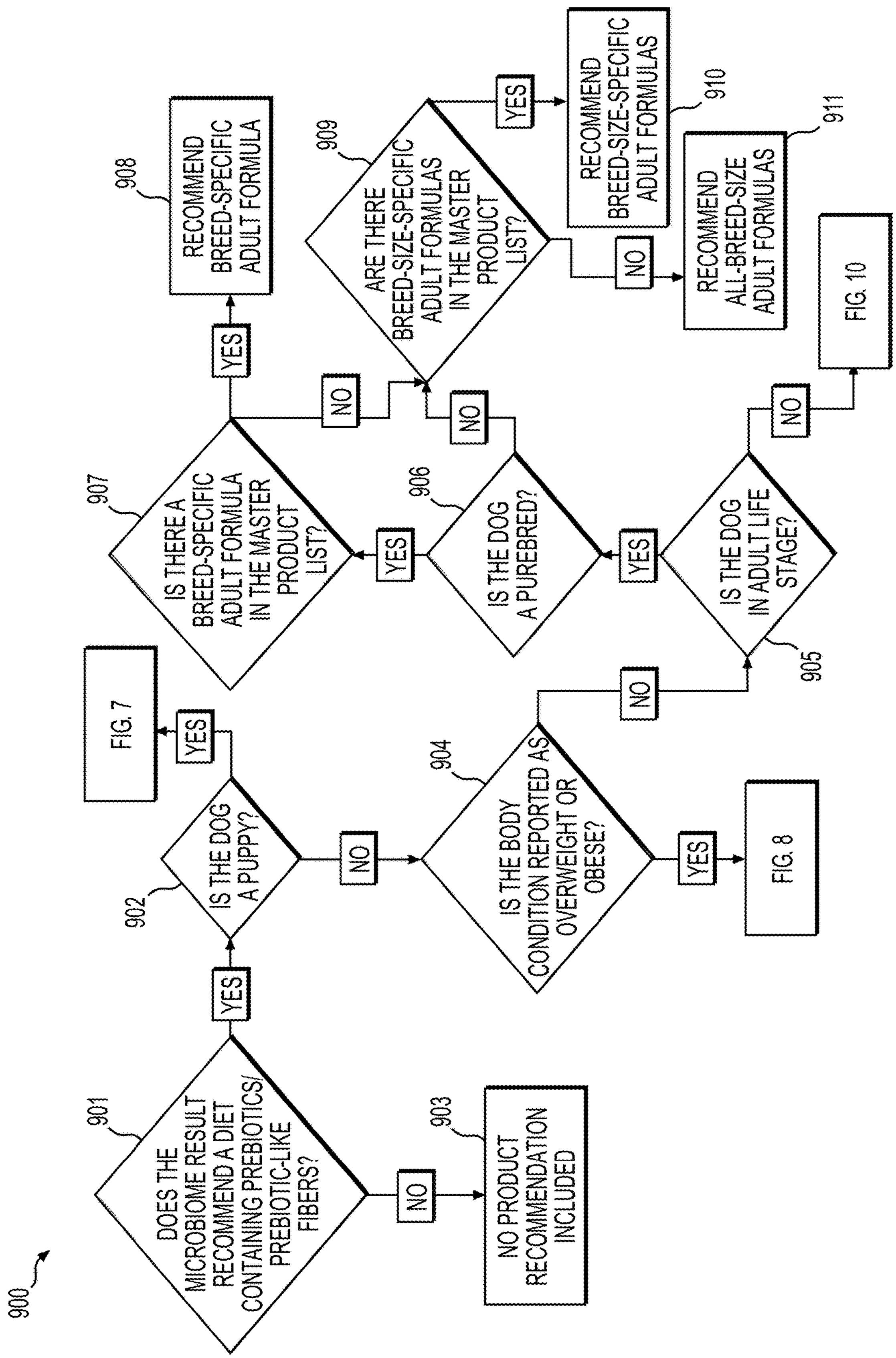

900
901
DOES THE MICROBIOME RESULT RECOMMEND A DIET CONTAINING PREBIOTICS/ PREBIOTIC-LIKE FIBERS?
NO
903
NO PRODUCT RECOMMENDATION INCLUDED
YES
902
IS THE DOG A PUPPY?
YES
FIG. 7
NO
904
IS THE BODY CONDITION REPORTED AS OVERWEIGHT OR OBESE?
YES
FIG. 8
NO
905
IS THE DOG IN ADULT LIFE STAGE?
NO
FIG. 10
YES
906
IS THE DOG A PUREBRED?
NO
YES
907
IS THERE A BREED-SPECIFIC ADULT FORMULA IN THE MASTER PRODUCT LIST?
YES
908
RECOMMEND BREED-SPECIFIC ADULT FORMULA
NO
909
ARE THERE BREED-SIZE-SPECIFIC ADULT FORMULAS IN THE MASTER PRODUCT LIST?
YES
910
RECOMMEND BREED-SIZE-SPECIFIC ADULT FORMULAS
NO
911
RECOMMEND ALL-BREED-SIZE ADULT FORMULAS

FIG. 9

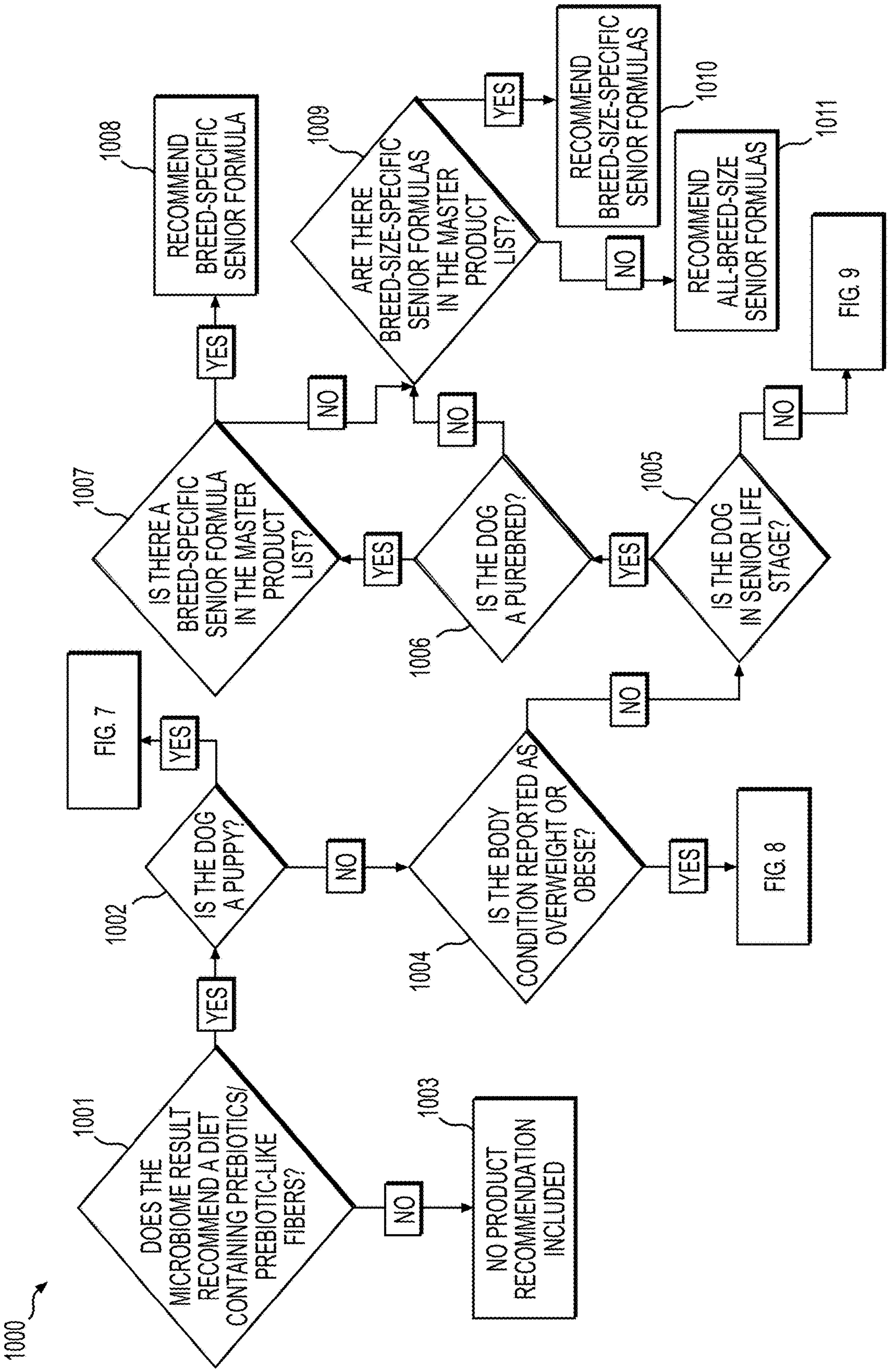

1000
1001
DOES THE MICROBIOME RESULT RECOMMEND A DIET CONTAINING PREBIOTICS/ PREBIOTIC-LIKE FIBERS?
YES
NO
1003
NO PRODUCT RECOMMENDATION INCLUDED
1002
IS THE DOG A PUPPY?
YES
FIG. 7
NO
1004
IS THE BODY CONDITION REPORTED AS OVERWEIGHT OR OBESE?
YES
FIG. 8
NO
1005
IS THE DOG IN SENIOR LIFE STAGE?
NO
FIG. 9
YES
1006
IS THE DOG A PUREBRED?
YES
NO
1007
IS THERE A BREED-SPECIFIC SENIOR FORMULA IN THE MASTER PRODUCT LIST?
YES
1008
RECOMMEND BREED-SPECIFIC SENIOR FORMULA
NO
1009
ARE THERE BREED-SIZE-SPECIFIC SENIOR FORMULAS IN THE MASTER PRODUCT LIST?
YES
1010
RECOMMEND BREED-SIZE-SPECIFIC SENIOR FORMULAS
NO
1011
RECOMMEND ALL-BREED-SIZE SENIOR FORMULAS

FIG. 10

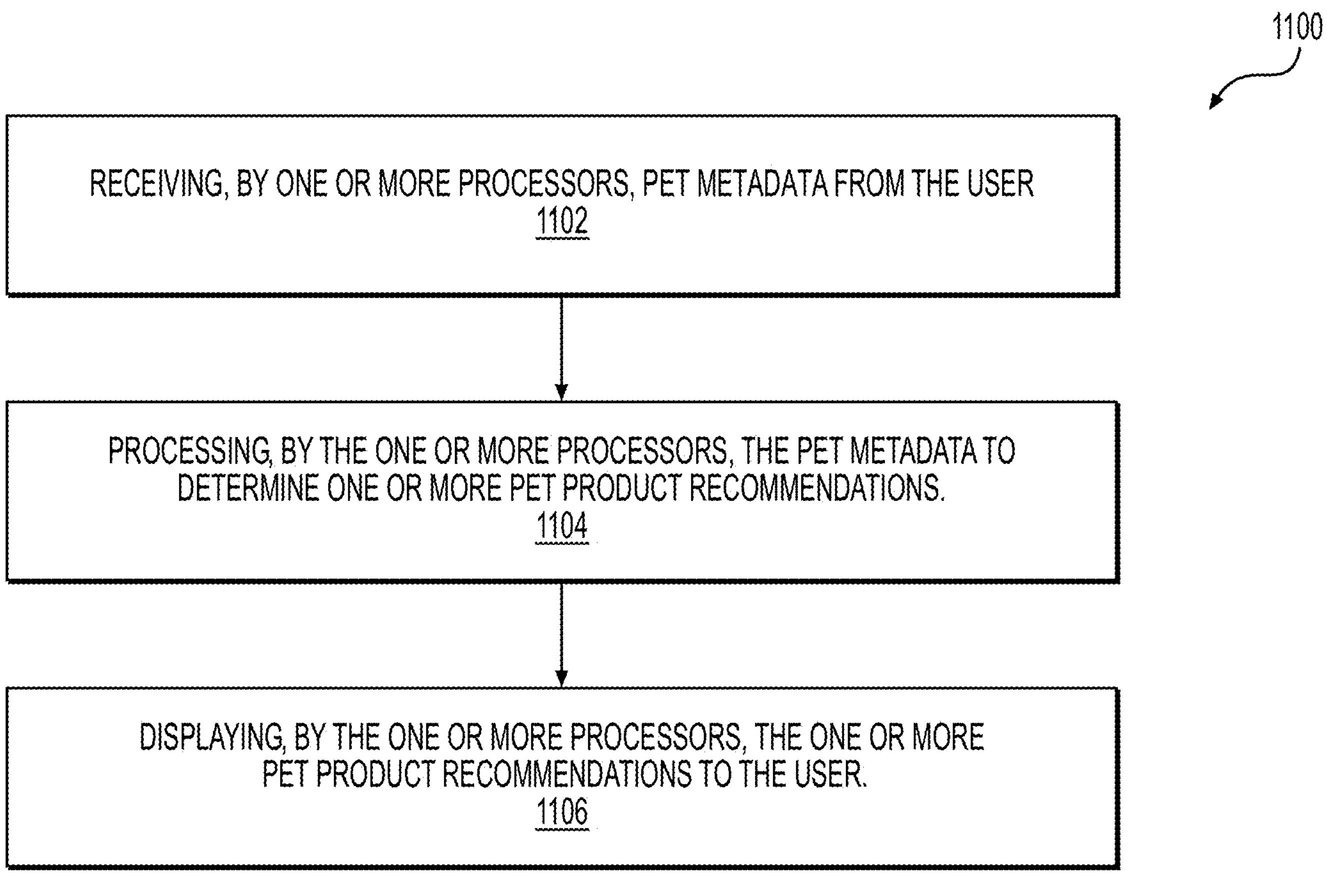

1100
RECEIVING, BY ONE OR MORE PROCESSORS, PET METADATA FROM THE USER
1102
PROCESSING, BY THE ONE OR MORE PROCESSORS, THE PET METADATA TO DETERMINE ONE OR MORE PET PRODUCT RECOMMENDATIONS.
1104
DISPLAYING, BY THE ONE OR MORE PROCESSORS, THE ONE OR MORE PET PRODUCT RECOMMENDATIONS TO THE USER.
1106

FIG. 11A

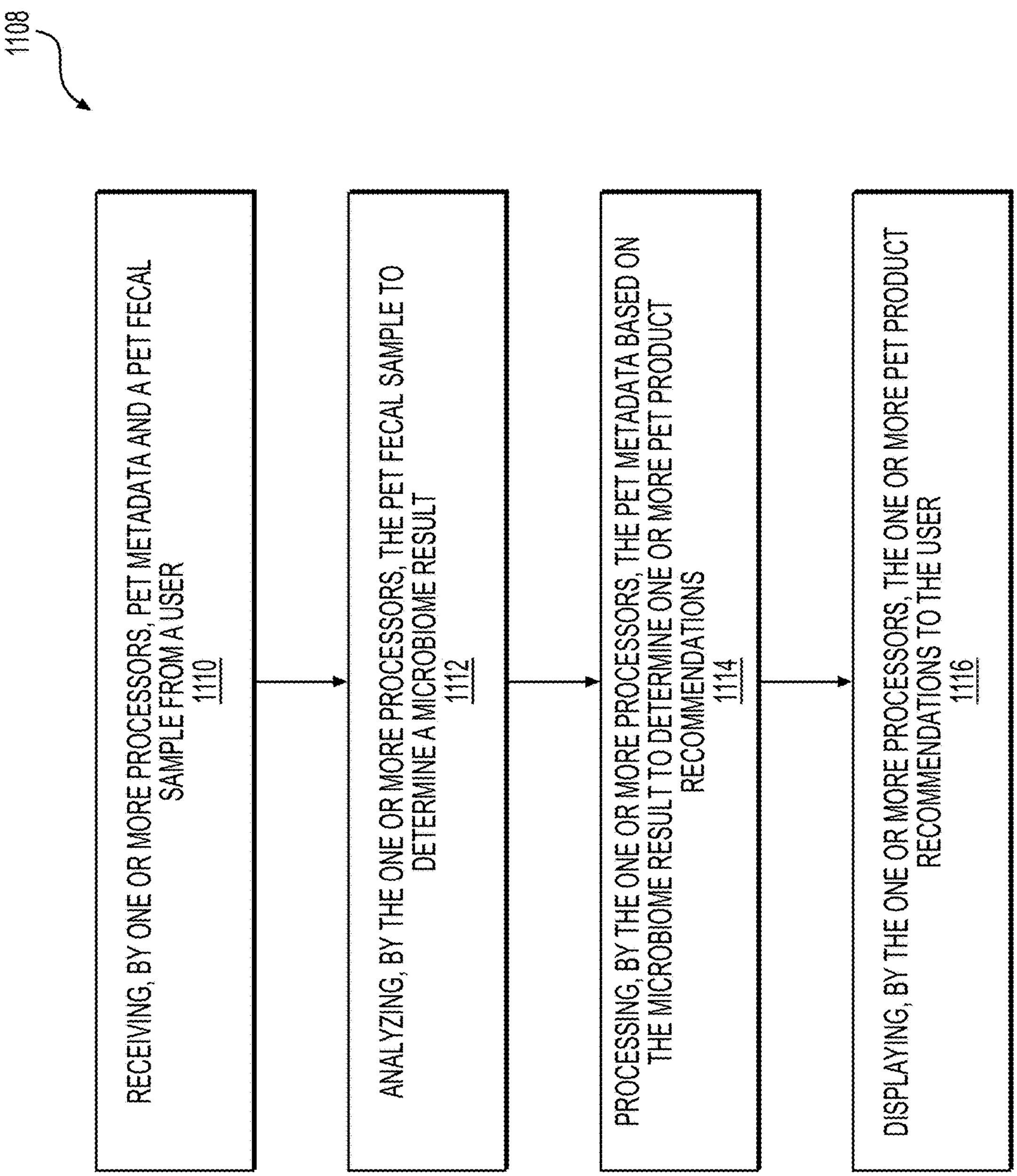

1108
RECEIVING, BY ONE OR MORE PROCESSORS, PET METADATA AND A PET FECAL SAMPLE FROM A USER
1110
ANALYZING, BY THE ONE OR MORE PROCESSORS, THE PET FECAL SAMPLE TO DETERMINE A MICROBIOME RESULT
1112
PROCESSING, BY THE ONE OR MORE PROCESSORS, THE PET METADATA BASED ON THE MICROBIOME RESULT TO DETERMINE ONE OR MORE PET PRODUCT RECOMMENDATIONS
1114
DISPLAYING, BY THE ONE OR MORE PROCESSORS, THE ONE OR MORE PET PRODUCT RECOMMENDATIONS TO THE USER
1116

FIG. 11B

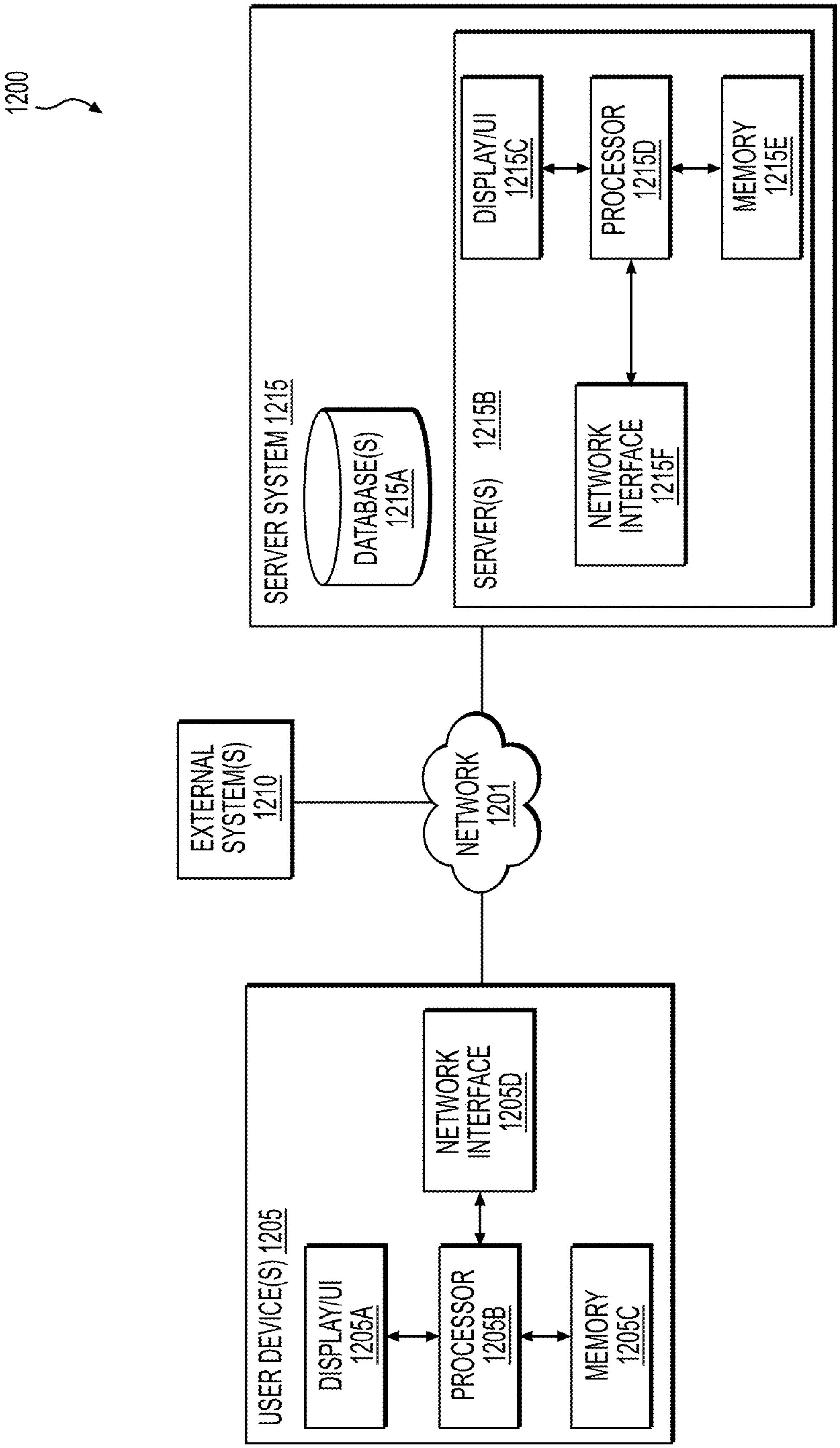

1200
USER DEVICE(S) 1205
DISPLAY/UI 1205A
PROCESSOR 1205B
MEMORY 1205C
NETWORK INTERFACE 1205D
EXTERNAL SYSTEM(S) 1210
NETWORK 1201
SERVER SYSTEM 1215
DATABASE(S) 1215A
SERVER(S) 1215B
DISPLAY/UI 1215C
PROCESSOR 1215D
MEMORY 1215E
NETWORK INTERFACE 1215F

FIG. 12

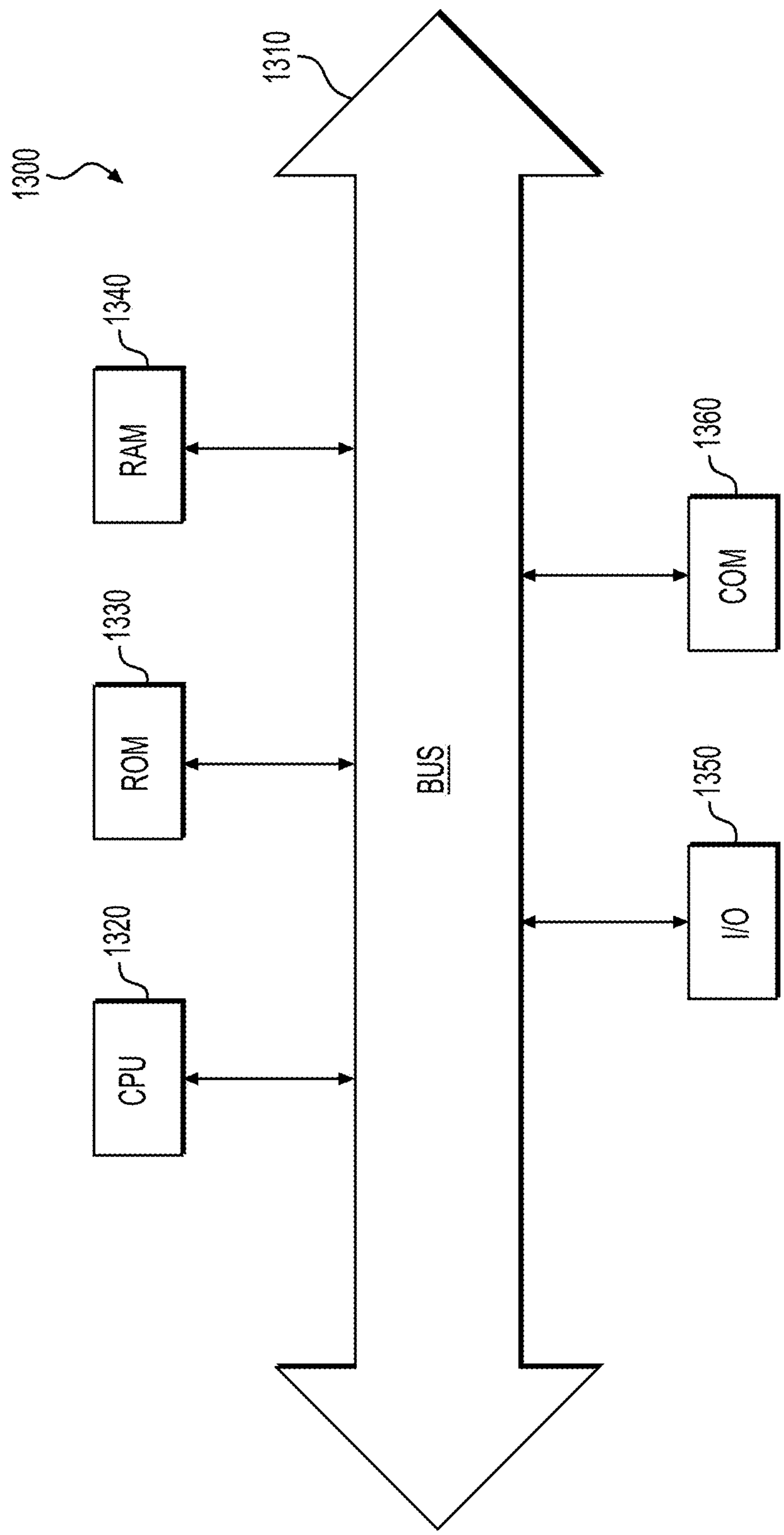


FIG. 13

SYSTEMS AND METHODS FOR RECOMMENDING A FOOD PRODUCT BASED ON CHARACTERISTICS OF AN ANIMAL

CROSS-REFERENCE TO RELATED APPLICATION(S)

[0001] This is a National Stage Application under 35 U.S.C. § 371 of International Application No. PCT/US2023/019106, filed on Apr. 19, 2023, which claims priority to U.S. Patent Application No. 63/363,551, filed on Apr. 25, 2022, the entireties of which are incorporated herein by reference.

TECHNICAL FIELD

[0002] Various embodiments of the present disclosure relate generally to providing one or more personalized pet product recommendations based on a microbiome result and pet metadata to a user. In some embodiments, the disclosure relates to systems and methods for using a microbiome result and pet metadata to determine at least one personalized pet product recommendation.

BACKGROUND

[0003] Pet product companies typically offer many different pet products (e.g., food, supplements, etc.) for pets (e.g., dogs, cats, birds, etc.). Given these many choices, a pet owner may have difficulty choosing an appropriate pet product for the owner's pet. The owner may choose a particular pet product without any external aid. In this case, the particular pet product may be incorrect, or inadequate, for the pet's needs. In another case, the owner may use a "product finder" application that provides the owner with a particular pet product recommendation based on some general input information, such as the breed of the pet, the age of the pet, etc. However, the application might select the recommended product based on overly-general characteristics of the pet, and might not account for latent, or specific, characteristics of the pet. In this case, the recommended pet product may also be incorrect, or inadequate, for the pet's particularized needs.

[0004] The pet's health may not improve, or may even deteriorate in the event that the pet is provided with incorrect, or inadequate, products. Accordingly, there is a need for a platform that can more accurately and holistically determine and recommend a particular pet product based on more particularized characteristics of a specific pet.

[0005] The present disclosure is directed to addressing the above-referenced challenges. The background description provided herein is for the purpose of generally presenting the context of the disclosure. Unless otherwise indicated herein, the materials described in the background section are not prior art to the claims in the present application and are not admitted to be prior art, or suggestions of the prior art, by inclusion in the background section.

SUMMARY OF THE DISCLOSURE

[0006] According to certain aspects of the disclosure, methods and systems are disclosed for providing one or more personalized pet product recommendations based on a microbiome result and pet metadata to a user.

[0007] In one aspect, an exemplary embodiment of a method for providing one or more personalized pet product recommendations based on pet metadata to a user includes receiving, by one or more processors, pet metadata from a user. The method may further include processing, by the one or more processors, the pet metadata to determine one or more pet product recommendations. The method may further include displaying, by the one or more processors, the one or more pet product recommendations to the user.

[0008] In a further aspect, an exemplary embodiment of a computer system for providing one or more personalized pet product recommendations based on a microbiome result and pet metadata to a user is disclosed, the computer system comprising at least one memory storing instructions, and at least one processor configured to execute the instructions to perform operations. The operations may include receiving pet metadata and a pet fecal sample from a user, analyzing the pet fecal sample to determine a microbiome result, processing the pet metadata based on the microbiome result to determine one or more pet product recommendations, and displaying the one or more pet product recommendations to the user.

[0009] In a further aspect, a non-transitory computer-readable medium containing instructions that, when executed by a processor, cause the processor to perform operations for providing one or more personalized pet product recommendations based on a microbiome result and pet metadata to a user is disclosed. The operations may include receiving pet metadata and a pet fecal sample from a user, analyzing the pet fecal sample to determine a microbiome result, processing the pet metadata based on the microbiome result to determine one or more pet product recommendations, and displaying the one or more pet product recommendations to the user.

[0010] In one aspect, an exemplary embodiment of a method for providing one or more personalized pet product recommendations based on a microbiome result and pet metadata to a user includes receiving, by one or more processors, pet metadata and a pet fecal sample from a user. The method may further include analyzing, by the one or more processors, the pet fecal sample to determine a microbiome result. The method may further include processing, by the one or more processors, the pet metadata based on the microbiome result to determine one or more pet product recommendations. The method may further include displaying, by the one or more processors, the one or more pet product recommendations to the user.

[0011] It is to be understood that both the foregoing general description and the following detailed description are exemplary and explanatory only and are not restrictive of the disclosed embodiments, as claimed.

BRIEF DESCRIPTION OF THE DRAWINGS

[0012] The accompanying drawings, which are incorporated in and constitute a part of the present specification, illustrate various exemplary embodiments and together with the description, serve to explain the principles of the disclosed embodiments.

[0013] FIGS. **1**A-**1**BB describe exemplary environments of a platform for analyzing a pet's fecal sample and displaying the result of the analysis, according to one or more embodiments.

[0014] FIGS. **2**A-**2**J also describe exemplary environments of a platform for analyzing a pet's fecal sample and displaying the result of the analysis, according to one or more embodiments.

[0015] FIGS. 3A-3D describe exemplary environments of a pet profile in a platform for gathering, analyzing, and displaying results corresponding to a pet's fecal sample, according to one or more embodiments.

[0016] FIGS. 4A-4J describe exemplary environments of a platform for gathering, analyzing, and displaying results corresponding to a pet's fecal sample, according to one or more embodiments.

[0017] FIGS. 5A-5JJ describe exemplary environments of a platform for analyzing a pet fecal sample and displaying corresponding results, according to one or more embodiments.

[0018] FIGS. 6A-6C describe exemplary environments of a platform for displaying pet product recommendations, according to one or more embodiments.

[0019] FIG. 7 depicts a flowchart of an exemplary method for determining a pet product recommendation for a puppy, according to one or more embodiments.

[0020] FIG. 8 depicts a flowchart of an exemplary method for determining a pet product recommendation for an overweight pet, according to one or more embodiments.

[0021] FIG. 9 depicts a flowchart of an exemplary method for determining a pet product recommendation for a normal weight adult pet, according to one or more embodiments.

[0022] FIG. 10 depicts a flowchart of an exemplary method for determining a pet product recommendation for a normal weight senior pet, according to one or more embodiments.

[0023] FIG. 11A depicts a flowchart of an exemplary method for providing one or more personalized pet product recommendations based on pet metadata to a user, according to one or more embodiments.

[0024] FIG. 11B depicts a flowchart of an exemplary method for providing one or more personalized pet product recommendations based on a microbiome result and pet metadata to a user, according to one or more embodiments.

[0025] FIG. 12 depicts an exemplary environment that may be utilized with techniques presented herein, according to one or more embodiments.

[0026] FIG. 13 depicts an example of a computing device that may execute the techniques described herein, according to one or more embodiments.

DETAILED DESCRIPTION OF EMBODIMENTS

[0027] As addressed above, some techniques for selecting a pet product might result in the selection of an incorrect, or inadequate, pet product. As a result, the pet's health may deteriorate, or worsen, based on an inappropriate diet. According to certain aspects of the disclosure, methods and systems are disclosed for providing one or more personalized pet product recommendations based on a microbiome result and pet metadata to a user. By determining a particular recommended pet product based on the microbiome condition of the pet, the embodiments herein provide an improvement in pet product selection technology by more accurately and holistically accounting for latent and particularized characteristics of a pet when determining recommended pet products. Further, and ultimately, the embodiments herein result in the improved health of pets by more accurately determining and recommending pet products in an individualized manner.

[0028] As will be discussed in more detail below, in various embodiments, systems and methods are described for providing one or more personalized pet product recommendations based on a microbiome result and pet metadata to a user. By receiving pet metadata and a pet fecal sample from a user, the systems and methods may be able to analyze the pet fecal sample to determine a microbiome result. The systems and methods may then use the microbiome result to process the pet metadata to determine one or more pet product recommendations. The systems and methods may then display the one or more pet product recommendations to the user.

[0029] As used herein, a "pet" may refer to any type of animal, such as a dog, a cat, a bird, a fish, etc., that an owner may feed using pet products. However, it should be understood that some embodiments herein are applicable to other types of animals that are less commonly used as "pets," such as cows, sheep, etc. In other words, some embodiments herein are applicable to any types of animals, including domesticated animals.

Exemplary Pet Fecal Sample Analysis Platform

[0030] FIGS. 1A-J describe exemplary environments of a platform for analyzing a pet's fecal sample and displaying the result of the analysis, according to one or more embodiments.

[0031] FIG. 1A illustrates a "tapp" page of an exemplary environment of a platform for gathering, analyzing, and displaying results corresponding to a pet's fecal sample. The platform may display a page allowing for a user to "Get started," "Sign in," or "Create an account."

[0032] FIG. 1B further illustrates a "Create an account" page of an exemplary environment of a platform for gathering, analyzing, and displaying results corresponding to a pet's fecal sample. The platform may prompt the user to provide an "email address" and "create a password." The platform may also prompt the user to confirm that the user is a particular age (e.g., "over 16"). The platform may also prompt the user to confirm that the user agrees to the "terms and conditions" of the system.

[0033] FIG. 1C further illustrates a "Before we dive in . . . " page of an exemplary environment of a platform for gathering, analyzing, and displaying results corresponding to a pet's fecal sample. The platform may ask the user to acknowledge and agree to particular terms and conditions (e.g., "This app is designed to help you monitor the well-being of your pet. It is not a diagnostic tool and does not replace regular consultations with your vet.") by clicking an "Ok, let's go!" button.

[0034] FIG. 1D further illustrates a "Healthy gut, healthy body" page of an exemplary environment of a platform for gathering, analyzing, and displaying results corresponding to a pet's fecal sample. The platform may emphasize the importance of a pet's healthy gut (e.g., "A healthy gut is important to your dog's overall health.") The platform may prompt the user to go to the next screen by pressing a "Next" button.

[0035] FIG. 1E further illustrates a "Breed" page of an exemplary environment of a platform for gathering, analyzing, and displaying results corresponding to a pet's fecal sample. The platform may prompt the user to enter at least one breed of the pet. In some embodiments, the user may enter more than one breed (e.g., "English Foxhound" and "Dachshund"). The platform may prompt the user to go to the next screen by pressing a "Next" button.

[0036] FIG. 1F further illustrates a "Human Food" page of an exemplary environment of a platform for gathering, analyzing, and displaying results corresponding to a pet's fecal sample. The platform may prompt the user to indicate if the pet eats human food, and if so, how frequently (e.g., "Every so often," "Fairly often," "Very often," or "Never").

[0037] FIG. 1G further illustrates a "Tracker" page of an exemplary environment of a platform for gathering, analyzing, and displaying results corresponding to a pet's fecal sample. The platform may prompt the user to provide additional information about the pet (e.g., "Tessie"). The platform may walk the user through how the platform works via a "How does this work?" button. The platform may provide buttons for test results (e.g., "Tests"), a fecal tracker (e.g., "Tracker"), the ability to capture an image of the fecal sample (e.g., "Capture"), a summary of the fecal results (e.g., "Summary"), and further information about the pet (e.g., "Pets").

[0038] FIG. 1H further illustrates a "Neutered Information" page of an exemplary environment of a platform for gathering, analyzing, and displaying results corresponding to a pet's fecal sample. The platform may prompt the user to select a button corresponding to whether the pet is neutered (e.g., "Yes," "No," or "I don't know"). The platform may provide an option to the user to "Skip" entering such information.

[0039] FIG. 1I further illustrates a "Food Type" page of an exemplary environment of a platform for gathering, analyzing, and displaying results corresponding to a pet's fecal sample. The platform may prompt the user to indicate the type of food (e.g., "Dry," "Wet," or "Mixed") that the pet is fed for each meal (e.g., "Meal 1," "Meal 2," or "Meal 3"). The platform may provide an option to the user to "Skip" entering such information.

[0040] FIG. 1J further illustrates an "Allergies" page of an exemplary environment of a platform for gathering, analyzing, and displaying results corresponding to a pet's fecal sample. The platform may prompt the user to indicate whether the pet has any allergies (e.g., "No, not that I know of" or "Yes, and they are: "). The platform may also allow the user to enter one or more allergies that the pet may have (e.g., "Lamb," "Mutton," "Anything"). The platform may provide an option to the user to "Skip" entering such information.

[0041] FIGS. 1K-M further illustrate a "Pets" page of an exemplary environment of a platform for gathering, analyzing, and displaying results corresponding to a pet's fecal sample. The platform may display the "Pets" page upon a user selecting the "Pets" button. The platform may display meal information corresponding to a particular pet. For example, the platform may display the pet's name (e.g., "Tessie"), the pet's gender (e.g. "Male"), or the pet's breed (e.g., "Mixed Breed"). The platform may also display the meals per day (e.g., "3"), the type(s) of dog food served for each meal (e.g., "Dry" and "Wet" and "Mixed"), how frequently the pet is given human treats (e.g., "Very often"), the current type of dry food (e.g., "Brand For Vitality—Adult"), the current wet food (e.g., "Brand—Adult"), whether the pet takes a diet supplement (e.g., "None"), the pet's age bracket (e.g., "6 to 12 months"), the pet's current weight (e.g., "10"), any known allergies of the pet (e.g., "Lamb, Mutton, Anything"), any existing conditions of the pet (e.g., "None"), or whether the pet is neutered (e.g., "No").

[0042] FIG. 1N further illustrates a "Known Allergies" page of an exemplary environment of a platform for gathering, analyzing, and displaying results corresponding to a pet's fecal sample. The platform may provide a user interface to allow the user to enter any allergies that the pet may have (e.g., "Lamb," "Mutton," "Anything").

[0043] FIG. 1O further illustrates a "Camera Roll" page of an exemplary environment of a platform for gathering, analyzing, and displaying results corresponding to a pet's fecal sample. The platform may allow a user to upload an image of the pet via a camera roll. The image may include a photo that was saved on the user's device. The image may also include a photo that was taken by the user's device. The user may modify the image via the mobile device (e.g., "Crop," "Adjust," etc.).

[0044] FIG. 1P further illustrates a "Pets" page with the pet's image of an exemplary environment of a platform for gathering, analyzing, and displaying results corresponding to a pet's fecal sample. Upon a user selecting the pet's image (as illustrated in FIG. 1O), the platform may display the image in the "Pets" page.

[0045] FIG. 1Q further illustrates a "Capture" page of an exemplary environment of a platform for gathering, analyzing, and displaying results corresponding to a pet's fecal sample. The platform may provide the opportunity for the user to capture one or more images of a pet's fecal sample. The user may digitally upload an image of the pet's fecal sample by selecting the "Capture new poop" option. The platform may also provide instructions and guidance to the user when the user selects the "How does this work?" button.

[0046] FIG. 1R further illustrates a "Fecal Capture" page of an exemplary environment of a platform for gathering, analyzing, and displaying results corresponding to a pet's fecal sample. The platform may allow a user to upload an image of the fecal sample via a camera roll. The image of the fecal sample may include a photo that was saved on the user's device. The image of the fecal sample may also include a photo that was taken by the user's device.

[0047] FIG. 1S further illustrates a "Result" page of an exemplary environment of a platform for gathering, analyzing, and displaying results corresponding to a pet's fecal sample. The platform may analyze the image of the fecal sample and display a result indicating where the fecal sample lands on a scale of "Too dry" to "Too wet." The platform may display one or more fecal results, where each fecal result corresponds to the analysis of a fecal sample (e.g., "Poop 1"). The platform may also display a timestamp of the fecal sample (e.g., "Poop 1 23:37") and a corresponding summary of the analysis (e.g., "A little wet, but ok"). The platform may also display future suggestions based on the analysis of the image of the fecal sample (e.g., "This could be normal for Tessie. Continue to monitor his poop and track his appetite and energy levels," and "If this is unusual for Tessie or he seems unwell, consult his vet for advice.").

[0048] FIG. 1T further illustrates a "Tracker" page of an exemplary environment of a platform for gathering, analyzing, and displaying results corresponding to a pet's fecal sample. The platform may display the page upon a user selecting the "Tracker" option. The platform may display one or more fecal records, where each fecal record corresponds to the analysis of a fecal sample (e.g., "Poop 1" and "Poop 2"). The platform may also display a timestamp of the fecal sample (e.g., "Poop 1 23:37" or "Poop 2 23:38") and a corresponding summary of the analysis (e.g., "A little wet, but ok"). The platform may also display the user's notes corresponding to a particular fecal record (e.g., "New food, Ate waste, Drinking less, Blood, Worms, Something odd, Straining, Flatulence, Low energy, High energy, Nervous energy"). The platform may also provide the option for the user to remove one or more of the fecal records.

[0049] FIG. **1**U further illustrates a "Delete Fecal Data" page of an exemplary environment of a platform for gathering, analyzing, and displaying results corresponding to a pet's fecal sample. The platform may allow a user to delete some or all of the fecal data. For example, the user may select the "Delete" button of one or more fecal records. Upon a user selecting the "Delete" button, the platform may display a confirmation regarding whether the user wants to delete the fecal data (e.g., "Are you sure you want to delete this poop data? This cannot be undone.")

[0050] FIG. **1**V further illustrates the "Tracker" page with the ability to see fecal details of an exemplary environment of a platform for gathering, analyzing, and displaying results corresponding to a pet's fecal sample. The platform may present the user an option to view fecal details (e.g., "Poop details") of the pet. The platform may also present the user an option to "share" the "Tracker" page via social media, email, and the like.

[0051] FIG. **1**W further illustrates a "Poop Details" page of an exemplary environment of a platform for gathering, analyzing, and displaying results corresponding to a pet's fecal sample. The platform may provide the option for the user to select a particular date range (e.g. "March 1 to March 16") for which the user would like feces details. Upon selecting the "Next" button, the platform may proceed to FIG. **1**X.

[0052] FIG. **1**X further illustrates a "Poop Photos" page of an exemplary environment of a platform for gathering, analyzing, and displaying results corresponding to a pet's fecal sample. The platform may ask the user if the user would like to include the previously submitted feces images (e.g., "Do you want to include the photos of the poops you have tracked?"). The platform may also provide the option for the user to select a particular date range (e.g. "March 1 to March 16") that the user would like feces details. Upon selecting the "Next" button, the platform may proceed to FIG. **1**Y.

[0053] FIG. **1**Y further illustrates a "Tracker Results" page of an exemplary environment of a platform for gathering, analyzing, and displaying results corresponding to a pet's fecal sample. The platform may provide the results of the fecal image analysis of the particular date range selected in FIG. **1**W. The results may include the date (e.g., "17 Mar. 2022"), the timestamp (e.g., "11:38 PM"), and the analysis results (e.g., "A little wet, but ok"). The platform may also provide the image of the pet, the name of the pet (e.g., "Tessie"), and/or the breed of the pet (e.g., "Mixed Breed (English, Foxhound, Dachshund)"). The platform may provide the option for the user to "Email" or "Download" the results (e.g., "Tessie's Poop Tracker Results").

[0054] FIG. **1**Z further illustrates an "Email Results" page of an exemplary environment of a platform for gathering, analyzing, and displaying results corresponding to a pet's fecal sample. The platform may provide the option for the user to enter one or more email addresses, where the platform may email the results of the fecal image analysis to the one or more email addresses.

[0055] FIG. **1**AA further illustrates a "Download" page of an exemplary environment of a platform for gathering, analyzing, and displaying results corresponding to a pet's fecal sample. The platform may display the "Download" page in response to the user selecting the "Download" option. The report may include the pet's name (e.g., "Tessie"), the pet's breed (e.g., "Mixed Breed (English Foxhound, Dachshund)"), the date (e.g., "17 Mar. 2022"), the timestamp (e.g., "11:38 PM"), the result of the fecal image analysis (e.g., "A little wet, but ok"), the fecal sample image, a graphic of a scale that indicates where the results land on a range of "Too dry" to "Too wet," and/or any custom notes (e.g., "This is a custom note added under a poop record."). The platform may provide the option for the user to enter one or more email addresses, where the platform may email the results of the fecal image analysis to the one or more email addresses.

[0056] FIG. **1**BB further illustrates the email from the "Email Results" page of an exemplary environment of a platform for gathering, analyzing, and displaying results corresponding to a pet's fecal sample. If a user selects the "Email" option in FIG. **1**Y, the platform may automatically generate an email to each of the one or more email addresses that were added in FIG. **1**Z. The platform may display the generated email to the user, where the user may decide to send the email. In some embodiments, the platform may automatically send the generated email to each of the one or more email addresses that were added.

[0057] FIGS. **2**A-D further illustrate a "Summary" page of an exemplary environment of a platform for gathering, analyzing, and displaying results corresponding to a pet's fecal sample. The "Summary" page may be presented to the user in response to the user selecting the "Summary" button. The platform may provide an "Overview" section that includes the average feces checked per day (e.g., "2"), the most common analysis results ("e.g., "A little wet, but ok"), and/or a most common note (e.g., "New food"). The platform may also display one or more links to one or more publications that may provide additional information (e.g., "Know Your Pet" or "Discover nutrition for puppies and dogs" or "Keep your dog at a healthy weight"). If the user selects one of the links, the platform may transfer the user to an external page that corresponds to the link, as shown in FIG. **2**D.

[0058] FIG. **2**E further illustrates an "Account Details" page of an exemplary environment of a platform for gathering, analyzing, and displaying results corresponding to a pet's fecal sample. The platform may display options for the user to "Manage account," provide "Help" (e.g., "How the poop scanner works," "FAQs," or "Contact us"), and/or provide a "Sign out" option.

[0059] FIGS. **2**F-H further illustrate a "FAQ" page of an exemplary environment of a platform for gathering, analyzing, and displaying results corresponding to a pet's fecal sample. The platform may provide answers to popular questions (e.g., "Can you tell me about my dog's gut microbiome"). The popular questions may be sectioned into questions regarding the "Tracker" and questions regarding the "Gut Report."

[0060] FIGS. **2**I-J further illustrate a "Delete Account" page of an exemplary environment of a platform for gathering, analyzing, and displaying results corresponding to a pet's fecal sample. The platform may provide the user an option to delete the user's data. The platform may confirm that the user would like to delete all of the user's data via an "Ok" button. The platform may further confirm (and clarify) that the user would like to delete the user's account via a "Delete account and all data" button.

[0061] FIG. **3**A further illustrates an "Add Another Dog" page of an exemplary environment of a platform for gathering, analyzing, and displaying results corresponding to a pet's fecal sample. The platform may display the current pet corresponding to the user (e.g., "Tessie"), as well as provide the user with an option to "Add another dog." Such an option may allow for the user to have one or more pet profiles associated with the user. This may allow for the user to more easily track the one or more pets. Upon the user selecting the "Add another dog" option, the platform may prompt the user to enter the pet's name (e.g., "Parker"), as illustrated in FIG. **3**B. Additionally, as illustrated in FIG. **3**C, the platform may further prompt the user to enter the pet's breed (e.g., "English Foxhound"), where the pet's breed may include one or more breeds. The platform may further prompt the user to fill in additional information, such as the pet's age, weight, and the like. Upon the user successfully adding all of the pet's information, the platform may associate the additional pet with the user. The additional association may be further illustrated in FIG. **3**D, where the platform display's a profile for "Parker" and a profile for "Tessie."

Exemplary Pet Fecal Sample Report Platform

[0062] FIGS. **4**A-J illustrate a "Tests" page of an exemplary environment of a platform for gathering, analyzing, and displaying results corresponding to a pet's fecal sample. The platform may begin with an introduction "Tests" page that outlines the gut microbiome, as illustrated in FIG. **4**A. Upon the user indicating that the user wishes to proceed (e.g., by pressing the arrow), the platform may first ask the user if the pet follows a prescription diet, as illustrated in FIG. **4**B. If the user presses "Yes," the platform may inform the user that reports are not provided for pets on prescription diets, as illustrated in FIG. **4**C.

[0063] If the user presses "No," indicating that the pet is not on a prescription diet, the platform may inform the user to proceed to the next step, as illustrated in FIG. **4**D. The platform may then ask the user to provide additional information to provide the proper pet personalization. For example, as illustrated in FIG. **4**E, the platform may ask the user to provide a unique identifier corresponding to the fecal sample (e.g., "BIOHO0776.339"). The platform may also ask the user for the date that the fecal sample was collected (e.g., "Mar. 17, 2022"). Upon entering such information, the user may then select the "Start" button. The platform may then ask the user to provide additional pet details (e.g., pet metadata). Such additional pet details, as illustrated in FIGS. **4**F and **4**G, may include, but may not be limited to, the size of the pet (e.g., "Medium"), the kind of pet (e.g., "English Foxhound, Dachshund"), the sex of the pet (e.g., "Male"), whether the pet has been neutered/spayed (e.g., "No"), how many meals the pet eats per day (e.g., "3"), the food type for each meal ("Dry, Wet, Mixed"), the type of dry food fed to the pet (e.g., "Brand—Adult"), and/or the type of wet food fed to the pet (e.g., "Brand—Adult").

[0064] Upon the user pressing the "Next" button, the platform may ask the user if the user is ready to submit the information submission (e.g., "survey"), as illustrated in FIG. **4**H. In response to the user pressing the "Ok" button, the platform may display a notification (e.g., "Thank you! We will notify you by email once your dog's report is ready."), as illustrated in FIG. **4**I. The platform may display an indicator stating that the report is being processed (e.g., "Your dog's report is being processed."), as illustrated in FIG. **4**J.

[0065] FIGS. **5**A-LL describe exemplary environments of a platform for gathering, analyzing, and displaying results corresponding to a pet's fecal sample, according to one or more embodiments.

[0066] As illustrated in FIG. **5**A, the platform may display particular disclaimers and/or terms and conditions to the user. Upon the user indicating the user's acknowledgment (e.g., by pressing the "Ok" button), the platform may display a "Gut Intelligence Report," as illustrated in FIGS. **5**B-LL.

[0067] FIGS. **5**C-E further illustrate a "The Headlines" page of an exemplary environment of a platform for gathering, analyzing, and displaying results corresponding to a pet's fecal sample. The platform may display the pet's "Diversity Score," indicating the diversity of the pet's microbiome in comparison to other pets that may share similar traits (e.g., similar age). The "Diversity Score" may be displayed as a scale, where the scale indicates the particular pet's "Diversity Score," as well as the "Diversity Score" of the other pets that may share one or more similar traits. The platform may also display a personalized analysis of the "Diversity Score" results, where the personalized analysis may be in a written form, as illustrated in FIGS. **5**C-E. Additionally, the platform may display the "Diversity Score" personalized analysis of the results in an easily digestible manner for the user, where common terms and words are used.

[0068] FIG. **5**F further illustrates an "Our Recommendations" page of an exemplary environment of a platform for gathering, analyzing, and displaying results corresponding to a pet's fecal sample. The platform may display one or more personalized recommendations for the pet, where the one or more recommendations (e.g., "Consider the amount of exercise Tessie gets" or "Ensure Tessie's diet contains natural fibers") are directed towards improving the pet's "Diversity Score." The platform may also provide an option for the user to receive one or more recommended diets (e.g., by the user selecting a "Discover your dog's recommended diets" button).

[0069] FIGS. **5**G-K further illustrate a "The Details" page of an exemplary environment of a platform for gathering, analyzing, and displaying results corresponding to a pet's fecal sample. The platform may display one or more areas or categories associated with the pet's health that the microbiome may provide clues about the pet's wellbeing. Such areas may include "Digestive Health," "Immunity," "Healthy Weight," "Exercise," "Mobility," and/or "Vitamin Production." The platform may display further details regarding each area, as described in further detail below.

[0070] FIGS. **5**H-N further illustrate a "Digestive Health" page of an exemplary environment of a platform for gathering, analyzing, and displaying results corresponding to a pet's fecal sample. The platform may display a notification (e.g., "Good") indicating if the pet's digestive health is on track. The platform may also display a personalized analysis of the "Digestive Health" of the pet, where the personalized analysis may be in a written form, as illustrated in FIGS. **5**H-J. Additionally, the platform may display the "Digestive Health" personalized analysis of the results in an easily digestible manner for the user, where common terms and words are used.

[0071] The platform may also display a notification (e.g., "Good") indicating if the pet's energy use is on track, as illustrated in FIG. **5**J. The platform may also display one or more sliding scales of which bugs/bacteria (e.g., "*Bacteroides,*" "*Bifidobacterium*," or "Ruminococcaceae") were found in the pet, as illustrated in FIG. **5**K. Additionally, the platform may display where the pet's results fall on one or more sliding scales by placing an indicator on the one or more scales. The platform may further display an analysis of each indicator (e.g., "High," "In Range," and the like).

[0072] The platform may also display a notification (e.g., "Good") indicating if the pet's gut defense is on track, as illustrated in FIG. **5**L. The platform may also display one or more sliding scales of which bugs (e.g., "*Lactobacillus*") was found in the pet, as illustrated in FIG. **5**L. Additionally, the platform may display where the pet's results fall on one or more sliding scales by placing an indicator on the one or more scales. The platform may further display an analysis of each indicator (e.g., "In Range").

[0073] The platform may also display a notification (e.g., "Good") indicating if the pet's gut wall support is on track, as illustrated in FIG. **5**M. The platform may also display one or more sliding scales of which bugs (e.g., "*Roseburia*," "Ruminococcaceae," and the like) were found in the pet, as illustrated in FIG. **5**M. Additionally, the platform may display where the pet's results fall on one or more sliding scales by placing an indicator on the one or more scales. The platform may further display an analysis of each indicator (e.g., "High").

[0074] The platform may also display a notification (e.g., "Good") indicating if the pet has any "tummy troublers" bacteria, as illustrated in FIG. **5**N. The platform may also display one or more sliding scales of which bugs (e.g., "*Clostridium perfringens*") were found in the pet, as illustrated in FIG. **5**N. Additionally, the platform may display where the pet's results fall on one or more sliding scales by placing an indicator on the one or more scales. The platform may further display an analysis of each indicator (e.g., "In Range").

[0075] FIGS. **5**O-R further illustrate an "Immunity" page of an exemplary environment of a platform for gathering, analyzing, and displaying results corresponding to a pet's fecal sample. The platform may display a notification (e.g., "Good") indicating if the pet's immune system is on track. The platform may also display a personalized analysis of the "Immunity" of the pet, where the personalized analysis may be in a written form, as illustrated in FIGS. **5**O-R. Additionally, the platform may display the "Immunity" personalized analysis of the results in an easily digestible manner for the user, where common terms and words are used.

[0076] The platform may also display a notification (e.g., "Good") indicating if the pet has any bugs for managing the pet's immune response, as illustrated in FIG. **5**Q. The platform may also display one or more sliding scales of which bugs (e.g., "*Faecalibacterium prausnitzii,*" "*Roseburia,*" and the like) were found in the pet, as illustrated in FIG. **5**Q. Additionally, the platform may display where the pet's results fall on one or more sliding scales by placing an indicator on the one or more scales. The platform may further display an analysis of each indicator (e.g., "High," "In Range," and the like).

[0077] The platform may also display a notification (e.g., "Good") indicating if the pet has any bugs for managing the pet's skin defense, as illustrated in FIG. **5**R. The platform may also display one or more sliding scales of which bugs (e.g., "*Lactobacillus*") were found in the pet, as illustrated in FIG. **5**R. Additionally, the platform may display where the pet's results fall on one or more sliding scales by placing an indicator on the one or more scales. The platform may further display an analysis of each indicator (e.g., "High," "In Range," and the like).

[0078] FIGS. **5**S-W further illustrate a "Healthy Weight" page of an exemplary environment of a platform for gathering, analyzing, and displaying results corresponding to a pet's fecal sample. The platform may display a notification (e.g., "Okay") indicating if the pet is able to maintain a healthy weight. The platform may also display a personalized analysis of the "Healthy Weight" of the pet, where the personalized analysis may be in a written form, as illustrated in FIGS. **5**S-W. Additionally, the platform may display the "Healthy Weight" personalized analysis of the results in an easily digestible manner for the user, where common terms and words are used.

[0079] The platform may also display a notification (e.g., "Okay") indicating if the pet has any bugs for managing the pet's weight, as illustrated in FIG. **5**V. The platform may also display one or more sliding scales of which bugs (e.g., "*Bacteroides,*" "*Clostridia,*" "*Lactobacillus,*" and the like) were found in the pet, as illustrated in FIG. **5**W. Additionally, the platform may display where the pet's results fall on one or more sliding scales by placing an indicator on the one or more scales. The platform may further display an analysis of each indicator (e.g., "High," "In Range," and the like).

[0080] FIGS. **5**X-AA further illustrate an "Exercise" page of an exemplary environment of a platform for gathering, analyzing, and displaying results corresponding to a pet's fecal sample. The platform may display a notification (e.g., "Okay") indicating if the pet has a good exercise level. The platform may also display a personalized analysis of the "Exercise" of the pet, where the personalized analysis may be in a written form, as illustrated in FIGS. **5**X-AA. Additionally, the platform may display the "Exercise" personalized analysis of the results in an easily digestible manner for the user, where common terms and words are used.

[0081] The platform may also display a notification (e.g., "Okay") corresponding to the pet's activity level, as illustrated in FIGS. **5**Y-Z. The platform may also display one or more sliding scales of which bugs (e.g., "*Bacteroides*," "Lachnospiraceae," "*Roseburia*," and the like) were found in the pet, as illustrated in FIG. **5**AA. Additionally, the platform may display where the pet's results fall on one or more sliding scales by placing an indicator on the one or more scales. The platform may further display an analysis of each indicator (e.g., "High," "In Range," and the like).

[0082] FIGS. **5**BB-EE further illustrate a "Mobility" page of an exemplary environment of a platform for gathering, analyzing, and displaying results corresponding to a pet's fecal sample. The platform may display a notification (e.g., "Okay") indicating if the pet has good mobility. The platform may also display a personalized analysis of the "Mobility" of the pet, where the personalized analysis may be in a written form, as illustrated in FIGS. **5**BB-EE. Additionally, the platform may display the "Mobility" personalized analysis of the results in an easily digestible manner for the user, where common terms and words are used.

[0083] The platform may also display a notification (e.g., "Okay") corresponding to the pet's joint support, as illustrated in FIGS. **5**CC-DD. The platform may also display one or more sliding scales of which bugs (e.g., "*Faecalibacterium Prausnitzii,*" "*Prevotella,*" and the like) were found in the pet, as illustrated in FIG. **5**EE. Additionally, the platform may display where the pet's results fall on one or more sliding scales by placing an indicator on the one or more scales. The platform may further display an analysis of each indicator (e.g., "High," "In Range," and the like).

[0084] FIGS. **5**FF-II further illustrate a "Vitamin Production" page of an exemplary environment of a platform for gathering, analyzing, and displaying results corresponding to a pet's fecal sample. The platform may display a notification (e.g., "Good") indicating if the pet is getting the right vitamins. The platform may also display a personalized analysis of the "Vitamin Production" of the pet, where the personalized analysis may be in a written form, as illustrated in FIGS. **5**FF-II. Additionally, the platform may display the "Vitamin Production" personalized analysis of the results in an easily digestible manner for the user, where common terms and words are used.

[0085] The platform may also display a notification (e.g., "Good") corresponding to the pet's vitamin levels, as illustrated in FIG. **5**II. The platform may also display one or more sliding scales of which bugs (e.g., "Bifidobacterium," "Lactobacillus," and the like) were found in the pet, as illustrated in FIG. **5**II. Additionally, the platform may display where the pet's results fall on one or more sliding scales by placing an indicator on the one or more scales. The platform may further display an analysis of each indicator (e.g., "High," "In Range," and the like).

[0086] FIG. **5**JJ further illustrates a "Tests" page of an exemplary environment of a platform for gathering, analyzing, and displaying results corresponding to a pet's fecal sample. The platform may display a summary of the fecal sample testing process.

[0087] Notably, the pages or user interfaces displayed for each area/category pertaining to the pet's health (e.g., digestive health, immunity, healthy weight, exercise, mobility, and vitamin production) are organized in a user-friendly manner that may aid the user's understanding of the results and may also improve the user's convenience. For example, the platform may first display a general overview/summary of the results pertaining to the corresponding area/category (e.g., FIGS. **5**H, **5**O, **5**S-**5**T, **5**X, **5**BB, **5**FF-**5**GG), followed by a more detailed, in-depth discussion (e.g., FIGS. **5**I-**5**N, **5**P-**5**R, **5**U-**5**W, **5**Y-**5**AA, **5**CC-**5**EE, **5**HH-**5**II) including bacterial levels relevant to the area/category and what the bacterial levels mean.

[0088] FIGS. **6**A-C describe exemplary environments of a platform for recommending a particular diet based on an analysis of a pet's fecal sample, according to one or more embodiments. In some embodiments, the platform may display FIGS. **6**A-C in response to a user selecting the "Discover your dog's recommended diets" button in FIG. **5**F. The platform may display one or more "Recommended Diets" (e.g., pet products) to the user. Each of the one or more "Recommended Diets" may be personalized recommendations to the pet, where the personalized recommendations are based on an analysis of the pet's fecal sample and/or pet metadata provided by user. Additionally, FIGS. **7**-**11**B may provide additional detail regarding the process for determining one or more recommended diets.

Exemplary Methods for Determining Pet Product Recommendations Based on a Hierarchy of Factors

[0089] FIGS. **7**-**10** illustrates exemplary processes for determining a pet product recommendation, according to one or more embodiments.

[0090] FIG. **7** depicts a flowchart of an exemplary method **700** for determining a pet product recommendation for a puppy, according to one or more embodiments. A platform may determine whether a microbiome result recommends a diet containing prebiotics or prebiotic-like fibers (Step **701**). Based on determining that the microbiome result recommends a diet containing prebiotics or prebiotic-like fibers (Step **701**—YES), the platform may determine whether the dog is a puppy (Step **702**). Additionally, the recommendation that the diet contains prebiotics or prebiotic-like fibers may include a range of prebiotics or prebiotic-like fibers. For example, the range may include a number corresponding to a threshold amount of prebiotics or prebiotic-like fibers that should be included in the pet product recommendation. Alternatively, for example, the range may include a range of specific prebiotics or prebiotic-like fibers that should be included in the pet product recommendation.

[0091] Alternatively, based on determining that the microbiome result does not recommend a diet containing prebiotics or prebiotic-like fibers (Step **701**—NO), the platform may not provide a product recommendation (Step **703**). Additionally, in some embodiments, the platform may determine whether a microbiome test has been performed, or whether a microbiome result is available. If a microbiome test has not been performed, or a microbiome result does not exist or cannot be provided, the platform may skip Step **701** and begin at Step **702**. In other words, the method **700** may begin at step **702** and may be performed without a microbiome result.

[0092] Based on determining that the dog is a puppy (Step **702**—YES), the platform may determine whether the dog is a purebred (**704**). Based on determining that the dog is a purebred (Step **704**—YES), the platform may determine whether there is a breed-specific puppy formula in a master product list (Step **705**) containing a plurality of pet products. Based on determining that there is a breed-specific puppy formula in the master product list (Step **705**—YES), the platform may recommend the breed-specific puppy formula (Step **706**).

[0093] Alternatively, based on determining that the dog is not a purebred (Step **704**—NO) or determining that there is not a breed-specific puppy formula in the master product list (Step **705**—NO), the platform may determine whether there are breed-size-specific puppy formulas in the master product list (Step **707**).

[0094] Based on determining that there are breed-size-specific puppy formulas in the master product list (Step **707**—YES), the platform may recommend the breed-size-specific puppy formulas (Step **708**). Alternatively, based on determining that there are not breed-size-specific puppy formulas in the master product list (Step **707**—NO), the platform may recommend all-breed-size puppy formulas (Step **709**).

[0095] FIG. **8** depicts a flowchart of an exemplary method **800** for determining a pet product recommendation for an overweight pet, according to one or more embodiments.

[0096] A platform may determine whether a microbiome result recommends a diet containing prebiotics or prebiotic-like fibers (Step **801**). Based on determining that the microbiome results recommends a diet containing prebiotics or prebiotic-like fibers (Step **801**—YES), the platform may determine whether the dog is a puppy (Step **802**). Additionally, the recommendation that the diet contains prebiotics or prebiotic-like fibers may include a range of prebiotics or prebiotic-like fibers. For example, the range may include a number corresponding to a threshold amount of prebiotics or prebiotic-like fibers that should be included in the pet product recommendation. Alternatively, for example, the range may include a range of specific prebiotics or prebiotic-like fibers that should be included in the pet product recommendation.

[0097] Alternatively, based on determining that the microbiome result does not recommend a diet containing prebiotics or prebiotic-like fibers (Step **801**—NO), the platform may not provide a product recommendation (Step **803**). Additionally, in some embodiments, the platform may determine whether a microbiome test has been performed, or whether a microbiome result is available. If a microbiome test has not been performed, or a microbiome result does not exist or cannot be provided, the platform may skip Step **801** and begin at Step **802**. In other words, the method **800** may begin at step **802** and may be performed without a microbiome result.

[0098] Based on determining that the dog is a puppy (Step **802**—YES), the platform may proceed to perform one or more Steps of the method **700** of FIG. **7**. Alternatively, based on determining that the dog is not a puppy (Step **802**—NO), the platform may determine whether a body condition of the dog is reported as being overweight or obese (Step **804**). Based on determining that body condition of the dog is reported as being overweight or obese (Step **804**—YES), the platform may determine whether the dog is a purebred (Step **805**). Alternatively, based on determining that the body condition of the dog is not reported as being overweight or obese (Step **804**—NO), the platform may perform one or more Steps of the method **900** of FIG. **9**.

[0099] Based on determining that the dog is a purebred (Step **805**—YES), the platform may determine whether there is a breed-specific weight control formula in a master product list (Step **806**). Based on determining that there is a breed-specific weight control formula in the master product list (Step **806**—YES), the platform may recommend the breed-specific weight control formula (Step **807**).

[0100] Alternatively, based on determining that the dog is not a purebred (Step **805**—NO) or determining that there is not a breed-specific weight control formula in the master product list (Step **806**—NO), the platform may determine whether there are breed-size-specific weight control formulas in the master product list (Step **808**). Based on determining that there are breed-size-specific weight control formulas in the master product list (Step **808**—YES), the platform may recommend the breed-size-specific weight control formulas (Step **809**). Alternatively, based on determining that there are not breed-size-specific weight control formulas in the master product list (Step **808**—NO), the platform may recommend all-breed-size weight control formulas (Step **810**).

[0101] FIG. **9** depicts a flowchart of an exemplary method **900** for determining a pet product recommendation for a normal weight adult pet, according to one or more embodiments.

[0102] A platform may determine whether a microbiome result recommends a diet containing prebiotics or prebiotic-like fibers (Step **901**). Based on determining that the microbiome result recommends a diet containing prebiotics or prebiotic-like fibers (Step **901**—YES), the platform may determine whether the dog is a puppy (Step **902**). Additionally, the recommendation that the diet contains prebiotics or prebiotic-like fibers may include a range of prebiotics or prebiotic-like fibers. For example, the range may include a number corresponding to a threshold of prebiotics or prebiotic-like fibers that should be included in the pet product recommendation. Alternatively, for example, the range may include a range of specific prebiotics or prebiotic-like fibers that should be included in the pet product recommendation.

[0103] Alternatively, based on determining that the microbiome result does not recommend a diet containing prebiotics or prebiotic-like fibers (Step **901**—NO), the platform may not provide a product recommendation (Step **903**). Additionally, in some embodiments, the platform may determine whether a microbiome test has been performed, or whether a microbiome result is available. If a microbiome test has not been performed, or a microbiome result does not exist or cannot be provided, the platform may skip Step **901** and begin at Step **902**. In other words, the method **900** may begin at step **902** and may be performed without a microbiome result.

[0104] Based on determining that the dog is a puppy (Step **902**—YES), the platform may perform one or more Steps of the method **700** of FIG. **7**. Alternatively, based on determining that the dog is not a puppy (Step **902**—NO), the platform may determine whether a body condition of the dog is reported as being overweight or obese (Step **904**). Based on determining that the body condition of the dog is reported as being overweight or obese (Step **904**—YES), the platform may perform one or more Steps of the method **800** of FIG. **8**. Alternatively, based on determining that the body condition of the dog is not reported as being overweight or obese (Step **904**—NO), the platform may determine whether the dog is in an adult life stage (Step **905**).

[0105] Based on determining that that the dog is not in an adult life stage (Step **905**—NO), the platform may perform one or more Steps of the method **1000** of FIG. **10**. Alternatively, based on determining that the dog is in an adult life stage (Step **905**—YES), the platform may determine whether the dog is a purebred (Step **906**).

[0106] Based on determining that the dog is a purebred (Step **906**—YES), the platform may determine whether there is a breed-specific adult formula in a master product list (Step **907**). Based on determining that there is a breed-specific adult formula in the master product list (Step **907**—YES), the platform may recommend the breed-specific adult formula (Step **908**).

[0107] Alternatively, based on determining that the dog is not a purebred (Step **906**—NO) or determining that there is not a breed-specific adult formula in the master product list (Step **907**—NO), the platform may determine whether there are breed-size-specific adult formulas in the master product list (Step **909**).

[0108] Based on determining that there are breed-size-specific adult formulas in the master product list (Step **909**—YES), the platform may recommend the breed-size-specific adult formulas (Step **910**). Alternatively, based on determining that there are not breed-size-specific adult formulas in the master product list (Step **909**—NO), the platform may recommend all-breed-size adult formulas (Step **911**).

[0109] FIG. **10** depicts a flowchart of an exemplary method **1000** for determining a pet product recommendation for a normal weight senior pet, according to one or more embodiments.

[0110] A platform may determine whether a microbiome result recommends a diet containing prebiotics or prebiotic-like fibers (Step **1001**). Based on determining that the microbiome result recommends a diet containing prebiotics or prebiotic-like fibers (Step **1001**—YES), the platform may determine whether the dog is a puppy (Step **1002**). Additionally, in some embodiments, the platform may determine whether a microbiome test has been performed, or whether a microbiome result is available. If a microbiome test has not been performed, the platform may skip Step **1001** and begin at Step **1002**. In other words, the method **1000** may begin at step **1002** and may be performed without a microbiome result. Alternatively, based on determining that the microbiome result does not recommend a diet containing prebiotics or prebiotic-like fibers (Step **1001**—NO), the platform may not provide a product recommendation (Step **1003**).

[0111] Based on determining that the dog is a puppy (Step **1002**—YES), the platform may perform one or more Steps of the method **700** of FIG. **7**. Alternatively, based on determining that the dog is not a puppy (Step **1002**—NO), the platform may determine whether a body condition of the dog is reported as being overweight or obese (Step **1004**).

[0112] Based on determining that the body condition of the dog is reported as being overweight or obese (Step **1004**—YES), the platform may perform one or more Steps of the method FIG. **8** of FIG. **8**. Alternatively, based on determining that the body condition of the dog is not reported as being overweight or obese (Step **1004**—NO), the platform may determine whether the dog is in a senior life stage (Step **1005**).

[0113] Based on determining that the dog is not in a senior life stage (Step **1005**—NO), the platform may perform one or more Steps of the method **900** of FIG. **9**. Alternatively, based on determining that the dog is in a senior life stage (Step **1005**—YES), the platform may determine whether the dog is a purebred (Step **1006**).

[0114] Based on determining that the dog is a purebred (Step **1006**—YES), the platform may determine whether there is a breed-specific senior formula in a master product list (Step **1007**). Based on determining that there is a breed-specific senior formula in a master product list (Step **1007**—YES), the platform may recommend the breed-specific senior formula (Step **1008**).

[0115] Alternatively, based on determining that the dog is not a purebred (Step **1006**—NO) or determining that there is not a breed-specific senior formula in the master product list (Step **1007**—NO), the platform may determine whether there are breed-size-specific senior formulas in the master product list (Step **1009**). Based on determining that there are breed-size-specific senior formulas in the master product list (Step **1009**—YES), the platform may recommend the breed-size-specific senior formulas (Step **1010**). Alternatively, based on determining that there are not breed-size-specific senior formulas in the master product list (Step **1009**—NO), the platform may recommend all-breed-size senior formulas (Step **1011**).

Exemplary Method for Providing Personalized Pet Product Recommendations

[0116] FIG. **11**A illustrates an exemplary process **1100** for providing one or more personalized pet product recommendations based on pet metadata to a user, according to one or more embodiments.

[0117] The method may include receiving, by one or more processors, pet metadata from a user (Step **1102**). The pet metadata may include at least one of: a pet name, a pet age, a pet breed, a pet life stage, a pet body condition, or a pet health condition. The pet breed may indicate one or more breeds that correspond to the pet. Additionally, the pet breed may indicate whether the pet is a purebred or a mixed breed pet. The pet life stage may indicate whether the pet is a puppy, adult, senior pet, and the like. The pet body condition may indicate whether the pet is underweight, overweight, obese, and the like. The pet health condition may indicate whether the pet has any medical conditions, such as diabetes, low blood pressure, and the like.

[0118] The method may further include processing, by the one or more processors, the pet metadata to determine one or more pet product recommendations (Step **1104**). The processing the pet metadata is based at least in part on a hierarchy of a plurality of factors. The hierarchy of the plurality of factors may include a pet life stage factor, a pet body condition factor, and/or a pet breed factor. The pet life stage factor may correspond to whether the pet is a puppy, an adult, or a senior. The pet body condition factor may correspond to whether the pet is underweight, overweight, or obese. The pet breed factor may correspond to whether the pet is a pure breed or a mixed breed. Such hierarchy of a plurality of factors may be further described in FIGS. **7-10**.

[0119] In some embodiments, the method may also include correlating, by the one or more processors, the processed pet metadata with a list of pet products that contain a minimum level of fiber, the correlating resulting in a list of the one or more pet product recommendations. For example, a list of pet products may be received from an external system. The method may include filtering such a list of pet products to determine which pet products meet a particular threshold regarding one or more ingredients. The processed pet metadata may then be correlated to the list of pet products to determine at least one pet product recommendation.

[0120] The method may further include displaying, by the one or more processors, the one or more pet product recommendations to the user (Step **1106**). An example embodiment of displaying the one or more pet product recommendations may be found in FIGS. **6A-6**C. The one or more pet product recommendations may include a link to facilitate a purchase of one or more pet products corresponding to the one or more pet product recommendations. Such link may be to an external service.

[0121] Although FIG. **11**A shows example blocks of exemplary method **1100**, in some implementations, the exemplary method **1100** may include additional blocks, fewer blocks, different blocks, or differently arranged blocks than those depicted in FIG. **11**A. Additionally, or alternatively, two or more of the blocks of the exemplary method **1100** may be performed in parallel.

[0122] FIG. **11**B illustrates an exemplary process **1108** for providing one or more personalized pet product recommendations based on a microbiome result and pet metadata to a user, according to one or more embodiments.

[0123] The method may include receiving, by one or more processors, pet metadata and a pet fecal sample from the user (Step **1110**). The pet metadata may include at least one of: a pet name, a pet age, a pet breed, a pet life stage, a pet body condition, or a pet health condition. The pet breed may indicate one or more breeds that correspond to the pet. Additionally, the pet breed may indicate whether the pet is a purebred or a mixed breed pet. The pet life stage may indicate whether the pet is a puppy, adult, senior pet, and the like. The pet body condition may indicate whether the pet is underweight, overweight, obese, and the like. The pet health condition may indicate whether the pet has any medical conditions, such as diabetes, low blood pressure, and the like. The pet fecal sample may include an image of the pet fecal sample and/or a physical pet fecal sample. The image of the pet fecal sample may be received from a user's device.

[0124] The method may further include analyzing, by the one or more processors, the pet fecal sample to determine a microbiome result (Step **1112**). The analyzing may be described in further detail regarding FIG. **5**. The microbiome result may include at least one recommendation for a pet product containing prebiotics or prebiotic-like fibers. For example, the at least one recommendation may include pet food that has a minimum amount of fiber and/or prebiotics.

[0125] In some embodiments, analyzing the pet fecal sample to determine a microbiome result may include detecting and quantitating bacterial taxa. For example, techniques for detecting and quantitating bacterial taxa may include, for example, polymerase chain reaction (PCR), quantitative PCR, 16S rDNA amplicon sequencing, shotgun sequencing, metagenome sequencing, Illumina sequencing, and nanopore sequencing. The bacterial taxa may be determined by sequencing the 16s rDNA sequence. Other methods may include shotgun sequencing to determine characteristic non-16SrDNA gene sequences or other metabolites and biomarkers for identification of the species.

[0126] In some embodiments, the bacterial taxa may be determined by sequencing the V4-V6 region, for example using Illumina sequencing. These methods may use the primers 319F:

[0127] CAAGCAGAAGACGGCATACGAGATGTGACTGGAGTTCAGACGTGTGCTCTTCCGA TCT (SEQ ID NO: 1) and 806R:

[0128] AATGATACGGCGACCACCGAGATCTACACTCTTTCCCTACACG ACGCTCTTCCGATCT (SEQ ID NO: 2).

[0129] The bacterial species may also be detected by other methods such as, for example, RNA sequencing, protein sequence homology or other biological marker indicative of the bacterial species.

[0130] Additionally, in some embodiments, the sequencing data can then be used to determine the presence or absence of different bacterial taxa in the sample. For example, the sequences may be clustered at about 98%, about 99% or 100% identity and abundant taxa (e.g., those representing more than 0.001 of the total sequences) may then be assessed for their relative proportions. Suitable techniques may include, for example, logistic regression, partial least squares discriminate analysis (PLSDA) or random forest analysis and other multivariate methods.

[0131] In some embodiments, if the pet fecal sample includes an image of the pet fecal sample, the analyzing may include comparing the pet fecal sample to stool chart pictures, in order to indicate whether there is good bowel performance. In some embodiments, if the pet fecal sample includes a physical pet fecal sample, the analyzing may include specifics about the bacteria included in the gut and/or feces of the pet.

[0132] The method may also include determining, by the one or more processors, whether the microbiome result recommends a pet product containing prebiotics or prebiotic-like fibers. For example, in some embodiments, the microbiome result may indicate that a pet product recommendation is not needed. This may be the case where the result has not met a particular threshold, or alternatively, where the result has surpassed a particular threshold. Additionally, in response to determining that the microbiome result recommends the pet product containing prebiotics or prebiotic-like fibers, determining, by the one or more processors, that a pet product recommendation is needed.

[0133] The method may further include processing, by the one or more processors, the pet metadata based on the microbiome result to determine one or more pet product recommendations (Step **1114**). The processing the pet metadata is based at least in part on a hierarchy of a plurality of factors. The hierarchy of the plurality of factors may include a pet life stage factor, a pet body condition factor, and/or a pet breed factor. The pet life stage factor may correspond to whether the pet is a puppy, an adult, or a senior. The pet body condition factor may correspond to whether the pet is underweight, overweight, or obese. The pet breed factor may correspond to whether the pet is a pure breed or a mixed breed. Such hierarchy of a plurality of factors may be further described in FIGS. **7-10**.

[0134] Additionally, the method may include correlating, by the one or more processors, the processed pet metadata with a list of pet products that contain a minimum level of fiber, the correlating resulting in a list of the one or more pet product recommendations. For example, a list of pet products may be received from an external system. The method may include filtering such a list of pet products to determine which pet products meet a particular threshold regarding one or more ingredients. The processed pet metadata may then be correlated to the list of pet products to determine at least one pet product recommendation.

[0135] The method may further include displaying, by the one or more processors, the one or more pet product recommendations to the user (Step **1116**). An example embodiment of displaying the one or more pet product recommendations may be found in FIGS. **6**A-**6**C. The one or more pet product recommendations may include a link to facilitate a purchase of one or more pet products corresponding to the one or more pet product recommendations. Such link may be to an external service.

[0136] Although FIG. **11**B shows example blocks of exemplary method **1108**, in some implementations, the exemplary method **1108** may include additional blocks, fewer blocks, different blocks, or differently arranged blocks than those depicted in FIG. **11**B. Additionally, or alternatively, two or more of the blocks of the exemplary method **1108** may be performed in parallel.

Exemplary Environment and Exemplary Device

[0137] FIG. **12** depicts an exemplary environment **1200** that may be utilized with techniques presented herein. One or more user device(s) **1205**, one or more external system(s) **1210**, and one or more server system(s) **1215** may communicate across a network **1201**. As will be discussed in further detail below, one or more server system(s) **1215** may communicate with one or more of the other components of the environment **1200** across network **1201**. The one or more user device(s) **1205** may be associated with a user.

[0138] In some embodiments, the components of the environment **1200** are associated with a common entity, e.g., a veterinarian, clinic, animal specialist, research center, or the like. In some embodiments, one or more of the components of the environment is associated with a different entity than another. The systems and devices of the environment **1200** may communicate in any arrangement.

[0139] The user device **1205** may be configured to enable the user to access and/or interact with other systems in the environment **1200**. For example, the user device **1205** may be a computer system such as, for example, a desktop computer, a mobile device, a tablet, etc. In some embodiments, the user device **1205** may include one or more electronic application(s), e.g., a program, plugin, browser extension, etc., installed on a memory of the user device **1205**.

[0140] The user device **1205** may include a display/user interface (UI) **1205**A, a processor **1205**B, a memory **1205**C, and/or a network interface **1205**D. The user device **1205** may execute, by the processor **1205**B, an operating system (O/S) and at least one electronic application (each stored in memory **1205**C). The electronic application may be a desktop program, a browser program, a web client, or a mobile application program (which may also be a browser program in a mobile O/S), an applicant specific program, system control software, system monitoring software, software development tools, or the like. For example, environment **1200** may extend information on a web client that may be accessed through a web browser. In some embodiments, the electronic application(s) may be associated with one or more of the other components in the environment **1200**. The application may manage the memory **1205**C, such as a database, to transmit streaming data to network **1201**. The display/UI **1205**A may be a touch screen or a display with other input systems (e.g., mouse, keyboard, etc.) so that the user(s) may interact with the application and/or the O/S. The network interface **1205**D may be a TCP/IP network interface for, e.g., Ethernet or wireless communications with the network **1201**. The processor **1205**B, while executing the application, may generate data and/or receive user inputs from the display/UI **1205**A and/or receive/transmit messages to the server system **1215**, and may further perform one or more operations prior to providing an output to the network **1201**.

[0141] External systems **1210** may be, for example, one or more third party and/or auxiliary systems that integrate and/or communicate with the server system **1215** in performing various product recommendation and/or gut microbiome testing tasks. External systems **1210** may be in communication with other device(s) or system(s) in the environment **1200** over the one or more networks **1201**. For example, external systems **1210** may communicate with the server system **1215** via API (application programming interface) access over the one or more networks **1201**, and also communicate with the user device(s) **1205** via web browser access over the one or more networks **1201**.

[0142] In various embodiments, the network **1201** may be a wide area network ("WAN"), a local area network ("LAN"), a personal area network ("PAN"), or the like. In some embodiments, network **1201** includes the Internet, and information and data provided between various systems occurs online. "Online" may mean connecting to or accessing source data or information from a location remote from other devices or networks coupled to the Internet. Alternatively, "online" may refer to connecting or accessing a network (wired or wireless) via a mobile communications network or device. The Internet is a worldwide system of computer networks—a network of networks in which a party at one computer or other device connected to the network can obtain information from any other computer and communicate with parties of other computers or devices. The most widely used part of the Internet is the World Wide Web (often-abbreviated "WWW" or called "the Web"). A "website page" generally encompasses a location, data store, or the like that is, for example, hosted and/or operated by a computer system so as to be accessible online, and that may include data configured to cause a program such as a web browser to perform operations such as send, receive, or process data, generate a visual display and/or an interactive interface, or the like.

[0143] The server system **1215** may include an electronic data system, e.g., a computer-readable memory such as a hard drive, flash drive, disk, etc. In some embodiments, the server system **1215** includes and/or interacts with an application programming interface for exchanging data to other systems, e.g., one or more of the other components of the environment.

[0144] The server system **1215** may include a database **1215**A and at least one server **1215**B. The server system **1215** may be a computer, system of computers (e.g., rack server(s)), and/or or a cloud service computer system. The server system may store or have access to database **1215**A (e.g., hosted on a third party server or in memory **1215**E). The server(s) may include a display/UI **1215**C, a processor **1215**D, a memory **1215**E, and/or a network interface **1215**F. The display/UI **1215**C may be a touch screen or a display with other input systems (e.g., mouse, keyboard, etc.) for an operator of the server **1215**B to control the functions of the server **1215**B. The server system **1215** may execute, by the processor **1215**D, an operating system (O/S) and at least one instance of a servlet program (each stored in memory **1215**E).

[0145] Although depicted as separate components in FIG. **12**, it should be understood that a component or portion of a component in the environment **1200** may, in some embodiments, be integrated with or incorporated into one or more other components. For example, a portion of the display **1215**C may be integrated into the user device **1205** or the like. In some embodiments, operations or aspects of one or more of the components discussed above may be distributed amongst one or more other components. Any suitable arrangement and/or integration of the various systems and devices of the environment **1200** may be used.

[0146] In the previous and following methods, various acts may be described as performed or executed by a component from FIG. **12**, such as the server system **1215**, the user device **1205**, or components thereof. However, it should be understood that in various embodiments, various components of the environment **1200** discussed above may execute instructions or perform acts including the acts discussed above. An act performed by a device may be considered to be performed by a processor, actuator, or the like associated with that device. Further, it should be understood that in various embodiments, various steps may be added, omitted, and/or rearranged in any suitable manner.

[0147] In general, any process or operation discussed in this disclosure that is understood to be computer-implementable, such as the processes illustrated in FIGS. **1-11**B, may be performed by one or more processors of a computer system, such any of the systems or devices in the environment **1200** of FIG. **12**, as described above. A process or process step performed by one or more processors may also be referred to as an operation. The one or more processors may be configured to perform such processes by having access to instructions (e.g., software or computer-readable code) that, when executed by the one or more processors, cause the one or more processors to perform the processes. The instructions may be stored in a memory of the computer system. A processor may be a central processing unit (CPU), a graphics processing unit (GPU), or any suitable types of processing unit.

[0148] A computer system, such as a system or device implementing a process or operation in the examples above, may include one or more computing devices, such as one or more of the systems or devices in FIG. **12**. One or more processors of a computer system may be included in a single computing device or distributed among a plurality of computing devices. A memory of the computer system may include the respective memory of each computing device of the plurality of computing devices.

[0149] FIG. **13** is a simplified functional block diagram of a computer **1300** that may be configured as a device for executing the environments and/or the methods of FIGS. **1-11**B, according to exemplary embodiments of the present disclosure. For example, device **1300** may include a central processing unit (CPU) **1320**. CPU **1320** may be any type of processor device including, for example, any type of special purpose or a general-purpose microprocessor device. As will be appreciated by persons skilled in the relevant art, CPU **1320** also may be a single processor in a multi-core/multiprocessor system, such system operating alone, or in a cluster of computing devices operating in a cluster or server farm. CPU **1320** may be connected to a data communication infrastructure **1310**, for example, a bus, message queue, network, or multi-core message-passing scheme.

[0150] Device **1300** also may include a main memory **1340**, for example, random access memory (RAM), and also may include a secondary memory **1330**. Secondary memory **1330**, e.g., a read-only memory (ROM), may be, for example, a hard disk drive or a removable storage drive. Such a removable storage drive may comprise, for example, a floppy disk drive, a magnetic tape drive, an optical disk drive, a flash memory, or the like. The removable storage drive in this example reads from and/or writes to a removable storage unit in a well-known manner. The removable storage unit may comprise a floppy disk, magnetic tape, optical disk, etc., which is read by and written to by the removable storage drive. As will be appreciated by persons skilled in the relevant art, such a removable storage unit generally includes a computer usable storage medium having stored therein computer software and/or data.

[0151] In alternative implementations, secondary memory **1330** may include other similar means for allowing computer programs or other instructions to be loaded into device **1300**. Examples of such means may include a program cartridge and cartridge interface (such as that found in video game devices), a removable memory chip (such as an EPROM, or PROM) and associated socket, and other removable storage units and interfaces, which allow software and data to be transferred from a removable storage unit to device **1300**.

[0152] Device **1300** also may include a communications interface ("COM") **1360**. Communications interface **1360** allows software and data to be transferred between device **1300** and external devices. Communications interface **1360** may include a modem, a network interface (such as an Ethernet card), a communications port, a PCMCIA slot and card, or the like. Software and data transferred via communications interface **1360** may be in the form of signals, which may be electronic, electromagnetic, optical, or other signals capable of being received by communications interface **1360**. These signals may be provided to communications interface **1360** via a communications path of device **1300**, which may be implemented using, for example, wire or cable, fiber optics, a phone line, a cellular phone link, an RF link or other communications channels.

[0153] The hardware elements, operating systems and programming languages of such equipment are conventional in nature, and it is presumed that those skilled in the art are adequately familiar therewith. Device **1300** also may include input and output ports **1350** to connect with input and output devices such as keyboards, mice, touchscreens, monitors, displays, etc. Of course, the various server functions may be implemented in a distributed fashion on a number of similar platforms, to distribute the processing load. Alternatively, the servers may be implemented by appropriate programming of one computer hardware platform.

[0154] Program aspects of the technology may be thought of as "products" or "articles of manufacture" typically in the form of executable code and/or associated data that is carried on or embodied in a type of machine-readable medium. "Storage" type media include any or all of the tangible memory of the computers, processors or the like, or associated modules thereof, such as various semiconductor memories, tape drives, disk drives and the like, which may provide non-transitory storage at any time for the software programming. All or portions of the software may at times be communicated through the Internet or various other telecommunication networks. Such communications, for example, may enable loading of the software from one computer or processor into another, for example, from a management server or host computer of the mobile communication network into the computer platform of a server and/or from a server to the mobile device. Thus, another type of media that may bear the software elements includes optical, electrical and electromagnetic waves, such as used across physical interfaces between local devices, through wired and optical landline networks and over various airlinks. The physical elements that carry such waves, such as wired or wireless links, optical links, or the like, also may be considered as media bearing the software. As used herein, unless restricted to non-transitory, tangible "storage" media, terms such as computer or machine "readable medium" refer to any medium that participates in providing instructions to a processor for execution.

[0155] Reference to any particular activity is provided in this disclosure only for convenience and not intended to limit the disclosure. A person of ordinary skill in the art would recognize that the concepts underlying the disclosed devices and methods may be utilized in any suitable activity.

The disclosure may be understood with reference to the following description and the appended drawings, wherein like elements are referred to with the same reference numerals.

[0156] The terminology used above may be interpreted in its broadest reasonable manner, even though it is being used in conjunction with a detailed description of certain specific examples of the present disclosure. Indeed, certain terms may even be emphasized above; however, any terminology intended to be interpreted in any restricted manner will be overtly and specifically defined as such in this Detailed Description section. Both the general description and the detailed description are exemplary and explanatory only and are not restrictive of the features, as claimed.

[0157] In this disclosure, the term “based on” means “based at least in part on.” The singular forms “a,” “an,” and “the” include plural referents unless the context dictates otherwise. The term “exemplary” is used in the sense of “example” rather than “ideal.” The terms “comprises,” “comprising,” “includes,” “including,” or other variations thereof, are intended to cover a non-exclusive inclusion such that a process, method, or product that comprises a list of elements does not necessarily include only those elements, but may include other elements not expressly listed or inherent to such a process, method, article, or apparatus. The term “or” is used disjunctively, such that “at least one of A or B” includes, (A), (B), (A and A), (A and B), etc. Relative terms, such as, “substantially” and “generally,” are used to indicate a possible variation of ±10% of a stated or understood value.

[0158] As used herein, a term such as “user” or the like generally encompasses a pet parent and/or pet parents. A term such as “pet” or the like generally encompasses a user’s pet, where the term may encompass multiple pets. Also, the term “pet” refers to any type of animal, including domesticated animals. A term such as “provider” or the like generally encompasses a pet care business

[0159] It should be appreciated that in the above description of exemplary embodiments of the invention, various features of the invention are sometimes grouped together in a single embodiment, figure, or description thereof for the purpose of streamlining the disclosure and aiding in the understanding of one or more of the various inventive aspects. This method of disclosure, however, is not to be interpreted as reflecting an intention that the claimed invention requires more features than are expressly recited in each claim. Rather, as the following claims reflect, inventive aspects lie in less than all features of a single foregoing disclosed embodiment. Thus, the claims following the Detailed Description are hereby expressly incorporated into this Detailed Description, with each claim standing on its own as a separate embodiment of this invention.

[0160] Furthermore, while some embodiments described herein include some but not other features included in other embodiments, combinations of features of different embodiments are meant to be within the scope of the invention, and form different embodiments, as would be understood by those skilled in the art. For example, in the following claims, any of the claimed embodiments can be used in any combination.

[0161] Thus, while certain embodiments have been described, those skilled in the art will recognize that other and further modifications may be made thereto without departing from the spirit of the invention, and it is intended to claim all such changes and modifications as falling within the scope of the invention. For example, functionality may be added or deleted from the block diagrams and operations may be interchanged among functional blocks. Steps may be added or deleted to methods described within the scope of the present invention.

[0162] The above disclosed subject matter is to be considered illustrative, and not restrictive, and the appended claims are intended to cover all such modifications, enhancements, and other implementations, which fall within the true spirit and scope of the present disclosure. Thus, to the maximum extent allowed by law, the scope of the present disclosure is to be determined by the broadest permissible interpretation of the following claims and their equivalents, and shall not be restricted or limited by the foregoing detailed description. While various implementations of the disclosure have been described, it will be apparent to those of ordinary skill in the art that many more implementations are possible within the scope of the disclosure. Accordingly, the disclosure is not to be restricted except in light of the attached claims and their equivalents.

What is claimed is:

1. A computer-implemented method for providing one or more personalized pet product recommendations based on pet metadata to a user, the method including:

receiving, by one or more processors, pet metadata from a user;

processing, by the one or more processors, the pet metadata to determine one or more pet product recommendations; and

displaying, by the one or more processors, the one or more pet product recommendations to the user.

2. The computer-implemented method of claim **1**, wherein the pet metadata includes at least one of: a pet name, a pet age, a pet breed, a pet life stage, a pet body condition, or a pet health condition.

3. The computer-implemented method of claim **1**, wherein the processing the pet metadata is based at least in part on a hierarchy of a plurality of factors.

4. The computer-implemented method of claim **3**, wherein the hierarchy of the plurality of factors includes a pet life stage factor, a pet body condition factor, and a pet breed factor.

5. The computer-implemented method of claim **1**, the method further comprising:

correlating, by the one or more processors, the processed pet metadata with a list of pet products that contain a minimum level of fiber, the correlating resulting in a list of the one or more pet product recommendations.

6. The computer-implemented method of claim **1**, wherein the one or more pet product recommendations include a link to facilitate a purchase of one or more pet products corresponding to the one or more pet product recommendations.

7. The computer-implemented method of claim **1**, further comprising:

obtaining, by the one or more processors, a microbiome result associated with a pet fecal sample; and

processing, by the one or more processors, the pet metadata further based on the microbiome result to determine one or more pet product recommendations.

8. The computer-implemented method of claim **7**, wherein, based on the microbiome result, the one or more pet product recommendations include at least one recommendation for a pet product containing prebiotics or prebiotic-like fibers.

9. The computer-implemented method of claim **7**, further comprising:

determining, by the one or more processors, a diet containing prebiotics or prebiotic-like fibers is recommended based on the microbiome result; and

in response to the determination, determining, by the one or more processors, the one or more pet product recommendations based on the processing of the pet metadata.

10. The computer-implemented method of claim **7**, wherein the pet fecal sample includes one or more of a physical pet fecal sample or an image of the pet fecal sample.

11. A computer system for providing one or more personalized pet product recommendations based on pet metadata to a user, the computer system comprising:

at least one memory storing instructions; and

at least one processor configured to execute the instructions to perform operations comprising:

receiving pet metadata from a user;

processing the pet metadata to determine one or more pet product recommendations; and

displaying the one or more pet product recommendations to the user.

12. The computer system of claim **11**, wherein the pet metadata includes at least one of: a pet name, a pet age, a pet breed, a pet life stage, a pet body condition, or a pet health condition.

13. The computer system of claim **11**, the operations further comprising:

correlating the processed pet metadata with a list of pet products that contain a minimum level of fiber, the correlating resulting in a list of the one or more pet product recommendations.

14. The computer system of claim **11**, wherein the processing the pet metadata is based at least in part on a hierarchy of a plurality of factors.

15. The computer system of claim **14**, wherein the hierarchy of the plurality of factors includes a pet life stage factor, a pet body condition factor, and a pet breed factor.

16. The computer system of claim **11**, wherein the one or more pet product recommendations include a link to facilitate a purchase of one or more pet products corresponding to the one or more pet product recommendations.

17. A non-transitory computer-readable medium containing instructions that, when executed by a processor, cause the processor to perform operations for providing one or more personalized pet product recommendations based on pet metadata to a user, the operations comprising:

receiving pet metadata from a user;

processing the pet metadata to determine one or more pet product recommendations; and

displaying the one or more pet product recommendations to the user.

18. The non-transitory computer-readable medium of claim **17**, wherein the one or more pet product recommendations include a link to facilitate a purchase of one or more pet products corresponding to the one or more pet product recommendations.

19. The non-transitory computer-readable medium of claim **17**, the operations further comprising:

correlating the processed pet metadata with a list of pet products that contain a minimum level of fiber, the correlating resulting in a list of the one or more pet product recommendations.

20. The non-transitory computer-readable medium of claim **19**, wherein the pet metadata includes at least one of: a pet name, a pet age, a pet breed, a pet life stage, a pet body condition, or a pet health condition.

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